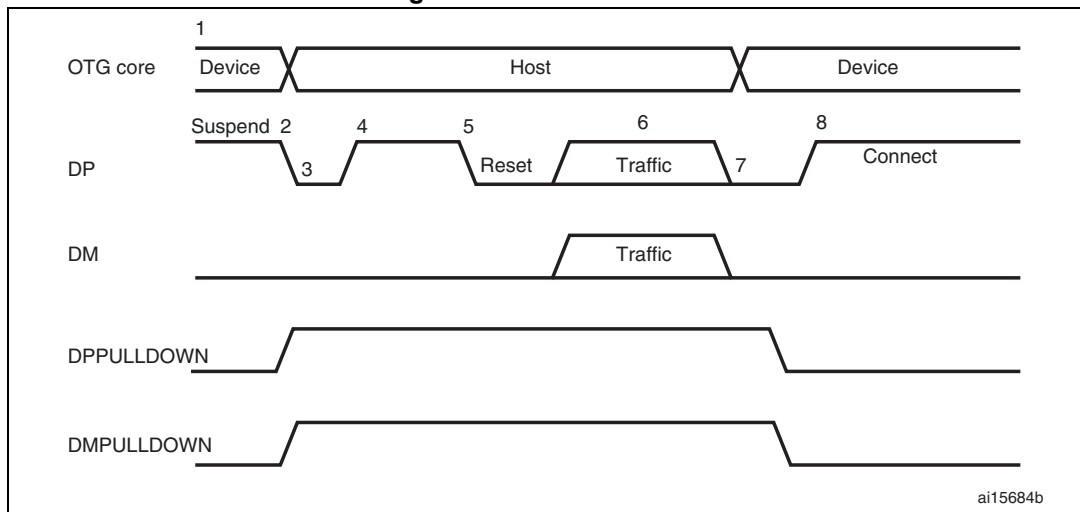


Figure 433. B-device HNP



1. DPPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DP line inside the PHY.  
 DMPULLDOWN = signal from core to PHY to enable/disable the pull-down on the DM line inside the PHY.

- The A-device sends the SetFeature b\_hnp\_enable descriptor to enable HNP support. The OTG\_HS controller's ACK response indicates that it supports HNP. The application must set the Device HNP enable bit in the OTG Control and status register to indicate HNP support.

The application sets the HNP request bit in the OTG Control and status register to indicate to the OTG\_HS controller to initiate HNP.

- When it has finished using the bus, the A-device suspends by writing the Port suspend bit in the host port control and status register.

The OTG\_HS controller sets the Early suspend bit in the Core interrupt register after 3 ms of bus idleness. Following this, the OTG\_HS controller sets the USB suspend bit in the Core interrupt register.

The OTG\_HS controller disconnects and the A-device detects SE0 on the bus, indicating HNP. The OTG\_HS controller asserts the DP pull down and DM pull down in the PHY to indicate its assumption of the host role.

The A-device responds by activating its OTG\_HS\_DP pull-up resistor within 3 ms of detecting SE0. The OTG\_HS controller detects this as a connect.

The OTG\_HS controller sets the host negotiation success status change interrupt in the OTG Interrupt status register, indicating the HNP status. The application must read the host negotiation success bit in the OTG Control and status register to determine

host negotiation success. The application must read the current Mode bit in the Core interrupt register (OTG\_HS\_GINTSTS) to determine host mode operation.

3. The application sets the reset bit (PRST in OTG\_HS\_HPRT) and the OTG\_HS controller issues a USB reset and enumerates the A-device for data traffic.
4. The OTG\_HS controller continues the host role of initiating traffic, and when done, suspends the bus by writing the Port suspend bit in the host port control and status register.
5. In Negotiated mode, when the A-device detects a suspend, it disconnects and switches back to the host role. The OTG\_HS controller deasserts the DP pull down and DM pull down in the PHY to indicate the assumption of the device role.
6. The application must read the current mode bit in the Core interrupt (OTG\_HS\_GINTSTS) register to determine the host mode operation.
7. The OTG\_HS controller connects, completing the HNP process.

## 36 Flexible static memory controller (FSMC)

This section applies to the whole STM32F40x/41x family only.

### 36.1 FSMC main features

The FSMC block is able to interface with synchronous and asynchronous memories and 16-bit PC memory cards. Its main purpose is to:

- Translate the AHB transactions into the appropriate external device protocol
- Meet the access timing requirements of the external devices

All external memories share the addresses, data and control signals with the controller. Each external device is accessed by means of a unique chip select. The FSMC performs only one access at a time to an external device.

The FSMC has the following main features:

- Interfaces with static memory-mapped devices including:
  - Static random access memory (SRAM)
  - NOR/OneNAND flash memory
  - PSRAM (4 memory banks)
- Two banks of NAND Flash with ECC hardware that checks up to 8 Kbytes of data
- 16-bit PC Card compatible devices
- Supports burst mode access to synchronous devices (NOR flash and PSRAM)
- 8- or 16-bit wide databus
- Independent chip select control for each memory bank
- Independent configuration for each memory bank
- Programmable timings to support a wide range of devices, in particular:
  - Programmable wait states (up to 15)
  - Programmable bus turnaround cycles (up to 15)
  - Programmable output enable and write enable delays (up to 15)
  - Independent read and write timings and protocol, so as to support the widest variety of memories and timings
- Write enable and byte lane select outputs for use with PSRAM and SRAM devices
- Translation of 32-bit wide AHB transactions into consecutive 16-bit or 8-bit accesses to external 16-bit or 8-bit devices
- A Write FIFO, 2-word long (16-word long for STM32F42x and STM32F43x), each word is 32 bits wide, only stores data and not the address. Therefore, this FIFO only buffers AHB write burst transactions. This makes it possible to write to slow memories and free the AHB quickly for other operations. Only one burst at a time is buffered: if a new AHB burst or single transaction occurs while an operation is in progress, the FIFO is drained. The FSMC inserts wait states until the current memory access is complete.
- External asynchronous wait control

The FSMC registers that define the external device type and associated characteristics are usually set at boot time and do not change until the next reset or power-up. However, it is possible to change the settings at any time.

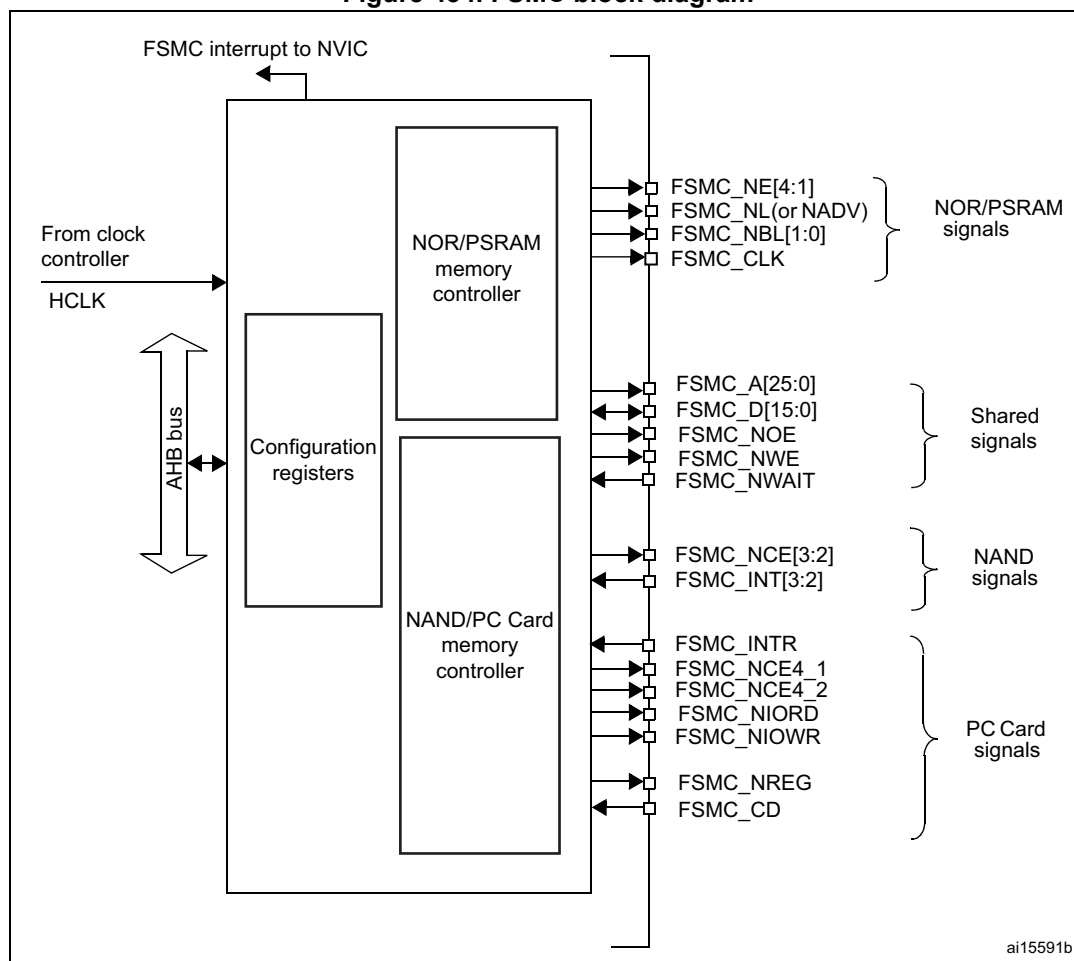
## 36.2 Block diagram

The FSMC consists of four main blocks:

- The AHB interface (including the FSMC configuration registers)
- The NOR flash/PSRAM controller
- The NAND Flash/PC Card controller
- The external device interface

The block diagram is shown in [Figure 434](#).

**Figure 434. FSMC block diagram**



## 36.3 AHB interface

The AHB slave interface enables internal CPUs and other bus master peripherals to access the external static memories.

AHB transactions are translated into the external device protocol. In particular, if the selected external memory is 16 or 8 bits wide, 32-bit wide transactions on the AHB are split into consecutive 16- or 8-bit accesses. The FSMC Chip Select (FSMC\_NEx) does not

toggle between consecutive accesses except when performing accesses in mode D with the extended mode enabled.

The FSMC generates an AHB error in the following conditions:

- When reading or writing to an FSMC bank which is not enabled
- When reading or writing to the NOR flash bank while the FACCEN bit is reset in the FSMC\_BCRx register.
- When reading or writing to the PC Card banks while the input pin FSMC\_CD (Card Presence Detection) is low.

The effect of this AHB error depends on the AHB master which has attempted the R/W access:

- If it is the Cortex<sup>®</sup>-M4 with FPU CPU, a hard fault interrupt is generated
- If is a DMA, a DMA transfer error is generated and the corresponding DMA channel is automatically disabled.

The AHB clock (HCLK) is the reference clock for the FSMC.

### 36.3.1 Supported memories and transactions

#### General transaction rules

The requested AHB transaction data size can be 8-, 16- or 32-bit wide whereas the accessed external device has a fixed data width. This may lead to inconsistent transfers.

Therefore, some simple transaction rules must be followed:

- AHB transaction size and memory data size are equal  
There is no issue in this case.
- AHB transaction size is greater than the memory size  
In this case, the FSMC splits the AHB transaction into smaller consecutive memory accesses in order to meet the external data width.
- AHB transaction size is smaller than the memory size  
Asynchronous transfers may or not be consistent depending on the type of external device.
  - Asynchronous accesses to devices that have the byte select feature (SRAM, ROM, PSRAM).
    - a) FSMC allows write transactions accessing the right data through its byte lanes NBL[1:0]
    - b) Read transactions are allowed. All memory bytes are read and the useless ones are discarded. The NBL[1:0] are kept low during read transactions.
  - Asynchronous accesses to devices that do not have the byte select feature (NOR and NAND Flash 16-bit).  
This situation occurs when a byte access is requested to a 16-bit wide flash memory. Clearly, the device cannot be accessed in byte mode (only 16-bit words can be read from/written to the flash memory) therefore:
    - a) Write transactions are not allowed
    - b) Read transactions are allowed. All memory bytes are read and the useless ones are discarded. The NBL[1:0] are set to 0 during read transactions.

Configuration registers

The FSMC can be configured using a register set. See [Section 36.5.6](#), for a detailed description of the NOR flash/PSRAM control registers. See [Section 36.6.8](#), for a detailed description of the NAND Flash/PC Card registers.

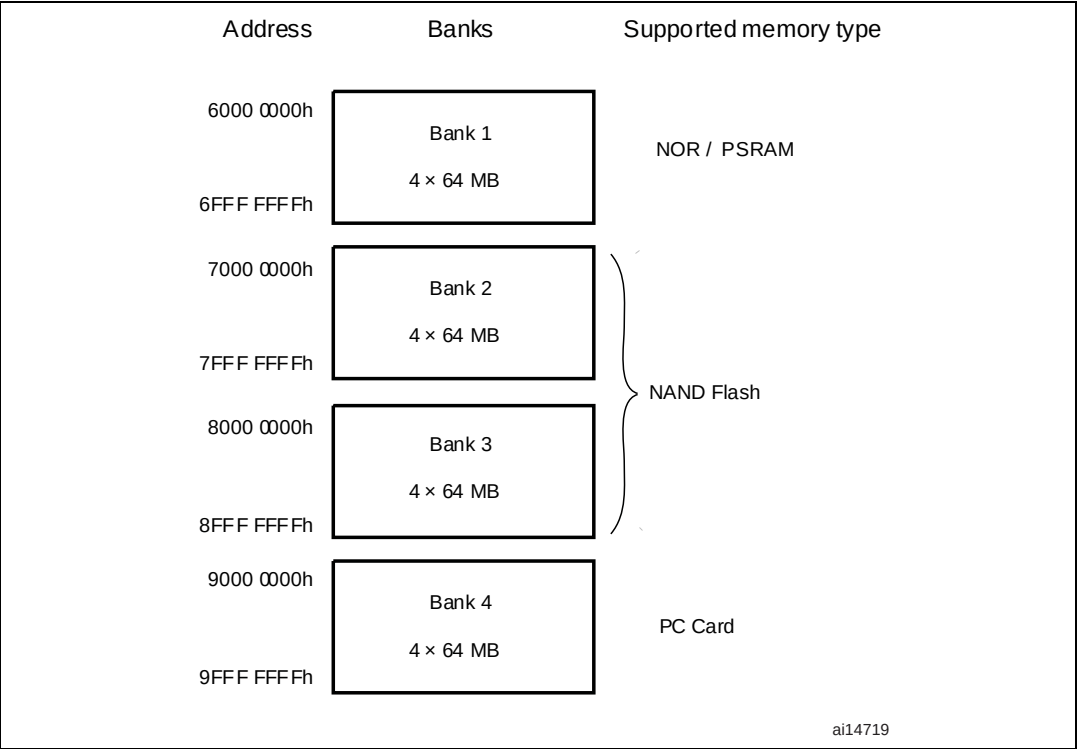
36.4 External device address mapping

From the FSMC point of view, the external memory is divided into 4 fixed-size banks of 256 Mbytes each (Refer to [Figure 435](#)):

- Bank 1 used to address up to four NOR flash or PSRAM memory devices. This bank is split into 4 NOR/PSRAM subbanks with four dedicated chip selects, as follows:
  - Bank 1 - NOR/PSRAM 1
  - Bank 1 - NOR/PSRAM 2
  - Bank 1 - NOR/PSRAM 3
  - Bank 1 - NOR/PSRAM 4
- Banks 2 and 3 used to address NAND Flash devices (1 device per bank)
- Bank 4 used to address a PC Card device

For each bank the type of memory to be used is user-defined in the Configuration register.

Figure 435. FSMC memory banks



36.4.1 NOR/PSRAM address mapping

HADDR[27:26] bits are used to select one of the four memory banks as shown in [Table 217](#).

**Table 217. NOR/PSRAM bank selection**

| HADDR[27:26] <sup>(1)</sup> | Selected bank        |
|-----------------------------|----------------------|
| 00                          | Bank 1 - NOR/PSRAM 1 |
| 01                          | Bank 1 - NOR/PSRAM 2 |
| 10                          | Bank 1 - NOR/PSRAM 3 |
| 11                          | Bank 1 - NOR/PSRAM 4 |

1. HADDR are internal AHB address lines that are translated to external memory.

HADDR[25:0] contain the external memory address. Since HADDR is a byte address whereas the memory is addressed in words, the address actually issued to the memory varies according to the memory data width, as shown in the following table.

**Table 218. External memory address**

| Memory width <sup>(1)</sup> | Data address issued to the memory | Maximum memory capacity (bits) |
|-----------------------------|-----------------------------------|--------------------------------|
| 8-bit                       | HADDR[25:0]                       | 64 Mbyte x 8 = 512 Mbit        |
| 16-bit                      | HADDR[25:1] >> 1                  | 64 Mbyte/2 x 16 = 512 Mbit     |

1. In case of a 16-bit external memory width, the FSMC internally uses HADDR[25:1] to generate the address for external memory FSMC\_A[24:0].  
Whatever the external memory width (16-bit or 8-bit), FSMC\_A[0] should be connected to external memory address A[0].

### Wrap support for NOR flash/PSRAM

Wrap burst mode for synchronous memories is not supported. The memories must be configured in linear burst mode of undefined length.

## 36.4.2 NAND/PC Card address mapping

In this case, three banks are available, each of them divided into memory spaces as indicated in [Table 219](#).

**Table 219. Memory mapping and timing registers**

| Start address | End address | FSMC Bank           | Memory space | Timing register   |
|---------------|-------------|---------------------|--------------|-------------------|
| 0x9C00 0000   | 0x9FFF FFFF | Bank 4 - PC card    | I/O          | FSMC_PIO4 (0xB0)  |
| 0x9800 0000   | 0x9BFF FFFF |                     | Attribute    | FSMC_PATT4 (0xAC) |
| 0x9000 0000   | 0x93FF FFFF |                     | Common       | FSMC_PMEM4 (0xA8) |
| 0x8800 0000   | 0x8BFF FFFF | Bank 3 - NAND Flash | Attribute    | FSMC_PATT3 (0x8C) |
| 0x8000 0000   | 0x83FF FFFF |                     | Common       | FSMC_PMEM3 (0x88) |
| 0x7800 0000   | 0x7BFF FFFF | Bank 2- NAND Flash  | Attribute    | FSMC_PATT2 (0x6C) |
| 0x7000 0000   | 0x73FF FFFF |                     | Common       | FSMC_PMEM2 (0x68) |

For NAND Flash memory, the common and attribute memory spaces are subdivided into three sections (see in [Table 220](#) below) located in the lower 256 Kbytes:

- Data section (first 64 Kbytes in the common/attribute memory space)
- Command section (second 64 Kbytes in the common / attribute memory space)
- Address section (next 128 Kbytes in the common / attribute memory space)

**Table 220. NAND bank selections**

| Section name    | HADDR[17:16] | Address range     |
|-----------------|--------------|-------------------|
| Address section | 1X           | 0x020000-0x03FFFF |
| Command section | 01           | 0x010000-0x01FFFF |
| Data section    | 00           | 0x000000-0x0FFFFF |

The application software uses the 3 sections to access the NAND Flash memory:

- **To send a command to NAND Flash memory:** the software must write the command value to any memory location in the command section.
- **To specify the NAND Flash address that must be read or written:** the software must write the address value to any memory location in the address section. Since an address can be 4 or 5 bytes long (depending on the actual memory size), several consecutive writes to the address section are needed to specify the full address.
- **To read or write data:** the software reads or writes the data value from or to any memory location in the data section.

Since the NAND Flash memory automatically increments addresses, there is no need to increment the address of the data section to access consecutive memory locations.

## 36.5 NOR flash/PSRAM controller

The FSMC generates the appropriate signal timings to drive the following types of memories:

- Asynchronous SRAM and ROM
  - 8-bit
  - 16-bit
  - 32-bit
- PSRAM (Cellular RAM)
  - Asynchronous mode
  - Burst mode for synchronous accesses
  - Multiplexed or nonmultiplexed
- NOR flash
  - Asynchronous mode
  - Burst mode for synchronous accesses
  - Multiplexed or nonmultiplexed

The FSMC outputs a unique chip select signal NE[4:1] per bank. All the other signals (addresses, data and control) are shared.

For synchronous accesses, the FSMC issues the clock (CLK) to the selected external device only during the read/write transactions. This clock is a submultiple of the HCLK clock. The size of each bank is fixed and equal to 64 Mbytes.

Each bank is configured by means of dedicated registers (see [Section 36.5.6](#)).

The programmable memory parameters include access timings (see [Table 221](#)) and support for wait management (for PSRAM and NOR flash accessed in burst mode).

**Table 221. Programmable NOR/PSRAM access parameters**

| Parameter          | Function                                                                         | Access mode                             | Unit                     | Min. | Max. |
|--------------------|----------------------------------------------------------------------------------|-----------------------------------------|--------------------------|------|------|
| Address setup      | Duration of the address setup phase                                              | Asynchronous                            | AHB clock cycle (HCLK)   | 0    | 15   |
| Address hold       | Duration of the address hold phase                                               | Asynchronous, muxed I/Os                | AHB clock cycle (HCLK)   | 1    | 15   |
| Data setup         | Duration of the data setup phase                                                 | Asynchronous                            | AHB clock cycle (HCLK)   | 1    | 256  |
| Bus turn           | Duration of the bus turnaround phase                                             | Asynchronous and synchronous read/write | AHB clock cycle (HCLK)   | 0    | 15   |
| Clock divide ratio | Number of AHB clock cycles (HCLK) to build one memory clock cycle (CLK)          | Synchronous                             | AHB clock cycle (HCLK)   | 2    | 16   |
| Data latency       | Number of clock cycles to issue to the memory before the first data of the burst | Synchronous                             | Memory clock cycle (CLK) | 2    | 17   |

### 36.5.1 External memory interface signals

[Table 222](#), [Table 223](#) and [Table 224](#) list the signals that are typically used to interface NOR flash, SRAM and PSRAM.

*Note:* Prefix “N”. specifies the associated signal as active low.

#### NOR flash, nonmultiplexed I/Os

**Table 222. Nonmultiplexed I/O NOR flash**

| FSMC signal name | I/O | Function                                                                            |
|------------------|-----|-------------------------------------------------------------------------------------|
| CLK              | O   | Clock (for synchronous access)                                                      |
| A[25:0]          | O   | Address bus                                                                         |
| D[15:0]          | I/O | Bidirectional data bus                                                              |
| NE[x]            | O   | Chip select, x = 1..4                                                               |
| NOE              | O   | Output enable                                                                       |
| NWE              | O   | Write enable                                                                        |
| NL(=NADV)        | O   | Latch enable (this signal is called address valid, NADV, by some NOR flash devices) |
| NWAIT            | I   | NOR flash wait input signal to the FSMC                                             |

NOR flash memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

### NOR flash, multiplexed I/Os

**Table 223. Multiplexed I/O NOR flash**

| FSMC signal name | I/O | Function                                                                            |
|------------------|-----|-------------------------------------------------------------------------------------|
| CLK              | O   | Clock (for synchronous access)                                                      |
| A[25:16]         | O   | Address bus                                                                         |
| AD[15:0]         | I/O | 16-bit multiplexed, bidirectional address/data bus                                  |
| NE[x]            | O   | Chip select, x = 1..4                                                               |
| NOE              | O   | Output enable                                                                       |
| NWE              | O   | Write enable                                                                        |
| NL(=NADV)        | O   | Latch enable (this signal is called address valid, NADV, by some NOR flash devices) |
| NWAIT            | I   | NOR flash wait input signal to the FSMC                                             |

NOR-flash memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

### PSRAM/SRAM, nonmultiplexed I/Os

**Table 224. Nonmultiplexed I/Os PSRAM/SRAM**

| FSMC signal name | I/O | Function                                                             |
|------------------|-----|----------------------------------------------------------------------|
| CLK              | O   | Clock (only for PSRAM synchronous access)                            |
| A[25:0]          | O   | Address bus                                                          |
| D[15:0]          | I/O | Data bidirectional bus                                               |
| NE[x]            | O   | Chip select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CRAM)) |
| NOE              | O   | Output enable                                                        |
| NWE              | O   | Write enable                                                         |
| NL(= NADV)       | O   | Address valid only for PSRAM input (memory signal name: NADV)        |
| NWAIT            | I   | PSRAM wait input signal to the FSMC                                  |
| NBL[1]           | O   | Upper byte enable (memory signal name: NUB)                          |
| NBL[0]           | O   | Lowest byte enable (memory signal name: NLB)                         |

PSRAM memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

PSRAM, multiplexed I/Os

Table 225. Multiplexed I/O PSRAM

| FSMC signal name | I/O | Function                                                             |
|------------------|-----|----------------------------------------------------------------------|
| CLK              | O   | Clock (for synchronous access)                                       |
| A[25:16]         | O   | Address bus                                                          |
| AD[15:0]         | I/O | 16-bit multiplexed, bidirectional address/data bus                   |
| NE[x]            | O   | Chip select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CRAM)) |
| NOE              | O   | Output enable                                                        |
| NWE              | O   | Write enable                                                         |
| NL(= NADV)       | O   | Address valid PSRAM input (memory signal name: NADV)                 |
| NWAIT            | I   | PSRAM wait input signal to the FSMC                                  |
| NBL[1]           | O   | Upper byte enable (memory signal name: NUB)                          |
| NBL[0]           | O   | Lower byte enable (memory signal name: NLB)                          |

PSRAM memories are addressed in 16-bit words. The maximum capacity is 512 Mbit (26 address lines).

### 36.5.2 Supported memories and transactions

[Table 226](#) below displays an example of the supported devices, access modes and transactions when the memory data bus is 16-bit for NOR, PSRAM and SRAM. Transactions not allowed (or not supported) by the FSMC in this example appear in gray.

Table 226. NOR flash/PSRAM controller: example of supported memories and transactions

| Device                                      | Mode              | R/W | AHB data size | Memory data size | Allowed/ not allowed | Comments                     |
|---------------------------------------------|-------------------|-----|---------------|------------------|----------------------|------------------------------|
| NOR flash<br>(muxed I/Os and nonmuxed I/Os) | Asynchronous      | R   | 8             | 16               | Y                    | -                            |
|                                             | Asynchronous      | W   | 8             | 16               | N                    | -                            |
|                                             | Asynchronous      | R   | 16            | 16               | Y                    | -                            |
|                                             | Asynchronous      | W   | 16            | 16               | Y                    | -                            |
|                                             | Asynchronous      | R   | 32            | 16               | Y                    | Split into two FSMC accesses |
|                                             | Asynchronous      | W   | 32            | 16               | Y                    | Split into two FSMC accesses |
|                                             | Asynchronous page | R   | -             | 16               | N                    | Mode is not supported        |
|                                             | Synchronous       | R   | 8             | 16               | N                    | -                            |
|                                             | Synchronous       | R   | 16            | 16               | Y                    | -                            |
|                                             | Synchronous       | R   | 32            | 16               | Y                    | -                            |

Table 226. NOR flash/PSRAM controller: example of supported memories and transactions

| Device                                      | Mode              | R/W | AHB data size | Memory data size | Allowed/ not allowed | Comments                                                 |
|---------------------------------------------|-------------------|-----|---------------|------------------|----------------------|----------------------------------------------------------|
| PSRAM (multiplexed and nonmultiplexed I/Os) | Asynchronous      | R   | 8             | 16               | Y                    | -                                                        |
|                                             | Asynchronous      | W   | 8             | 16               | Y                    | Use of byte lanes NBL[1:0]                               |
|                                             | Asynchronous      | R   | 16            | 16               | Y                    | -                                                        |
|                                             | Asynchronous      | W   | 16            | 16               | Y                    | -                                                        |
|                                             | Asynchronous      | R   | 32            | 16               | Y                    | Split into two FSMC accesses                             |
|                                             | Asynchronous      | W   | 32            | 16               | Y                    | Split into two FSMC accesses                             |
|                                             | Asynchronous page | R   | -             | 16               | N                    | Mode is not supported                                    |
|                                             | Synchronous       | R   | 8             | 16               | N                    | -                                                        |
|                                             | Synchronous       | R   | 16            | 16               | Y                    | -                                                        |
|                                             | Synchronous       | R   | 32            | 16               | Y                    | -                                                        |
|                                             | Synchronous       | W   | 8             | 16               | Y                    | Use of byte lanes NBL[1:0]                               |
|                                             | Synchronous       | W   | 16 / 32       | 16               | Y                    | -                                                        |
| SRAM and ROM                                | Asynchronous      | R   | 8 / 16        | 16               | Y                    | -                                                        |
|                                             | Asynchronous      | W   | 8 / 16        | 16               | Y                    | Use of byte lanes NBL[1:0]                               |
|                                             | Asynchronous      | R   | 32            | 16               | Y                    | Split into two FSMC accesses                             |
|                                             | Asynchronous      | W   | 32            | 16               | Y                    | Split into two FSMC accesses. Use of byte lanes NBL[1:0] |

### 36.5.3 General timing rules

#### Signals synchronization

- All controller output signals change on the rising edge of the internal clock (HCLK)
- In synchronous mode (read or write), all output signals change on the rising edge of HCLK. Whatever the CLKDIV value, all outputs change as follows:
  - NOEL/NWEL/ NEL/NADV L/ NADV H/ NBLL/ Address valid outputs change on the falling edge of FSMC\_CLK clock.
  - NOEH/ NWEH / NEH/ NOEH/NBLH/ Address invalid outputs change on the rising edge of FSMC\_CLK clock.

### 36.5.4 NOR flash/PSRAM controller asynchronous transactions

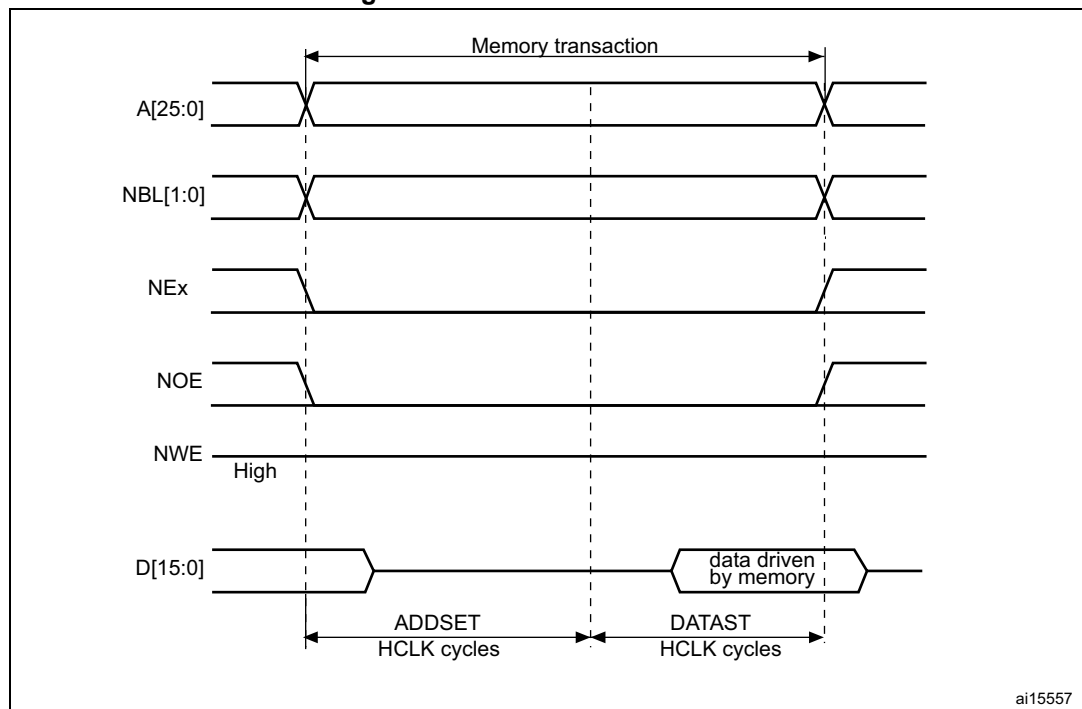
#### Asynchronous static memories (NOR flash memory, PSRAM, SRAM)

- Signals are synchronized by the internal clock HCLK. This clock is not issued to the memory
- The FSMC always samples the data before de-asserting the NOE signals. This guarantees that the memory data-hold timing constraint is met (chip enable high to data transition, usually 0 ns min.)
- If the extended mode is enabled (EXTMOD bit is set in the FSMC\_BCRx register), up to four extended modes (A, B, C and D) are available. It is possible to mix A, B, C and D modes for read and write operations. For example, read operation can be performed in mode A and write in mode B.
- If the extended mode is disabled (EXTMOD bit is reset in the FSMC\_BCRx register), the FSMC can operate in Mode1 or Mode2 as follows:
  - Mode 1 is the default mode when SRAM/PSRAM memory type is selected (MTYP[0:1] = 0x0 or 0x01 in the FSMC\_BCRx register)
  - Mode 2 is the default mode when NOR memory type is selected (MTYP[0:1] = 0x10 in the FSMC\_BCRx register).

#### Mode 1 - SRAM/PSRAM (CRAM)

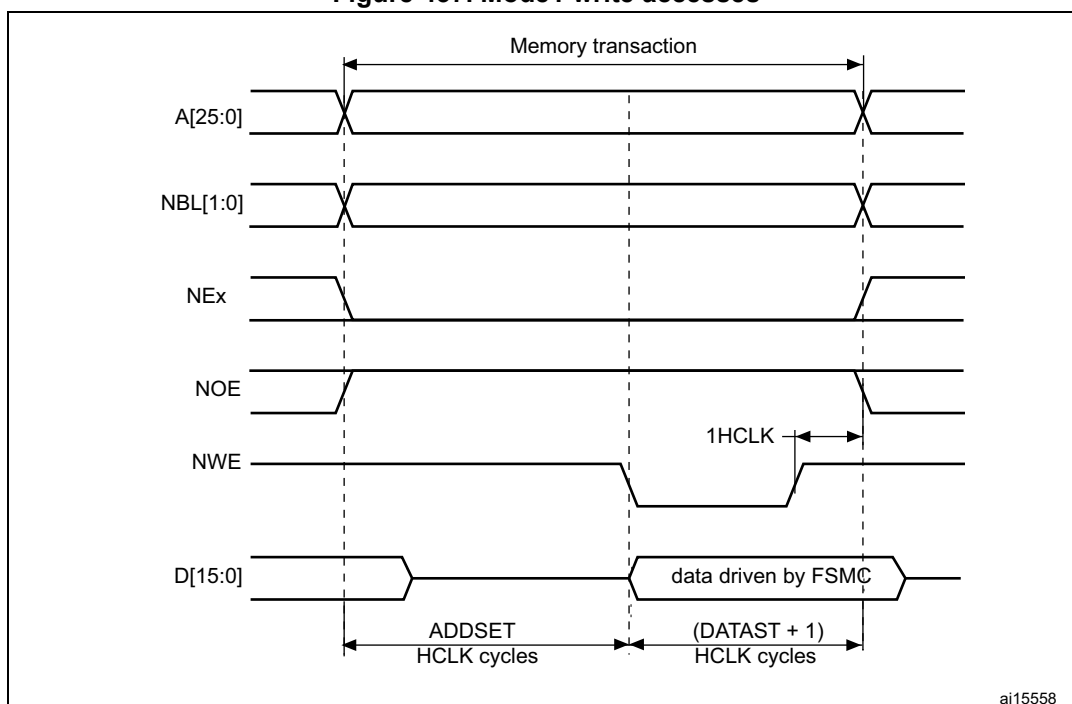
The next figures show the read and write transactions for the supported modes followed by the required configuration of FSMC\_BCRx, and FSMC\_BTRx/FSMC\_BWTRx registers.

**Figure 436. Mode1 read accesses**



1. NBL[1:0] are driven low during read access.

Figure 437. Mode1 write accesses



ai15558

The one HCLK cycle at the end of the write transaction helps guarantee the address and data hold time after the NWE rising edge. Due to the presence of this one HCLK cycle, the DATAST value must be greater than zero ( $\text{DATAST} > 0$ ).

Table 227. FSMC\_BCRx bit fields

| Bit number | Bit name  | Value to set                                                       |
|------------|-----------|--------------------------------------------------------------------|
| 31-20      | Reserved  | 0x000                                                              |
| 19         | CBURSTRW  | 0x0 (no effect on asynchronous mode)                               |
| 18:16      | CPSIZE    | 0x0 (no effect on asynchronous mode)                               |
| 15         | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD    | 0x0                                                                |
| 13         | WAITEN    | 0x0 (no effect on asynchronous mode)                               |
| 12         | WREN      | As needed                                                          |
| 11         | WAITCFG   | Don't care                                                         |
| 10         | WRAPMOD   | 0x0                                                                |
| 9          | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN   | 0x0                                                                |
| 7          | Reserved  | 0x1                                                                |
| 6          | FACCEN    | Don't care                                                         |
| 5-4        | MWID      | As needed                                                          |
| 3-2        | MTYP[0:1] | As needed, exclude 0x2 (NOR flash)                                 |

Table 227. FSMC\_BCRx bit fields (continued)

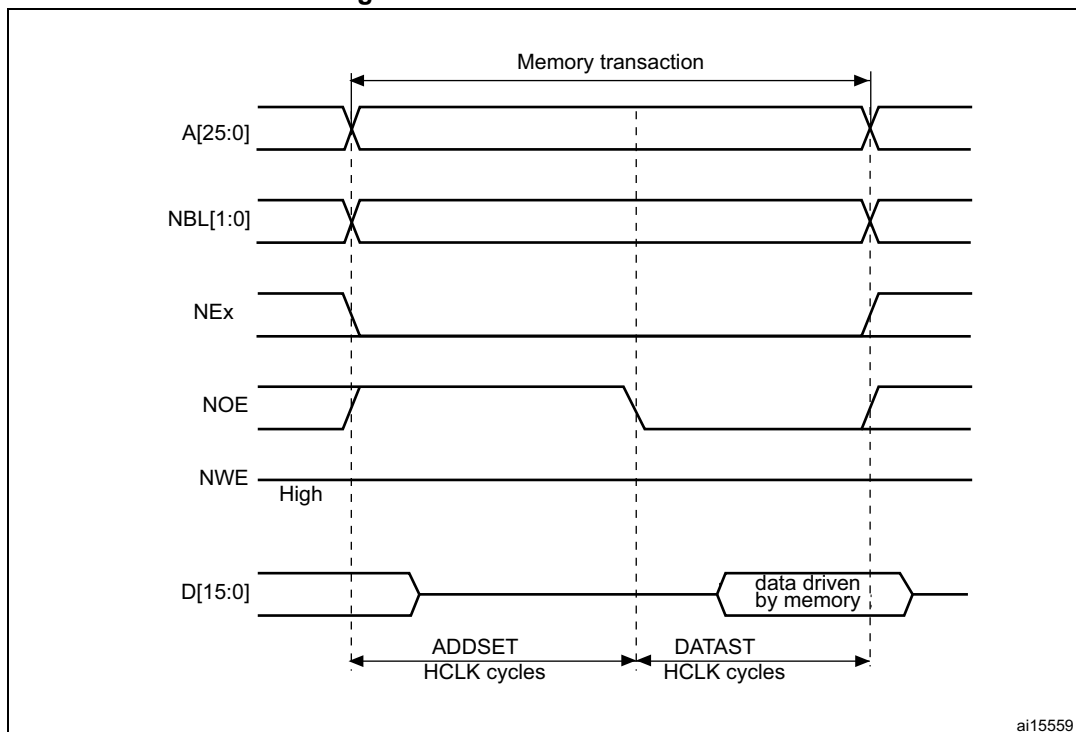
| Bit number | Bit name | Value to set |
|------------|----------|--------------|
| 1          | MUXE     | 0x0          |
| 0          | MBKEN    | 0x1          |

Table 228. FSMC\_BTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                         |
|------------|-------------|----------------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                                  |
| 29-28      | ACCMOD      | Don't care                                                                                                           |
| 27-24      | DATLAT      | Don't care                                                                                                           |
| 23-20      | CLKDIV      | Don't care                                                                                                           |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                                      |
| 15-8       | DATAST      | Duration of the second access phase (DATAST+1 HCLK cycles for write accesses, DATAST HCLK cycles for read accesses). |
| 7-4        | ADDHLD      | Don't care                                                                                                           |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles).<br>Minimum value for ADDSET is 0.                           |

### Mode A - SRAM/PSRAM (CRAM) OE toggling

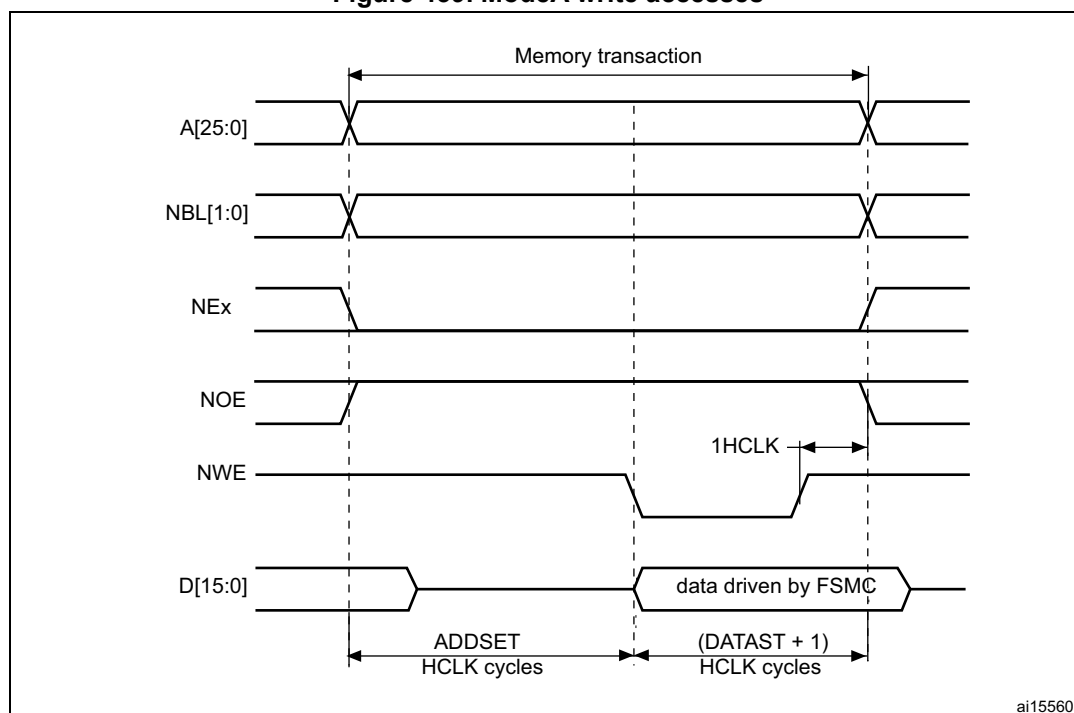
Figure 438. ModeA read accesses



ai15559

1. NBL[1:0] are driven low during read access.

Figure 439. ModeA write accesses



The differences compared with mode1 are the toggling of NOE and the independent read and write timings.

Table 229. FSMC\_BCRx bit fields

| Bit number | Bit name  | Value to set                                                       |
|------------|-----------|--------------------------------------------------------------------|
| 31-20      | Reserved  | 0x000                                                              |
| 19         | CBURSTRW  | 0x0 (no effect on asynchronous mode)                               |
| 18:16      | CPSIZE    | 0x0 (no effect on asynchronous mode)                               |
| 15         | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD    | 0x1                                                                |
| 13         | WAITEN    | 0x0 (no effect on asynchronous mode)                               |
| 12         | WREN      | As needed                                                          |
| 11         | WAITCFG   | Don't care                                                         |
| 10         | WRAPMOD   | 0x0                                                                |
| 9          | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN   | 0x0                                                                |
| 7          | Reserved  | 0x1                                                                |
| 6          | FACCEN    | Don't care                                                         |
| 5-4        | MWID      | As needed                                                          |
| 3-2        | MTYP[0:1] | As needed, exclude 0x2 (NOR flash)                                 |

Table 229. FSMC\_BCRx bit fields (continued)

| Bit number | Bit name | Value to set |
|------------|----------|--------------|
| 1          | MUXEN    | 0x0          |
| 0          | MBKEN    | 0x1          |

Table 230. FSMC\_BTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                 |
|------------|-------------|--------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                          |
| 29-28      | ACCMOD      | 0x0                                                                                                          |
| 27-24      | DATLAT      | Don't care                                                                                                   |
| 23-20      | CLKDIV      | Don't care                                                                                                   |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                              |
| 15-8       | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                                  |
| 7-4        | ADDHLD      | Don't care                                                                                                   |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses.<br>Minimum value for ADDSET is 0. |

Table 231. FSMC\_BWTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                  |
|------------|-------------|---------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                           |
| 29-28      | ACCMOD      | 0x0                                                                                                           |
| 27-24      | DATLAT      | Don't care                                                                                                    |
| 23-20      | CLKDIV      | Don't care                                                                                                    |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                               |
| 15-8       | DATAST      | Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,                                 |
| 7-4        | ADDHLD      | Don't care                                                                                                    |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses.<br>Minimum value for ADDSET is 0. |

Mode 2/B - NOR flash

Figure 440. Mode2 and mode B read accesses

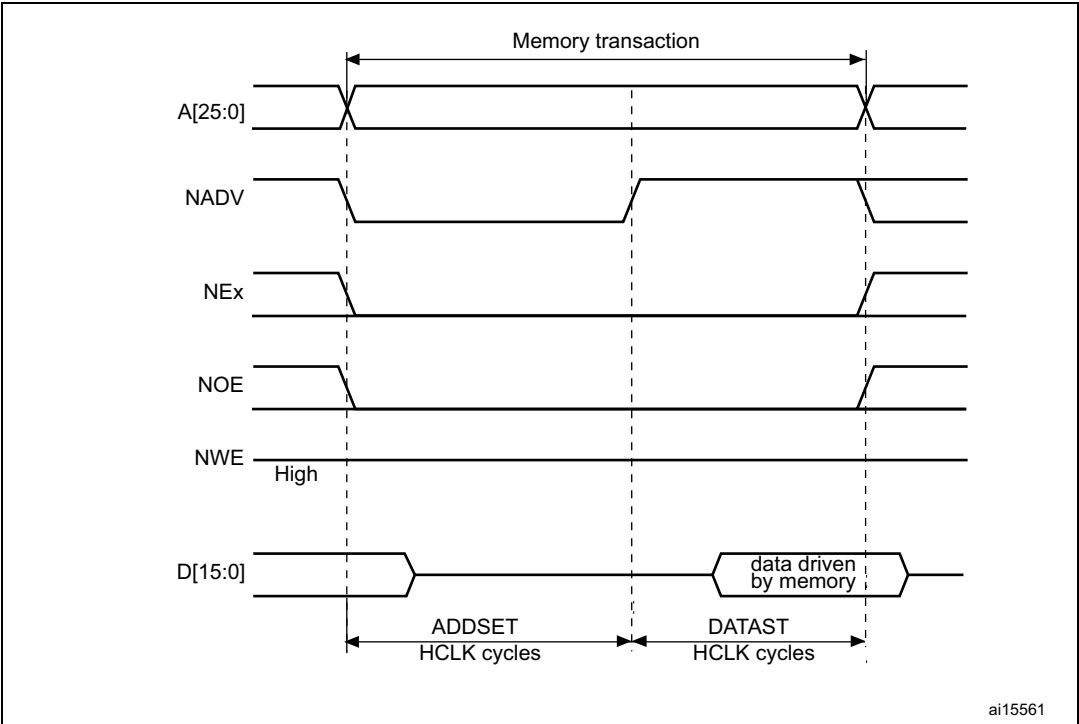


Figure 441. Mode2 write accesses

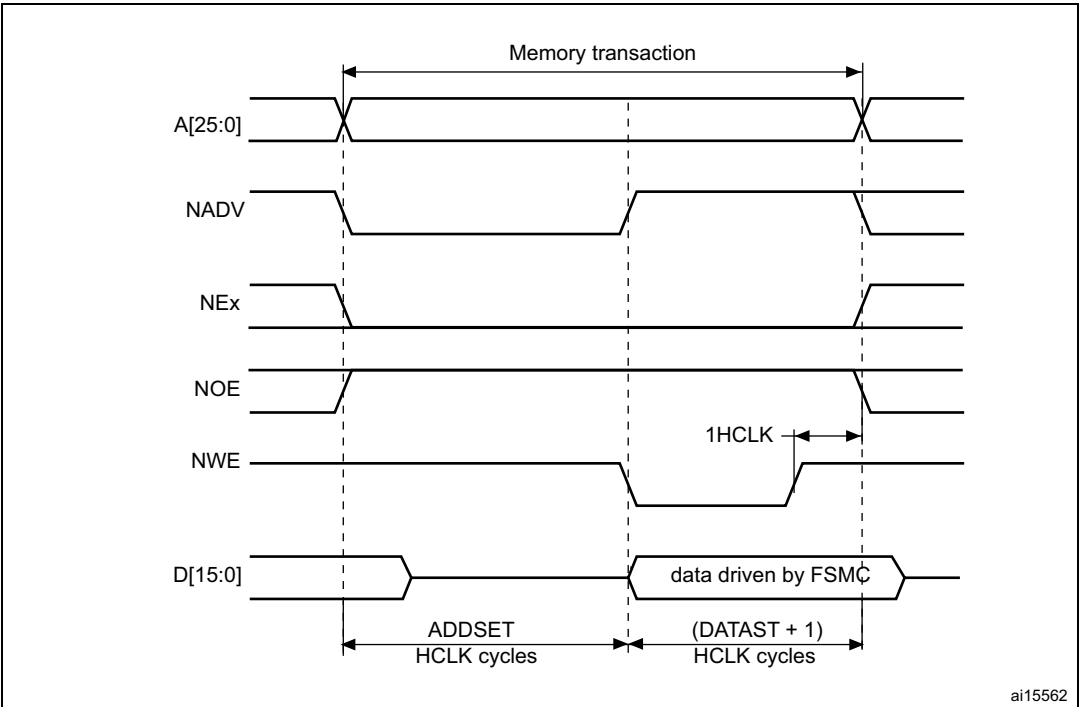
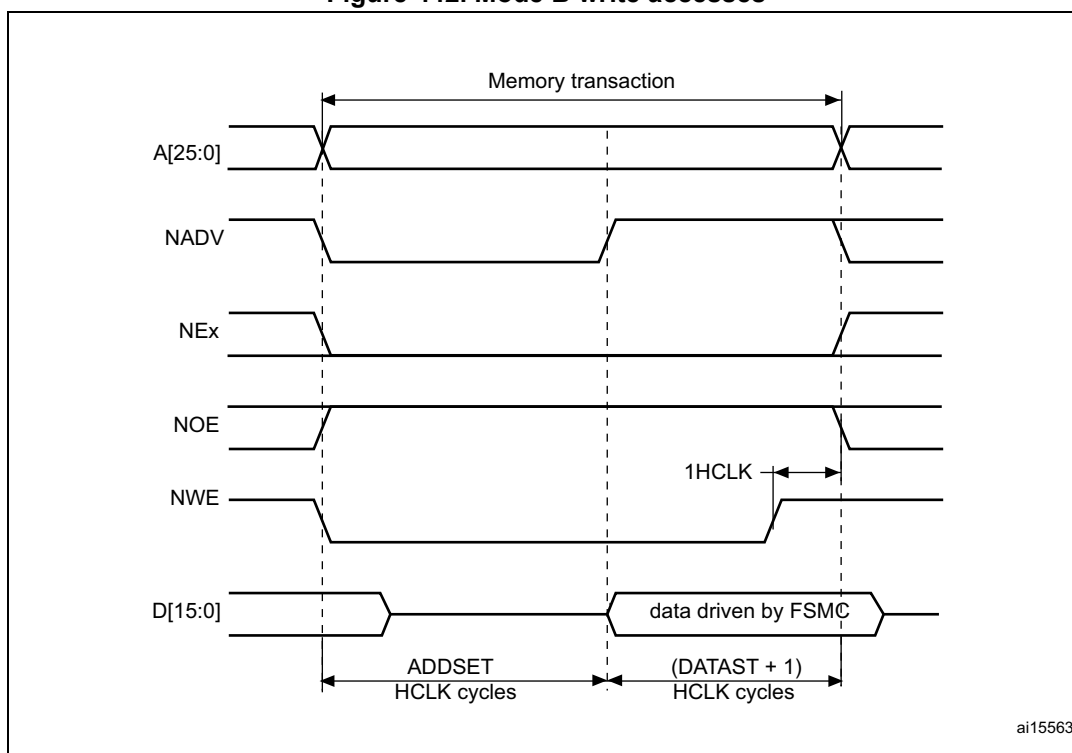


Figure 442. Mode B write accesses



ai15563

The differences with mode1 are the toggling of NWE and the independent read and write timings when extended mode is set (Mode B).

Table 232. FSMC\_BCRx bit fields

| Bit number | Bit name   | Value to set                                                       |
|------------|------------|--------------------------------------------------------------------|
| 31-20      | Reserved   | 0x000                                                              |
| 19         | CBURSTRW   | 0x0 (no effect on asynchronous mode)                               |
| 18:16      | Reserved   | 0x0 (no effect on asynchronous mode)                               |
| 15         | ASYNCAWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD     | 0x1 for mode B, 0x0 for mode 2                                     |
| 13         | WAITEN     | 0x0 (no effect on asynchronous mode)                               |
| 12         | WREN       | As needed                                                          |
| 11         | WAITCFG    | Don't care                                                         |
| 10         | WRAPMOD    | 0x0                                                                |
| 9          | WAITPOL    | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN    | 0x0                                                                |
| 7          | Reserved   | 0x1                                                                |
| 6          | FACCEN     | 0x1                                                                |
| 5-4        | MWID       | As needed                                                          |

Table 232. FSMC\_BCRx bit fields (continued)

| Bit number | Bit name  | Value to set           |
|------------|-----------|------------------------|
| 3-2        | MTYP[0:1] | 0x2 (NOR flash memory) |
| 1          | MUXEN     | 0x0                    |
| 0          | MBKEN     | 0x1                    |

Table 233. FSMC\_BTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                 |
|------------|-------------|--------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                          |
| 29-28      | ACCMOD      | 0x1                                                                                                          |
| 27-24      | DATLAT      | Don't care                                                                                                   |
| 23-20      | CLKDIV      | Don't care                                                                                                   |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                              |
| 15-8       | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                                  |
| 7-4        | ADDHLD      | Don't care                                                                                                   |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses.<br>Minimum value for ADDSET is 0. |

Table 234. FSMC\_BWTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                  |
|------------|-------------|---------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                           |
| 29-28      | ACCMOD      | 0x1                                                                                                           |
| 27-24      | DATLAT      | Don't care                                                                                                    |
| 23-20      | CLKDIV      | Don't care                                                                                                    |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                               |
| 15-8       | DATAST      | Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,                                 |
| 7-4        | ADDHLD      | Don't care                                                                                                    |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses.<br>Minimum value for ADDSET is 0. |

**Note:** The FSMC\_BWTRx register is valid only if extended mode is set (mode B), otherwise all its content is don't care.

## Mode C - NOR flash - OE toggling

Figure 443. Mode C read accesses

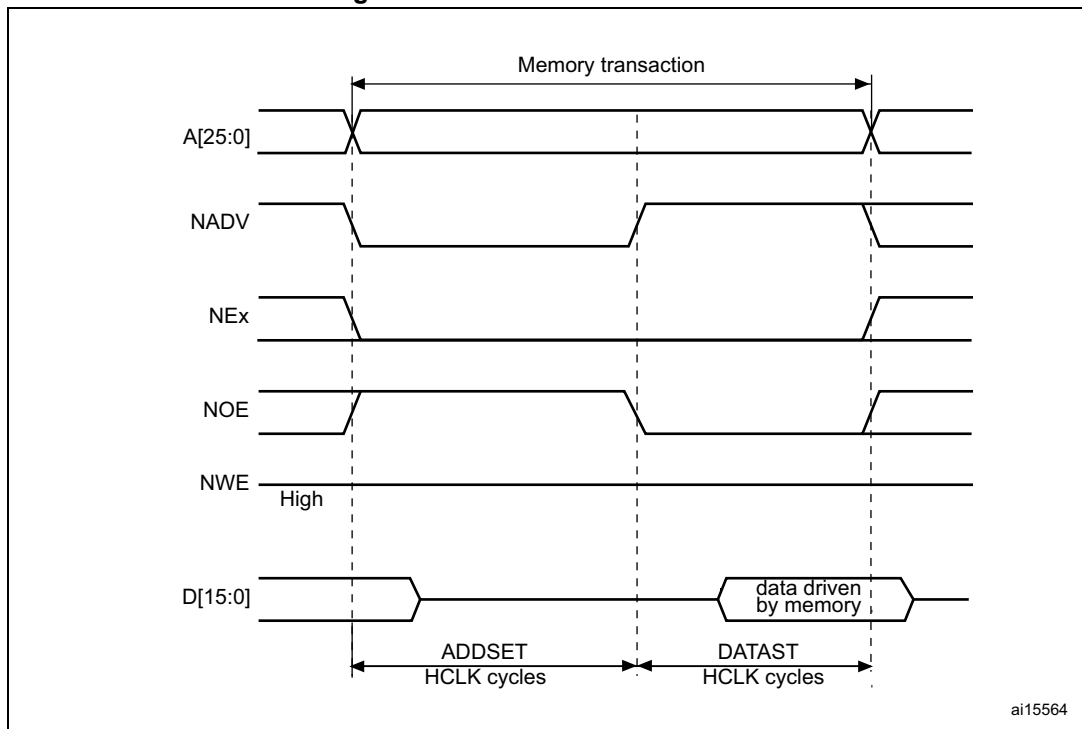
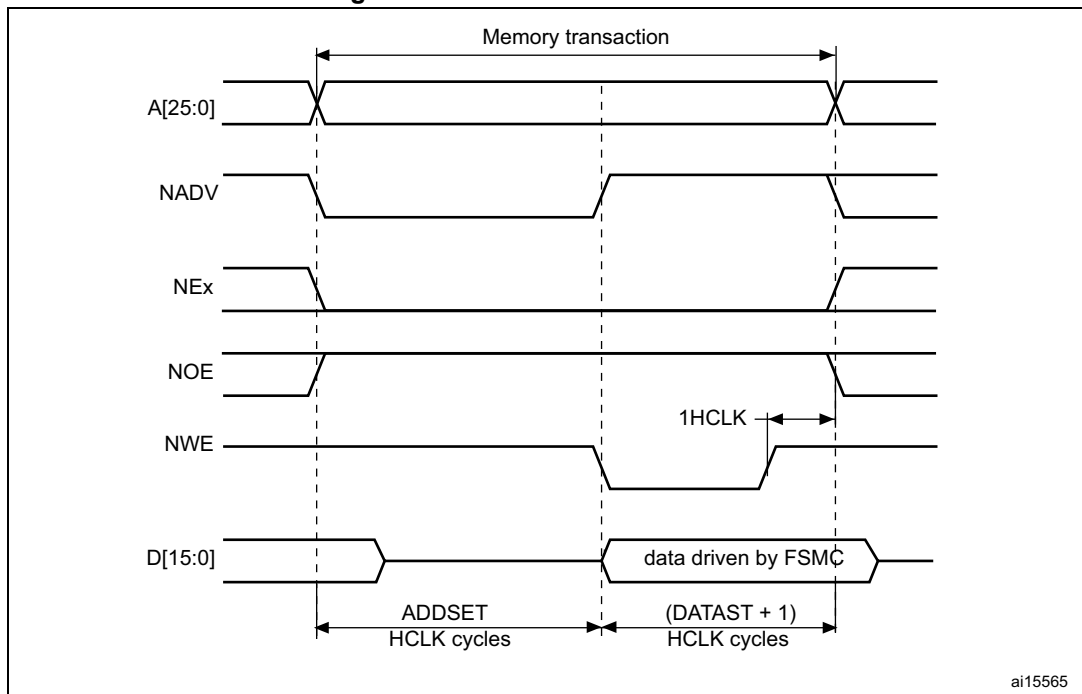


Figure 444. Mode C write accesses



The differences compared with mode1 are the toggling of  $NOE$  and the independent read and write timings.

Table 235. FSMC\_BCRx bit fields

| Bit No. | Bit name   | Value to set                                                       |
|---------|------------|--------------------------------------------------------------------|
| 31-20   | Reserved   | 0x000                                                              |
| 19      | CBURSTRW   | 0x0 (no effect on asynchronous mode)                               |
| 18:16   | CPSIZE     | 0x0 (no effect on asynchronous mode)                               |
| 15      | ASYNCAWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD     | 0x1                                                                |
| 13      | WAITEN     | 0x0 (no effect on asynchronous mode)                               |
| 12      | WREN       | As needed                                                          |
| 11      | WAITCFG    | Don't care                                                         |
| 10      | WRAPMOD    | 0x0                                                                |
| 9       | WAITPOL    | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN    | 0x0                                                                |
| 7       | Reserved   | 0x1                                                                |
| 6       | FACCEN     | 0x1                                                                |
| 5-4     | MWID       | As needed                                                          |
| 3-2     | MTYP[0:1]  | 0x2 (NOR flash memory)                                             |
| 1       | MUXEN      | 0x0                                                                |
| 0       | MBKEN      | 0x1                                                                |

Table 236. FSMC\_BTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                 |
|------------|-------------|--------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                          |
| 29-28      | ACCMOD      | 0x2                                                                                                          |
| 27-24      | DATLAT      | 0x0                                                                                                          |
| 23-20      | CLKDIV      | 0x0                                                                                                          |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                              |
| 15-8       | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                                  |
| 7-4        | ADDHLD      | Don't care                                                                                                   |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses.<br>Minimum value for ADDSET is 0. |

Table 237. FSMC\_BWTRx bit fields

| Bit number | Bit name    | Value to set                                                                                                  |
|------------|-------------|---------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                           |
| 29-28      | ACCMOD      | 0x2                                                                                                           |
| 27-24      | DATLAT      | Don't care                                                                                                    |
| 23-20      | CLKDIV      | Don't care                                                                                                    |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                               |
| 15-8       | DATAST      | Duration of the second access phase (DATAST+1 HCLK cycles for write accesses,                                 |
| 7-4        | ADDHLD      | Don't care                                                                                                    |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses.<br>Minimum value for ADDSET is 0. |

**Mode D - asynchronous access with extended address**

Figure 445. Mode D read accesses

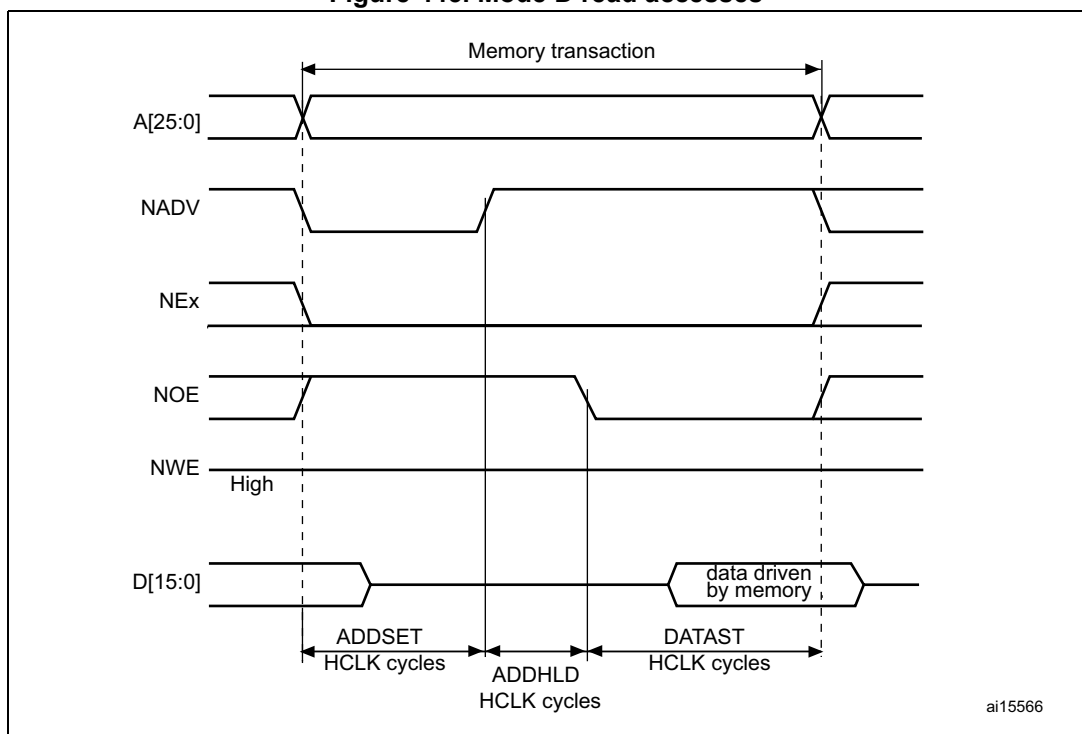
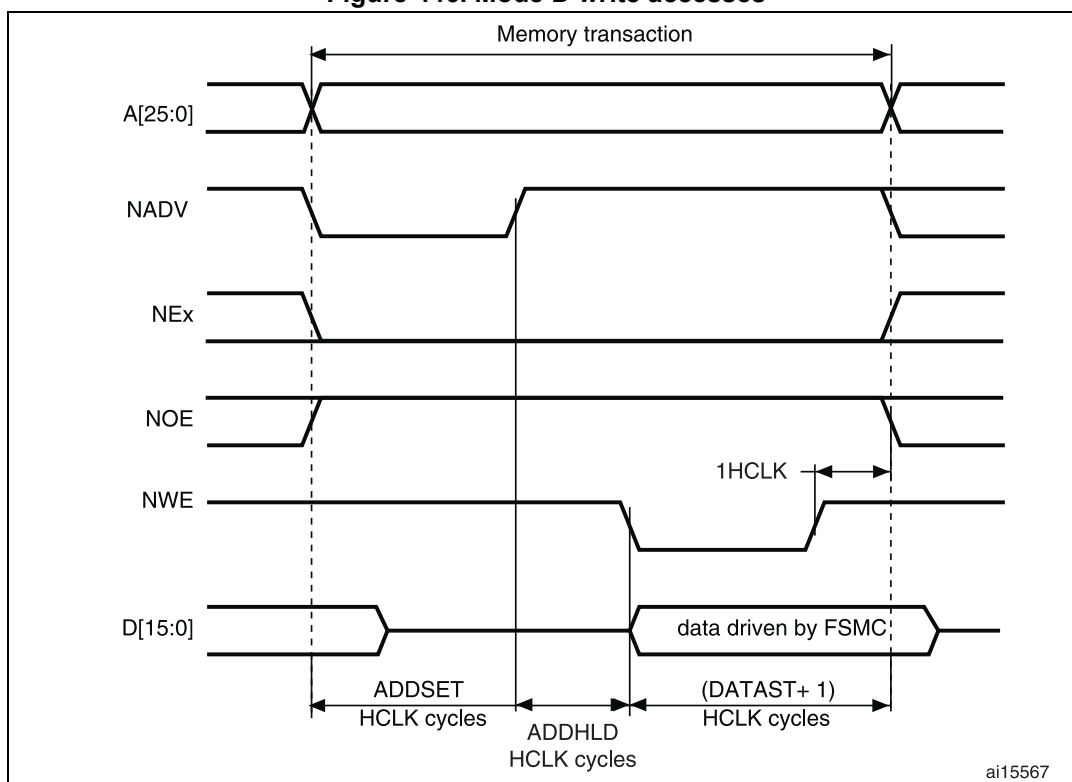


Figure 446. Mode D write accesses



The differences with mode1 are the toggling of NOE that goes on toggling after NADV changes and the independent read and write timings.

Table 238. FSMC\_BCRx bit fields

| Bit No. | Bit name  | Value to set                                                       |
|---------|-----------|--------------------------------------------------------------------|
| 31-20   | Reserved  | 0x000                                                              |
| 19      | CBURSTRW  | 0x0 (no effect on asynchronous mode)                               |
| 18:16   | CPSIZE    | 0x0 (no effect on asynchronous mode)                               |
| 15      | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD    | 0x1                                                                |
| 13      | WAITEN    | 0x0 (no effect on asynchronous mode)                               |
| 12      | WREN      | As needed                                                          |
| 11      | WAITCFG   | Don't care                                                         |
| 10      | WRAPMOD   | 0x0                                                                |
| 9       | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN   | 0x0                                                                |
| 7       | Reserved  | 0x1                                                                |
| 6       | FACCEN    | Set according to memory support                                    |
| 5-4     | MWID      | As needed                                                          |

Table 238. FSMC\_BCRx bit fields (continued)

| Bit No. | Bit name  | Value to set |
|---------|-----------|--------------|
| 3-2     | MTYP[0:1] | As needed    |
| 1       | MUXEN     | 0x0          |
| 0       | MBKEN     | 0x1          |

Table 239. FSMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                              |
|---------|-------------|-----------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                       |
| 29-28   | ACCMOD      | 0x3                                                                                                       |
| 27-24   | DATLAT      | Don't care                                                                                                |
| 23-20   | CLKDIV      | Don't care                                                                                                |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                           |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                               |
| 7-4     | ADDHLD      | Duration of the middle phase of the read access (ADDHLD HCLK cycles)                                      |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0. |

Table 240. FSMC\_BWTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                                 |
|---------|-------------|--------------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                          |
| 29-28   | ACCMOD      | 0x3                                                                                                          |
| 27-24   | DATLAT      | 0x0                                                                                                          |
| 23-20   | CLKDIV      | 0x0                                                                                                          |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                              |
| 15-8    | DATAST      | Duration of the second access phase (DATAST+1 HCLK cycles) for write accesses                                |
| 7-4     | ADDHLD      | Duration of the middle phase of the write access (ADDHLD HCLK cycles)                                        |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET+1 HCLK cycles) for write accesses. Minimum value for ADDSET is 0. |

## Muxed mode - multiplexed asynchronous access to NOR flash memory

Figure 447. Multiplexed read accesses

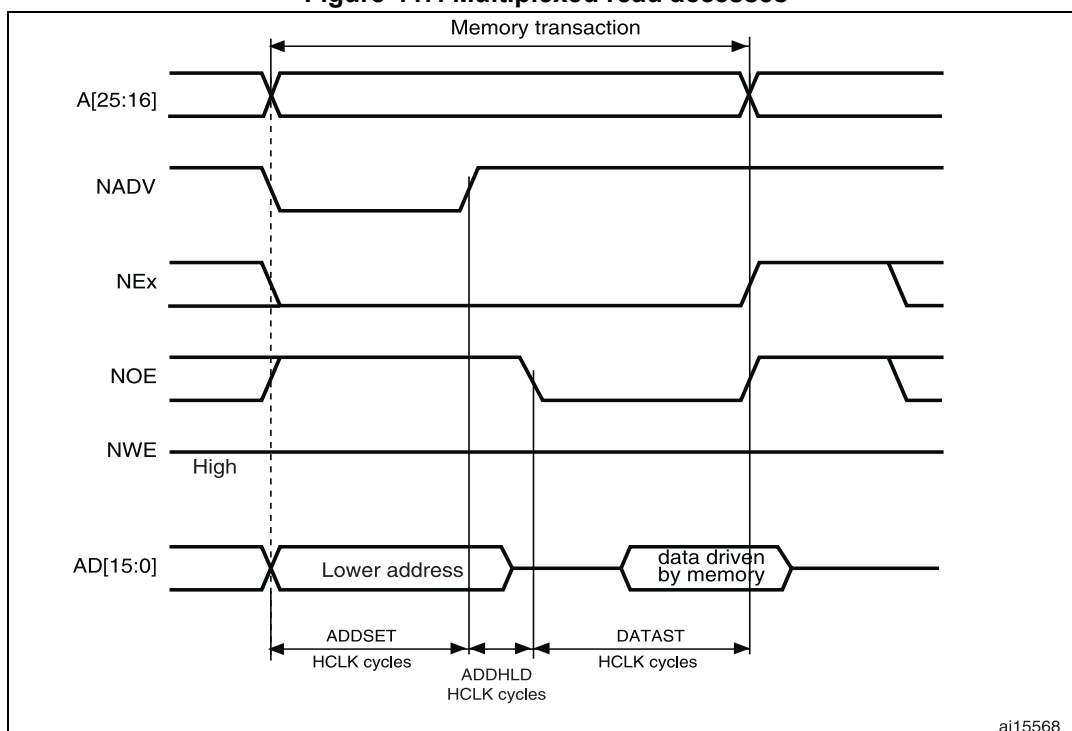
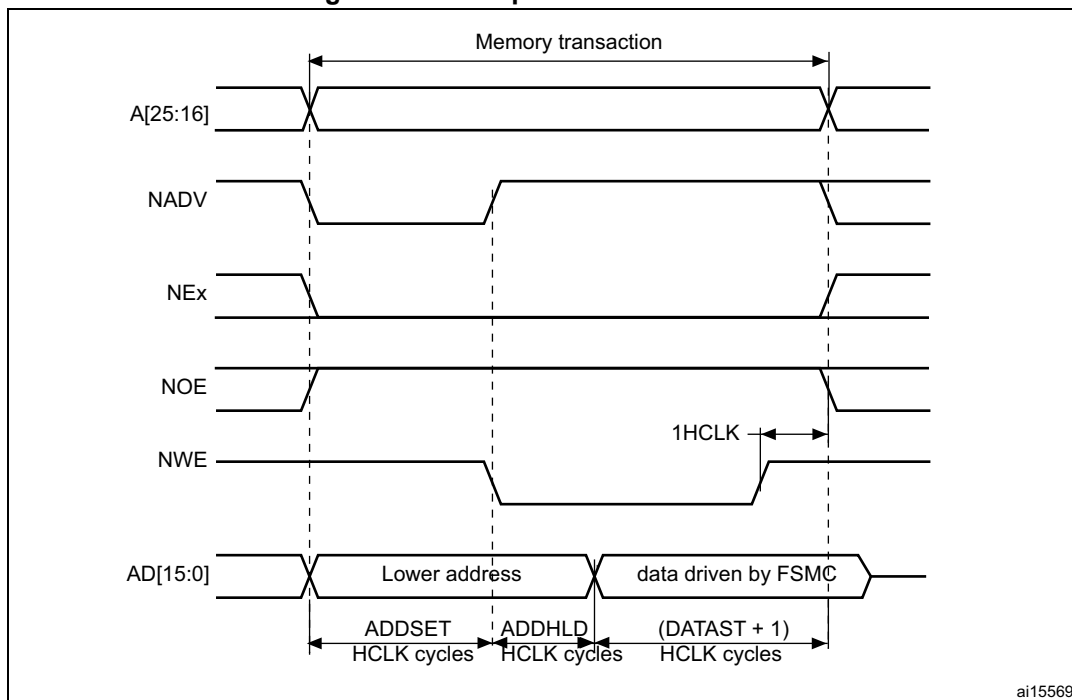


Figure 448. Multiplexed write accesses



The difference with mode D is the drive of the lower address byte(s) on the databus.

Table 241. FSMC\_BCRx bit fields

| Bit No. | Bit name  | Value to set                                                       |
|---------|-----------|--------------------------------------------------------------------|
| 31-21   | Reserved  | 0x000                                                              |
| 19      | CBURSTRW  | 0x0 (no effect on asynchronous mode)                               |
| 18:16   | CPSIZE    | 0x0 (no effect on asynchronous mode)                               |
| 15      | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD    | 0x0                                                                |
| 13      | WAITEN    | 0x0 (no effect on asynchronous mode)                               |
| 12      | WREN      | As needed                                                          |
| 11      | WAITCFG   | Don't care                                                         |
| 10      | WRAPMOD   | 0x0                                                                |
| 9       | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN   | 0x0                                                                |
| 7       | Reserved  | 0x1                                                                |
| 6       | FACCEN    | 0x1                                                                |
| 5-4     | MWID      | As needed                                                          |
| 3-2     | MTYP[0:1] | 0x2 (NOR flash memory)                                             |
| 1       | MUXEN     | 0x1                                                                |
| 0       | MBKEN     | 0x1                                                                |

Table 242. FSMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                                            |
|---------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                                     |
| 29-28   | ACCMOD      | 0x0                                                                                                                     |
| 27-24   | DATLAT      | Don't care                                                                                                              |
| 23-20   | CLKDIV      | Don't care                                                                                                              |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                                         |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles for read accesses and DATAST+1 HCLK cycles for write accesses). |
| 7-4     | ADDHLD      | Duration of the middle phase of the access (ADDHLD HCLK cycles).                                                        |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles).<br>Minimum value for ADDSET is 1.                              |

### WAIT management in asynchronous accesses

If the asynchronous memory asserts a WAIT signal to indicate that it is not yet ready to accept or to provide data, the ASYNCWAIT bit has to be set in FSMC\_BCRx register.

If the WAIT signal is active (high or low depending on the WAITPOL bit), the second access phase (Data setup phase) programmed by the DATAST bits, is extended until WAIT becomes inactive. Unlike the data setup phase, the first access phases (Address setup and Address hold phases), programmed by the ADDSET[3:0] and ADDHLD bits, are not WAIT sensitive and so they are not prolonged.

The data setup phase (DATAST in the FSMC\_BTRx register) must be programmed so that WAIT can be detected 4 HCLK cycles before the end of memory transaction. The following cases must be considered:

1. DATAST in FSMC\_BTRx register) Memory asserts the WAIT signal aligned to NOE/NWE which toggles:

$$\text{DATAST} \geq (4 \times \text{HCLK}) + \text{max\_wait\_assertion\_time}$$

2. Memory asserts the WAIT signal aligned to NEx (or NOE/NWE not toggling):  
if

$$\text{max\_wait\_assertion\_time} > \text{address\_phase} + \text{hold\_phase}$$

then

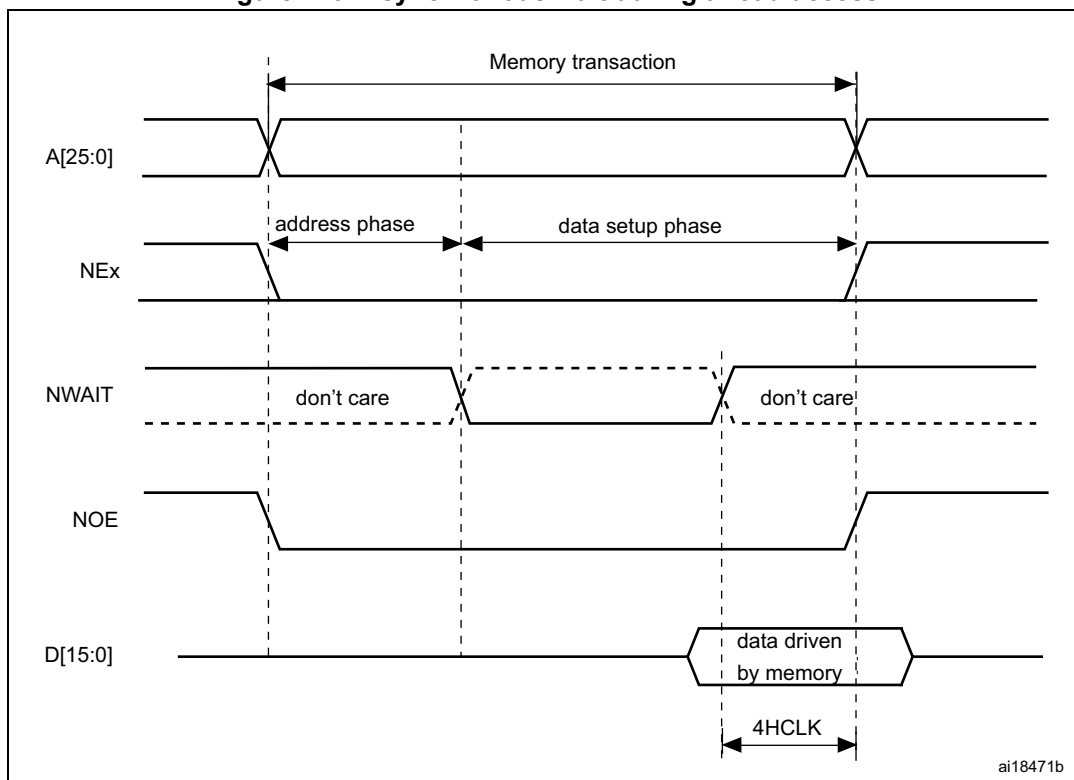
$$\text{DATAST} \geq (4 \times \text{HCLK}) + (\text{max\_wait\_assertion\_time} - \text{address\_phase} - \text{hold\_phase})$$

otherwise

$$\text{DATAST} \geq 4 \times \text{HCLK}$$

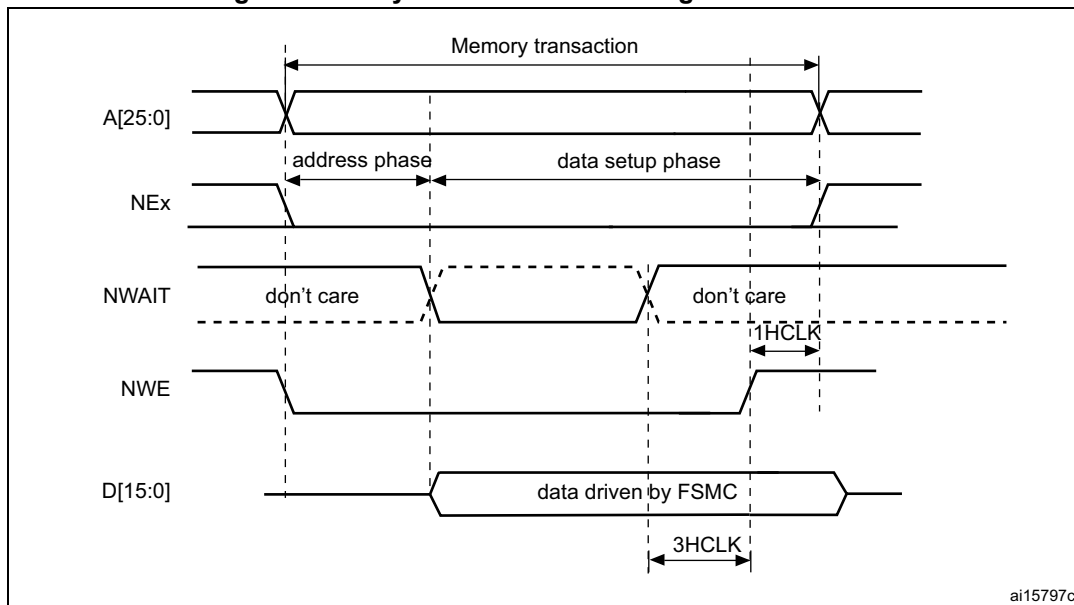
where max\_wait\_assertion\_time is the maximum time taken by the memory to assert the WAIT signal once NEx/NOE/NWE is low.

[Figure 449](#) and [Figure 450](#) show the number of HCLK clock cycles that are added to the memory access after WAIT is released by the asynchronous memory (independently of the above cases).

**Figure 449. Asynchronous wait during a read access**

ai18471b

1. NWAIT polarity depends on WAITPOL bit setting in FSMC\_BCRx register.

**Figure 450. Asynchronous wait during a write access**

ai15797c

1. NWAIT polarity depends on WAITPOL bit setting in FSMC\_BCRx register.

### 36.5.5 Synchronous transactions

The memory clock, CLK, is a submultiple of HCLK according to the value of parameter CLKDIV.

NOR flash memories specify a minimum time from NADV assertion to CLK high. To meet this constraint, the FSMC does not issue the clock to the memory during the first internal clock cycle of the synchronous access (before NADV assertion). This guarantees that the rising edge of the memory clock occurs *in the middle* of the NADV low pulse.

#### Data latency versus NOR flash latency

The data latency is the number of cycles to wait before sampling the data. The DATLAT value must be consistent with the latency value specified in the NOR flash configuration register. The FSMC does not include the clock cycle when NADV is low in the data latency count.

**Caution:** Some NOR flash memories include the NADV Low cycle in the data latency count, so the exact relation between the latency and the FSMC DATLAT parameter can be either of:

- NOR flash latency = (DATLAT + 2) CLK clock cycles
- NOR flash latency = (DATLAT + 3) CLK clock cycles

Some recent memories assert NWAIT during the latency phase. In such cases DATLAT can be set to its minimum value. As a result, the FSMC samples the data and waits long enough to evaluate if the data are valid. Thus the FSMC detects when the memory exits latency and real data are taken.

Other memories do not assert NWAIT during latency. In this case the latency must be set correctly for both the FSMC and the memory, otherwise invalid data are mistaken for good data, or valid data are lost in the initial phase of the memory access.

#### Single-burst transfer

When the selected bank is configured in burst mode for synchronous accesses, if for example an AHB single-burst transaction is requested on 16-bit memories, the FSMC performs a burst transaction of length 1 (if the AHB transfer is 16-bit), or length 2 (if the AHB transfer is 32-bit) and de-assert the chip select signal when the last data is strobed.

Clearly, such a transfer is not the most efficient in terms of cycles (compared to an asynchronous read). Nevertheless, a random asynchronous access would first require to re-program the memory access mode, which would altogether last longer.

#### Cross boundary page for Cellular RAM 1.5

Cellular RAM 1.5 does not allow burst access to cross the page boundary. The FSMC controller allows to split automatically the burst access when the memory page size is reached by configuring the CPSIZE bits in the FSMC\_BCR1 register following the memory page size.

#### Wait management

For synchronous NOR flash memories, NWAIT is evaluated after the programmed latency period, (DATLAT+2) CLK clock cycles.

If NWAIT is sensed active (low level when WAITPOL = 0, high level when WAITPOL = 1), wait states are inserted until NWAIT is sensed inactive (high level when WAITPOL = 0, low level when WAITPOL = 1).

When NWAIT is inactive, the data is considered valid either immediately (bit WAITCFG = 1) or on the next clock edge (bit WAITCFG = 0).

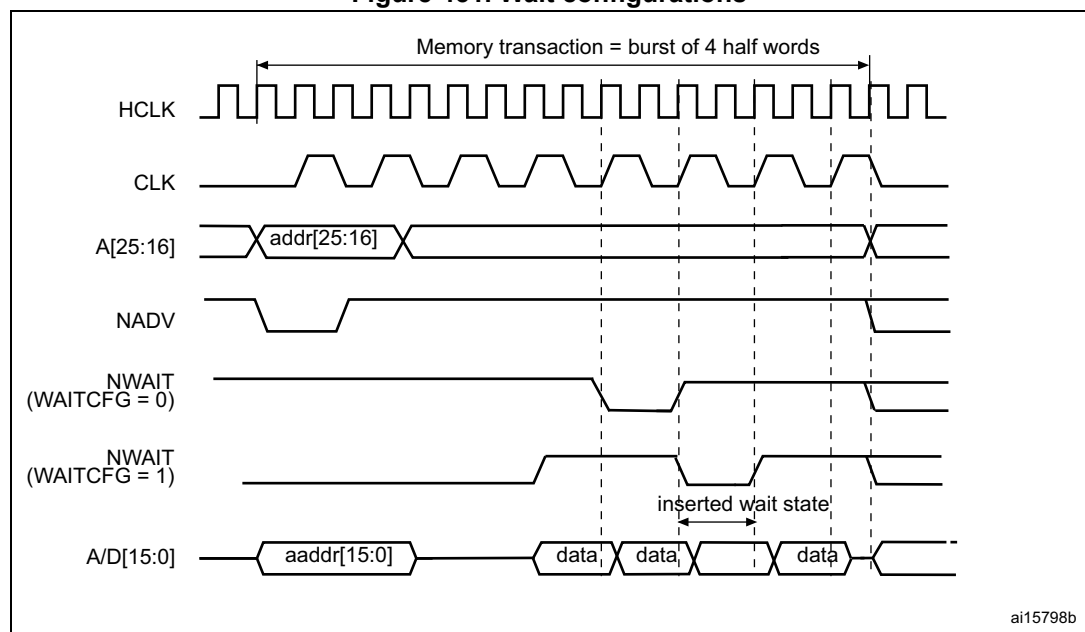
During wait-state insertion via the NWAIT signal, the controller continues to send clock pulses to the memory, keeping the chip select and output enable signals valid, and does not consider the data valid.

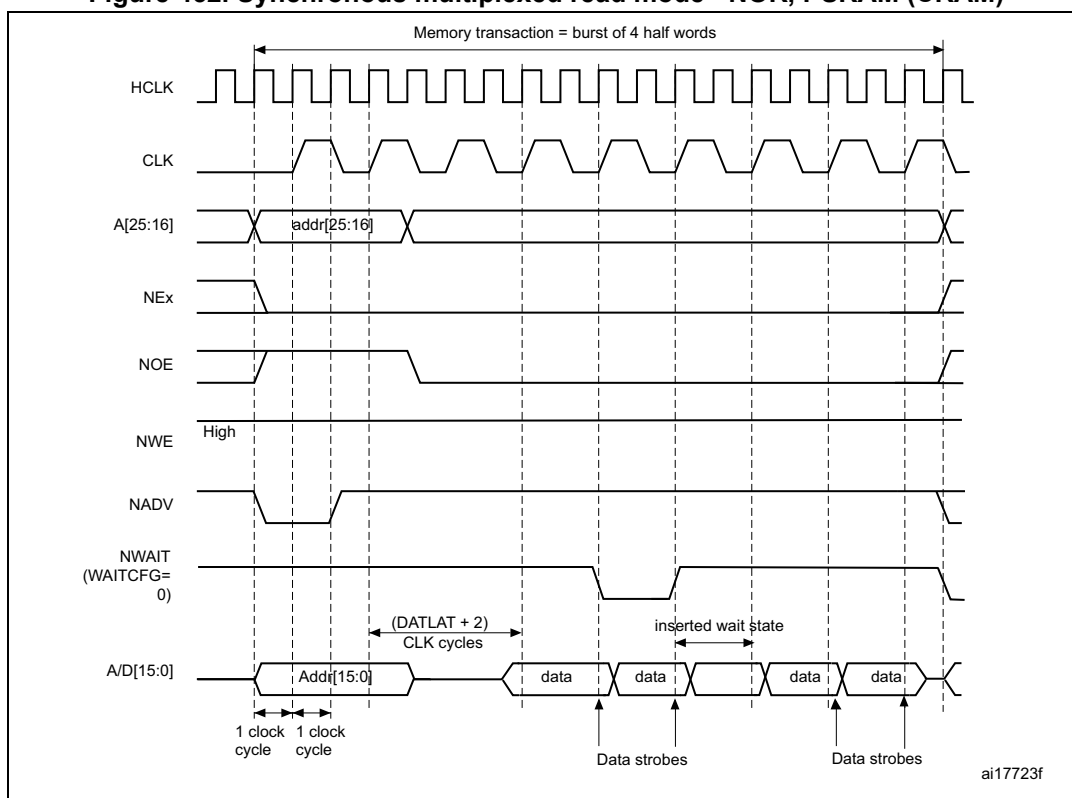
There are two timing configurations for the NOR flash NWAIT signal in burst mode:

- Flash memory asserts the NWAIT signal one data cycle before the wait state (default after reset)
- Flash memory asserts the NWAIT signal during the wait state

These two NOR flash wait state configurations are supported by the FSMC, individually for each chip select, thanks to the WAITCFG bit in the FSMC\_BCRx registers (x = 0..3).

**Figure 451. Wait configurations**



**Figure 452. Synchronous multiplexed read mode - NOR, PSRAM (CRAM)**

1. Byte lane outputs BL are not shown; for NOR access, they are held high, and, for PSRAM (CRAM) access, they are held low.
2. NWAIT polarity is set to 0.

**Table 243. FSMC\_BCRx bit fields**

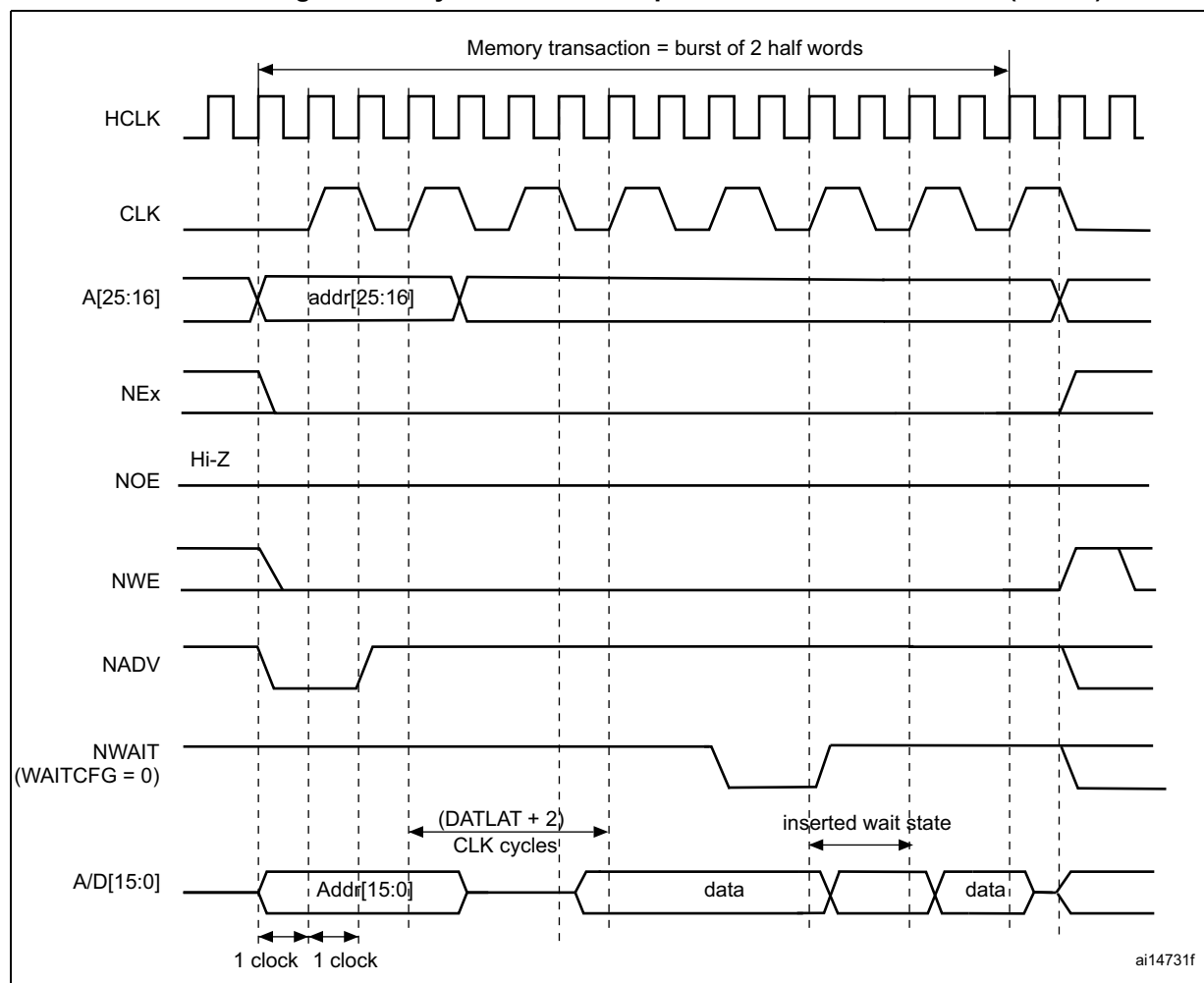
| Bit No. | Bit name  | Value to set                                                       |
|---------|-----------|--------------------------------------------------------------------|
| 31-20   | Reserved  | 0x000                                                              |
| 19      | CBURSTRW  | No effect on synchronous read                                      |
| 18-16   | CPSIZE    | As needed (0x1 for CRAM 1.5)                                       |
| 15      | ASCYCWAIT | 0x0                                                                |
| 14      | EXTMOD    | 0x0                                                                |
| 13      | WAITEN    | Set to 1 if the memory supports this feature, otherwise keep at 0. |
| 12      | WREN      | no effect on synchronous read                                      |
| 11      | WAITCFG   | to be set according to memory                                      |
| 10      | WRAPMOD   | 0x0                                                                |
| 9       | WAITPOL   | to be set according to memory                                      |
| 8       | BURSTEN   | 0x1                                                                |
| 7       | Reserved  | 0x1                                                                |
| 6       | FACCEN    | Set according to memory support (NOR flash memory)                 |
| 5-4     | MWID      | As needed                                                          |

Table 243. FSMC\_BCRx bit fields (continued)

| Bit No. | Bit name  | Value to set |
|---------|-----------|--------------|
| 3-2     | MTYP[0:1] | 0x1 or 0x2   |
| 1       | MUXEN     | As needed    |
| 0       | MBKEN     | 0x1          |

Table 244. FSMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                             |
|---------|-------------|--------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                      |
| 29:28   | ACCMOD      | 0x0                                                                      |
| 27-24   | DATLAT      | Data latency                                                             |
| 23-20   | CLKDIV      | 0x0 to get CLK = HCLK (not supported)<br>0x1 to get CLK = 2 × HCLK<br>.. |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                          |
| 15-8    | DATAST      | Don't care                                                               |
| 7-4     | ADDHLD      | Don't care                                                               |
| 3-0     | ADDSET[3:0] | Don't care                                                               |

**Figure 453. Synchronous multiplexed write mode - PSRAM (CRAM)**

1. Memory must issue NWAIT signal one cycle in advance, accordingly WAITCFG must be programmed to 0.
2. NWAIT polarity is set to 0.
3. Byte Lane (NBL) outputs are not shown, they are held low while NEx is active.

**Table 245. FSMC\_BCRx bit fields**

| Bit No. | Bit name  | Value to set                                                       |
|---------|-----------|--------------------------------------------------------------------|
| 31-20   | Reserved  | 0x000                                                              |
| 19      | CBURSTRW  | 0x1                                                                |
| 18-16   | CPSIZE    | As needed (0x1 for CRAM 1.5)                                       |
| 15      | ASCYCWAIT | 0x0                                                                |
| 14      | EXTMOD    | 0x0                                                                |
| 13      | WAITEN    | Set to 1 if the memory supports this feature, otherwise keep at 0. |
| 12      | WREN      | 0x1                                                                |
| 11      | WAITCFG   | 0x0                                                                |
| 10      | WRAPMOD   | 0x0                                                                |

**Table 245. FSMC\_BCRx bit fields (continued)**

| Bit No. | Bit name  | Value to set                    |
|---------|-----------|---------------------------------|
| 9       | WAITPOL   | to be set according to memory   |
| 8       | BURSTEN   | no effect on synchronous write  |
| 7       | Reserved  | 0x1                             |
| 6       | FACCEN    | Set according to memory support |
| 5-4     | MWID      | As needed                       |
| 3-2     | MTYP[0:1] | 0x1                             |
| 1       | MUXEN     | As needed                       |
| 0       | MBKEN     | 0x1                             |

**Table 246. FSMC\_BTRx bit fields**

| Bit No. | Bit name    | Value to set                                                       |
|---------|-------------|--------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                |
| 29:28   | ACCMOD      | 0x0                                                                |
| 27-24   | DATLAT      | Data latency                                                       |
| 23-20   | CLKDIV      | 0x0 to get CLK = HCLK (not supported)<br>0x1 to get CLK = 2 × HCLK |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                    |
| 15-8    | DATAST      | Don't care                                                         |
| 7-4     | ADDHLD      | Don't care                                                         |
| 3-0     | ADDSET[3:0] | Don't care                                                         |

### 36.5.6 NOR/PSRAM control registers

The NOR/PSRAM control registers have to be accessed by words (32 bits).

#### SRAM/NOR-flash chip-select control registers 1..4 (FSMC\_BCR1..4)

Address offset:  $0xA000\ 0000 + 8 * (x - 1)$ ,  $x = 1...4$

Reset value: 0x0000 30DB for Bank1 and 0x0000 30D2 for Bank 2 to 4

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR flash memories.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19       | 18          | 17 | 16 | 15        | 14     | 13     | 12   | 11      | 10      | 9       | 8       | 7        | 6      | 5         | 4  | 3         | 2  | 1     | 0     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----------|-------------|----|----|-----------|--------|--------|------|---------|---------|---------|---------|----------|--------|-----------|----|-----------|----|-------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    | CBURSTRW | CPSIZE[2:0] |    |    | ASCYCWAIT | EXTMOD | WAITEN | WREN | WAITCFG | WRAPMOD | WAITPOL | BURSTEN | Reserved | FACCEN | MWID[1:0] |    | MTYP[1:0] |    | MUXEN | MBKEN |
|          |    |    |    |    |    |    |    |    |    |    |    | RW       |             |    |    | RW        | RW     | RW     | RW   | RW      | RW      | RW      | RW      |          |        | RW        | RW | RW        | RW | RW    | RW    |
|          |    |    |    |    |    |    |    |    |    |    |    |          |             |    |    |           |        |        |      |         |         |         |         |          |        |           |    |           |    |       |       |

Bits 31: 20 Reserved, must be kept at reset value.

Bit 19 **CBURSTRW**: Write burst enable.

For Cellular RAM (PSRAM) memories, this bit enables the synchronous burst protocol during write operations. The enable bit for synchronous read accesses is the BURSTEN bit in the FSMC\_BCRx register.

0: Write operations are always performed in asynchronous mode

1: Write operations are performed in synchronous mode.

Bits 18: 16 **CPSIZE[2:0]**: CRAM page size.

These are used for Cellular RAM 1.5 which does not allow burst access to cross the address boundaries between pages. When these bits are configured, the FSMC controller splits automatically the burst access when the memory page size is reached (refer to memory datasheet for page size).

000: No burst split when crossing page boundary (default after reset)

001: 128 bytes

010: 256 bytes

011: 512 bytes

100: 1024 bytes

Others: reserved.

Bit 15 **ASYNCAWAIT**: Wait signal during asynchronous transfers

This bit enables/disables the FSMC to use the wait signal even during an asynchronous protocol.

0: NWAIT signal is not taken into account when running an asynchronous protocol (default after reset)

1: NWAIT signal is taken into account when running an asynchronous protocol

**Bit 14 EXTMOD:** Extended mode enable.

This bit enables the FSMC to program the write timings for non-multiplexed asynchronous accesses inside the FSMC\_BWTR register, thus resulting in different timings for read and write operations.

0: values inside FSMC\_BWTR register are not taken into account (default after reset)

1: values inside FSMC\_BWTR register are taken into account

*Note: When the extended mode is disabled, the FSMC can operate in Mode1 or Mode2 as follows:*

- *Mode 1 is the default mode when the SRAM/PSRAM memory type is selected (MTYP [0:1]=0x0 or 0x01)*
- *Mode 2 is the default mode when the NOR memory type is selected (MTYP [0:1]= 0x10).*

**Bit 13 WAITEN:** Wait enable bit.

This bit enables/disables wait-state insertion via the NWAIT signal when accessing the flash memory in synchronous mode.

0: NWAIT signal is disabled (its level not taken into account, no wait state inserted after the programmed flash latency period)

1: NWAIT signal is enabled (its level is taken into account after the programmed flash latency period to insert wait states if asserted) (default after reset)

**Bit 12 WREN:** Write enable bit.

This bit indicates whether write operations are enabled/disabled in the bank by the FSMC:

0: Write operations are disabled in the bank by the FSMC, an AHB error is reported,

1: Write operations are enabled for the bank by the FSMC (default after reset).

**Bit 11 WAITCFG:** Wait timing configuration.

The NWAIT signal indicates whether the data from the memory are valid or if a wait state must be inserted when accessing the flash memory in synchronous mode. This configuration bit determines if NWAIT is asserted by the memory one clock cycle before the wait state or during the wait state:

0: NWAIT signal is active one data cycle before wait state (default after reset),

1: NWAIT signal is active during wait state (not used for PRAM).

**Bit 10 WRAPMOD:** Wrapped burst mode support.

Defines whether the controller splits or not an AHB burst wrap access into two linear accesses. Valid only when accessing memories in burst mode

0: Direct wrapped burst is not enabled (default after reset),

1: Direct wrapped burst is enabled.

*Note: This bit has no effect as the CPU and DMA cannot generate wrapping burst transfers.*

**Bit 9 WAITPOL:** Wait signal polarity bit.

Defines the polarity of the wait signal from memory. Valid only when accessing the memory in burst mode:

0: NWAIT active low (default after reset),

1: NWAIT active high.

**Bit 8 BURSTEN:** Burst enable bit.

This bit enables/disables synchronous accesses during read operations. It is valid only for synchronous memories operating in burst mode:

0: Burst mode disabled (default after reset). Read accesses are performed in asynchronous mode.

1: Burst mode enable. Read accesses are performed in synchronous mode.

Bit 7 Reserved, must be kept at reset value.

Bit 6 **FACCEN**: Flash access enable

Enables NOR flash memory access operations.

0: Corresponding NOR flash memory access is disabled

1: Corresponding NOR flash memory access is enabled (default after reset)

Bits 5:4 **MWID[1:0]**: Memory databus width.

Defines the external memory device width, valid for all type of memories.

00: 8 bits,

01: 16 bits (default after reset),

10: reserved, do not use,

11: reserved, do not use.

Bits 3:2 **MTYP[1:0]**: Memory type.

Defines the type of external memory attached to the corresponding memory bank:

00: SRAM (default after reset for Bank 2...4)

01: PSRAM (CRAM)

10: NOR flash/OneNAND flash (default after reset for Bank 1)

11: reserved

Bit 1 **MUXEN**: Address/data multiplexing enable bit.

When this bit is set, the address and data values are multiplexed on the databus, valid only with NOR and PSRAM memories:

0: Address/Data nonmultiplexed

1: Address/Data multiplexed on databus (default after reset)

Bit 0 **MBKEN**: Memory bank enable bit.

Enables the memory bank. After reset Bank1 is enabled, all others are disabled.

Accessing a disabled bank causes an ERROR on AHB bus.

0: Corresponding memory bank is disabled

1: Corresponding memory bank is enabled

**SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC\_BTR1..4)**

Address offset:  $0xA000\ 0000 + 0x04 + 8 * (x - 1)$ ,  $x = 1..4$

Reset value:  $0x0FFF\ FFFF$

FSMC\_BTRx bits are written by software to add a delay at the end of a read /write transaction. This delay allows matching the minimum time between consecutive transactions ( $t_{EHEL}$  from NEx high to FSMC\_NEx low) and the maximum time required by the memory to free the data bus after a read access ( $t_{EHQZ}$ ).

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR flash memories. If the EXTMOD bit is set in the FSMC\_BCRx register, then this register is partitioned for write and read access, that is, 2 registers are available: one to configure read accesses (this register) and one to configure write accesses (FSMC\_BWTRx registers).

| 31       | 30 | 29          | 28 | 27          | 26 | 25 | 24 | 23          | 22 | 21 | 20 | 19           | 18 | 17 | 16 | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8           | 7  | 6  | 5  | 4           | 3  | 2  | 1  | 0 |
|----------|----|-------------|----|-------------|----|----|----|-------------|----|----|----|--------------|----|----|----|-------------|----|----|----|----|----|----|-------------|----|----|----|-------------|----|----|----|---|
| Reserved |    | ACCMOD[1:0] |    | DATLAT[3:0] |    |    |    | CLKDIV[3:0] |    |    |    | BUSTURN[3:0] |    |    |    | DATAST[7:0] |    |    |    |    |    |    | ADDHLD[3:0] |    |    |    | ADDSET[3:0] |    |    |    |   |
|          |    | rW          | rW | rW          | rW | rW | rW | rW          | rW | rW | rW | rW           | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW          | rW | rW | rW |   |

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:28 **ACCMOD[1:0]**: Access mode

Specifies the asynchronous access modes as shown in the timing diagrams. These bits are taken into account only when the EXTMOD bit in the FSMC\_BCRx register is 1.

- 00: access mode A
- 01: access mode B
- 10: access mode C
- 11: access mode D

Bits 27:24 **DATLAT[3:0]**: Data latency for synchronous memory (see note below bit description table)

For synchronous accesses with read/write burst mode enabled (BURSTEN / CBURSTRW bits set), this field defines the number of memory clock cycles (+2) to issue to the memory before reading/writing the first data. This timing parameter is not expressed in HCLK periods, but in FSMC\_CLK periods. For asynchronous accesses, this value is don't care.

- 0000: Data latency of 2 CLK clock cycles for first burst access
- 1111: Data latency of 17 CLK clock cycles for first burst access (default value after reset)

Bits 23:20 **CLKDIV[3:0]**: Clock divide ratio (for FSMC\_CLK signal)

Defines the period of FSMC\_CLK clock output signal, expressed in number of HCLK cycles:

- 0000: Reserved
- 0001: FSMC\_CLK period =  $2 \times$  HCLK periods
- 0010: FSMC\_CLK period =  $3 \times$  HCLK periods
- 1111: FSMC\_CLK period =  $16 \times$  HCLK periods (default value after reset)

In asynchronous NOR flash, SRAM or PSRAM accesses, this value is don't care.

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

These bits are written by software to add a delay at the end of a write-to-read (and read-to-write) transaction. The programmed bus turnaround delay is inserted between an asynchronous read (muxed or D mode) or a write transaction and any other asynchronous/synchronous read or write to/from a static bank (for a read operation, the bank can be the same or a different one; for a write operation, the bank can be different except in r muxed or D mode).

In some cases, the bus turnaround delay is fixed, whatever the programmed BUSTURN values:

- No bus turnaround delay is inserted between two consecutive asynchronous write transfers to the same static memory bank except in muxed and D mode.
- A bus turnaround delay of 1 FSMC clock cycle is inserted between:
  - Two consecutive asynchronous read transfers to the same static memory bank except for muxed and D modes.
  - An asynchronous read to an asynchronous or synchronous write to any static bank or dynamic bank except for muxed and D modes.
  - An asynchronous (modes 1, 2, A, B or C) read and a read operation from another static bank.
- A bus turnaround delay of 2 FSMC clock cycles is inserted between:
  - Two consecutive synchronous write accesses (in burst or single mode) to the same bank
  - A synchronous write (burst or single) access and an asynchronous write or read transfer to or from static memory bank (the bank can be the same or different in case of a read operation).
  - Two consecutive synchronous read accesses (in burst or single mode) followed by a any synchronous/asynchronous read or write from/to another static memory bank.
- A bus turnaround delay of 3 FSMC clock cycles is inserted between:
  - Two consecutive synchronous write operations (in burst or single mode) to different static banks.
  - A synchronous write access (in burst or single mode) and a synchronous read access from the same or to a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 × HCLK clock cycles (default value after reset)

Bits 15:8 **DATAST[7:0]**: Data-phase duration

These bits are written by software to define the duration of the data phase (refer to [Figure 436](#) to [Figure 448](#)), used in asynchronous accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

For each memory type and access mode data-phase duration, refer to the respective figure ([Figure 436](#) to [Figure 448](#)).

Example: Mode1, write access, DATAST=1: Data-phase duration= DATAST+1 = 2 HCLK clock cycles.

*Note: In synchronous accesses, this value is don't care.*

Bits 7:4 **ADDHLD[3:0]**: Address-hold phase duration

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 445](#) to [Figure 448](#)), used in mode D and multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address-hold phase duration, refer to the respective figure ([Figure 445](#) to [Figure 448](#)).

*Note: In synchronous accesses, this value is not used, the address hold phase is always 1 memory clock period duration.*

Bits 3:0 **ADDSET[3:0]**: Address setup phase duration

These bits are written by software to define the duration of the *address setup* phase (refer to [Figure 436](#) to [Figure 448](#)), used in SRAMs, ROMs and asynchronous NOR flash and PSRAM accesses:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address setup phase duration, refer to the respective figure (refer to [Figure 436](#) to [Figure 448](#)).

*Note: In synchronous NOR flash and PSRAM accesses, this value is don't care.*

**Note:** *PSRAMs (CRAMs) have a variable latency due to internal refresh. Therefore these memories issue the NWAIT signal during the whole latency phase to prolong the latency as needed.*

*With PSRAMs (CRAMs) the DATLAT field must be set to 0, so that the FSMC exits its latency phase soon and starts sampling NWAIT from memory, then starts to read or write when the memory is ready.*

*This method can be used also with the latest generation of synchronous flash memories that issue the NWAIT signal, unlike older flash memories (check the datasheet of the specific flash memory being used).*

**SRAM/NOR-Flash write timing registers 1..4 (FSMC\_BWTR1..4)**

Address offset:  $0xA000\ 0000 + 0x104 + 8 * (x - 1)$ ,  $x = 1...4$

Reset value: 0x0FFF FFFF

This register contains the control information of each memory bank, used for SRAMs, PSRAMs and NOR flash memories. This register is active for write asynchronous access only when the EXTMOD bit is set in the FSMC\_BCRx register.

|      |                 |    |          |    |    |    |    |    |    |    |    |              |    |    |    |             |    |    |    |    |    |    |    |             |    |    |    |             |    |    |    |
|------|-----------------|----|----------|----|----|----|----|----|----|----|----|--------------|----|----|----|-------------|----|----|----|----|----|----|----|-------------|----|----|----|-------------|----|----|----|
| 31   | 30              | 29 | 28       | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19           | 18 | 17 | 16 | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3           | 2  | 1  | 0  |
| Res. | ACCM<br>OD[2:0] |    | Reserved |    |    |    |    |    |    |    |    | BUSTURN[3:0] |    |    |    | DATAST[7:0] |    |    |    |    |    |    |    | ADDHLD[3:0] |    |    |    | ADDSET[3:0] |    |    |    |
|      | rW              | rW |          |    |    |    |    |    |    |    |    | rW           | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW          | rW | rW | rW |

Bits 31:30 Reserved, must be kept at reset value.

Bits 29:28 **ACCMOD[2:0]**: Access mode.

Specifies the asynchronous access modes as shown in the next timing diagrams. These bits are taken into account only when the EXTMOD bit in the FSMC\_BCRx register is 1.

- 00: access mode A
- 01: access mode B
- 10: access mode C
- 11: access mode D

Bits 27:20 Reserved, must be kept at reset value.

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

The programmed bus turnaround delay is inserted between a an asynchronous write transfer and any other asynchronous/synchronous read or write transfer to/from a static bank (for a read operation, the bank can be the same or a different one; for a write operation, the bank can be different except in r muxed or D mode).

In some cases, the bus turnaround delay is fixed, whatever the programmed BUSTURN values:

- No bus turnaround delay is inserted between two consecutive asynchronous write transfers to the same static memory bank except in muxed and D mode.
- A bus turnaround delay of 2 FSMC clock cycles is inserted between:
  - Two consecutive synchronous write accesses (in burst or single mode) to the same bank.
  - A synchronous write transfer (in burst or single mode) and an asynchronous write or read transfer to/from static a memory bank.
- A bus turnaround delay of 3 FSMC clock cycles is inserted between:
  - Two consecutive synchronous write accesses (in burst or single mode) to different static banks.
  - A synchronous write transfer (in burst or single mode) and a synchronous read from the same or from a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 HCLK clock cycles added (default value after reset)

Bits 15:8 **DATAST[7:0]**: Data-phase duration.

These bits are written by software to define the duration of the data phase (refer to [Figure 436](#) to [Figure 448](#)), used in asynchronous SRAM, PSRAM and NOR flash memory accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

*Note: In synchronous accesses, this value is don't care.*

Bits 7:4 **ADDHLD[3:0]**: Address-hold phase duration.

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 445](#) to [Figure 448](#)), used in asynchronous multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

*Note: In synchronous NOR flash accesses, this value is not used, the address hold phase is always 1 flash clock period duration.*

Bits 3:0 **ADDSET[3:0]**: Address setup phase duration.

These bits are written by software to define the duration of the *address setup* phase in HCLK cycles (refer to [Figure 445](#) to [Figure 448](#)), used in asynchronous accessed:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

*Note: In synchronous NOR flash and PSRAM accesses, this value is don't care.*

## 36.6 NAND Flash/PC Card controller

The FSMC generates the appropriate signal timings to drive the following types of device:

- NAND Flash
  - 8-bit
  - 16-bit
- 16-bit PC Card compatible devices

The NAND/PC Card controller can control three external banks. Bank 2 and bank 3 support NAND Flash devices. Bank 4 supports PC Card devices.

Each bank is configured by means of dedicated registers ([Section 36.6.8](#)). The programmable memory parameters include access timings (shown in [Table 247](#)) and ECC configuration.

Table 247. Programmable NAND/PC Card access parameters

| Parameter             | Function                                                                                                                  | Access mode | Unit                   | Min. | Max. |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------|-------------|------------------------|------|------|
| Memory setup time     | Number of clock cycles (HCLK) to set up the address before the command assertion                                          | Read/Write  | AHB clock cycle (HCLK) | 1    | 255  |
| Memory wait           | Minimum duration (HCLK clock cycles) of the command assertion                                                             | Read/Write  | AHB clock cycle (HCLK) | 2    | 256  |
| Memory hold           | Number of clock cycles (HCLK) to hold the address (and the data in case of a write access) after the command de-assertion | Read/Write  | AHB clock cycle (HCLK) | 1    | 254  |
| Memory databus high-Z | Number of clock cycles (HCLK) during which the databus is kept in high-Z state after the start of a write access          | Write       | AHB clock cycle (HCLK) | 0    | 255  |

### 36.6.1 External memory interface signals

The following tables list the signals that are typically used to interface NAND Flash and PC Card.

Note: Prefix "N". specifies the associated signal as active low.

#### 8-bit NAND Flash

Table 248. 8-bit NAND Flash

| FSMC signal name | I/O | Function                                             |
|------------------|-----|------------------------------------------------------|
| A[17]            | O   | NAND Flash address latch enable (ALE) signal         |
| A[16]            | O   | NAND Flash command latch enable (CLE) signal         |
| D[7:0]           | I/O | 8-bit multiplexed, bidirectional address/data bus    |
| NCE[x]           | O   | Chip select, x = 2, 3                                |
| NOE(= NRE)       | O   | Output enable (memory signal name: read enable, NRE) |
| NWE              | O   | Write enable                                         |
| NWAIT/INT[3:2]   | I   | NAND Flash ready/busy input signal to the FSMC       |

There is no theoretical capacity limitation as the FSMC can manage as many address cycles as needed.

**16-bit NAND Flash****Table 249. 16-bit NAND Flash**

| FSMC signal name | I/O | Function                                             |
|------------------|-----|------------------------------------------------------|
| A[17]            | O   | NAND Flash address latch enable (ALE) signal         |
| A[16]            | O   | NAND Flash command latch enable (CLE) signal         |
| D[15:0]          | I/O | 16-bit multiplexed, bidirectional address/data bus   |
| NCE[x]           | O   | Chip select, x = 2, 3                                |
| NOE(= NRE)       | O   | Output enable (memory signal name: read enable, NRE) |
| NWE              | O   | Write enable                                         |
| NWAIT/INT[3:2]   | I   | NAND Flash ready/busy input signal to the FSMC       |

*There is no theoretical capacity limitation as the FSMC can manage as many address cycles as needed.*

**16-bit PC Card****Table 250. 16-bit PC Card**

| FSMC signal name | I/O | Function                                                                                                                                                                                 |
|------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A[10:0]          | O   | Address bus                                                                                                                                                                              |
| NIORD            | O   | Output enable for I/O space                                                                                                                                                              |
| NIOWR            | O   | Write enable for I/O space                                                                                                                                                               |
| NREG             | O   | Register signal indicating if access is in Common or Attribute space                                                                                                                     |
| D[15:0]          | I/O | Bidirectional databus                                                                                                                                                                    |
| NCE4_1           | O   | Chip select 1                                                                                                                                                                            |
| NCE4_2           | O   | Chip select 2 (indicates if access is 16-bit or 8-bit)                                                                                                                                   |
| NOE              | O   | Output enable in Common and in Attribute space                                                                                                                                           |
| NWE              | O   | Write enable in Common and in Attribute space                                                                                                                                            |
| NWAIT            | I   | PC Card wait input signal to the FSMC (memory signal name IORDY)                                                                                                                         |
| INTR             | I   | PC Card interrupt to the FSMC (only for PC Cards that can generate an interrupt)                                                                                                         |
| CD               | I   | PC Card presence detection. Active high. If an access is performed to the PC Card banks while CD is low, an AHB error is generated. Refer to <a href="#">Section 36.3: AHB interface</a> |

### 36.6.2 NAND Flash / PC Card supported memories and transactions

*Table 251 below shows the supported devices, access modes and transactions. Transactions not allowed (or not supported) by the NAND Flash / PC Card controller appear in gray.*

**Table 251. Supported memories and transactions**

| Device      | Mode         | R/W | AHB data size | Memory data size | Allowed/not allowed | Comments                   |
|-------------|--------------|-----|---------------|------------------|---------------------|----------------------------|
| NAND 8-bit  | Asynchronous | R   | 8             | 8                | Y                   | -                          |
|             | Asynchronous | W   | 8             | 8                | Y                   | -                          |
|             | Asynchronous | R   | 16            | 8                | Y                   | Split into 2 FSMC accesses |
|             | Asynchronous | W   | 16            | 8                | Y                   | Split into 2 FSMC accesses |
|             | Asynchronous | R   | 32            | 8                | Y                   | Split into 4 FSMC accesses |
|             | Asynchronous | W   | 32            | 8                | Y                   | Split into 4 FSMC accesses |
| NAND 16-bit | Asynchronous | R   | 8             | 16               | Y                   | -                          |
|             | Asynchronous | W   | 8             | 16               | N                   | -                          |
|             | Asynchronous | R   | 16            | 16               | Y                   | -                          |
|             | Asynchronous | W   | 16            | 16               | Y                   | -                          |
|             | Asynchronous | R   | 32            | 16               | Y                   | Split into 2 FSMC accesses |
|             | Asynchronous | W   | 32            | 16               | Y                   | Split into 2 FSMC accesses |

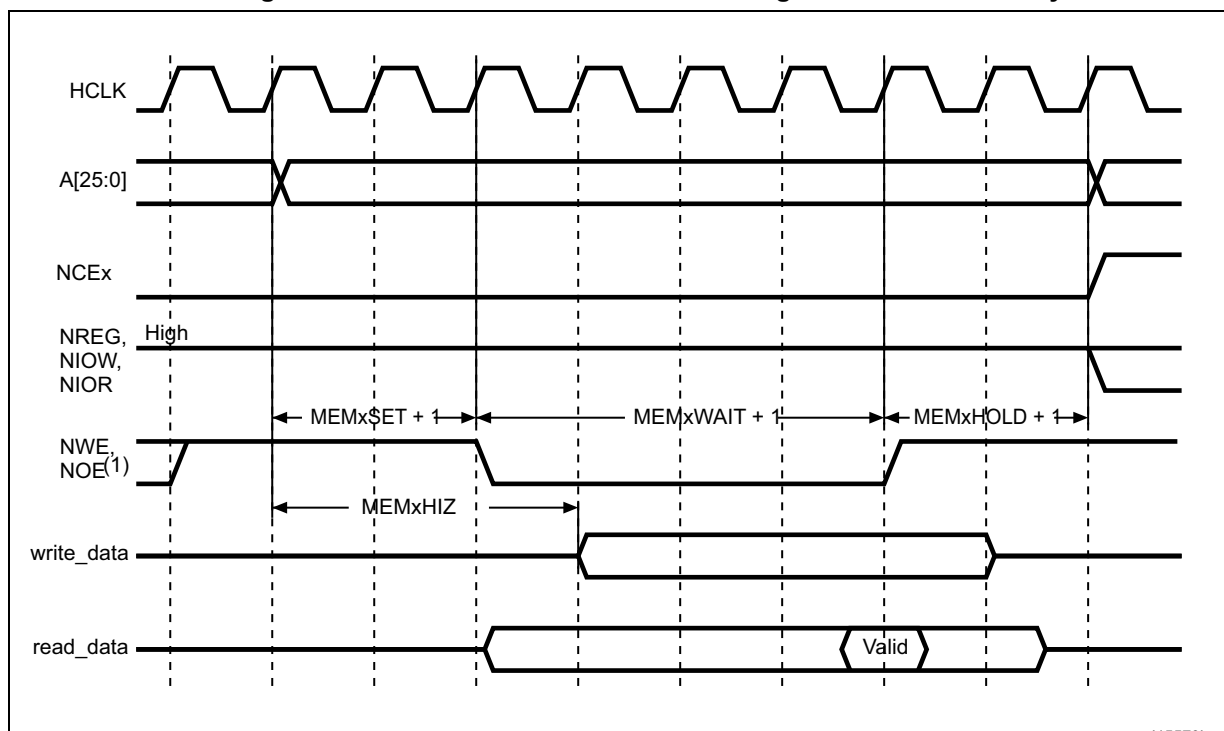
### 36.6.3 Timing diagrams for NAND and PC Card

Each PC Card/CompactFlash and NAND Flash memory bank is managed through a set of registers:

- Control register: FSMC\_PCRx
- Interrupt status register: FSMC\_SRx
- ECC register: FSMC\_ECCRx
- Timing register for Common memory space: FSMC\_PMEMx
- Timing register for Attribute memory space: FSMC\_PATTx
- Timing register for I/O space: FSMC\_PIOx

Each timing configuration register contains three parameters used to define number of HCLK cycles for the three phases of any PC Card/CompactFlash or NAND Flash access, plus one parameter that defines the timing for starting driving the databus in the case of a write. [Figure 454](#) shows the timing parameter definitions for common memory accesses, knowing that Attribute and I/O (only for PC Card) memory space access timings are similar.

Figure 454. NAND/PC Card controller timing for common memory access



1. NOE remains high (inactive) during write access. NWE remains high (inactive) during read access.
2. For write accesses, the hold phase delay is  $(MEMxHOLD) \times HCLK$  cycles, while it is  $(MEMxHOLD + 2) \times HCLK$  cycles for read accesses.

### 36.6.4 NAND Flash operations

The command latch enable (CLE) and address latch enable (ALE) signals of the NAND Flash device are driven by some address signals of the FSMC controller. This means that to send a command or an address to the NAND Flash memory, the CPU has to perform a write to a certain address in its memory space.

A typical page read operation from the NAND Flash device is as follows:

1. Program and enable the corresponding memory bank by configuring the FSMC\_PCRx and FSMC\_PMEMx (and for some devices, FSMC\_PATTx, see [Section 36.6.5](#)) registers according to the characteristics of the NAND Flash (PWID bits for the databus width of the NAND Flash, PTYP = 1, PWAITEN = 0 or 1 as needed, see [Common memory space timing register 2.4 \(FSMC\\_PMEM2.4\)](#) for timing configuration).
2. The CPU performs a byte write in the common memory space, with data byte equal to one Flash command byte (for example 0x00 for Samsung NAND Flash devices). The CLE input of the NAND Flash is active during the write strobe (low pulse on NWE), thus the written byte is interpreted as a command by the NAND Flash. Once the command is latched by the NAND Flash device, it does not need to be written for the following page read operations.
3. The CPU can send the start address (STARTAD) for a read operation by writing the required bytes (for example four bytes or three for smaller capacity devices), STARTAD[7:0], STARTAD[15:8], STARTAD[23:16] and finally STARTAD[25:24] for 64 Mb x 8 bit NAND Flash) in the common memory or attribute space. The ALE input of the NAND Flash device is active during the write strobe (low pulse on NWE), thus the

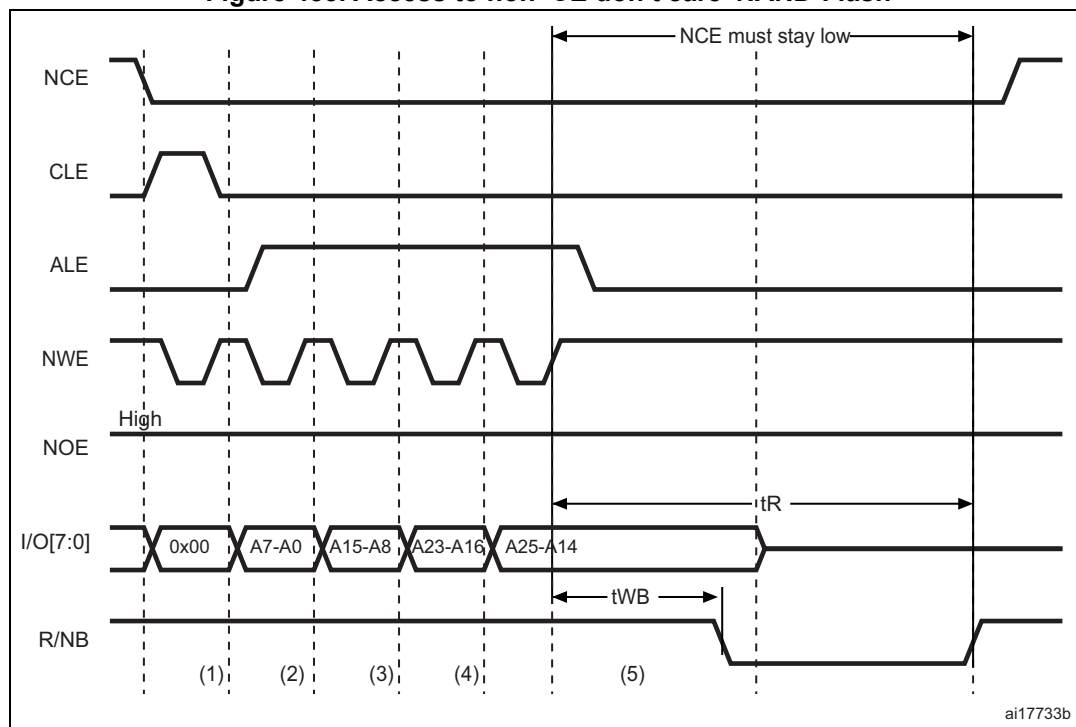
written bytes are interpreted as the start address for read operations. Using the attribute memory space makes it possible to use a different timing configuration of the FSMC, which can be used to implement the prewait functionality needed by some NAND Flash memories (see details in [Section 36.6.5](#)).

4. The controller waits for the NAND Flash to be ready (R/NB signal high) to become active, before starting a new access (to same or another memory bank). While waiting, the controller maintains the NCE signal active (low).
5. The CPU can then perform byte read operations in the common memory space to read the NAND Flash page (data field + Spare field) byte by byte.
6. The next NAND Flash page can be read without any CPU command or address write operation, in three different ways:
  - by simply performing the operation described in step 5
  - a new random address can be accessed by restarting the operation at step 3
  - a new command can be sent to the NAND Flash device by restarting at step 2

### 36.6.5 NAND Flash prewait functionality

Some NAND Flash devices require that, after writing the last part of the address, the controller wait for the R/NB signal to go low as shown in [Figure 455](#).

**Figure 455. Access to non 'CE don't care' NAND-Flash**



1. CPU wrote byte 0x00 at address 0x7001 0000.
2. CPU wrote byte A7-A0 at address 0x7002 0000.
3. CPU wrote byte A15-A8 at address 0x7002 0000.
4. CPU wrote byte A23-A16 at address 0x7002 0000.
5. CPU wrote byte A25-A24 at address 0x7802 0000: FSMC performs a write access using FSMC\_PATT2 timing definition, where  $ATTHOLD \geq 7$  (providing that  $(7+1) \times HCLK = 112 \text{ ns} > t_{WB \text{ max}}$ ). This guarantees that NCE remains low until R/NB goes low and high again (only requested for NAND Flash memories where NCE is not don't care).

When this functionality is needed, it can be guaranteed by programming the MEMHOLD value to meet the  $t_{\text{WB}}$  timing. However CPU read accesses to the NAND Flash memory has a hold delay of  $(\text{MEMHOLD} + 2) \times \text{HCLK}$  cycles, while CPU write accesses have a hold delay of  $(\text{MEMHOLD}) \times \text{HCLK}$  cycles.

To overcome this timing constraint, the attribute memory space can be used by programming its timing register with an ATTHOLD value that meets the  $t_{\text{WB}}$  timing, and leaving the MEMHOLD value at its minimum. Then, the CPU must use the common memory space for all NAND Flash read and write accesses, except when writing the last address byte to the NAND Flash device, where the CPU must write to the attribute memory space.

### 36.6.6 Computation of the error correction code (ECC) in NAND Flash memory

The FSMC PC-Card controller includes two error correction code computation hardware blocks, one per memory bank. They are used to reduce the host CPU workload when processing the error correction code by software in the system.

These two registers are identical and associated with bank 2 and bank 3, respectively. As a consequence, no hardware ECC computation is available for memories connected to bank 4.

The error correction code (ECC) algorithm implemented in the FSMC can perform 1-bit error correction and 2-bit error detection per 256, 512, 1 024, 2 048, 4 096 or 8 192 bytes read from or written to NAND Flash memory. It is based on the Hamming coding algorithm and consists in calculating the row and column parity.

The ECC modules monitor the NAND Flash databus and read/write signals (NCE and NWE) each time the NAND Flash memory bank is active.

The functional operations are:

- When access to NAND Flash is made to bank 2 or bank 3, the data present on the D[15:0] bus is latched and used for ECC computation.
- When access to NAND Flash occurs at any other address, the ECC logic is idle, and does not perform any operation. Thus, write operations for defining commands or addresses to NAND Flash are not taken into account for ECC computation.

Once the desired number of bytes has been read from/written to the NAND Flash by the host CPU, the FSMC\_ECCR2/3 registers must be read in order to retrieve the computed value. Once read, they should be cleared by resetting the ECCEN bit to zero. To compute a new data block, the ECCEN bit must be set to one in the FSMC\_PCR2/3 registers.

To perform an ECC computation:

1. Enable the ECCEN bit in the FSMC\_PCR2/3 register.
2. Write data to the NAND Flash memory page. While the NAND page is written, the ECC block computes the ECC value.
3. Read the ECC value available in the FSMC\_ECCR2/3 register and store it in a variable.
4. Clear the ECCEN bit and then enable it in the FSMC\_PCR2/3 register before reading back the written data from the NAND page. While the NAND page is read, the ECC block computes the ECC value.
5. Read the new ECC value available in the FSMC\_ECCR2/3 register.
6. If the two ECC values are the same, no correction is required, otherwise there is an ECC error and the software correction routine returns information on whether the error can be corrected or not.

### 36.6.7 PC Card/CompactFlash operations

#### Address spaces and memory accesses

The FSMC supports Compact Flash storage or PC Cards in Memory Mode and I/O Mode (True IDE mode is not supported).

The Compact Flash storage and PC Cards are made of 3 memory spaces:

- Common Memory Space
- Attribute Space
- I/O Memory Space

The nCE2 and nCE1 pins (FSMC\_NCE4\_2 and FSMC\_NCE4\_1 respectively) select the card and indicate whether a byte or a word operation is being performed: nCE2 accesses the odd byte on D15-8 and nCE1 accesses the even byte on D7-0 if A0=0 or the odd byte on D7-0 if A0=1. The full word is accessed on D15-0 if both nCE2 and nCE1 are low.

The memory space is selected by asserting low nOE for read accesses or nWE for write accesses, combined with the low assertion of nCE2/nCE1 and nREG.

- If pin nREG=1 during the memory access, the common memory space is selected
- If pin nREG=0 during the memory access, the attribute memory space is selected

The I/O Space is selected by asserting low nIORD for read accesses or nIOWR for write accesses [instead of nOE/nWE for memory Space], combined with nCE2/nCE1. Note that nREG must also be asserted low during accesses to I/O Space.

Three type of accesses are allowed for a 16-bit PC Card:

- Accesses to Common Memory Space for data storage can be either 8-bit accesses at even addresses or 16 bit AHB accesses.  
Note that 8-bit accesses at odd addresses are not supported and do not lead to the low assertion of nCE2. A 32-bit AHB request is translated into two 16-bit memory accesses.
- Accesses to Attribute Memory Space where the PC Card stores configuration information are limited to 8-bit AHB accesses at even addresses.  
Note that a 16-bit AHB access is converted into a single 8-bit memory transfer: nCE1 is asserted low, nCE2 is asserted high and only the even Byte on D7-D0 is valid. Instead

a 32-bit AHB access is converted into two 8-bit memory transfers at even addresses: nCE1 is asserted low, NCE2 is asserted high and only the even bytes are valid.

- Accesses to I/O Space can be performed either through AHB 8-bit or 16-bit accesses.

**Table 252. 16-bit PC-Card signals and access type**

| nCE2 | nCE1 | nREG | nOE/nWE | nIORD /nIOWR | A10 | A9 | A7-1 | A0 | Space               | Access Type                                    | Allowed/not Allowed |
|------|------|------|---------|--------------|-----|----|------|----|---------------------|------------------------------------------------|---------------------|
| 1    | 0    | 1    | 0       | 1            | X   | X  | X-X  | X  | Common Memory Space | Read/Write byte on D7-D0                       | YES                 |
| 0    | 1    | 1    | 0       | 1            | X   | X  | X-X  | X  |                     | Read/Write byte on D15-D8                      | Not supported       |
| 0    | 0    | 1    | 0       | 1            | X   | X  | X-X  | 0  |                     | Read/Write word on D15-D0                      | YES                 |
| X    | 0    | 0    | 0       | 1            | 0   | 1  | X-X  | 0  | Attribute Space     | Read or Write Configuration registers          | YES                 |
| X    | 0    | 0    | 0       | 1            | 0   | 0  | X-X  | 0  |                     | Read or Write CIS (Card Information Structure) | YES                 |
| 1    | 0    | 0    | 0       | 1            | X   | X  | X-X  | 1  | Attribute Space     | Invalid Read or Write (odd address)            | YES                 |
| 0    | 1    | 0    | 0       | 1            | X   | X  | X-X  | x  |                     | Invalid Read or Write (odd address)            | YES                 |
| 1    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 0  | I/O space           | Read Even Byte on D7-0                         | YES                 |
| 1    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 1  |                     | Read Odd Byte on D7-0                          | YES                 |
| 1    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 0  |                     | Write Even Byte on D7-0                        | YES                 |
| 1    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 1  |                     | Write Odd Byte on D7-0                         | YES                 |
| 0    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 0  |                     | Read Word on D15-0                             | YES                 |
| 0    | 0    | 0    | 1       | 0            | X   | X  | X-X  | 0  |                     | Write word on D15-0                            | YES                 |
| 0    | 1    | 0    | 1       | 0            | X   | X  | X-X  | X  |                     | Read Odd Byte on D15-8                         | Not supported       |
| 0    | 1    | 0    | 1       | 0            | X   | X  | X-X  | X  |                     | Write Odd Byte on D15-8                        | Not supported       |

The FSMC Bank 4 gives access to those 3 memory spaces as described in [Section 36.4.2: NAND/PC Card address mapping](#) and [Table 219: Memory mapping and timing registers](#).

### Wait Feature

The CompactFlash Storage or PC Card may request the FSMC to extend the length of the access phase programmed by MEMWAITx/ATTWAITx/IOWAITx bits, asserting the nWAIT signal after nOE/nWE or nIORD/nIOWR activation if the wait feature is enabled through the PWAITEN bit in the FSMC\_PCRx register. In order to detect the nWAIT assertion correctly, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed as follows:

$xxWAITx \geq 4 + \text{max\_wait\_assertion\_time}/HCLK$

Where max\_wait\_assertion\_time is the maximum time taken by NWAIT to go low once nOE/nWE or nIORD/nIOWR is low.

After the de-assertion of nWAIT, the FSMC extends the WAIT phase for 4 HCLK clock cycles.

### 36.6.8 NAND Flash/PC Card control registers

The NAND Flash/PC Card control registers have to be accessed by words (32 bits).

#### PC Card/NAND Flash control registers 2..4 (FSMC\_PCR2..4)

Address offset:  $0xA0000000 + 0x40 + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0x0000 0018

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19         | 18  | 17  | 16       | 15  | 14  | 13  | 12        | 11  | 10  | 9   | 8    | 7     | 6         | 5   | 4    | 3     | 2       | 1        | 0   |
|----------|----|----|----|----|----|----|----|----|----|----|----|------------|-----|-----|----------|-----|-----|-----|-----------|-----|-----|-----|------|-------|-----------|-----|------|-------|---------|----------|-----|
| Reserved |    |    |    |    |    |    |    |    |    |    |    | ECCPS[2:0] |     |     | TAR[2:0] |     |     |     | TCLR[2:0] |     |     |     | Res. | ECCEN | PWID[1:0] |     | PTYP | PBKEN | PWAITEN | Reserved |     |
|          |    |    |    |    |    |    |    |    |    |    |    | r/w        | r/w | r/w | r/w      | r/w | r/w | r/w | r/w       | r/w | r/w | r/w |      | r/w   | r/w       | r/w | r/w  | r/w   | r/w     |          | r/w |

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:17 **ECCPS[2:0]**: ECC page size.

Defines the page size for the extended ECC:

000: 256 bytes  
 001: 512 bytes  
 010: 1024 bytes  
 011: 2048 bytes  
 100: 4096 bytes  
 101: 8192 bytes

Bits 16:13 **TAR[2:0]**: ALE to RE delay.

Sets time from ALE low to RE low in number of AHB clock cycles (HCLK).

Time is:  $t_{ar} = (TAR + SET + 2) \times THCLK$  where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)  
 1111: 16 HCLK cycles

*Note: SET is MEMSET or ATTSET according to the addressed space.*

Bits 12:9 **TCLR[2:0]**: CLE to RE delay.

Sets time from CLE low to RE low in number of AHB clock cycles (HCLK).

Time is:  $t_{clr} = (TCLR + SET + 2) \times THCLK$  where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)  
 1111: 16 HCLK cycles

*Note: SET is MEMSET or ATTSET according to the addressed space.*

Bits 8:7 Reserved, must be kept at reset value.

Bit 6 **ECCEN**: ECC computation logic enable bit

0: ECC logic is disabled and reset (default after reset),  
 1: ECC logic is enabled.

Bits 5:4 **PWID[1:0]**: Databus width.

Defines the external memory device width.

00: 8 bits

01: 16 bits (default after reset). This value is mandatory for PC Cards.

10: reserved, do not use

11: reserved, do not use

Bit 3 **PTYP**: Memory type.

Defines the type of device attached to the corresponding memory bank:

0: PC Card, CompactFlash, CF+ or PCMCIA

1: NAND Flash (default after reset)

Bit 2 **PBKEN**: PC Card/NAND Flash memory bank enable bit.

Enables the memory bank. Accessing a disabled memory bank causes an ERROR on AHB bus

0: Corresponding memory bank is disabled (default after reset)

1: Corresponding memory bank is enabled

Bit 1 **PWAITEN**: Wait feature enable bit.

Enables the Wait feature for the PC Card/NAND Flash memory bank:

0: disabled

1: enabled

*Note: For a PC Card, when the wait feature is enabled, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed to a value as follows:*

$xxWAITx \geq 4 + \frac{\text{max\_wait\_assertion\_time}}{HCLK}$

*Where max\_wait\_assertion\_time is the maximum time taken by NWAIT to go low once nOE/nWE or nIORD/nIOWR is low.*

Bit 0 Reserved, must be kept at reset value.

## FIFO status and interrupt register 2..4 (FSMC\_SR2..4)

Address offset:  $0xA000\ 0000 + 0x44 + 0x20 * (x-1)$ ,  $x = 2..4$

Reset value: 0x0000 0040

This register contains information about FIFO status and interrupt. The FSMC has a FIFO that is used when writing to memories to store up to 16 words of data from the AHB.

This is used to quickly write to the AHB and free it for transactions to peripherals other than the FSMC, while the FSMC is draining its FIFO into the memory. This register has one of its bits that indicates the status of the FIFO, for ECC purposes.

The ECC is calculated while the data are written to the memory, so in order to read the correct ECC the software must wait until the FIFO is empty.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18    | 17 | 16   | 15   | 14   | 13  | 12  | 11  | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|-------|----|------|------|------|-----|-----|-----|----|---|---|---|---|---|---|---|---|---|---|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    | FEMPT |    | IFEN | ILEN | IREN | IFS | ILS | IRS |    |   |   |   |   |   |   |   |   |   |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    | r     | rw | rw   | rw   | rw   | rw  | rw  | rw  |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:7 Reserved, must be kept at reset value.

Bit 6 **FEMPT**: FIFO empty.

Read-only bit that provides the status of the FIFO  
0: FIFO not empty  
1: FIFO empty

Bit 5 **IFEN**: Interrupt falling edge detection enable bit

0: Interrupt falling edge detection request disabled  
1: Interrupt falling edge detection request enabled

Bit 4 **ILEN**: Interrupt high-level detection enable bit

0: Interrupt high-level detection request disabled  
1: Interrupt high-level detection request enabled

Bit 3 **IREN**: Interrupt rising edge detection enable bit

0: Interrupt rising edge detection request disabled  
1: Interrupt rising edge detection request enabled

Bit 2 **IFS**: Interrupt falling edge status

The flag is set by hardware and reset by software.  
0: No interrupt falling edge occurred  
1: Interrupt falling edge occurred

*Note: This bit is set by programming it to 1 by software.*

Bit 1 **ILS**: Interrupt high-level status

The flag is set by hardware and reset by software.  
0: No Interrupt high-level occurred  
1: Interrupt high-level occurred

Bit 0 **IRS**: Interrupt rising edge status

The flag is set by hardware and reset by software.  
0: No interrupt rising edge occurred  
1: Interrupt rising edge occurred

*Note: This bit is set by programming it to 1 by software.*

### Common memory space timing register 2..4 (FSMC\_PMEM2..4)

Address offset: Address:  $0xA000\ 0000 + 0x48 + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0xFCFC FCFC

Each FSMC\_PMEMx ( $x = 2..4$ ) read/write register contains the timing information for PC Card or NAND Flash memory bank x, used for access to the common memory space of the 16-bit PC Card/CompactFlash, or to access the NAND Flash for command, address write access and data read/write access.

|             |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| MEMHIZ[7:0] |    |    |    |    |    |    |    | MEMHOLD[7:0] |    |    |    |    |    |    |    | MEMWAIT[7:0] |    |    |    |    |    |    |    | MEMSET[7:0] |    |    |    |    |    |    |    |
| rW          | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW |

Bits 31:24 **MEMHIZx[7:0]**: Common memory x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC Card/NAND Flash write access to common memory space on socket x. Only valid for write transaction:

0000 0000: 1 HCLK cycle  
 1111 1110: 255 HCLK cycles  
 1111 1111: Reserved

Bits 23:16 **MEMHOLDx[7:0]**: Common memory x hold time

For NAND Flash read accesses to the common memory space, these bits define the number of (HCLK+2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For NAND Flash write accesses to the common memory space, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: Reserved  
 0000 0001: 1 HCLK cycle for write accesses, 3 HCLK cycles for read accesses  
 1111 1110: 254 HCLK cycle for write accesses, 256 HCLK cycles for read accesses  
 1111 1111: Reserved

Bits 15:8 **MEMWAITx[7:0]**: Common memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND Flash read or write access to common memory space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: Reserved  
 0000 0001: 2 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)  
 1111 1110: 255 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)  
 1111 1111: Reserved.

Bits 7:0 **MEMSETx[7:0]**: Common memory x setup time

Defines the number of HCLK ( ) clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card/NAND Flash read or write access to common memory space on socket x:

0000 0000: 1 HCLK cycle  
 1111 1110: 255 HCLK cycles  
 1111 1111: Reserved

## Attribute memory space timing registers 2..4 (FSMC\_PATT2..4)

Address offset:  $0xA000\ 0000 + 0x4C + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0xFCFC FCFC

Each FSMC\_PATTx ( $x = 2..4$ ) read/write register contains the timing information for PC Card/CompactFlash or NAND Flash memory bank x. It is used for 8-bit accesses to the attribute memory space of the PC Card/CompactFlash or to access the NAND Flash for the last address write access if the timing must differ from that of previous accesses (for Ready/Busy management, refer to [Section 36.6.5: NAND Flash prewait functionality](#)).

|             |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| ATTHIZ[7:0] |    |    |    |    |    |    |    | ATTHOLD[7:0] |    |    |    |    |    |    |    | ATTWAIT[7:0] |    |    |    |    |    |    |    | ATTSET[7:0] |    |    |    |    |    |    |    |
| rW          | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW |

Bits 31:24 **ATT Hiz[7:0]**: Attribute memory x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC CARD/NAND Flash write access to attribute memory space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

Bits 23:16 **ATT HOLD[7:0]**: Attribute memory x hold time

For PC Card/NAND Flash read accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles (HCLK +2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For PC Card/NAND Flash write accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: reserved

0000 0001: 1 HCLK cycle for write access, 3 HCLK cycles for read accesses

1111 1110: 254 HCLK cycle for write access, 256 HCLK cycles for read accesses

1111 1111: Reserved

Bits 15:8 **ATT WAIT[7:0]**: Attribute memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND Flash read or write access to attribute memory space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: Reserved

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1111: 255 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)

1111 1111: Reserved.

Bits 7:0 **ATT SET[7:0]**: Attribute memory x setup time

Defines the number of HCLK (+1) clock cycles to set up address before the command assertion (NWE, NOE), for PC CARD/NAND Flash read or write access to attribute memory space on socket x:

0000 0000: 1 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

**I/O space timing register 4 (FSMC\_PIO4)**

Address offset: 0xA000 0000 + 0xB0

Reset value: 0xFCFCFCFC

The FSMC\_PIO4 read/write registers contain the timing information used to gain access to the I/O space of the 16-bit PC Card/CompactFlash.

|            |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |            |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23          | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| IOHIZ[7:0] |    |    |    |    |    |    |    | IOHOLD[7:0] |    |    |    |    |    |    |    | IOWAIT[7:0] |    |    |    |    |    |    |    | IOSET[7:0] |    |    |    |    |    |    |    |
| rW         | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW | rW         | rW | rW | rW | rW | rW | rW | rW |

Bits 31:24 **IOHIZ[7:0]**: I/O x databus HiZ time

Defines the number of HCLK clock cycles during which the databus is kept in HiZ after the start of a PC Card write access to I/O space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1111: 255 HCLK cycles (default value after reset)

Bits 23:16 **IOHOLD[7:0]**: I/O x hold time

Defines the number of HCLK clock cycles to hold address (and data for write access) after the command deassertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: reserved

0000 0001: 1 HCLK cycle

1111 1111: 255 HCLK cycles (default value after reset)

Bits 15:8 **IOWAIT[7:0]**: I/O x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (SMNWE, SMNOE), for PC Card read or write access to I/O space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: reserved, do not use this value

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1111: 256 HCLK cycles (+ wait cycle introduced by the Card deasserting NWAIT) (default value after reset)

Bits 7:0 **IOSET[7:0]**: I/O x setup time

Defines the number of HCLK (+1) clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: 1 HCLK cycle

1111 1111: 256 HCLK cycles (default value after reset)

**ECC result registers 2/3 (FSMC\_ECCR2/3)**

Address offset:  $0xA000\ 0000 + 0x54 + 0x20 * (x - 1)$ ,  $x = 2$  or  $3$

Reset value: 0x0000 0000

These registers contain the current error correction code value computed by the ECC computation modules of the FSMC controller (one module per NAND Flash memory bank). When the CPU reads the data from a NAND Flash memory page at the correct address (refer to [Section 36.6.6: Computation of the error correction code \(ECC\) in NAND Flash memory](#)), the data read from or written to the NAND Flash are processed automatically by ECC computation module. At the end of X bytes read (according to the ECCPS field in the FSMC\_PCRx registers), the CPU must read the computed ECC value from the FSMC\_ECCx registers, and then verify whether these computed parity data are the same as the parity value recorded in the spare area, to determine whether a page is valid, and, to correct it if applicable. The FSMC\_ECCRx registers should be cleared after being read by setting the ECCEN bit to zero. For computing a new data block, the ECCEN bit must be set to one.

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| ECCx[31:0] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:0 **ECCx[31:0]**: ECC result

This field provides the value computed by the ECC computation logic. [Table 253](#) hereafter describes the contents of these bit fields.

**Table 253. ECC result relevant bits**

| <b>ECCPS[2:0]</b> | <b>Page size in bytes</b> | <b>ECC bits</b> |
|-------------------|---------------------------|-----------------|
| 000               | 256                       | ECC[21:0]       |
| 001               | 512                       | ECC[23:0]       |
| 010               | 1024                      | ECC[25:0]       |
| 011               | 2048                      | ECC[27:0]       |
| 100               | 4096                      | ECC[29:0]       |
| 101               | 8192                      | ECC[31:0]       |

### 36.6.9 FSMC register map

The following table summarizes the FSMC registers.

**Table 254. FSMC register map**

| Offset | Register               | 31       | 30                  | 29 | 28 | 27          | 26 | 25 | 24          | 23 | 22 | 21           | 20           | 19       | 18          | 17          | 16 | 15         | 14     | 13     | 12   | 11          | 10          | 9       | 8           | 7           | 6      | 5         | 4 | 3         | 2 | 1     | 0     |
|--------|------------------------|----------|---------------------|----|----|-------------|----|----|-------------|----|----|--------------|--------------|----------|-------------|-------------|----|------------|--------|--------|------|-------------|-------------|---------|-------------|-------------|--------|-----------|---|-----------|---|-------|-------|
| 0000   | FSMC_BCR1              | Reserved |                     |    |    |             |    |    |             |    |    |              |              | CBURSTRW | CPSIZE[2:0] |             |    | ASYNCAWAIT | EXTMOD | WAITEN | WREN | WAITCFG     | WRAPMOD     | WAITPOL | BURSTEN     | Reserved    | FACCEN | MWID[1:0] |   | MTYP[0:1] |   | MUXEN | MBKEN |
| 0008   | FSMC_BCR2              | Reserved |                     |    |    |             |    |    |             |    |    |              |              | CBURSTRW | CPSIZE[2:0] |             |    | ASYNCAWAIT | EXTMOD | WAITEN | WREN | WAITCFG     | WRAPMOD     | WAITPOL | BURSTEN     | Reserved    | FACCEN | MWID[1:0] |   | MTYP[0:1] |   | MUXEN | MBKEN |
| 0010   | FSMC_BCR3              | Reserved |                     |    |    |             |    |    |             |    |    |              |              | CBURSTRW | CPSIZE[2:0] |             |    | ASYNCAWAIT | EXTMOD | WAITEN | WREN | WAITCFG     | WRAPMOD     | WAITPOL | BURSTEN     | Reserved    | FACCEN | MWID[1:0] |   | MTYP[0:1] |   | MUXEN | MBKEN |
| 0018   | FSMC_BCR4              | Reserved |                     |    |    |             |    |    |             |    |    |              |              | CBURSTRW | CPSIZE[2:0] |             |    | ASYNCAWAIT | EXTMOD | WAITEN | WREN | WAITCFG     | WRAPMOD     | WAITPOL | BURSTEN     | Reserved    | FACCEN | MWID[1:0] |   | MTYP[0:1] |   | MUXEN | MBKEN |
| 0004   | FSMC_BTR1              | Res.     | ACCMOD[1:0]         |    |    | DATLAT[3:0] |    |    | CLKDIV[3:0] |    |    | BUSTURN[3:0] |              |          | DATAST[7:0] |             |    |            |        |        |      | ADDHLD[3:0] |             |         | ADDSET[3:0] |             |        |           |   |           |   |       |       |
| 000C   | FSMC_BTR2              | Res.     | ACCMOD[1:0]         |    |    | DATLAT[3:0] |    |    | CLKDIV[3:0] |    |    | BUSTURN[3:0] |              |          | DATAST[7:0] |             |    |            |        |        |      | ADDHLD[3:0] |             |         | ADDSET[3:0] |             |        |           |   |           |   |       |       |
| 0014   | FSMC_BTR3              | Res.     | ACCMOD[1:0]         |    |    | DATLAT[3:0] |    |    | CLKDIV[3:0] |    |    | BUSTURN[3:0] |              |          | DATAST[7:0] |             |    |            |        |        |      | ADDHLD[3:0] |             |         | ADDSET[3:0] |             |        |           |   |           |   |       |       |
| 001C   | FSMC_BTR4              | Res.     | ACCMOD[1:0]         |    |    | DATLAT[3:0] |    |    | CLKDIV[3:0] |    |    | BUSTURN[3:0] |              |          | DATAST[7:0] |             |    |            |        |        |      | ADDHLD[3:0] |             |         | ADDSET[3:0] |             |        |           |   |           |   |       |       |
| 0104   | FSMC_BWTR <sub>1</sub> | Res.     | ACC<br>MOD<br>[1:0] |    |    | Res.        |    |    |             |    |    |              | BUSTURN[3:0] |          |             | DATAST[7:0] |    |            |        |        |      |             | ADDHLD[3:0] |         |             | ADDSET[3:0] |        |           |   |           |   |       |       |
| 010C   | FSMC_BWTR <sub>2</sub> | Res.     | ACC<br>MOD<br>[1:0] |    |    | Res.        |    |    |             |    |    |              | BUSTURN[3:0] |          |             | DATAST[7:0] |    |            |        |        |      |             | ADDHLD[3:0] |         |             | ADDSET[3:0] |        |           |   |           |   |       |       |

Table 254. FSMC register map (continued)

| Offset         | Register       | 31          | 30 | 29                  | 28 | 27   | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19           | 18 | 17       | 16 | 15           | 14        | 13 | 12 | 11   | 10 | 9     | 8           | 7           | 6    | 5           | 4       | 3        | 2   | 1   | 0 |
|----------------|----------------|-------------|----|---------------------|----|------|----|----|----|--------------|----|----|----|--------------|----|----------|----|--------------|-----------|----|----|------|----|-------|-------------|-------------|------|-------------|---------|----------|-----|-----|---|
| 0114           | FSMC_BWTR<br>3 | Res.        |    | ACC<br>MOD<br>[1:0] |    | Res. |    |    |    |              |    |    |    | BUSTURN[3:0] |    |          |    | DATAST[7:0]  |           |    |    |      |    |       | ADDHLD[3:0] |             |      | ADDSET[3:0] |         |          |     |     |   |
| 011C           | FSMC_BWTR<br>4 | Res.        |    | ACC<br>MOD<br>[1:0] |    | Res. |    |    |    |              |    |    |    | BUSTURN[3:0] |    |          |    | DATAST[7:0]  |           |    |    |      |    |       | ADDHLD[3:0] |             |      | ADDSET[3:0] |         |          |     |     |   |
| 0xA000<br>0060 | FSMC_PCR2      | Reserved    |    |                     |    |      |    |    |    |              |    |    |    | ECCPS[2:0]   |    | TAR[2:0] |    |              | TCLR[2:0] |    |    | Res. |    | ECCEN | PWID[1:0]   |             | PTYP | PBKEN       | PWAITEN | Reserved |     |     |   |
| 0xA000<br>0080 | FSMC_PCR3      | Reserved    |    |                     |    |      |    |    |    |              |    |    |    | ECCPS[2:0]   |    | TAR[2:0] |    |              | TCLR[2:0] |    |    | Res. |    | ECCEN | PWID[1:0]   |             | PTYP | PBKEN       | PWAITEN | Reserved |     |     |   |
| 0xA000<br>00A0 | FSMC_PCR4      | Reserved    |    |                     |    |      |    |    |    |              |    |    |    | ECCPS[2:0]   |    | TAR[2:0] |    |              | TCLR[2:0] |    |    | Res. |    | ECCEN | PWID[1:0]   |             | PTYP | PBKEN       | PWAITEN | Reserved |     |     |   |
| 0xA000<br>0064 | FSMC_SR2       | Reserved    |    |                     |    |      |    |    |    |              |    |    |    |              |    |          |    |              |           |    |    |      |    |       |             | FEMPT       | IFEN | ILEN        | IREN    | IFS      | ILS | IRS |   |
| 0xA000<br>0084 | FSMC_SR3       | Reserved    |    |                     |    |      |    |    |    |              |    |    |    |              |    |          |    |              |           |    |    |      |    |       |             | FEMPT       | IFEN | ILEN        | IREN    | IFS      | ILS | IRS |   |
| 0xA000<br>00A4 | FSMC_SR4       | Reserved    |    |                     |    |      |    |    |    |              |    |    |    |              |    |          |    |              |           |    |    |      |    |       |             | FEMPT       | IFEN | ILEN        | IREN    | IFS      | ILS | IRS |   |
| 0xA000<br>0068 | FSMC_PMEM<br>2 | MEMHIZ[7:0] |    |                     |    |      |    |    |    | MEMHOLD[7:0] |    |    |    |              |    |          |    | MEMWAIT[7:0] |           |    |    |      |    |       |             | MEMSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>0088 | FSMC_PMEM<br>3 | MEMHIZ[7:0] |    |                     |    |      |    |    |    | MEMHOLD[7:0] |    |    |    |              |    |          |    | MEMWAIT[7:0] |           |    |    |      |    |       |             | MEMSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>00A8 | FSMC_PMEM<br>4 | MEMHIZ[7:0] |    |                     |    |      |    |    |    | MEMHOLD[7:0] |    |    |    |              |    |          |    | MEMWAIT[7:0] |           |    |    |      |    |       |             | MEMSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>006C | FSMC_PATT2     | ATTHIZ[7:0] |    |                     |    |      |    |    |    | ATTHOLD[7:0] |    |    |    |              |    |          |    | ATTWAIT[7:0] |           |    |    |      |    |       |             | ATTSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>008C | FSMC_PATT3     | ATTHIZ[7:0] |    |                     |    |      |    |    |    | ATTHOLD[7:0] |    |    |    |              |    |          |    | ATTWAIT[7:0] |           |    |    |      |    |       |             | ATTSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>00AC | FSMC_PATT4     | ATTHIZ[7:0] |    |                     |    |      |    |    |    | ATTHOLD[7:0] |    |    |    |              |    |          |    | ATTWAIT[7:0] |           |    |    |      |    |       |             | ATTSET[7:0] |      |             |         |          |     |     |   |
| 0xA000<br>00B0 | FSMC_PIO4      | IOHIZ[7:0]  |    |                     |    |      |    |    |    | IOHOLD[7:0]  |    |    |    |              |    |          |    | IOWAIT[7:0]  |           |    |    |      |    |       |             | IOSET[7:0]  |      |             |         |          |     |     |   |
| 0xA000<br>0074 | FSMC_ECCR2     | ECC[31:0]   |    |                     |    |      |    |    |    |              |    |    |    |              |    |          |    |              |           |    |    |      |    |       |             |             |      |             |         |          |     |     |   |
| 0xA000<br>0094 | FSMC_ECCR3     | ECC[31:0]   |    |                     |    |      |    |    |    |              |    |    |    |              |    |          |    |              |           |    |    |      |    |       |             |             |      |             |         |          |     |     |   |

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.

## 37 Flexible memory controller (FMC)

The Flexible memory controller (FMC) includes three memory controllers:

- The NOR/PSRAM memory controller
- The NAND/PC Card memory controller
- The Synchronous DRAM (SDRAM/Mobile LPDDR SDRAM) controller

This section applies to STM32F42xxx and STM32F43xxx only.

### 37.1 FMC main features

The FMC functional block makes the interface with synchronous and asynchronous static memories, SDRAM memories, and 16-bit PC memory cards. Its main purposes are:

- to translate AHB transactions into the appropriate external device protocol
- to meet the access time requirements of the external memory devices

All external memories share the addresses, data and control signals with the controller. Each external device is accessed by means of a unique Chip Select. The FMC performs only one access at a time to an external device.

The main features of the FMC controller are the following:

- Interface with static-memory mapped devices including:
  - Static random access memory (SRAM)
  - NOR Flash memory/OneNAND Flash memory
  - PSRAM (4 memory banks)
  - 16-bit PC Card compatible devices
  - Two banks of NAND Flash memory with ECC hardware to check up to 8 Kbytes of data
- Interface with synchronous DRAM (SDRAM/Mobile LPDDR SDRAM) memories
- Burst mode support for faster access to synchronous devices such as NOR Flash memory, PSRAM and SDRAM)
- Programmable continuous clock output for asynchronous and synchronous accesses
- 8-, 16- or 32-bit wide data bus
- Independent Chip Select control for each memory bank
- Independent configuration for each memory bank
- Write enable and byte lane select outputs for use with PSRAM, SRAM and SDRAM devices
- External asynchronous wait control
- Write Data FIFO with 16 x 32-bit depth
- Write Address FIFO with 16 x 30-bit depth
- Cacheable Read FIFO with 6 x 32-bit depth (6 x 14-bit address tag) for SDRAM controller.

The FMC embeds two Write FIFOs: a Write Data FIFO with a 16x33-bit depth and a Write Address FIFO with a 16x30-bit depth.

- The Write Data FIFO stores the AHB data to be written to the memory (up to 32 bits) plus one bit for the AHB transfer (burst or not sequential mode)
- The Write Address FIFO stores the AHB address (up to 28 bits) plus the AHB data size (up to 2 bits). When operating in burst mode, only the start address is stored except when crossing a page boundary (for PSRAM and SDRAM). In this case, the AHB burst is broken into two FIFO entries.

At startup the FMC pins must be configured by the user application. The FMC I/O pins which are not used by the application can be used for other purposes.

The FMC registers that define the external device type and associated characteristics are usually set at boot time and do not change until the next reset or power-up. However, the settings can be changed at any time.

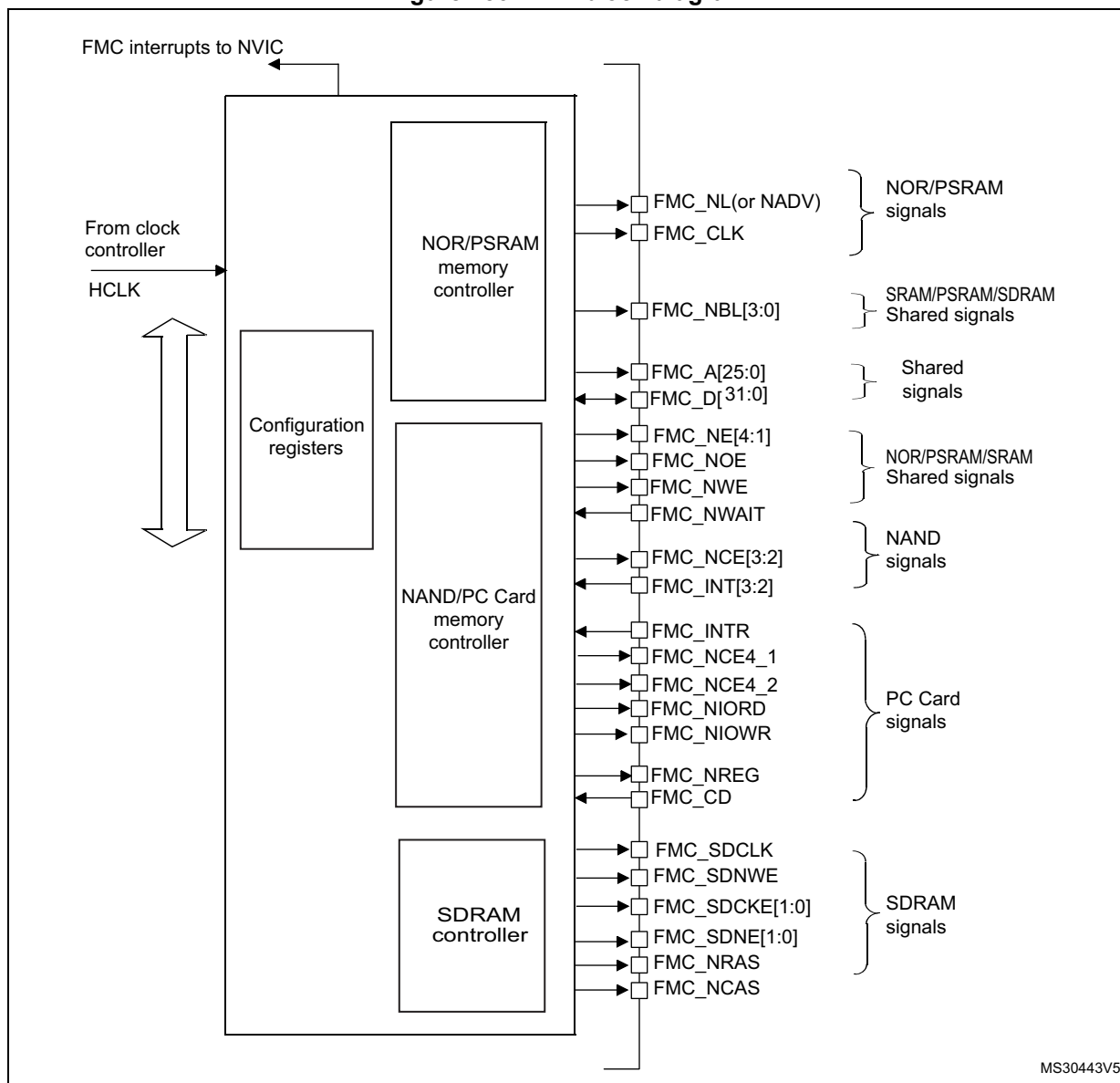
## 37.2 Block diagram

The FMC consists of five main blocks:

- The AHB interface (including the FMC configuration registers)
- The NOR Flash/PSRAM/SRAM controller
- The NAND Flash/PC Card controller
- The SDRAM controller
- The external device interface

The block diagram is shown in [Figure 456](#).

Figure 456. FMC block diagram



### 37.3 AHB interface

The AHB slave interface allows internal CPUs and other bus master peripherals to access the external memories.

AHB transactions are translated into the external device protocol. In particular, if the selected external memory is 16- or 8-bit wide, 32-bit wide transactions on the AHB are split into consecutive 16- or 8-bit accesses. The FMC Chip Select (FMC\_NEx) does not toggle between consecutive accesses except when performing accesses in mode D with the extended mode enabled.

The FMC generates an AHB error in the following conditions:

- When reading or writing to an FMC bank (Bank 1 to 4) which is not enabled.
- When reading or writing to the NOR Flash bank while the FACCEN bit is reset in the FMC\_BCRx register.
- When reading or writing to the PC Card banks while the FMC\_CD input pin (Card Presence Detection) is low.
- When writing to a write protected SDRAM bank (WP bit set in the SDRAM\_SDCRx register).
- When the SDRAM address range is violated (access to reserved address range)

The effect of an AHB error depends on the AHB master which has attempted the R/W access:

- If the access has been attempted by the Cortex<sup>®</sup>-M4 with FPU CPU, a hard fault interrupt is generated.
- If the access has been performed by a DMA controller, a DMA transfer error is generated and the corresponding DMA channel is automatically disabled.

The AHB clock (HCLK) is the reference clock for the FMC.

### 37.3.1 Supported memories and transactions

#### General transaction rules

The requested AHB transaction data size can be 8-, 16- or 32-bit wide whereas the accessed external device has a fixed data width. This may lead to inconsistent transfers.

Therefore, some simple transaction rules must be followed:

- AHB transaction size and memory data size are equal  
There is no issue in this case.
- AHB transaction size is greater than the memory size:  
In this case, the FMC splits the AHB transaction into smaller consecutive memory accesses to meet the external data width. The FMC Chip Select (FMC\_NEx) does not toggle between the consecutive accesses.
- AHB transaction size is smaller than the memory size:  
The transfer may or not be consistent depending on the type of external device:
  - Accesses to devices that have the byte select feature (SRAM, ROM, PSRAM, SDRAM)  
In this case, the FMC allows read/write transactions and accesses the right data through its byte lanes BL[3:0].  
byte to be written are addressed by NBL[3:0].  
All memory byte are read (NBL[3:0] are driven low during read transaction) and the useless ones are discarded.
  - Accesses to devices that do not have the byte select feature (16-bit NOR and NAND Flash memories)  
This situation occurs when a byte access is requested to a 16-bit wide Flash memory. Since the device cannot be accessed in byte mode (only 16-bit words can be read/written from/to the Flash memory), Write transactions and Read transactions are allowed (the controller reads the entire 16-bit memory word and uses only the required byte).

### Configuration registers

The FMC can be configured through a set of registers. Refer to [Section 37.5.6](#), for a detailed description of the NOR Flash/PSRAM controller registers. Refer to [Section 37.6.8](#), for a detailed description of the NAND Flash/PC Card registers and to [Section 37.7.5](#) for a detailed description of the SDRAM controller registers.

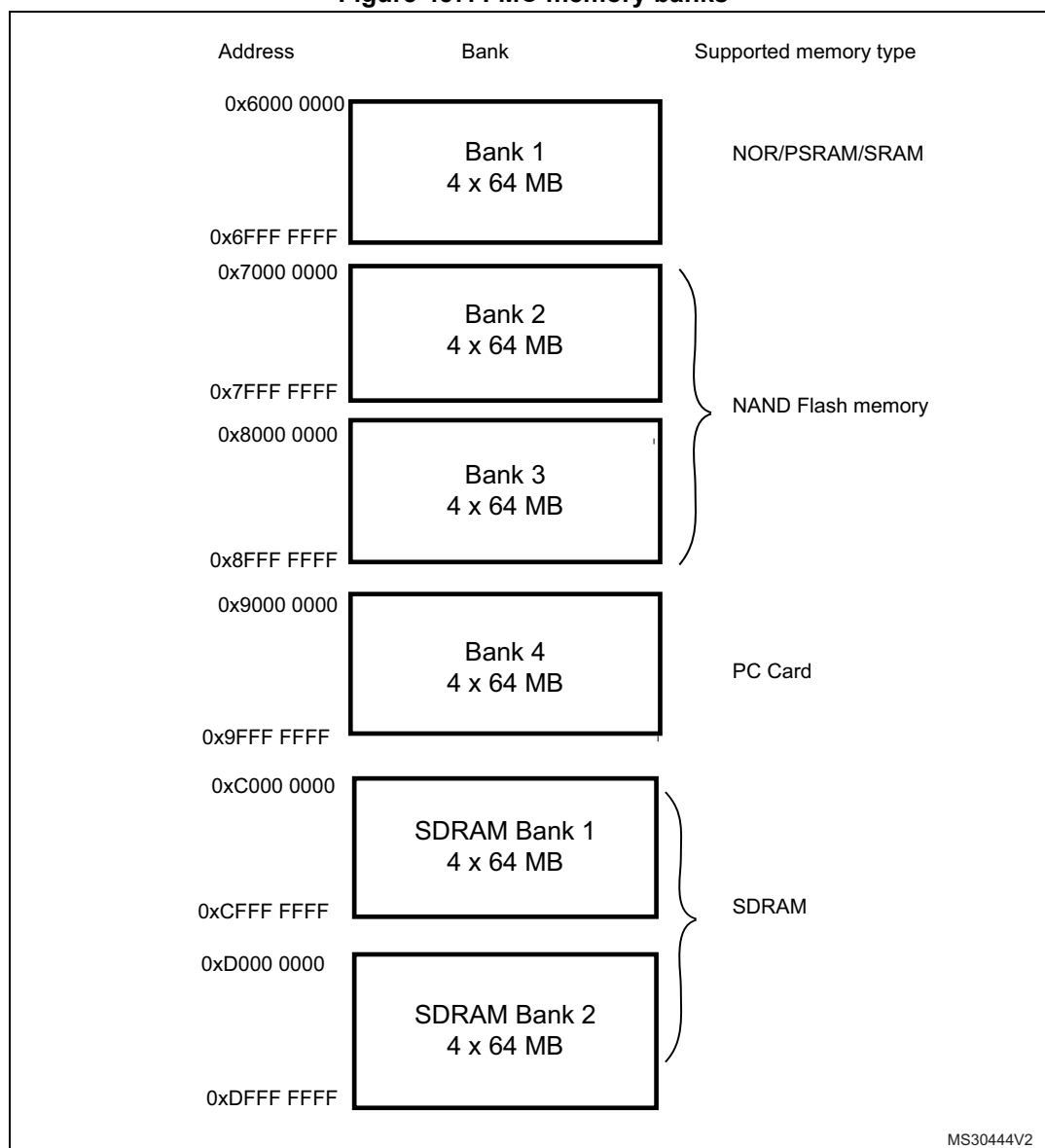
## 37.4 External device address mapping

From the FMC point of view, the external memory is divided into 6 fixed-size banks of 256 Mbyte each (see [Figure 457](#)):

- Bank 1 used to address up to 4 NOR Flash memory or PSRAM devices. This bank is split into 4 NOR/PSRAM subbanks with 4 dedicated Chip Selects, as follows:
  - Bank 1 - NOR/PSRAM 1
  - Bank 1 - NOR/PSRAM 2
  - Bank 1 - NOR/PSRAM 3
  - Bank 1 - NOR/PSRAM 4
- Banks 2 and 3 used to address NAND Flash memory devices (1 device per bank)
- Bank 4 used to address a PC Card
- Bank 5 and 6 used to address SDRAM devices (1 device per bank).

For each bank the type of memory to be used can be configured by the user application through the Configuration register.

Figure 457. FMC memory banks



### 37.4.1 NOR/PSRAM address mapping

HADDR[27:26] bits are used to select one of the four memory banks as shown in [Table 255](#).

Table 255. NOR/PSRAM bank selection

| HADDR[27:26] <sup>(1)</sup> | Selected bank        |
|-----------------------------|----------------------|
| 00                          | Bank 1 - NOR/PSRAM 1 |
| 01                          | Bank 1 - NOR/PSRAM 2 |
| 10                          | Bank 1 - NOR/PSRAM 3 |
| 11                          | Bank 1 - NOR/PSRAM 4 |

1. HADDR are internal AHB address lines that are translated to external memory.

The HADDR[25:0] bits contain the external memory address. Since HADDR is a byte address whereas the memory is addressed at word level, the address actually issued to the memory varies according to the memory data width, as shown in the following table.

**Table 256. NOR/PSRAM External memory address**

| Memory width <sup>(1)</sup> | Data address issued to the memory | Maximum memory capacity (bits) |
|-----------------------------|-----------------------------------|--------------------------------|
| 8-bit                       | HADDR[25:0]                       | 64 Mbyte x 8 = 512 Mbit        |
| 16-bit                      | HADDR[25:1] >> 1                  | 64 Mbyte/2 x 16 = 512 Mbit     |
| 32-bit                      | HADDR[25:2] >> 2                  | 64 Mbyte/4 x 32 = 512 Mbit     |

1. In case of a 16-bit external memory width, the FMC internally uses HADDR[25:1] to generate the address for external memory FMC\_A[24:0]. In case of a 32-bit memory width, the FMC internally uses HADDR[25:2] to generate the external address.  
Whatever the external memory width, FMC\_A[0] should be connected to external memory address A[0].

### Wrap support for NOR Flash/PSRAM

Wrap burst mode for synchronous memories is not supported. The memories must be configured in linear burst mode of undefined length.

## 37.4.2 NAND Flash memory/PC Card address mapping

In this case, three banks are available, each of them being divided into memory areas as indicated in [Table 257](#).

**Table 257. NAND/PC Card memory mapping and timing registers**

| Start address | End address | FMC bank            | Memory space | Timing register  |
|---------------|-------------|---------------------|--------------|------------------|
| 0x9C00 0000   | 0x9FFF FFFF | Bank 4 - PC card    | I/O          | FMC_PIO4 (0xB0)  |
| 0x9800 0000   | 0x9BFF FFFF |                     | Attribute    | FMC_PATT4 (0xAC) |
| 0x9000 0000   | 0x93FF FFFF |                     | Common       | FMC_PMEM4 (0xA8) |
| 0x8800 0000   | 0x8BFF FFFF | Bank 3 - NAND Flash | Attribute    | FMC_PATT3 (0x8C) |
| 0x8000 0000   | 0x83FF FFFF |                     | Common       | FMC_PMEM3 (0x88) |
| 0x7800 0000   | 0x7BFF FFFF | Bank 2- NAND Flash  | Attribute    | FMC_PATT2 (0x6C) |
| 0x7000 0000   | 0x73FF FFFF |                     | Common       | FMC_PMEM2 (0x68) |

For NAND Flash memory, the common and attribute memory spaces are subdivided into three sections (see in [Table 258](#) below) located in the lower 256 Kbytes:

- Data section (first 64 Kbytes in the common/attribute memory space)
- Command section (second 64 Kbytes in the common / attribute memory space)
- Address section (next 128 Kbytes in the common / attribute memory space)

Table 258. NAND bank selection

| Section name    | HADDR[17:16] | Address range     |
|-----------------|--------------|-------------------|
| Address section | 1X           | 0x020000-0x03FFFF |
| Command section | 01           | 0x010000-0x01FFFF |
| Data section    | 00           | 0x000000-0x0FFFFF |

The application software uses the 3 sections to access the NAND Flash memory:

- **To send a command to NAND Flash memory**, the software must write the command value to any memory location in the command section.
- **To specify the NAND Flash address that must be read or written**, the software must write the address value to any memory location in the address section. Since an address can be 4 or 5 byte long (depending on the actual memory size), several consecutive write operations to the address section are required to specify the full address.
- **To read or write data**, the software reads or writes the data from/to any memory location in the data section.

Since the NAND Flash memory automatically increments addresses, there is no need to increment the address of the data section to access consecutive memory locations.

### 37.4.3 SDRAM address mapping

The HADDR[28] bit (internal AHB address line 28) is used to select one of the two memory banks as indicated in [Table 259](#).

Table 259. SDRAM bank selection

| HADDR[28] | Selected bank | Control register | Timing register |
|-----------|---------------|------------------|-----------------|
| 0         | SDRAM Bank1   | FMC_SDCR1        | FMC_SDTR1       |
| 1         | SDRAM Bank2   | FMC_SDCR2        | FMC_SDTR2       |

The following table shows SDRAM mapping for an 13-bit row ,a 11-bit column and 4 internal bank configurations.

Table 260. SDRAM address mapping

| Memory width <sup>(1)</sup> | Internal bank | Row address  | Column address <sup>(2)</sup> | Maximum memory capacity (Mbyte) |
|-----------------------------|---------------|--------------|-------------------------------|---------------------------------|
| 8-bit                       | HADDR[25:24]  | HADDR[23:11] | HADDR[10:0]                   | 64 Mbyte:<br>4 x 8K x 2K        |
| 16-bit                      | HADDR[26:25]  | HADDR[24:12] | HADDR[11:1]                   | 128 Mbyte:<br>4 x 8K x 2K x 2   |
| 32-bit                      | HADDR[27:26]  | HADDR[25:13] | HADDR[12:2]                   | 256 Mbyte:<br>4 x 8K x 2K x 4   |

1. When interfacing with a 16-bit memory, the FMC internally uses the HADDR[11:1] internal AHB address lines to generate the external address. When interfacing with a 32-bit memory, the FMC internally uses HADDR[12:2] lines to generate the external address. Whatever the memory width, FMC\_A[0] has to be connected to the external memory address A[0].
2. The AutoPrecharge is not supported. FMC\_A[10] must be connected to the external memory address A[10] but it is always driven 'low'.

The HADDR[27:0] bits are translated to external SDRAM address depending on the SDRAM controller configuration:

- Data size: 8, 16 or 32 bits
- Row size: 11, 12 or 13 bits
- Column size: 8, 9, 10 or 11 bits
- Number of internal banks: two or four internal banks

[Table 261](#) to [Table 263](#) shows the SDRAM address mapping versus the SDRAM controller configuration.

**Table 261. SDRAM address mapping with 8-bit data bus width<sup>(1)(2)</sup>**

| Row size configuration        | HADDR(AHB Internal Address Lines) |    |            |            |            |            |            |            |           |           |    |    |    |    |              |              |              |             |             |             |   |   |   |   |   |   |   |   |  |  |
|-------------------------------|-----------------------------------|----|------------|------------|------------|------------|------------|------------|-----------|-----------|----|----|----|----|--------------|--------------|--------------|-------------|-------------|-------------|---|---|---|---|---|---|---|---|--|--|
|                               | 27                                | 26 | 25         | 24         | 23         | 22         | 21         | 20         | 19        | 18        | 17 | 16 | 15 | 14 | 13           | 12           | 11           | 10          | 9           | 8           | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |  |  |
| 11-bit row size configuration | Res.                              |    |            |            |            |            |            | Bank [1:0] |           | Row[10:0] |    |    |    |    |              |              |              |             |             | Column[7:0] |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            |            |            | Bank [1:0] |            | Row[10:0] |           |    |    |    |    |              |              |              |             | Column[8:0] |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            |            | Bank [1:0] |            | Row[10:0]  |           |           |    |    |    |    |              |              |              | Column[9:0] |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            | Bank [1:0] |            | Row[10:0]  |            |           |           |    |    |    |    |              |              | Column[10:0] |             |             |             |   |   |   |   |   |   |   |   |  |  |
| 12-bit row size configuration | Res.                              |    |            |            |            |            | Bank [1:0] |            | Row[11:0] |           |    |    |    |    |              |              |              |             | Column[7:0] |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            |            | Bank [1:0] |            | Row[11:0]  |           |           |    |    |    |    |              |              |              | Column[8:0] |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            | Bank [1:0] |            | Row[11:0]  |            |           |           |    |    |    |    |              |              | Column[9:0]  |             |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            | Bank [1:0] |            | Row[11:0]  |            |            |           |           |    |    |    |    |              | Column[10:0] |              |             |             |             |   |   |   |   |   |   |   |   |  |  |
| 13-bit row size configuration | Res.                              |    |            |            |            | Bank [1:0] |            | Row[12:0]  |           |           |    |    |    |    |              |              |              | Column[7:0] |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            |            | Bank [1:0] |            | Row[12:0]  |            |           |           |    |    |    |    |              |              | Column[8:0]  |             |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    |            | Bank [1:0] |            | Row[12:0]  |            |            |           |           |    |    |    |    |              | Column[9:0]  |              |             |             |             |   |   |   |   |   |   |   |   |  |  |
|                               | Res.                              |    | Bank [1:0] |            | Row[12:0]  |            |            |            |           |           |    |    |    |    | Column[10:0] |              |              |             |             |             |   |   |   |   |   |   |   |   |  |  |

1. BANK[1:0] are the Bank Address BA[1:0]. When only 2 internal banks are used, BA1 must always be set to '0'.
2. Access to Reserved (Res.) address range generates an AHB error.

Table 262. SDRAM address mapping with 16-bit data bus width<sup>(1)(2)</sup>

| Row size Configuration        | HADDR(AHB address Lines) |            |            |            |            |            |            |           |           |    |    |    |    |              |              |              |             |             |             |   |     |     |     |     |     |                    |   |   |
|-------------------------------|--------------------------|------------|------------|------------|------------|------------|------------|-----------|-----------|----|----|----|----|--------------|--------------|--------------|-------------|-------------|-------------|---|-----|-----|-----|-----|-----|--------------------|---|---|
|                               | 27                       | 26         | 25         | 24         | 23         | 22         | 21         | 20        | 19        | 18 | 17 | 16 | 15 | 14           | 13           | 12           | 11          | 10          | 9           | 8 | 7   | 6   | 5   | 4   | 3   | 2                  | 1 | 0 |
| 11-bit row size configuration | Res.                     |            |            |            |            |            | Bank [1:0] |           | Row[10:0] |    |    |    |    |              |              |              |             |             | Column[7:0] |   |     |     |     |     |     | BM0 <sup>(3)</sup> |   |   |
|                               | Res.                     |            |            |            |            | Bank [1:0] |            | Row[10:0] |           |    |    |    |    |              |              |              |             | Column[8:0] |             |   |     |     |     |     | BM0 |                    |   |   |
|                               | Res.                     |            |            |            | Bank [1:0] |            | Row[10:0]  |           |           |    |    |    |    |              |              |              | Column[9:0] |             |             |   |     |     |     | BM0 |     |                    |   |   |
|                               | Res.                     |            |            | Bank [1:0] |            | Row[10:0]  |            |           |           |    |    |    |    |              |              | Column[10:0] |             |             |             |   |     |     | BM0 |     |     |                    |   |   |
| 12-bit row size configuration | Res.                     |            |            |            |            | Bank [1:0] |            | Row[11:0] |           |    |    |    |    |              |              |              |             | Column[7:0] |             |   |     |     |     |     | BM0 |                    |   |   |
|                               | Res.                     |            |            |            | Bank [1:0] |            | Row[11:0]  |           |           |    |    |    |    |              |              |              | Column[8:0] |             |             |   |     |     |     | BM0 |     |                    |   |   |
|                               | Res.                     |            |            | Bank [1:0] |            | Row[11:0]  |            |           |           |    |    |    |    |              |              | Column[9:0]  |             |             |             |   |     |     | BM0 |     |     |                    |   |   |
|                               | Res.                     |            | Bank [1:0] |            | Row[11:0]  |            |            |           |           |    |    |    |    |              | Column[10:0] |              |             |             |             |   |     | BM0 |     |     |     |                    |   |   |
| 13-bit row size configuration | Res.                     |            |            |            | Bank [1:0] |            | Row[12:0]  |           |           |    |    |    |    |              |              |              | Column[7:0] |             |             |   |     |     |     | BM0 |     |                    |   |   |
|                               | Res.                     |            |            | Bank [1:0] |            | Row[12:0]  |            |           |           |    |    |    |    |              |              | Column[8:0]  |             |             |             |   |     |     | BM0 |     |     |                    |   |   |
|                               | Res.                     |            | Bank [1:0] |            | Row[12:0]  |            |            |           |           |    |    |    |    |              | Column[9:0]  |              |             |             |             |   |     | BM0 |     |     |     |                    |   |   |
|                               | Res. s.                  | Bank [1:0] |            | Row[12:0]  |            |            |            |           |           |    |    |    |    | Column[10:0] |              |              |             |             |             |   | BM0 |     |     |     |     |                    |   |   |

1. BANK[1:0] are the Bank Address BA[1:0]. When only 2 internal banks are used, BA1 must always be set to '0'.

2. Access to Reserved space (Res.) generates an AHB error.

3. BM0: is the byte mask for 16-bit access.

Table 263. SDRAM address mapping with 32-bit data bus width<sup>(1)(2)</sup>

| Row size configuration        | HADDR(AHB address Lines) |            |            |            |            |            |           |    |    |    |    |    |    |    |    |              |              |             |             |   |   |   |         |         |         |                        |   |   |
|-------------------------------|--------------------------|------------|------------|------------|------------|------------|-----------|----|----|----|----|----|----|----|----|--------------|--------------|-------------|-------------|---|---|---|---------|---------|---------|------------------------|---|---|
|                               | 27                       | 26         | 25         | 24         | 23         | 22         | 21        | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12           | 11           | 10          | 9           | 8 | 7 | 6 | 5       | 4       | 3       | 2                      | 1 | 0 |
| 11-bit row size configuration | Res.                     |            |            |            |            | Bank [1:0] | Row[10:0] |    |    |    |    |    |    |    |    |              |              |             | Column[7:0] |   |   |   |         |         |         | BM[1:0] <sup>(3)</sup> |   |   |
|                               | Res.                     |            |            |            | Bank [1:0] | Row[10:0]  |           |    |    |    |    |    |    |    |    |              |              | Column[8:0] |             |   |   |   |         |         | BM[1:0] |                        |   |   |
|                               | Res.                     |            |            | Bank [1:0] | Row[10:0]  |            |           |    |    |    |    |    |    |    |    |              |              | Column[9:0] |             |   |   |   |         |         | BM[1:0] |                        |   |   |
|                               | Res.                     |            | Bank [1:0] | Row[10:0]  |            |            |           |    |    |    |    |    |    |    |    |              | Column[10:0] |             |             |   |   |   |         | BM[1:0] |         |                        |   |   |
| 12-bit row size configuration | Res.                     |            |            |            | Bank [1:0] | Row[11:0]  |           |    |    |    |    |    |    |    |    |              |              | Column[7:0] |             |   |   |   |         |         | BM[1:0] |                        |   |   |
|                               | Res.                     |            |            | Bank [1:0] | Row[11:0]  |            |           |    |    |    |    |    |    |    |    |              |              | Column[8:0] |             |   |   |   |         |         | BM[1:0] |                        |   |   |
|                               | Res.                     |            | Bank [1:0] | Row[11:0]  |            |            |           |    |    |    |    |    |    |    |    |              | Column[9:0]  |             |             |   |   |   |         | BM[1:0] |         |                        |   |   |
|                               | Res.                     | Bank [1:0] | Row[11:0]  |            |            |            |           |    |    |    |    |    |    |    |    | Column[10:0] |              |             |             |   |   |   | BM[1:0] |         |         |                        |   |   |

**Table 263. SDRAM address mapping with 32-bit data bus width<sup>(1)(2)</sup> (continued)**

| Row size configuration        | HADDR(AHB address Lines) |            |            |    |           |    |    |    |    |    |    |    |    |    |    |              |             |    |             |   |   |   |   |         |         |   |         |   |
|-------------------------------|--------------------------|------------|------------|----|-----------|----|----|----|----|----|----|----|----|----|----|--------------|-------------|----|-------------|---|---|---|---|---------|---------|---|---------|---|
|                               | 27                       | 26         | 25         | 24 | 23        | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12           | 11          | 10 | 9           | 8 | 7 | 6 | 5 | 4       | 3       | 2 | 1       | 0 |
| 13-bit row size configuration | Res.                     |            | Bank [1:0] |    | Row[12:0] |    |    |    |    |    |    |    |    |    |    |              |             |    | Column[7:0] |   |   |   |   |         |         |   | BM[1:0] |   |
|                               | Res.                     |            | Bank [1:0] |    | Row[12:0] |    |    |    |    |    |    |    |    |    |    |              |             |    | Column[8:0] |   |   |   |   |         |         |   | BM[1:0] |   |
|                               | Res.                     | Bank [1:0] | Row[12:0]  |    |           |    |    |    |    |    |    |    |    |    |    |              | Column[9:0] |    |             |   |   |   |   |         | BM[1:0] |   |         |   |
|                               | Bank [1:0]               | Row[12:0]  |            |    |           |    |    |    |    |    |    |    |    |    |    | Column[10:0] |             |    |             |   |   |   |   | BM[1:0] |         |   |         |   |

1. BANK[1:0] are the Bank Address BA[1:0]. When only 2 internal banks are used, BA1 must always be set to '0'.
2. Access to Reserved space (Res.) generates an AHB error.
3. BM[1:0]: is the byte mask for 32-bit access.

## 37.5 NOR Flash/PSRAM controller

The FMC generates the appropriate signal timings to drive the following types of memories:

- Asynchronous SRAM and ROM
  - 8 bits
  - 16 bits
  - 32 bits
- PSRAM (Cellular RAM)
  - Asynchronous mode
  - Burst mode for synchronous accesses
  - Multiplexed or non-multiplexed
- NOR Flash memory
  - Asynchronous mode
  - Burst mode for synchronous accesses
  - Multiplexed or non-multiplexed

The FMC outputs a unique Chip Select signal, NE[4:1], per bank. All the other signals (addresses, data and control) are shared.

The FMC supports a wide range of devices through a programmable timings among which:

- Programmable wait states (up to 15)
- Programmable bus turnaround cycles (up to 15)
- Programmable output enable and write enable delays (up to 15)
- Independent read and write timings and protocol to support the widest variety of memories and timings
- Programmable continuous clock (FMC\_CLK) output.

The FMC Clock (FMC\_CLK) is a submultiple of the HCLK clock. It can be delivered to the selected external device either during synchronous accesses only or during asynchronous

and synchronous accesses depending on the CCKEN bit configuration in the FMC\_BCR1 register:

- If the CCKEN bit is reset, the FMC generates the clock (CLK) only during synchronous accesses (Read/write transactions).
- If the CCKEN bit is set, the FMC generates a continuous clock during asynchronous and synchronous accesses. To generate the FMC\_CLK continuous clock, Bank 1 must be configured in synchronous mode (see [Section 37.5.6: NOR/PSRAM controller registers](#)). Since the same clock is used for all synchronous memories, when a continuous output clock is generated and synchronous accesses are performed, the AHB data size has to be the same as the memory data width (MWID) otherwise the FMC\_CLK frequency is changed depending on AHB data transaction (refer to [Section 37.5.5: Synchronous transactions](#) for FMC\_CLK divider ratio formula).

The size of each bank is fixed and equal to 64 Mbyte. Each bank is configured through dedicated registers (see [Section 37.5.6: NOR/PSRAM controller registers](#)).

The programmable memory parameters include access times (see [Table 264](#)) and support for wait management (for PSRAM and NOR Flash accessed in burst mode).

**Table 264. Programmable NOR/PSRAM access parameters**

| Parameter          | Function                                                                         | Access mode                             | Unit                     | Min. | Max. |
|--------------------|----------------------------------------------------------------------------------|-----------------------------------------|--------------------------|------|------|
| Address setup      | Duration of the address setup phase                                              | Asynchronous                            | AHB clock cycle (HCLK)   | 0    | 15   |
| Address hold       | Duration of the address hold phase                                               | Asynchronous, muxed I/Os                | AHB clock cycle (HCLK)   | 1    | 15   |
| Data setup         | Duration of the data setup phase                                                 | Asynchronous                            | AHB clock cycle (HCLK)   | 1    | 256  |
| Bust turn          | Duration of the bus turnaround phase                                             | Asynchronous and synchronous read/write | AHB clock cycle (HCLK)   | 0    | 15   |
| Clock divide ratio | Number of AHB clock cycles (HCLK) to build one memory clock cycle (CLK)          | Synchronous                             | AHB clock cycle (HCLK)   | 2    | 16   |
| Data latency       | Number of clock cycles to issue to the memory before the first data of the burst | Synchronous                             | Memory clock cycle (CLK) | 2    | 17   |

### 37.5.1 External memory interface signals

[Table 265](#), [Table 266](#) and [Table 267](#) list the signals that are typically used to interface with NOR Flash memory, SRAM and PSRAM.

*Note:* The prefix “N” identifies the signals which are active low.

**NOR Flash memory, non-multiplexed I/Os****Table 265. Non-multiplexed I/O NOR Flash memory**

| FMC signal name | I/O | Function                                                                            |
|-----------------|-----|-------------------------------------------------------------------------------------|
| CLK             | O   | Clock (for synchronous access)                                                      |
| A[25:0]         | O   | Address bus                                                                         |
| D[31:0]         | I/O | Bidirectional data bus                                                              |
| NE[x]           | O   | Chip Select, x = 1..4                                                               |
| NOE             | O   | Output enable                                                                       |
| NWE             | O   | Write enable                                                                        |
| NL(=NADV)       | O   | Latch enable (this signal is called address valid, NADV, by some NOR Flash devices) |
| NWAIT           | I   | NOR Flash wait input signal to the FMC                                              |

The maximum capacity is 512 Mbits (26 address lines).

**NOR Flash memory, 16-bit multiplexed I/Os****Table 266. 16-bit multiplexed I/O NOR Flash memory**

| FMC signal name | I/O | Function                                                                                                                        |
|-----------------|-----|---------------------------------------------------------------------------------------------------------------------------------|
| CLK             | O   | Clock (for synchronous access)                                                                                                  |
| A[25:16]        | O   | Address bus                                                                                                                     |
| AD[15:0]        | I/O | 16-bit multiplexed, bidirectional address/data bus (the 16-bit address A[15:0] and data D[15:0] are multiplexed on the databus) |
| NE[x]           | O   | Chip Select, x = 1..4                                                                                                           |
| NOE             | O   | Output enable                                                                                                                   |
| NWE             | O   | Write enable                                                                                                                    |
| NL(=NADV)       | O   | Latch enable (this signal is called address valid, NADV, by some NOR Flash devices)                                             |
| NWAIT           | I   | NOR Flash wait input signal to the FMC                                                                                          |

The maximum capacity is 512 Mbits.

**PSRAM/SRAM, non-multiplexed I/Os****Table 267. Non-multiplexed I/Os PSRAM/SRAM**

| FMC signal name | I/O | Function                                  |
|-----------------|-----|-------------------------------------------|
| CLK             | O   | Clock (only for PSRAM synchronous access) |
| A[25:0]         | O   | Address bus                               |
| D[31:0]         | I/O | Data bidirectional bus                    |

**Table 267. Non-multiplexed I/Os PSRAM/SRAM (continued)**

| FMC signal name | I/O | Function                                                             |
|-----------------|-----|----------------------------------------------------------------------|
| NE[x]           | O   | Chip Select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CRAM)) |
| NOE             | O   | Output enable                                                        |
| NWE             | O   | Write enable                                                         |
| NL(= NADV)      | O   | Address valid only for PSRAM input (memory signal name: NADV)        |
| NWAIT           | I   | PSRAM wait input signal to the FMC                                   |
| NBL[3]          | O   | Byte3 Upper byte enable (memory signal name: NUB)                    |
| NBL[2]          | O   | Byte2 Lowed byte enable (memory signal name: NLB)                    |
| NBL[1]          | O   | Byte1 Upper byte enable (memory signal name: NLB)                    |
| NBL[0]          | O   | Byte0 Lower byte enable (memory signal name: NLB)                    |

The maximum capacity is 512 Mbits.

### PSRAM, 16-bit multiplexed I/Os

**Table 268. 16-Bit multiplexed I/O PSRAM**

| FMC signal name | I/O | Function                                                                                                                        |
|-----------------|-----|---------------------------------------------------------------------------------------------------------------------------------|
| CLK             | O   | Clock (for synchronous access)                                                                                                  |
| A[25:16]        | O   | Address bus                                                                                                                     |
| AD[15:0]        | I/O | 16-bit multiplexed, bidirectional address/data bus (the 16-bit address A[15:0] and data D[15:0] are multiplexed on the databus) |
| NE[x]           | O   | Chip Select, x = 1..4 (called NCE by PSRAM (Cellular RAM i.e. CRAM))                                                            |
| NOE             | O   | Output enable                                                                                                                   |
| NWE             | O   | Write enable                                                                                                                    |
| NL(= NADV)      | O   | Address valid PSRAM input (memory signal name: NADV)                                                                            |
| NWAIT           | I   | PSRAM wait input signal to the FMC                                                                                              |
| NBL[1]          | O   | Upper byte enable (memory signal name: NUB)                                                                                     |
| NBL[0]          | O   | Lowod byte enable (memory signal name: NLB)                                                                                     |

The maximum capacity is 512 Mbits (26 address lines).

## 37.5.2 Supported memories and transactions

[Table 269](#) below shows an example of the supported devices, access modes and transactions when the memory data bus is 16-bit wide for NOR Flash memory, PSRAM and SRAM. The transactions not allowed (or not supported) by the FMC are shown in gray in this example.

Table 269. NOR Flash/PSRAM: Example of supported memories and transactions

| Device                                               | Mode              | R/W | AHB data size | Memory data size | Allowed/not allowed | Comments                                                |
|------------------------------------------------------|-------------------|-----|---------------|------------------|---------------------|---------------------------------------------------------|
| NOR Flash<br>(muxed I/Os and nonmuxed I/Os)          | Asynchronous      | R   | 8             | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | W   | 8             | 16               | N                   | -                                                       |
|                                                      | Asynchronous      | R   | 16            | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | W   | 16            | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | R   | 32            | 16               | Y                   | Split into 2 FMC accesses                               |
|                                                      | Asynchronous      | W   | 32            | 16               | Y                   | Split into 2 FMC accesses                               |
|                                                      | Asynchronous page | R   | -             | 16               | N                   | Mode is not supported                                   |
|                                                      | Synchronous       | R   | 8             | 16               | N                   | -                                                       |
|                                                      | Synchronous       | R   | 16            | 16               | Y                   | -                                                       |
|                                                      | Synchronous       | R   | 32            | 16               | Y                   | -                                                       |
| PSRAM<br>(multiplexed I/Os and non-multiplexed I/Os) | Asynchronous      | R   | 8             | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | W   | 8             | 16               | Y                   | Use of byte lanes NBL[1:0]                              |
|                                                      | Asynchronous      | R   | 16            | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | W   | 16            | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | R   | 32            | 16               | Y                   | Split into 2 FMC accesses                               |
|                                                      | Asynchronous      | W   | 32            | 16               | Y                   | Split into 2 FMC accesses                               |
|                                                      | Asynchronous page | R   | -             | 16               | N                   | Mode is not supported                                   |
|                                                      | Synchronous       | R   | 8             | 16               | N                   | -                                                       |
|                                                      | Synchronous       | R   | 16            | 16               | Y                   | -                                                       |
|                                                      | Synchronous       | R   | 32            | 16               | Y                   | -                                                       |
|                                                      | Synchronous       | W   | 8             | 16               | Y                   | Use of byte lanes NBL[1:0]                              |
|                                                      | Synchronous       | W   | 16/32         | 16               | Y                   | -                                                       |
| SRAM and ROM                                         | Asynchronous      | R   | 8 / 16        | 16               | Y                   | -                                                       |
|                                                      | Asynchronous      | W   | 8 / 16        | 16               | Y                   | Use of byte lanes NBL[1:0]                              |
|                                                      | Asynchronous      | R   | 32            | 16               | Y                   | Split into 2 FMC accesses                               |
|                                                      | Asynchronous      | W   | 32            | 16               | Y                   | Split into 2 FMC accesses<br>Use of byte lanes NBL[1:0] |

### 37.5.3 General timing rules

#### Signals synchronization

- All controller output signals change on the rising edge of the internal clock (HCLK)
- In synchronous mode (read or write), all output signals change on the rising edge of HCLK. Whatever the CLKDIV value, all outputs change as follows:
  - NOEL/NWEL/ NEL/NADV L/ NADV H /NBLL/ Address valid outputs change on the falling edge of FMC\_CLK clock.
  - NOEH/ NWEH / NEH/ NOEH/NBLH/ Address invalid outputs change on the rising edge of FMC\_CLK clock.

### 37.5.4 NOR Flash/PSRAM controller asynchronous transactions

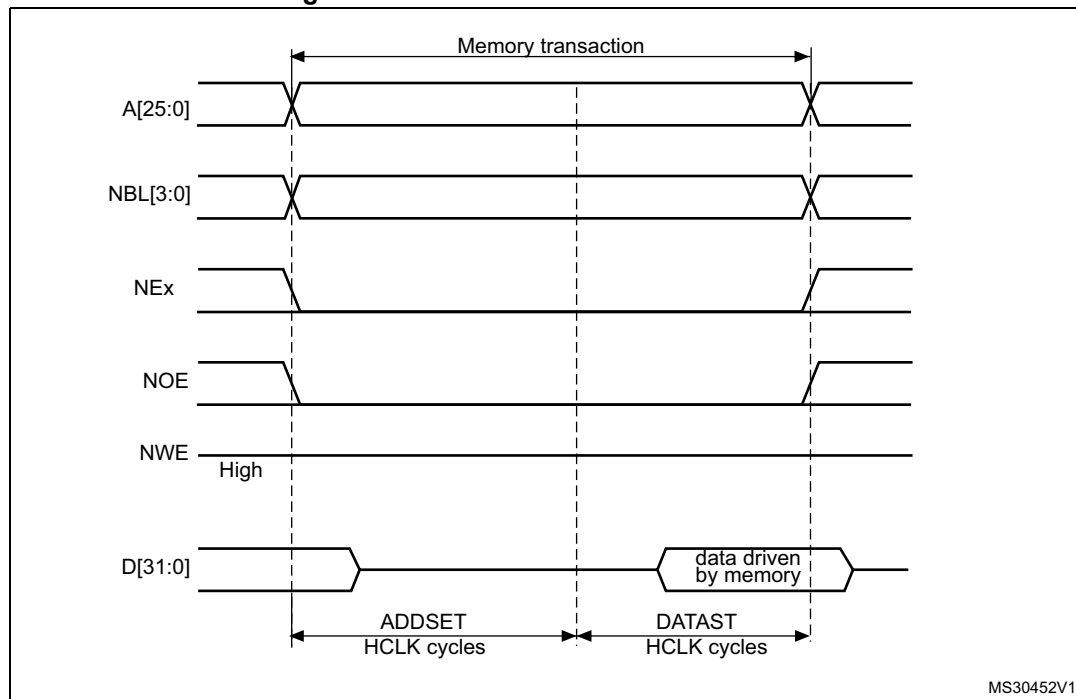
#### Asynchronous static memories (NOR Flash, PSRAM, SRAM)

- Signals are synchronized by the internal clock HCLK. This clock is not issued to the memory
- The FMC always samples the data before de-asserting the NOE signal. This guarantees that the memory data hold timing constraint is met (minimum Chip Enable high to data transition is usually 0 ns)
- If the extended mode is enabled (EXTMOD bit is set in the FMC\_BCRx register), up to four extended modes (A, B, C and D) are available. It is possible to mix A, B, C and D modes for read and write operations. For example, read operation can be performed in mode A and write in mode B.
- If the extended mode is disabled (EXTMOD bit is reset in the FMC\_BCRx register), the FMC can operate in Mode1 or Mode2 as follows:
  - Mode 1 is the default mode when SRAM/PSRAM memory type is selected (MTYP[1:0] = 0x0 or 0x01 in the FMC\_BCRx register)
  - Mode 2 is the default mode when NOR memory type is selected (MTYP[1:0] = 0x10 in the FMC\_BCRx register).

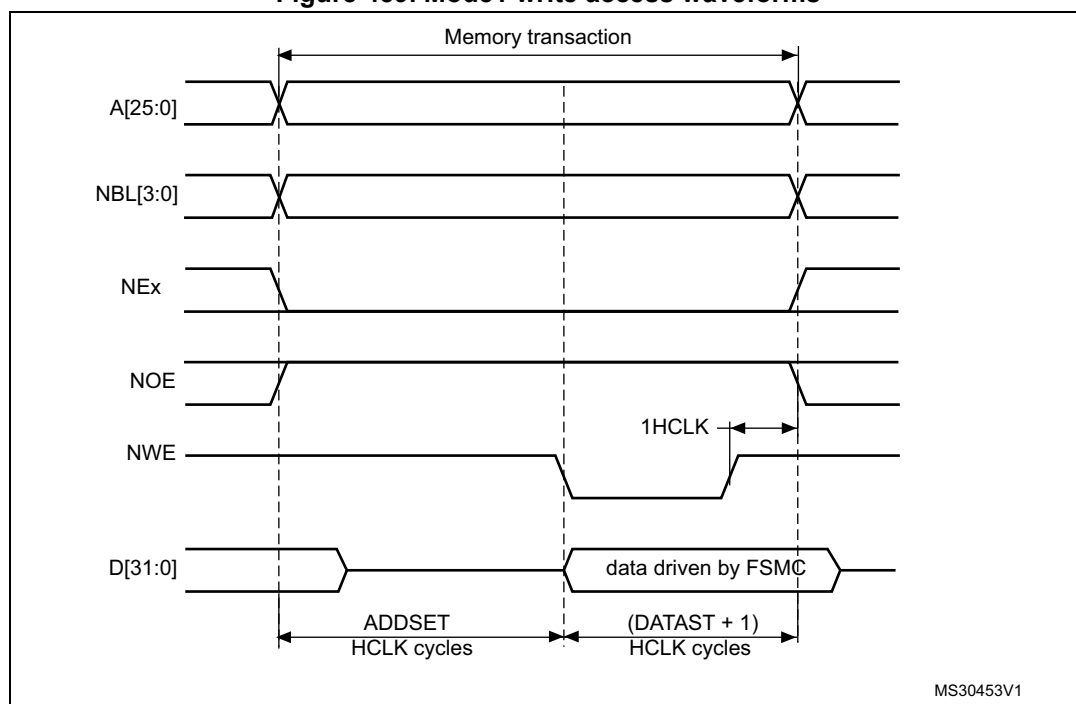
### Mode 1 - SRAM/PSRAM (CRAM)

The next figures show the read and write transactions for the supported modes followed by the required configuration of FMC\_BCRx, and FMC\_BTRx/FMC\_BWTRx registers.

**Figure 458. Mode1 read access waveforms**



**Figure 459. Mode1 write access waveforms**



The one HCLK cycle at the end of the write transaction helps guarantee the address and data hold time after the NWE rising edge. Due to the presence of this HCLK cycle, the DATAST value must be greater than zero (DATAST > 0).

**Table 270. FMC\_BCRx bit fields**

| Bit number | Bit name   | Value to set                                                       |
|------------|------------|--------------------------------------------------------------------|
| 31-21      | Reserved   | 0x000                                                              |
| 20         | CCLKEN     | As needed                                                          |
| 19         | CBURSTRW   | 0x0 (no effect in asynchronous mode)                               |
| 18:16      | CPSIZE     | 0x0 (no effect in asynchronous mode)                               |
| 15         | ASYNCAWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD     | 0x0                                                                |
| 13         | WAITEN     | 0x0 (no effect in asynchronous mode)                               |
| 12         | WREN       | As needed                                                          |
| 11         | WAITCFG    | Don't care                                                         |
| 10         | WRAPMOD    | 0x0                                                                |
| 9          | WAITPOL    | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN    | 0x0                                                                |
| 7          | Reserved   | 0x1                                                                |
| 6          | FACCEN     | Don't care                                                         |
| 5-4        | MWID       | As needed                                                          |
| 3-2        | MTYP[1:0]  | As needed, exclude 0x2 (NOR Flash memory)                          |
| 1          | MUXE       | 0x0                                                                |
| 0          | MBKEN      | 0x1                                                                |

**Table 271. FMC\_BTRx bit fields**

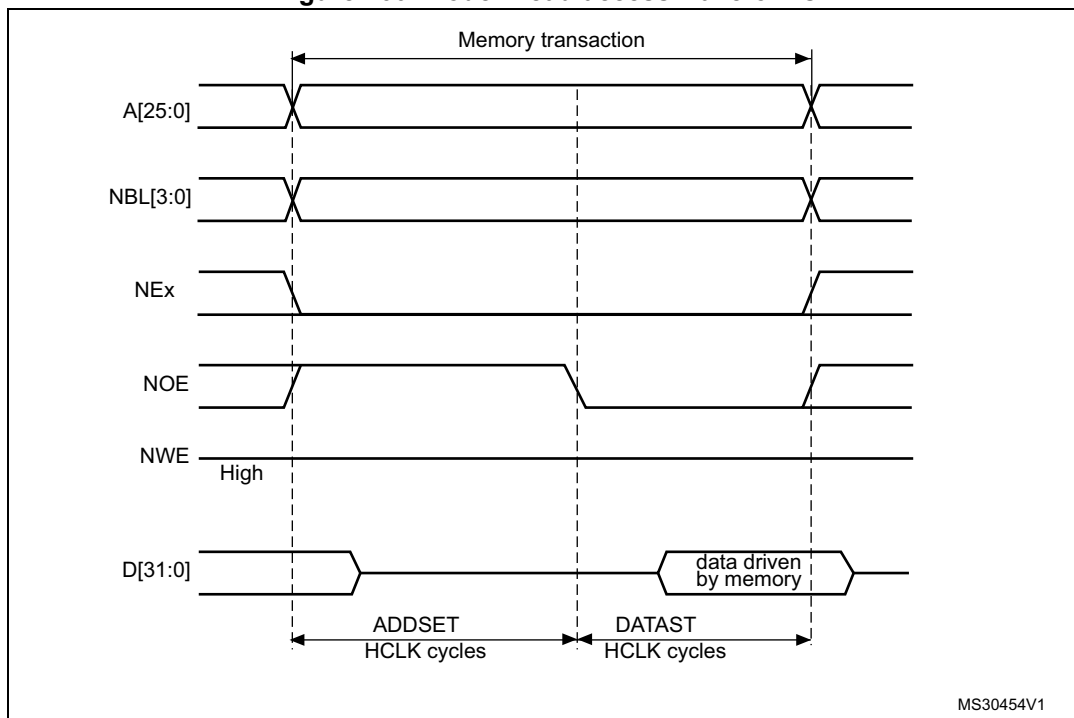
| Bit number | Bit name | Value to set                                                                                                         |
|------------|----------|----------------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved | 0x0                                                                                                                  |
| 29-28      | ACCMOD   | Don't care                                                                                                           |
| 27-24      | DATLAT   | Don't care                                                                                                           |
| 23-20      | CLKDIV   | Don't care                                                                                                           |
| 19-16      | BUSTURN  | Time between NEx high to NEx low (BUSTURN HCLK)                                                                      |
| 15-8       | DATAST   | Duration of the second access phase (DATAST+1 HCLK cycles for write accesses, DATAST HCLK cycles for read accesses). |

Table 271. FMC\_BTRx bit fields (continued)

| Bit number | Bit name | Value to set                                                                               |
|------------|----------|--------------------------------------------------------------------------------------------|
| 7-4        | ADDHLD   | Don't care                                                                                 |
| 3-0        | ADDSET   | Duration of the first access phase (ADDSET HCLK cycles).<br>Minimum value for ADDSET is 0. |

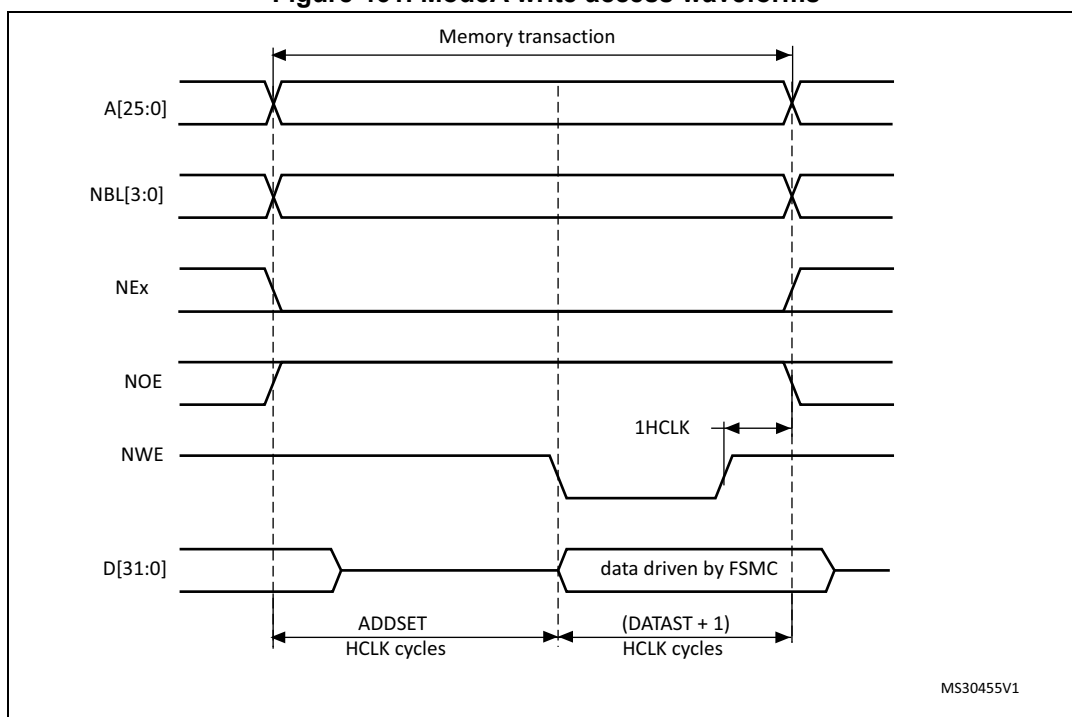
**Mode A - SRAM/PSRAM (CRAM) OE toggling**

Figure 460. ModeA read access waveforms



1. NBL[3:0] are driven low during the read access

Figure 461. ModeA write access waveforms



The differences compared with mode1 are the toggling of NOE and the independent read and write timings.

Table 272. FMC\_BCRx bit fields

| Bit number | Bit name  | Value to set                                                       |
|------------|-----------|--------------------------------------------------------------------|
| 31-21      | Reserved  | 0x000                                                              |
| 20         | CCLKEN    | As needed                                                          |
| 19         | CBURSTRW  | 0x0 (no effect in asynchronous mode)                               |
| 18:16      | CPSIZE    | 0x0 (no effect in asynchronous mode)                               |
| 15         | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD    | 0x1                                                                |
| 13         | WAITEN    | 0x0 (no effect in asynchronous mode)                               |
| 12         | WREN      | As needed                                                          |
| 11         | WAITCFG   | Don't care                                                         |
| 10         | WRAPMOD   | 0x0                                                                |
| 9          | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN   | 0x0                                                                |
| 7          | Reserved  | 0x1                                                                |
| 6          | FACCEN    | Don't care                                                         |

**Table 272. FMC\_BCRx bit fields (continued)**

| Bit number | Bit name  | Value to set                              |
|------------|-----------|-------------------------------------------|
| 5-4        | MWID      | As needed                                 |
| 3-2        | MTYP[1:0] | As needed, exclude 0x2 (NOR Flash memory) |
| 1          | MUXEN     | 0x0                                       |
| 0          | MBKEN     | 0x1                                       |

**Table 273. FMC\_BTRx bit fields**

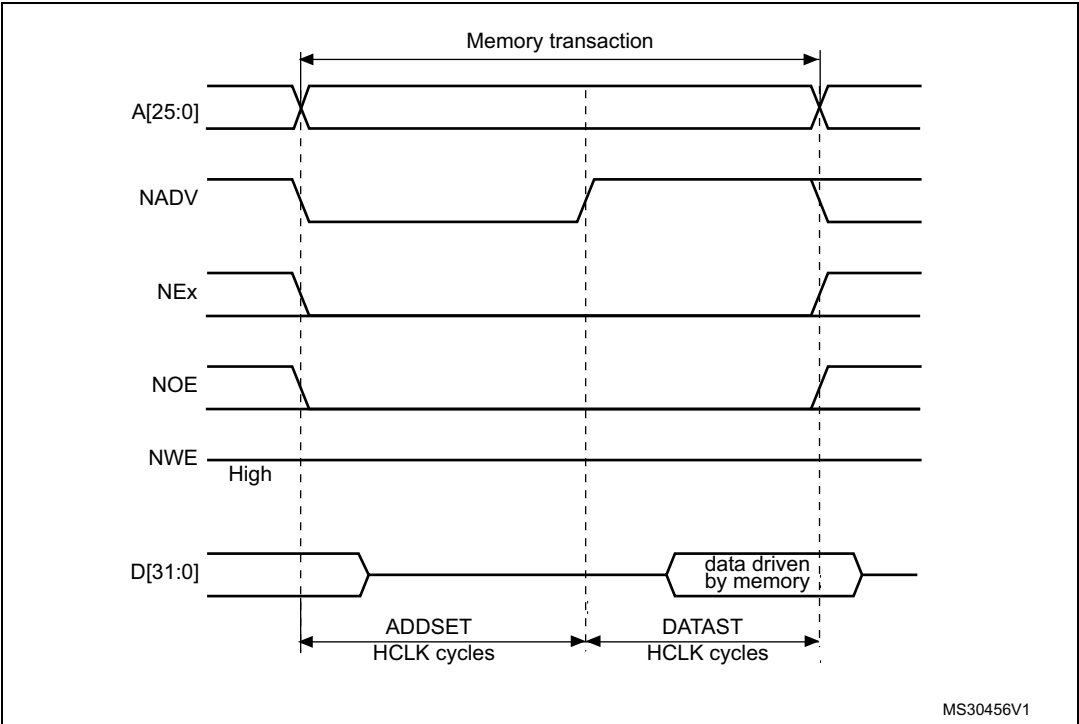
| Bit number | Bit name    | Value to set                                                                                                 |
|------------|-------------|--------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                          |
| 29-28      | ACCMOD      | 0x0                                                                                                          |
| 27-24      | DATLAT      | Don't care                                                                                                   |
| 23-20      | CLKDIV      | Don't care                                                                                                   |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                              |
| 15-8       | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                                  |
| 7-4        | ADDHLD      | Don't care                                                                                                   |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses.<br>Minimum value for ADDSET is 0. |

**Table 274. FMC\_BWTRx bit fields**

| Bit number | Bit name    | Value to set                                                                                                  |
|------------|-------------|---------------------------------------------------------------------------------------------------------------|
| 31:30      | Reserved    | 0x0                                                                                                           |
| 29-28      | ACCMOD      | 0x0                                                                                                           |
| 27-24      | DATLAT      | Don't care                                                                                                    |
| 23-20      | CLKDIV      | Don't care                                                                                                    |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                               |
| 15-8       | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for write accesses.                                  |
| 7-4        | ADDHLD      | Don't care                                                                                                    |
| 3-0        | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses.<br>Minimum value for ADDSET is 0. |

Mode 2/B - NOR Flash

Figure 462. Mode2 and mode B read access waveforms



- 1. NBL[3:0] are driven low during the read access

Figure 463. Mode2 write access waveforms

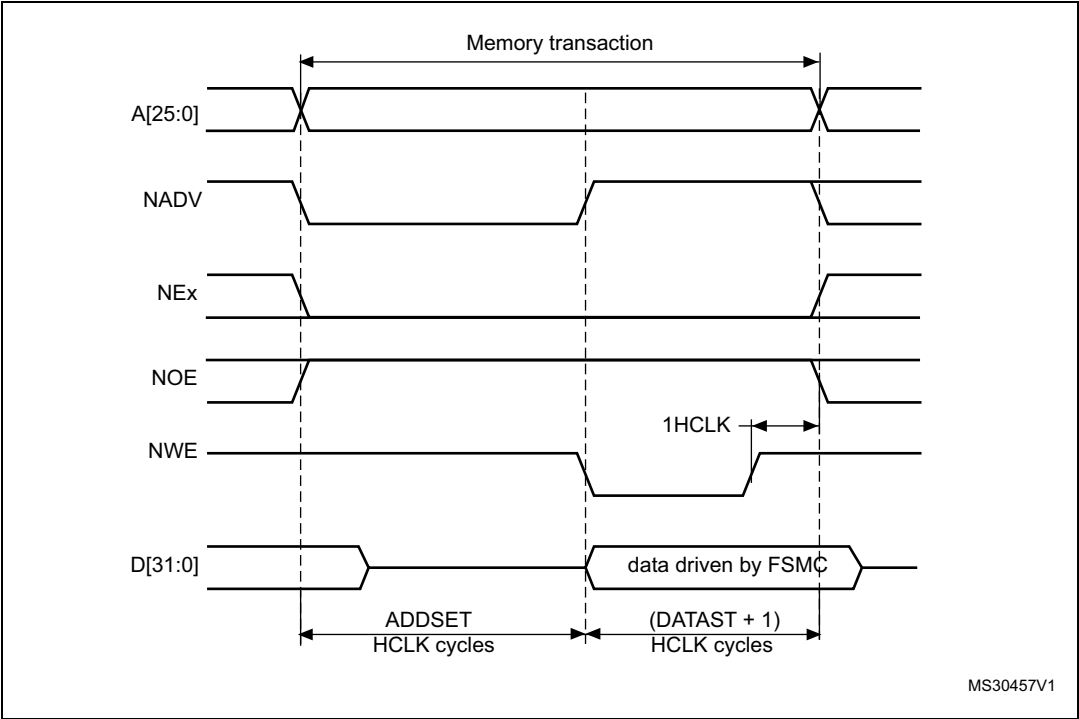
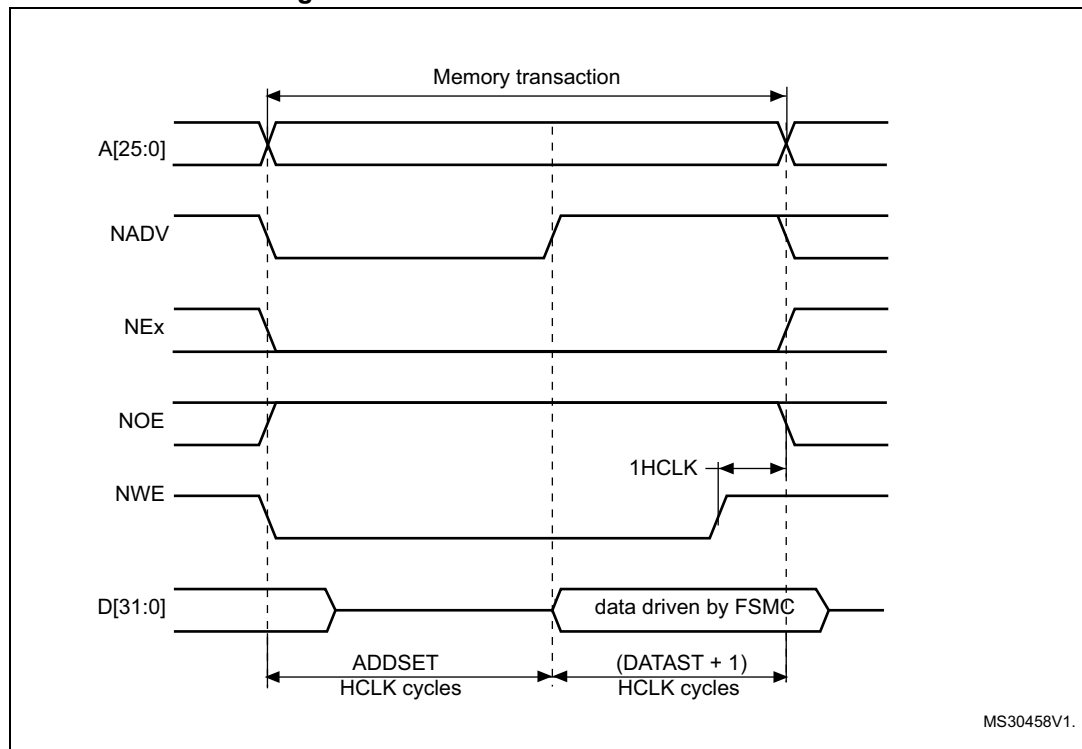


Figure 464. ModeB write access waveforms



The differences with mode1 are the toggling of NWE and the independent read and write timings when extended mode is set (Mode B).

Table 275. FMC\_BCRx bit fields

| Bit number | Bit name  | Value to set                                                       |
|------------|-----------|--------------------------------------------------------------------|
| 31-21      | Reserved  | 0x000                                                              |
| 20         | CCLKEN    | As needed                                                          |
| 19         | CBURSTRW  | 0x0 (no effect in asynchronous mode)                               |
| 18:16      | CPSIZE    | 0x0 (no effect in asynchronous mode)                               |
| 15         | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14         | EXTMOD    | 0x1 for mode B, 0x0 for mode 2                                     |
| 13         | WAITEN    | 0x0 (no effect in asynchronous mode)                               |
| 12         | WREN      | As needed                                                          |
| 11         | WAITCFG   | Don't care                                                         |
| 10         | WRAPMOD   | 0x0                                                                |
| 9          | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8          | BURSTEN   | 0x0                                                                |
| 7          | Reserved  | 0x1                                                                |
| 6          | FACCEN    | 0x1                                                                |

Table 275. FMC\_BCRx bit fields (continued)

| Bit number | Bit name  | Value to set           |
|------------|-----------|------------------------|
| 5-4        | MWID      | As needed              |
| 3-2        | MTYP[1:0] | 0x2 (NOR Flash memory) |
| 1          | MUXEN     | 0x0                    |
| 0          | MBKEN     | 0x1                    |

Table 276. FMC\_BTRx bit fields

| Bit number | Bit name    | Value to set                                                                                              |
|------------|-------------|-----------------------------------------------------------------------------------------------------------|
| 31-30      | Reserved    | 0x0                                                                                                       |
| 29-28      | ACCMOD      | 0x1 if extended mode is set                                                                               |
| 27-24      | DATLAT      | Don't care                                                                                                |
| 23-20      | CLKDIV      | Don't care                                                                                                |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                           |
| 15-8       | DATAST      | Duration of the access second phase (DATAST HCLK cycles) for read accesses.                               |
| 7-4        | ADDHLD      | Don't care                                                                                                |
| 3-0        | ADDSET[3:0] | Duration of the access first phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0. |

Table 277. FMC\_BWTRx bit fields

| Bit number | Bit name    | Value to set                                                                                               |
|------------|-------------|------------------------------------------------------------------------------------------------------------|
| 31-30      | Reserved    | 0x0                                                                                                        |
| 29-28      | ACCMOD      | 0x1 if extended mode is set                                                                                |
| 27-24      | DATLAT      | Don't care                                                                                                 |
| 23-20      | CLKDIV      | Don't care                                                                                                 |
| 19-16      | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                            |
| 15-8       | DATAST      | Duration of the access second phase (DATAST HCLK cycles) for write accesses.                               |
| 7-4        | ADDHLD      | Don't care                                                                                                 |
| 3-0        | ADDSET[3:0] | Duration of the access first phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 0. |

**Note:** The FMC\_BWTRx register is valid only if the extended mode is set (mode B), otherwise its content is don't care.

## Mode C - NOR Flash - OE toggling

Figure 465. ModeC read access waveforms

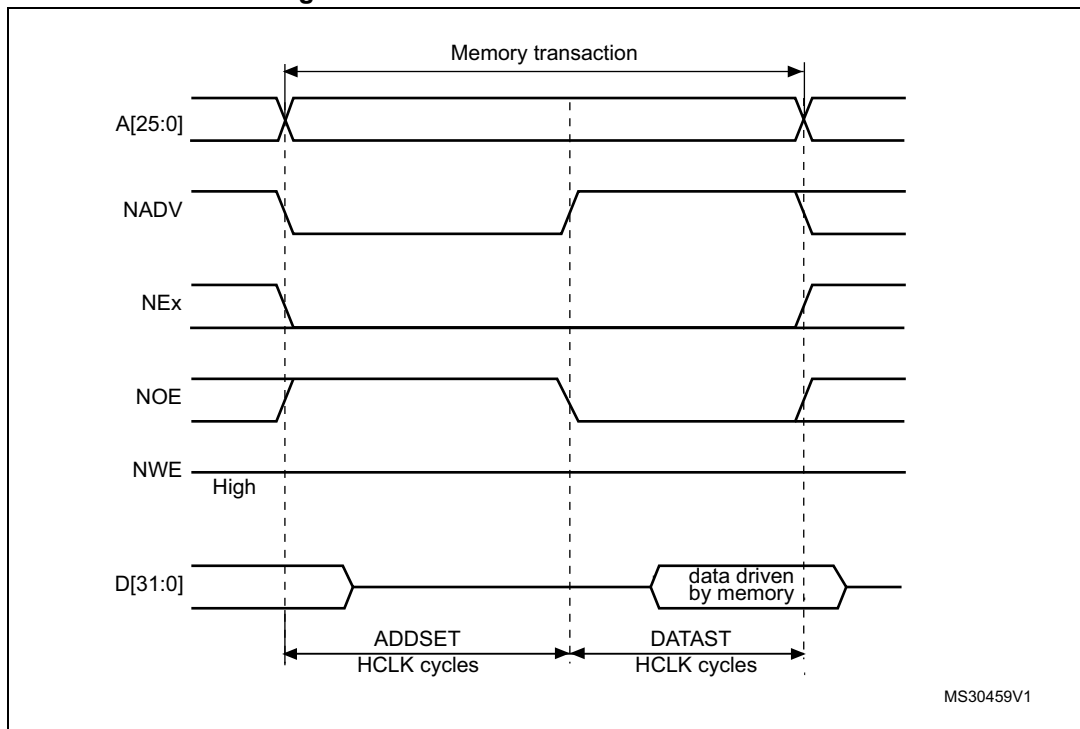
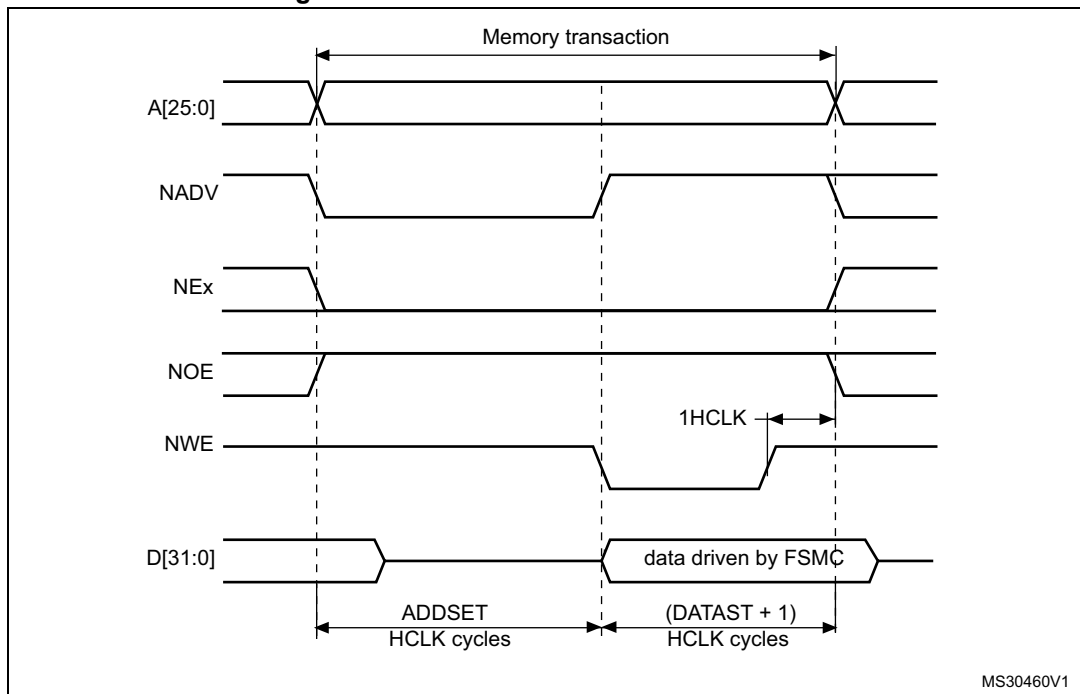


Figure 466. ModeC write access waveforms



The differences compared with mode1 are the toggling of NOE and the independent read and write timings.

Table 278. FMC\_BCRx bit fields

| Bit No. | Bit name   | Value to set                                                       |
|---------|------------|--------------------------------------------------------------------|
| 31-21   | Reserved   | 0x000                                                              |
| 20      | CCLKEN     | As needed                                                          |
| 19      | CBURSTRW   | 0x0 (no effect in asynchronous mode)                               |
| 18:16   | Reserved   | 0x0 (no effect in asynchronous mode)                               |
| 15      | ASYNCAWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD     | 0x1                                                                |
| 13      | WAITEN     | 0x0 (no effect in asynchronous mode)                               |
| 12      | WREN       | As needed                                                          |
| 11      | WAITCFG    | Don't care                                                         |
| 10      | WRAPMOD    | 0x0                                                                |
| 9       | WAITPOL    | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN    | 0x0                                                                |
| 7       | Reserved   | 0x1                                                                |
| 6       | FACCEN     | 0x1                                                                |
| 5-4     | MWID       | As needed                                                          |
| 3-2     | MTYP[1:0]  | 0x02 (NOR Flash memory)                                            |
| 1       | MUXEN      | 0x0                                                                |
| 0       | MBKEN      | 0x1                                                                |

Table 279. FMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                              |
|---------|-------------|-----------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                       |
| 29-28   | ACCMOD      | 0x2                                                                                                       |
| 27-24   | DATLAT      | 0x0                                                                                                       |
| 23-20   | CLKDIV      | 0x0                                                                                                       |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                           |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                               |
| 7-4     | ADDHLD      | Don't care                                                                                                |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 0. |

Table 280. FMC\_BWTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                               |
|---------|-------------|------------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                        |
| 29-28   | ACCMOD      | 0x2                                                                                                        |
| 27-24   | DATLAT      | Don't care                                                                                                 |
| 23-20   | CLKDIV      | Don't care                                                                                                 |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                            |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for write accesses.                               |
| 7-4     | ADDHLD      | Don't care                                                                                                 |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 0. |

**Mode D - asynchronous access with extended address**

Figure 467. ModeD read access waveforms

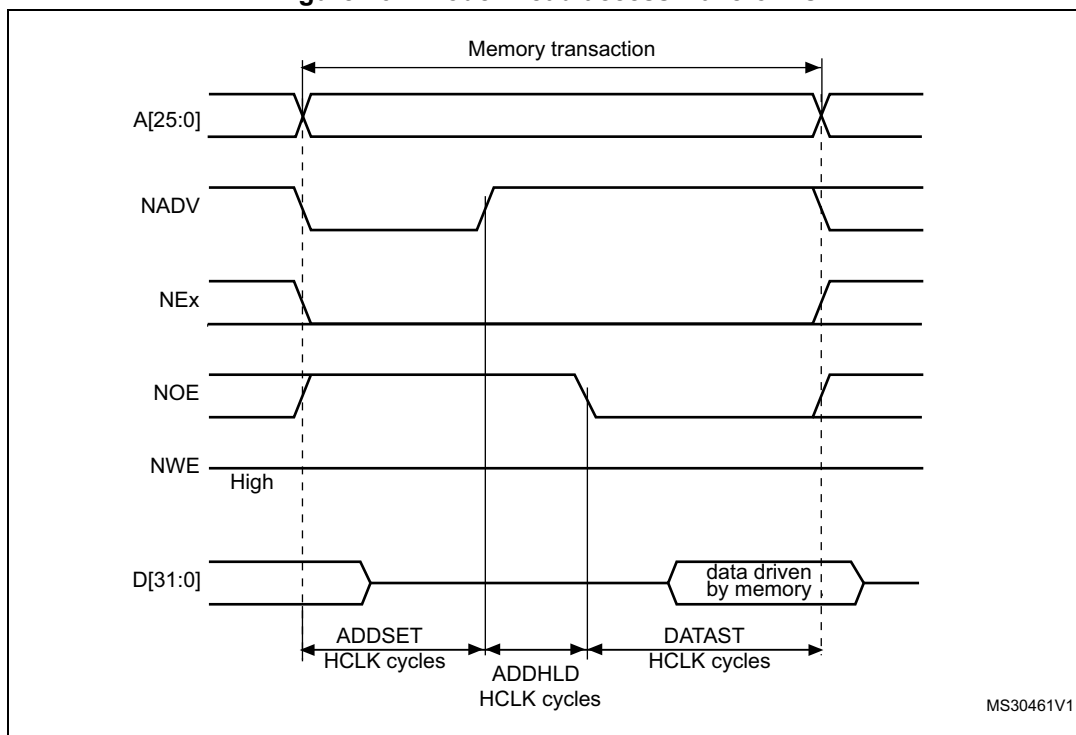
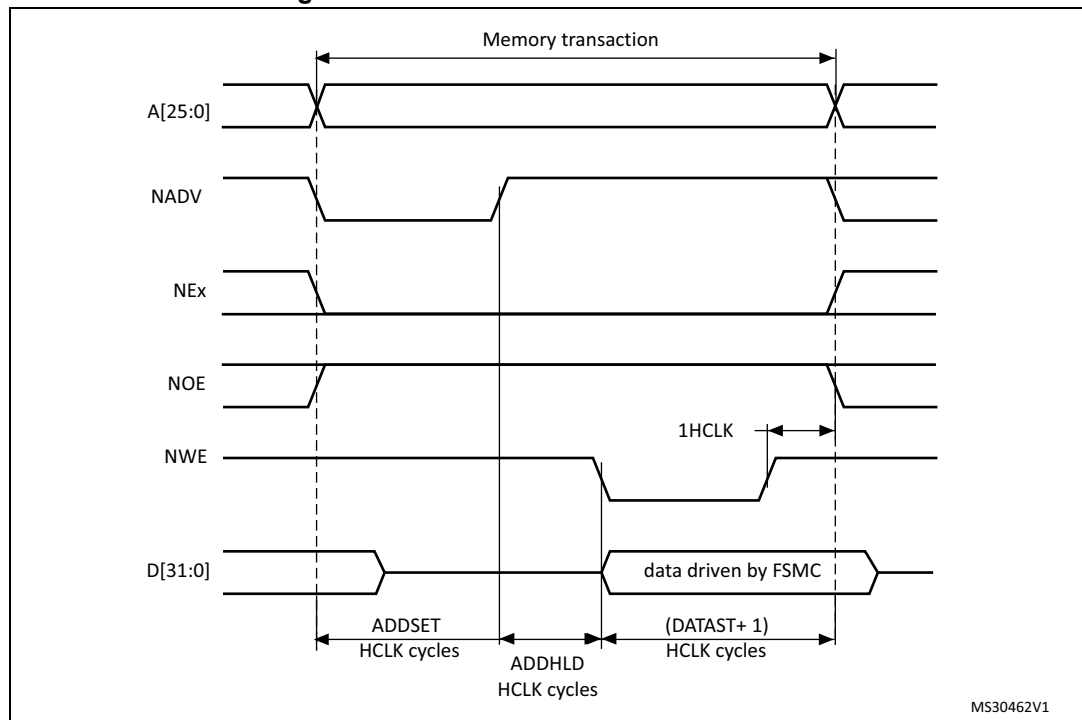


Figure 468. ModeD write access waveforms



MS30462V1

The differences with mode1 are the toggling of NOE that goes on toggling after NADV changes and the independent read and write timings.

Table 281. FMC\_BCRx bit fields

| Bit No. | Bit name   | Value to set                                                       |
|---------|------------|--------------------------------------------------------------------|
| 31-21   | Reserved   | 0x000                                                              |
| 20      | CCLKEN     | As needed                                                          |
| 19      | CBURSTRW   | 0x0 (no effect in asynchronous mode)                               |
| 18:16   | CPSIZE     | 0x0 (no effect in asynchronous mode)                               |
| 15      | ASYNCAWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD     | 0x1                                                                |
| 13      | WAITEN     | 0x0 (no effect in asynchronous mode)                               |
| 12      | WREN       | As needed                                                          |
| 11      | WAITCFG    | Don't care                                                         |
| 10      | WRAPMOD    | 0x0                                                                |
| 9       | WAITPOL    | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN    | 0x0                                                                |
| 7       | Reserved   | 0x1                                                                |
| 6       | FACCEN     | Set according to memory support                                    |
| 5-4     | MWID       | As needed                                                          |

Table 281. FMC\_BCRx bit fields (continued)

| Bit No. | Bit name  | Value to set |
|---------|-----------|--------------|
| 3-2     | MTYP[1:0] | As needed    |
| 1       | MUXEN     | 0x0          |
| 0       | MBKEN     | 0x1          |

Table 282. FMC\_BTRx bit fields

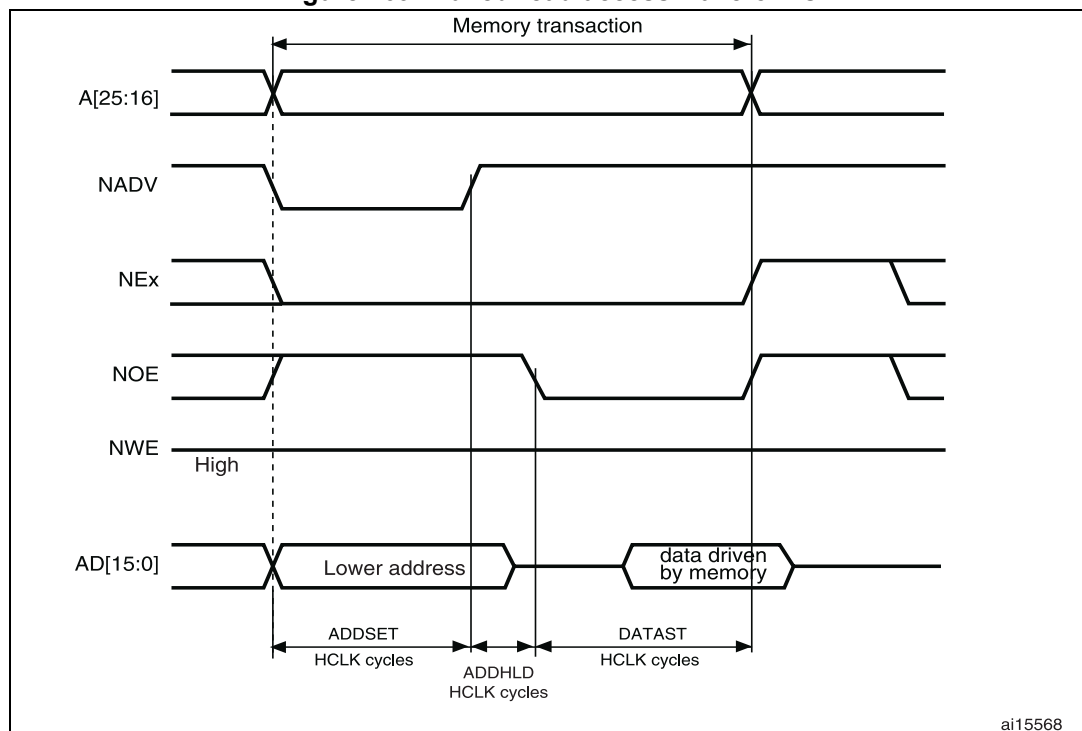
| Bit No. | Bit name    | Value to set                                                                                              |
|---------|-------------|-----------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                       |
| 29-28   | ACCMOD      | 0x3                                                                                                       |
| 27-24   | DATLAT      | Don't care                                                                                                |
| 23-20   | CLKDIV      | Don't care                                                                                                |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                           |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles) for read accesses.                               |
| 7-4     | ADDHLD      | Duration of the middle phase of the read access (ADDHLD HCLK cycles)                                      |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for read accesses. Minimum value for ADDSET is 1. |

Table 283. FMC\_BWTRx bit fields

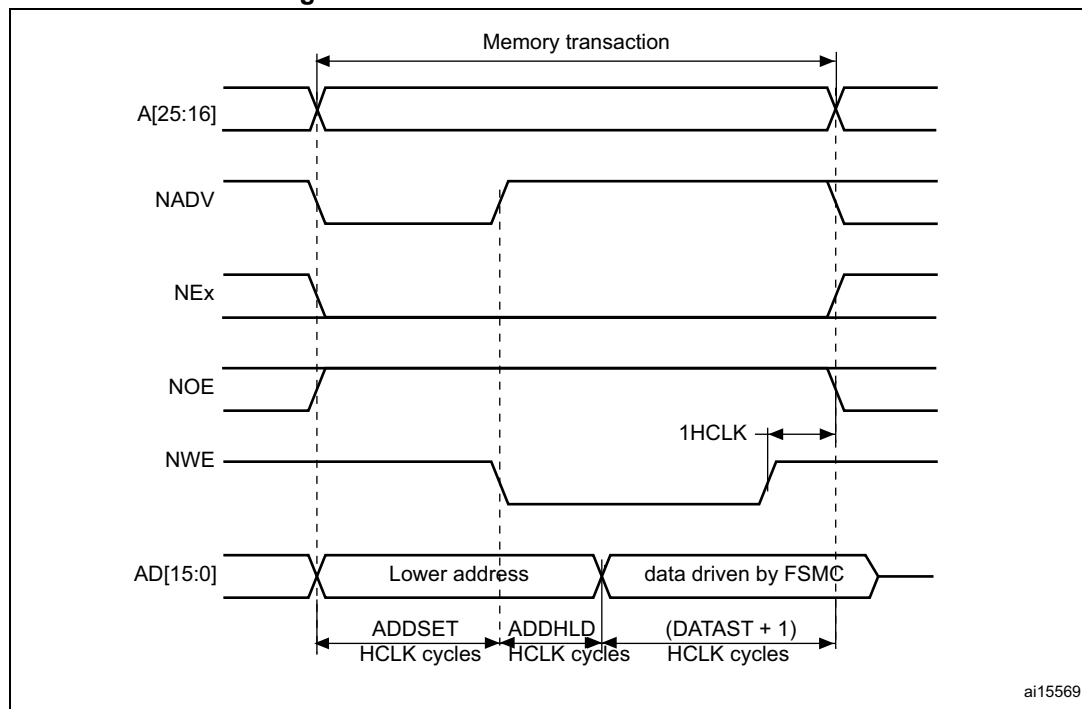
| Bit No. | Bit name    | Value to set                                                                                               |
|---------|-------------|------------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                        |
| 29-28   | ACCMOD      | 0x3                                                                                                        |
| 27-24   | DATLAT      | Don't care                                                                                                 |
| 23-20   | CLKDIV      | Don't care                                                                                                 |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                            |
| 15-8    | DATAST      | Duration of the second access phase (DATAST + 1 HCLK cycles) for write accesses.                           |
| 7-4     | ADDHLD      | Duration of the middle phase of the write access (ADDHLD HCLK cycles)                                      |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles) for write accesses. Minimum value for ADDSET is 1. |

## Muxed mode - multiplexed asynchronous access to NOR Flash memory

**Figure 469. Muxed read access waveforms**



**Figure 470. Muxed write access waveforms**



The difference with mode D is the drive of the lower address byte(s) on the data bus.

Table 284. FMC\_BCRx bit fields

| Bit No. | Bit name  | Value to set                                                       |
|---------|-----------|--------------------------------------------------------------------|
| 31-21   | Reserved  | 0x000                                                              |
| 20      | CCLKEN    | As needed                                                          |
| 19      | CBURSTRW  | 0x0 (no effect in asynchronous mode)                               |
| 18:16   | CPSIZE    | 0x0 (no effect in asynchronous mode)                               |
| 15      | ASYNCWAIT | Set to 1 if the memory supports this feature. Otherwise keep at 0. |
| 14      | EXTMOD    | 0x0                                                                |
| 13      | WAITEN    | 0x0 (no effect in asynchronous mode)                               |
| 12      | WREN      | As needed                                                          |
| 11      | WAITCFG   | Don't care                                                         |
| 10      | WRAPMOD   | 0x0                                                                |
| 9       | WAITPOL   | Meaningful only if bit 15 is 1                                     |
| 8       | BURSTEN   | 0x0                                                                |
| 7       | Reserved  | 0x1                                                                |
| 6       | FACCEN    | 0x1                                                                |
| 5-4     | MWID      | As needed                                                          |
| 3-2     | MTYP[1:0] | 0x2 (NOR Flash memory)                                             |
| 1       | MUXEN     | 0x1                                                                |
| 0       | MBKEN     | 0x1                                                                |

Table 285. FMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                                                                            |
|---------|-------------|-------------------------------------------------------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                                                                     |
| 29-28   | ACCMOD      | 0x0                                                                                                                     |
| 27-24   | DATLAT      | Don't care                                                                                                              |
| 23-20   | CLKDIV      | Don't care                                                                                                              |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                                                                         |
| 15-8    | DATAST      | Duration of the second access phase (DATAST HCLK cycles for read accesses and DATAST+1 HCLK cycles for write accesses). |
| 7-4     | ADDHLD      | Duration of the middle phase of the access (ADDHLD HCLK cycles).                                                        |
| 3-0     | ADDSET[3:0] | Duration of the first access phase (ADDSET HCLK cycles). Minimum value for ADDSET is 1.                                 |

### WAIT management in asynchronous accesses

If the asynchronous memory asserts the WAIT signal to indicate that it is not yet ready to accept or to provide data, the ASYNCWAIT bit has to be set in FMC\_BCRx register.

If the WAIT signal is active (high or low depending on the WAITPOL bit), the second access phase (Data setup phase), programmed by the DATAST bits, is extended until WAIT becomes inactive. Unlike the data setup phase, the first access phases (Address setup and Address hold phases), programmed by the ADDSET[3:0] and ADDHLD bits, are not WAIT sensitive and so they are not prolonged.

The data setup phase must be programmed so that WAIT can be detected 4 HCLK cycles before the end of the memory transaction. The following cases must be considered:

1. The memory asserts the WAIT signal aligned to NOE/NWE which toggles:

$$\text{DATAST} \geq (4 \times \text{HCLK}) + \text{max\_wait\_assertion\_time}$$

2. The memory asserts the WAIT signal aligned to NEx (or NOE/NWE not toggling):  
if

$$\text{max\_wait\_assertion\_time} > \text{address\_phase} + \text{hold\_phase}$$

then:

$$\text{DATAST} \geq (4 \times \text{HCLK}) + (\text{max\_wait\_assertion\_time} - \text{address\_phase} - \text{hold\_phase})$$

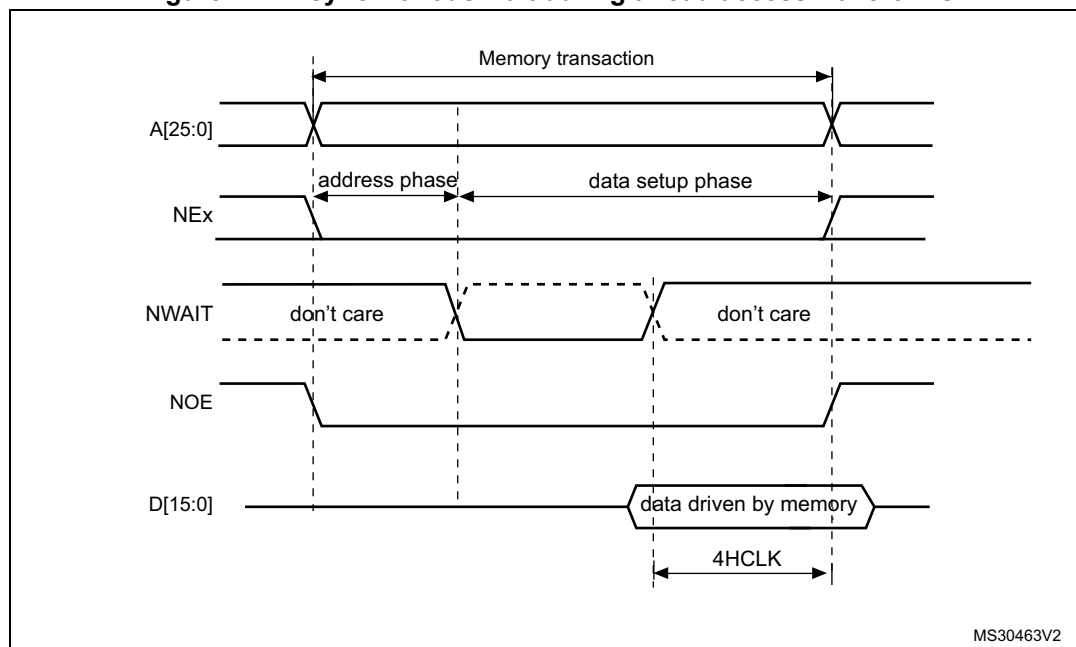
otherwise

$$\text{DATAST} \geq 4 \times \text{HCLK}$$

where max\_wait\_assertion\_time is the maximum time taken by the memory to assert the WAIT signal once NEx/NOE/NWE is low.

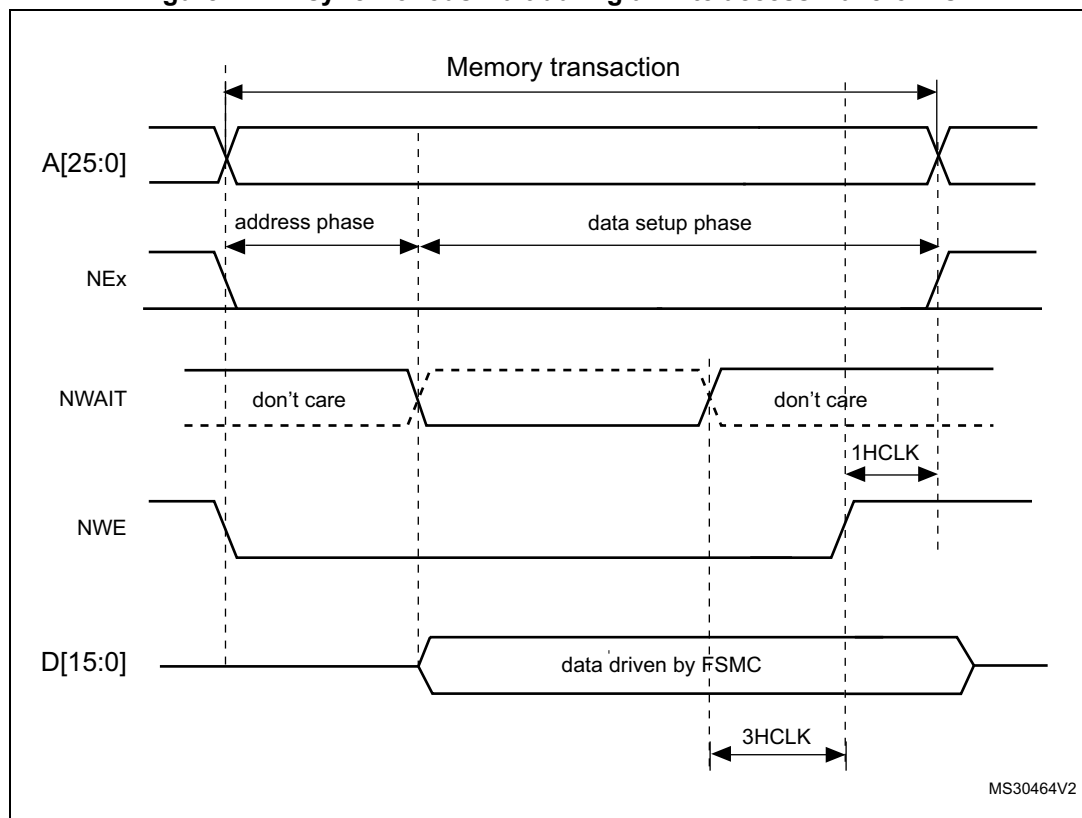
Figure 471 and Figure 472 show the number of HCLK clock cycles that are added to the memory access phase after WAIT is released by the asynchronous memory (independently of the above cases).

**Figure 471. Asynchronous wait during a read access waveforms**



1. N WAIT polarity depends on WAITPOL bit setting in FMC\_BCRx register.

Figure 472. Asynchronous wait during a write access waveforms



1. NWAIT polarity depends on WAITPOL bit setting in FMC\_BCRx register.

### 37.5.5 Synchronous transactions

The memory clock, FMC\_CLK, is a submultiple of HCLK. It depends on the value of CLKDIV and the MWID/ AHB data size, following the formula given below:

$$\text{FMC\_CLK divider ratio} = \max(\text{CLKDIV} + 1, \text{MWID}(\text{AHB data size}))$$

If MWID is 16 or 8 bits, the FMC\_CLK divider ratio is always defined by the programmed CLKDIV value.

If MWID is 32 bits, the FMC\_CLK divider ratio depends also on AHB data size.

Example:

- If CLKDIV=1, MWID=32 bits, AHB data size=8 bits, FMC\_CLK=HCLK/4.
- If CLKDIV=1, MWID=16 bits, AHB data size=8 bits, FMC\_CLK=HCLK/2.

NOR Flash memories specify a minimum time from NADV assertion to CLK high. To meet this constraint, the FMC does not issue the clock to the memory during the first internal clock cycle of the synchronous access (before NADV assertion). This guarantees that the rising edge of the memory clock occurs in the middle of the NADV low pulse.

#### Data latency versus NOR memory latency

The data latency is the number of cycles to wait before sampling the data. The DATLAT value must be consistent with the latency value specified in the NOR Flash configuration

register. The FMC does not include the clock cycle when NADV is low in the data latency count.

**Caution:** Some NOR Flash memories include the NADV Low cycle in the data latency count, so that the exact relation between the NOR Flash latency and the FMC DATLAT parameter can be either:

- NOR Flash latency = (DATLAT + 2) CLK clock cycles
- or NOR Flash latency = (DATLAT + 3) CLK clock cycles

Some recent memories assert NWAIT during the latency phase. In such cases DATLAT can be set to its minimum value. As a result, the FMC samples the data and waits long enough to evaluate if the data are valid. Thus the FMC detects when the memory exits latency and real data are processed.

Other memories do not assert NWAIT during latency. In this case the latency must be set correctly for both the FMC and the memory, otherwise invalid data are mistaken for good data, or valid data are lost in the initial phase of the memory access.

### Single-burst transfer

When the selected bank is configured in burst mode for synchronous accesses, if for example an AHB single-burst transaction is requested on 16-bit memories, the FMC performs a burst transaction of length 1 (if the AHB transfer is 16 bits), or length 2 (if the AHB transfer is 32 bits) and de-assert the Chip Select signal when the last data is strobed.

Such transfers are not the most efficient in terms of cycles compared to asynchronous read operations. Nevertheless, a random asynchronous access would first require to re-program the memory access mode, which would altogether last longer.

### Cross boundary page for Cellular RAM 1.5

Cellular RAM 1.5 does not allow burst access to cross the page boundary. The FMC controller allows to split automatically the burst access when the memory page size is reached by configuring the CPSIZE bits in the FMC\_BCR1 register following the memory page size.

### Wait management

For synchronous NOR Flash memories, NWAIT is evaluated after the programmed latency period, which corresponds to (DATLAT+2) CLK clock cycles.

If NWAIT is active (low level when WAITPOL = 0, high level when WAITPOL = 1), wait states are inserted until NWAIT is inactive (high level when WAITPOL = 0, low level when WAITPOL = 1).

When NWAIT is inactive, the data is considered valid either immediately (bit WAITCFG = 1) or on the next clock edge (bit WAITCFG = 0).

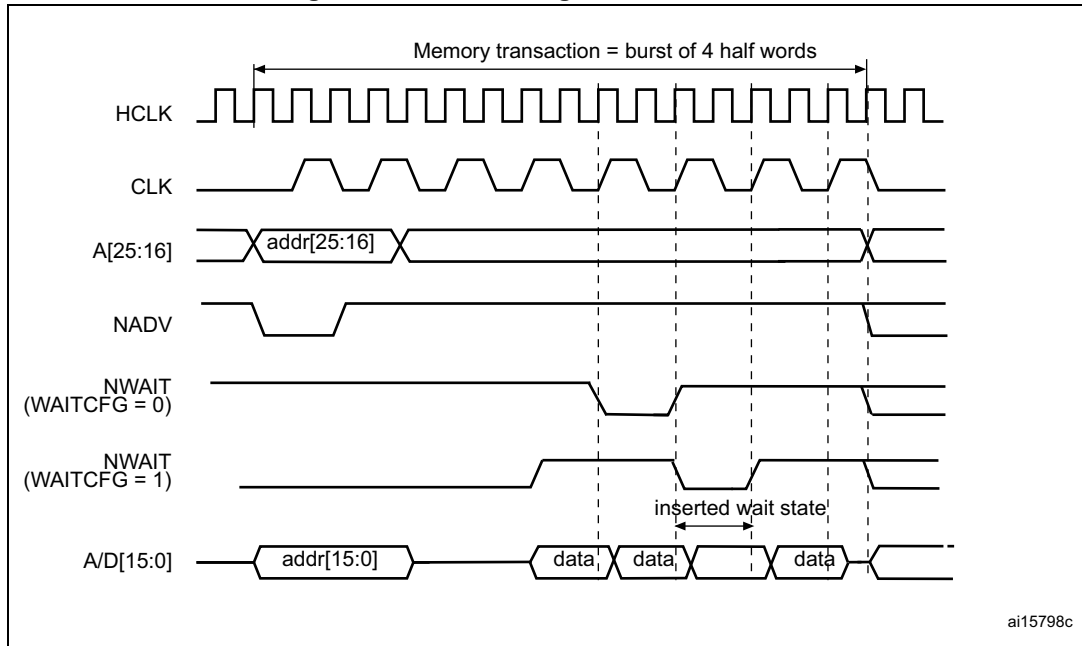
During wait-state insertion via the NWAIT signal, the controller continues to send clock pulses to the memory, keeping the Chip Select and output enable signals valid. It does not consider the data as valid.

In burst mode, there are two timing configurations for the NOR Flash NWAIT signal:

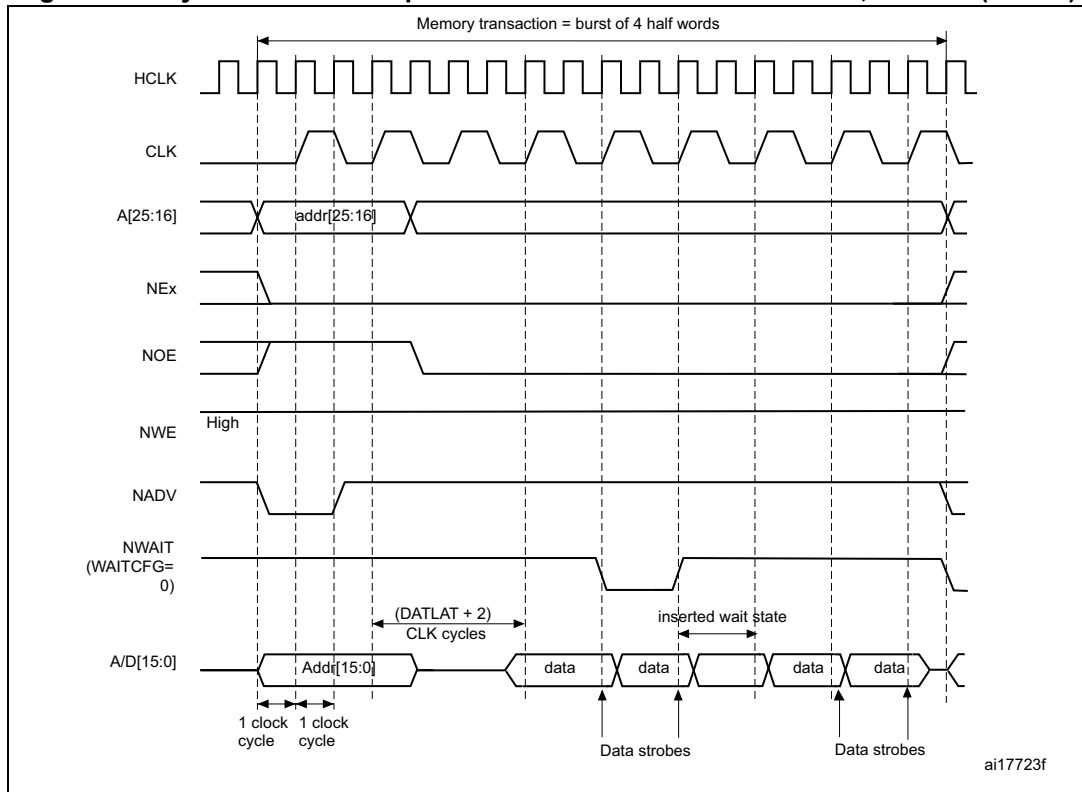
- The Flash memory asserts the NWAIT signal one data cycle before the wait state (default after reset).
- The Flash memory asserts the NWAIT signal during the wait state

The FMC supports both NOR Flash wait state configurations, for each Chip Select, thanks to the WAITCFG bit in the FMC\_BCRx registers ( $x = 0..3$ ).

**Figure 473. Wait configuration waveforms**



**Figure 474. Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM)**



1. Byte lane outputs BL are not shown; for NOR access, they are held high, and, for PSRAM (CRAM) access,

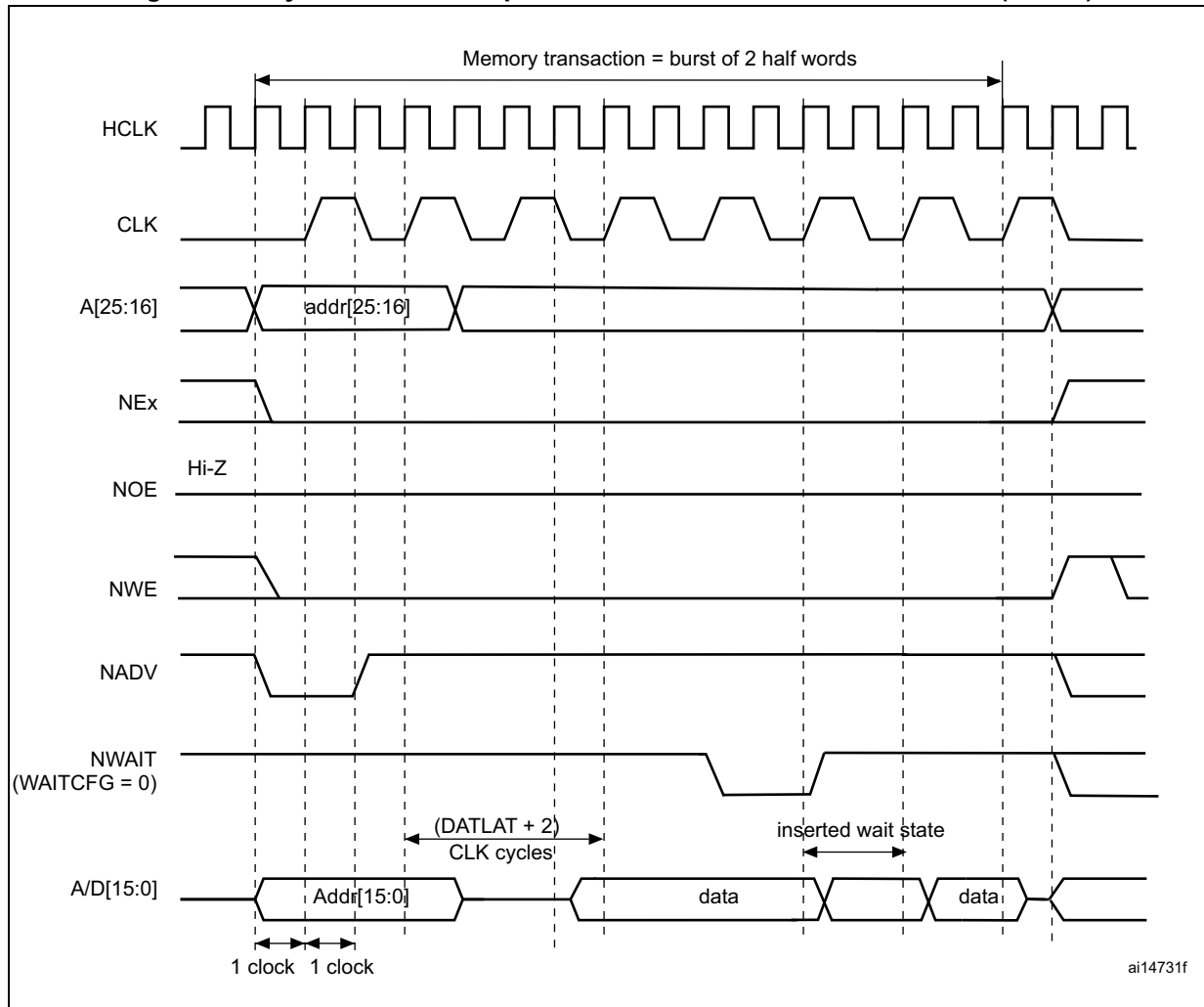
they are held low.

**Table 286. FMC\_BCRx bit fields**

| Bit No. | Bit name   | Value to set                                                                  |
|---------|------------|-------------------------------------------------------------------------------|
| 31-21   | Reserved   | 0x000                                                                         |
| 20      | CCLKEN     | As needed                                                                     |
| 19      | CBURSTRW   | No effect on synchronous read                                                 |
| 18-16   | CPSIZE     | As needed (0x1 for CRAM 1.5)                                                  |
| 15      | ASYNCAWAIT | 0x0                                                                           |
| 14      | EXTMOD     | 0x0                                                                           |
| 13      | WAITEN     | to be set to 1 if the memory supports this feature, to be kept at 0 otherwise |
| 12      | WREN       | no effect on synchronous read                                                 |
| 11      | WAITCFG    | to be set according to memory                                                 |
| 10      | WRAPMOD    | 0x0                                                                           |
| 9       | WAITPOL    | to be set according to memory                                                 |
| 8       | BURSTEN    | 0x1                                                                           |
| 7       | Reserved   | 0x1                                                                           |
| 6       | FACCEN     | Set according to memory support (NOR Flash memory)                            |
| 5-4     | MWID       | As needed                                                                     |
| 3-2     | MTYP[1:0]  | 0x1 or 0x2                                                                    |
| 1       | MUXEN      | As needed                                                                     |
| 0       | MBKEN      | 0x1                                                                           |

**Table 287. FMC\_BTRx bit fields**

| Bit No. | Bit name    | Value to set                                                             |
|---------|-------------|--------------------------------------------------------------------------|
| 31:30   | Reserved    | 0x0                                                                      |
| 29:28   | ACCMOD      | 0x0                                                                      |
| 27-24   | DATLAT      | Data latency                                                             |
| 27-24   | DATLAT      | Data latency                                                             |
| 23-20   | CLKDIV      | 0x0 to get CLK = HCLK (Not supported)<br>0x1 to get CLK = 2 × HCLK<br>.. |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                          |
| 15-8    | DATAST      | Don't care                                                               |
| 7-4     | ADDHLD      | Don't care                                                               |
| 3-0     | ADDSET[3:0] | Don't care                                                               |

**Figure 475. Synchronous multiplexed write mode waveforms - PSRAM (CRAM)**

1. The memory must issue NWAIT signal one cycle in advance, accordingly WAITCFG must be programmed to 0.
2. Byte Lane (NBL) outputs are not shown, they are held low while NEx is active.

**Table 288. FMC\_BCRx bit fields**

| Bit No. | Bit name  | Value to set                                                                   |
|---------|-----------|--------------------------------------------------------------------------------|
| 31-20   | Reserved  | 0x000                                                                          |
| 20      | CCLKEN    | As needed                                                                      |
| 19      | CBURSTRW  | 0x1                                                                            |
| 18-16   | CPSIZE    | As needed (0x1 for CRAM 1.5)                                                   |
| 15      | ASYNCWAIT | 0x0                                                                            |
| 14      | EXTMOD    | 0x0                                                                            |
| 13      | WAITEN    | to be set to 1 if the memory supports this feature, to be kept at 0 otherwise. |
| 12      | WREN      | 0x1                                                                            |

Table 288. FMC\_BCRx bit fields (continued)

| Bit No. | Bit name  | Value to set                    |
|---------|-----------|---------------------------------|
| 11      | WAITCFG   | 0x0                             |
| 10      | WRAPMOD   | 0x0                             |
| 9       | WAITPOL   | to be set according to memory   |
| 8       | BURSTEN   | no effect on synchronous write  |
| 7       | Reserved  | 0x1                             |
| 6       | FACCEN    | Set according to memory support |
| 5-4     | MWID      | As needed                       |
| 3-2     | MTYP[1:0] | 0x1                             |
| 1       | MUXEN     | As needed                       |
| 0       | MBKEN     | 0x1                             |

Table 289. FMC\_BTRx bit fields

| Bit No. | Bit name    | Value to set                                                       |
|---------|-------------|--------------------------------------------------------------------|
| 31-30   | Reserved    | 0x0                                                                |
| 29:28   | ACCMOD      | 0x0                                                                |
| 27-24   | DATLAT      | Data latency                                                       |
| 23-20   | CLKDIV      | 0x0 to get CLK = HCLK (not supported)<br>0x1 to get CLK = 2 × HCLK |
| 19-16   | BUSTURN     | Time between NEx high to NEx low (BUSTURN HCLK)                    |
| 15-8    | DATAST      | Don't care                                                         |
| 7-4     | ADDHLD      | Don't care                                                         |
| 3-0     | ADDSET[3:0] | Don't care                                                         |

## 37.5.6 NOR/PSRAM controller registers

### SRAM/NOR-Flash chip-select control registers 1..4 (FMC\_BCR1..4)

Address offset:  $8 * (x - 1)$ ,  $x = 1...4$

Reset value: 0x0000 30DB for Bank1 and 0x0000 30D2 for Bank 2 to 4

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR Flash memories.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19     | 18          | 17       | 16       | 15     | 14     | 13   | 12      | 11      | 10      | 9       | 8        | 7      | 6         | 5  | 4  | 3         | 2  | 1     | 0     |
|----------|----|----|----|----|----|----|----|----|----|----|----|--------|-------------|----------|----------|--------|--------|------|---------|---------|---------|---------|----------|--------|-----------|----|----|-----------|----|-------|-------|
| Reserved |    |    |    |    |    |    |    |    |    |    |    | CCLKEN | CPSIZE[2:0] | CBURSTRW | ASCYWAIT | EXTMOD | WAITEN | WREN | WAITCFG | WRAPMOD | WAITPOL | BURSTEN | Reserved | FACCEN | MWID[1:0] |    |    | MTYP[1:0] |    | MUXEN | MBKEN |
|          |    |    |    |    |    |    |    |    |    |    |    | rw     |             |          |          |        |        |      |         |         |         |         |          | rw     | rw        | rw | rw | rw        | rw | rw    | rw    |

Bits 31: 21 Reserved, must be kept at reset value

Bit 20 **CCLKEN**: Continuous Clock Enable.

This bit enables the FMC\_CLK clock output to external memory devices.

0: The FMC\_CLK is only generated during the synchronous memory access (read/write transaction). The FMC\_CLK clock ratio is specified by the programmed CLKDIV value in the FMC\_BCRx register (default after reset) .

1: The FMC\_CLK is generated continuously during asynchronous and synchronous access. The FMC\_CLK clock is activated when the CCLKEN is set.

*Note: The CCLKEN bit of the FMC\_BCR2..4 registers is don't care. It is only enabled through the FMC\_BCR1 register. Bank 1 must be configured in synchronous mode to generate the FMC\_CLK continuous clock.*

*Note: If CCLKEN bit is set, the FMC\_CLK clock ratio is specified by CLKDIV value in the FMC\_BTR1 register. CLKDIV in FMC\_BWTR1 is don't care.*

*Note: If the synchronous mode is used and CCLKEN bit is set, the synchronous memories connected to other banks than Bank 1 are clocked by the same clock (the CLKDIV value in the FMC\_BTR2..4 and FMC\_BWTR2..4 registers for other banks has no effect.)*

Bit 19 **CBURSTRW**: Write burst enable.

For PSRAM (CRAM) operating in burst mode, the bit enables synchronous accesses during write operations. The enable bit for synchronous read accesses is the BURSTEN bit in the FMC\_BCRx register.

0: Write operations are always performed in asynchronous mode

1: Write operations are performed in synchronous mode.

Bits 18:16 **CPSIZE[2:0]**: CRAM page size.

These are used for Cellular RAM 1.5 which does not allow burst access to cross the address boundaries between pages. When these bits are configured, the FMC controller splits automatically the burst access when the memory page size is reached (refer to memory datasheet for page size).

000: No burst split when crossing page boundary (default after reset)

001: 128 bytes

010: 256 bytes

011: 512 bytes

100: 1024 bytes

Others: reserved

- Bit 15 **ASYNCWAIT**: Wait signal during asynchronous transfers  
 This bit enables/disables the FMC to use the wait signal even during an asynchronous protocol.  
 0: NWAIT signal is not taken in to account when running an asynchronous protocol (default after reset)  
 1: NWAIT signal is taken in to account when running an asynchronous protocol
- Bit 14 **EXTMOD**: Extended mode enable.  
 This bit enables the FMC to program the write timings for non-multiplexed asynchronous accesses inside the FMC\_BWTR register, thus resulting in different timings for read and write operations.  
 0: values inside FMC\_BWTR register are not taken into account (default after reset)  
 1: values inside FMC\_BWTR register are taken into account  
*Note: When the extended mode is disabled, the FMC can operate in Mode1 or Mode2 as follows:*  
 – Mode 1 is the default mode when the SRAM/PSRAM memory type is selected (MTYP[1:0] = 0x0 or 0x01)  
 – Mode 2 is the default mode when the NOR memory type is selected (MTYP[1:0] = 0x10).
- Bit 13 **WAITEN**: Wait enable bit.  
 This bit enables/disables wait-state insertion via the NWAIT signal when accessing the memory in synchronous mode.  
 0: NWAIT signal is disabled (its level not taken into account, no wait state inserted after the programmed Flash latency period)  
 1: NWAIT signal is enabled (its level is taken into account after the programmed latency period to insert wait states if asserted) (default after reset)
- Bit 12 **WREN**: Write enable bit.  
 This bit indicates whether write operations are enabled/disabled in the bank by the FMC:  
 0: Write operations are disabled in the bank by the FMC, an AHB error is reported,  
 1: Write operations are enabled for the bank by the FMC (default after reset).
- Bit 11 **WAITCFG**: Wait timing configuration.  
 The NWAIT signal indicates whether the data from the memory are valid or if a wait state must be inserted when accessing the memory in synchronous mode. This configuration bit determines if NWAIT is asserted by the memory one clock cycle before the wait state or during the wait state:  
 0: NWAIT signal is active one data cycle before wait state (default after reset),  
 1: NWAIT signal is active during wait state (not used for PSRAM).
- Bit 10 **WRAPMOD**: Wrapped burst mode support.  
 Defines whether the controller splits or not an AHB burst wrap access into two linear accesses. Valid only when accessing memories in burst mode  
 0: Direct wrapped burst is not enabled (default after reset),  
 1: Direct wrapped burst is enabled.  
*Note: This bit has no effect as the CPU and DMA cannot generate wrapping burst transfers.*
- Bit 9 **WAITPOL**: Wait signal polarity bit.  
 Defines the polarity of the wait signal from memory used for either in synchronous or asynchronous mode:  
 0: NWAIT active low (default after reset),  
 1: NWAIT active high.
- Bit 8 **BURSTEN**: Burst enable bit.  
 This bit enables/disables synchronous accesses during read operations. It is valid only for synchronous memories operating in burst mode:  
 0: Burst mode disabled (default after reset). Read accesses are performed in asynchronous mode.  
 1: Burst mode enable. Read accesses are performed in synchronous mode.
- Bit 7 Reserved, must be kept at reset value

- Bit 6 **FACCEN**: Flash access enable  
 Enables NOR Flash memory access operations.  
 0: Corresponding NOR Flash memory access is disabled  
 1: Corresponding NOR Flash memory access is enabled (default after reset)
- Bits 5:4 **MWID[1:0]**: Memory data bus width.  
 Defines the external memory device width, valid for all type of memories.  
 00: 8 bits,  
 01: 16 bits (default after reset),  
 10: 32 bits,  
 11: reserved, do not use.
- Bits 3:2 **MTYP[1:0]**: Memory type.  
 Defines the type of external memory attached to the corresponding memory bank:  
 00: SRAM (default after reset for Bank 2...4)  
 01: PSRAM (CRAM)  
 10: NOR Flash/OneNAND Flash (default after reset for Bank 1)  
 11: reserved
- Bit 1 **MUXEN**: Address/data multiplexing enable bit.  
 When this bit is set, the address and data values are multiplexed on the data bus, valid only with NOR and PSRAM memories:  
 0: Address/Data nonmultiplexed  
 1: Address/Data multiplexed on databus (default after reset)
- Bit 0 **MBKEN**: Memory bank enable bit.  
 Enables the memory bank. After reset Bank1 is enabled, all others are disabled. Accessing a disabled bank causes an ERROR on AHB bus.  
 0: Corresponding memory bank is disabled  
 1: Corresponding memory bank is enabled

### SRAM/NOR-Flash chip-select timing registers 1..4 (FMC\_BTR1..4)

Address offset:  $0x04 + 8 * (x - 1)$ ,  $x = 1..4$

Reset value: 0x0FFF FFFF

Reset value: 0x0FFF FFFF

FMC\_BTRx bits are written by software to add a delay at the end of a read /write transaction. This delay allows matching the minimum time between consecutive transactions ( $t_{EHEL}$  from NEx high to FMC\_NEx low) and the maximum time required by the memory to free the data bus after a read access ( $t_{EHQZ}$ ).

This register contains the control information of each memory bank, used for SRAMs, PSRAM and NOR Flash memories. If the EXTMOD bit is set in the FMC\_BCRx register, then this register is partitioned for write and read access, that is, 2 registers are available: one to configure read accesses (this register) and one to configure write accesses (FMC\_BWTRx registers).

| 31       | 30 | 29          | 28  | 27          | 26  | 25  | 24  | 23          | 22  | 21  | 20  | 19           | 18  | 17  | 16  | 15        | 14  | 13  | 12  | 11  | 10  | 9   | 8           | 7   | 6   | 5   | 4           | 3   | 2   | 1   | 0   |
|----------|----|-------------|-----|-------------|-----|-----|-----|-------------|-----|-----|-----|--------------|-----|-----|-----|-----------|-----|-----|-----|-----|-----|-----|-------------|-----|-----|-----|-------------|-----|-----|-----|-----|
| Reserved |    | ACCMOD[1:0] |     | DATLAT[3:0] |     |     |     | CLKDIV[3:0] |     |     |     | BUSTURN[3:0] |     |     |     | DATA[7:0] |     |     |     |     |     |     | ADDHLD[3:0] |     |     |     | ADDSET[3:0] |     |     |     |     |
|          |    | r/w         | r/w | r/w         | r/w | r/w | r/w | r/w         | r/w | r/w | r/w | r/w          | r/w | r/w | r/w | r/w       | r/w | r/w | r/w | r/w | r/w | r/w | r/w         | r/w | r/w | r/w | r/w         | r/w | r/w | r/w | r/w |

Bits 31:30 Reserved, must be kept at reset value

Bits 29:28 **ACCMOD[1:0]**: Access mode

Specifies the asynchronous access modes as shown in the timing diagrams. These bits are taken into account only when the EXTMOD bit in the FMC\_BCRx register is 1.

00: access mode A

01: access mode B

10: access mode C

11: access mode D

Bits 27:24 **DATLAT[3:0]**: Data latency for synchronous NOR Flash memory (see note below bit description table)

For synchronous accesses with read/write burst mode enabled (BURSTEN / CBURSTRW bits set), this field defines the number of memory clock cycles (+2) to issue to the memory before reading/writing the first data. This timing parameter is not expressed in HCLK periods, but in FMC\_CLK periods. For asynchronous accesses, this value is don't care.

0000: Data latency of 2 CLK clock cycles for first burst access

1111: Data latency of 17 CLK clock cycles for first burst access (default value after reset)

Bits 23:20 **CLKDIV[3:0]**: Clock divide ratio (for FMC\_CLK signal)

Defines the period of FMC\_CLK clock output signal, expressed in number of HCLK cycles:

0000: Reserved

0001: FMC\_CLK period = 2 × HCLK periods

0010: FMC\_CLK period = 3 × HCLK periods

1111: FMC\_CLK period = 16 × HCLK periods (default value after reset)

In asynchronous NOR Flash, SRAM or PSRAM accesses, this value is don't care.

*Note: Refer to section 37.5.5: Synchronous transactions for FMC\_CLK divider ratio formula)*

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

These bits are written by software to add a delay at the end of a write-to-read (and read-to-write) transaction. This delay allows to match the minimum time between consecutive transactions ( $t_{\text{EHEL}}$  from NEx high to NEx low) and the maximum time needed by the memory to free the data bus after a read access ( $t_{\text{EHQZ}}$ ). The programmed bus turnaround delay is inserted between an asynchronous read (muxed or mode D) or write transaction and any other asynchronous /synchronous read or write to or from a static bank. The bank can be the same or different in case of read, in case of write the bank can be different except for muxed or mode D.

In some cases, whatever the programmed BUSTURN values, the bus turnaround delay is fixed as follows:

- The bus turnaround delay is not inserted between two consecutive asynchronous write transfers to the same static memory bank except for modes muxed and D.
- There is a bus turnaround delay of 1 FMC clock cycle between:
  - Two consecutive asynchronous read transfers to the same static memory bank except for modes muxed and D.
  - An asynchronous read to an asynchronous or synchronous write to any static bank or dynamic bank except for modes muxed and D.
  - An asynchronous (modes 1, 2, A, B or C) read and a read from another static bank.
- There is a bus turnaround delay of 2 FMC clock cycle between:
  - Two consecutive synchronous writes (burst or single) to the same bank.
  - A synchronous write (burst or single) access and an asynchronous write or read transfer to or from static memory bank (the bank can be the same or different for the case of read).
  - Two consecutive synchronous reads (burst or single) followed by any synchronous/asynchronous read or write from/to another static memory bank.
- There is a bus turnaround delay of 3 FMC clock cycle between:
  - Two consecutive synchronous writes (burst or single) to different static bank.
  - A synchronous write (burst or single) access and a synchronous read from the same or a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 x HCLK clock cycles added (default value after reset)

**Bits 15:8 DATAST[7:0]:** Data-phase duration

These bits are written by software to define the duration of the data phase (refer to [Figure 458](#) to [Figure 470](#)), used in asynchronous accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

For each memory type and access mode data-phase duration, please refer to the respective figure ([Figure 458](#) to [Figure 470](#)).

Example: Mode1, write access, DATAST=1: Data-phase duration= DATAST+1 = 2 HCLK clock cycles.

*Note: In synchronous accesses, this value is don't care.*

**Bits 7:4 ADDHLD[3:0]:** Address-hold phase duration

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 467](#) to [Figure 470](#)), used in mode D or multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address-hold phase duration, please refer to the respective figure ([Figure 467](#) to [Figure 470](#)).

*Note: In synchronous accesses, this value is not used, the address hold phase is always 1 memory clock period duration.*

**Bits 3:0 ADDSET[3:0]:** Address setup phase duration

These bits are written by software to define the duration of the *address setup* phase (refer to [Figure 458](#) to [Figure 470](#)), used in SRAMs, ROMs and asynchronous NOR Flash and PSRAM accesses:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

For each access mode address setup phase duration, please refer to the respective figure (refer to [Figure 458](#) to [Figure 470](#)).

*Note: In synchronous accesses, this value is don't care.*

*In Muxed mode or Mode D, the minimum value for ADDSET is 1.*

**Note:** *PSRAMs (CRAMs) have a variable latency due to internal refresh. Therefore these memories issue the NWAIT signal during the whole latency phase to prolong the latency as needed.*

*With PSRAMs (CRAMs) the filled DATLAT must be set to 0, so that the FMC exits its latency phase soon and starts sampling NWAIT from memory, then starts to read or write when the memory is ready.*

*This method can be used also with the latest generation of synchronous Flash memories that issue the NWAIT signal, unlike older Flash memories (check the datasheet of the specific Flash memory being used).*

**SRAM/NOR-Flash write timing registers 1..4 (FMC\_BWTR1..4)**

Address offset:  $0x104 + 8 * (x - 1)$ ,  $x = 1...4$

Reset value: 0x0FFF FFFF

This register contains the control information of each memory bank. It is used for SRAMs, PSRAMs and NOR Flash memories. When the EXTMOD bit is set in the FMC\_BCRx register, then this register is active for write access.

| 31       | 30 | 29  | 28  | 27       | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19           | 18  | 17  | 16  | 15         | 14  | 13  | 12  | 11  | 10  | 9   | 8   | 7           | 6   | 5   | 4   | 3           | 2   | 1   | 0   |
|----------|----|-----|-----|----------|----|----|----|----|----|----|----|--------------|-----|-----|-----|------------|-----|-----|-----|-----|-----|-----|-----|-------------|-----|-----|-----|-------------|-----|-----|-----|
| Reserved |    |     |     | Reserved |    |    |    |    |    |    |    | BUSTURN[3:0] |     |     |     | DATAS[7:0] |     |     |     |     |     |     |     | ADDHLD[3:0] |     |     |     | ADDSET[3:0] |     |     |     |
|          |    |     |     |          |    |    |    |    |    |    |    |              |     |     |     |            |     |     |     |     |     |     |     |             |     |     |     |             |     |     |     |
|          |    | r/w | r/w |          |    |    |    |    |    |    |    | r/w          | r/w | r/w | r/w | r/w        | r/w | r/w | r/w | r/w | r/w | r/w | r/w | r/w         | r/w | r/w | r/w | r/w         | r/w | r/w | r/w |

Bits 31:30 Reserved, must be kept at reset value

Bits 29:28 **ACCMOD[1:0]**: Access mode.

Specifies the asynchronous access modes as shown in the next timing diagrams. These bits are taken into account only when the EXTMOD bit in the FMC\_BCRx register is 1.

- 00: access mode A
- 01: access mode B
- 10: access mode C
- 11: access mode D

Bits 27:20 Reserved, must be kept at reset value

Bits 19:16 **BUSTURN[3:0]**: Bus turnaround phase duration

The programmed bus turnaround delay is inserted between an asynchronous write transfer and any other asynchronous /synchronous read or write transfer to or from a static bank. The bank can be the same or different in case of read, in case of write the bank can be different except for muxed or mode D.

In some cases, whatever the programmed BUSTURN values, the bus turnaround delay is fixed as follows:

- The bus turnaround delay is not inserted between two consecutive asynchronous write transfers to the same static memory bank except for modes muxed and D.
- There is a bus turnaround delay of 2 FMC clock cycle between:
  - Two consecutive synchronous writes (burst or single) to the same bank.
  - A synchronous write (burst or single) transfer and an asynchronous write or read transfer to or from static memory bank.
- There is a bus turnaround delay of 3 FMC clock cycle between:
  - Two consecutive synchronous writes (burst or single) to different static bank.
  - A synchronous write (burst or single) transfer and a synchronous read from the same or a different bank.

0000: BUSTURN phase duration = 0 HCLK clock cycle added

...

1111: BUSTURN phase duration = 15 HCLK clock cycles added (default value after reset)

Bits 15:8 **DATAST[3:0]**: Data-phase duration.

These bits are written by software to define the duration of the data phase (refer to [Figure 458](#) to [Figure 470](#)), used in asynchronous SRAM, PSRAM and NOR Flash memory accesses:

0000 0000: Reserved

0000 0001: DATAST phase duration = 1 × HCLK clock cycles

0000 0010: DATAST phase duration = 2 × HCLK clock cycles

...

1111 1111: DATAST phase duration = 255 × HCLK clock cycles (default value after reset)

Bits 7:4 **ADDHLD[3:0]**: Address-hold phase duration.

These bits are written by software to define the duration of the *address hold* phase (refer to [Figure 467](#) to [Figure 470](#)), used in asynchronous multiplexed accesses:

0000: Reserved

0001: ADDHLD phase duration = 1 × HCLK clock cycle

0010: ADDHLD phase duration = 2 × HCLK clock cycle

...

1111: ADDHLD phase duration = 15 × HCLK clock cycles (default value after reset)

*Note: In synchronous NOR Flash accesses, this value is not used, the address hold phase is always 1 Flash clock period duration.*

Bits 3:0 **ADDSET[3:0]**: Address setup phase duration.

These bits are written by software to define the duration of the *address setup* phase in HCLK cycles (refer to [Figure 467](#) to [Figure 470](#)), used in asynchronous accesses:

0000: ADDSET phase duration = 0 × HCLK clock cycle

...

1111: ADDSET phase duration = 15 × HCLK clock cycles (default value after reset)

*Note: In synchronous NOR Flash and PSRAM accesses, this value is not used, the address setup phase is always 1 Flash clock period duration. In muxed mode, the minimum ADDSET value is 1.*

## 37.6 NAND Flash/PC Card controller

The FMC generates the appropriate signal timings to drive the following types of device:

- 8- and 16-bit NAND Flash memories
- 16-bit PC Card compatible devices

The NAND Flash/PC Card controller can control three external banks, Bank 2, 3 and 4:

- Bank 2 and Bank 3 support NAND Flash devices
- Bank 4 supports PC Card devices.

Each bank is configured through dedicated registers ([Section 37.6.8](#)). The programmable memory parameters include access timings (shown in [Table 290](#)) and ECC configuration.

**Table 290. Programmable NAND Flash/PC Card access parameters**

| Parameter             | Function                                                                                                                                                | Access mode | Unit                   | Min. | Max. |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------|------|------|
| Memory setup time     | Number of clock cycles (HCLK) required to set up the address before the command assertion                                                               | Read/Write  | AHB clock cycle (HCLK) | 1    | 256  |
| Memory wait           | Minimum duration (in HCLK clock cycles) of the command assertion                                                                                        | Read/Write  | AHB clock cycle (HCLK) | 2    | 255  |
| Memory hold           | Number of clock cycles (HCLK) during which the address must be held (as well as the data if a write access is performed) after the command de-assertion | Read/Write  | AHB clock cycle (HCLK) | 1    | 254  |
| Memory databus high-Z | Number of clock cycles (HCLK) during which the data bus is kept in high-Z state after a write access has started                                        | Write       | AHB clock cycle (HCLK) | 1    | 255  |

### 37.6.1 External memory interface signals

The following tables list the signals that are typically used to interface NAND Flash memory and PC Card.

Note: The prefix "N" identifies the signals which are active low.

#### 8-bit NAND Flash memory

**Table 291. 8-bit NAND Flash**

| FMC signal name | I/O | Function                                             |
|-----------------|-----|------------------------------------------------------|
| A[17]           | O   | NAND Flash address latch enable (ALE) signal         |
| A[16]           | O   | NAND Flash command latch enable (CLE) signal         |
| D[7:0]          | I/O | 8-bit multiplexed, bidirectional address/data bus    |
| NCE[x]          | O   | Chip Select, x = 2, 3                                |
| NOE(= NRE)      | O   | Output enable (memory signal name: read enable, NRE) |
| NWE             | O   | Write enable                                         |
| NWAIT/INT[3:2]  | I   | NAND Flash ready/busy input signal to the FMC        |

Theoretically, there is no capacity limitation as the FMC can manage as many address cycles as needed.

**16-bit NAND Flash memory****Table 292. 16-bit NAND Flash**

| FMC signal name | I/O | Function                                             |
|-----------------|-----|------------------------------------------------------|
| A[17]           | O   | NAND Flash address latch enable (ALE) signal         |
| A[16]           | O   | NAND Flash command latch enable (CLE) signal         |
| D[15:0]         | I/O | 16-bit multiplexed, bidirectional address/data bus   |
| NCE[x]          | O   | Chip Select, x = 2, 3                                |
| NOE(= NRE)      | O   | Output enable (memory signal name: read enable, NRE) |
| NWE             | O   | Write enable                                         |
| NWAIT/INT[3:2]  | I   | NAND Flash ready/busy input signal to the FMC        |

*Theoretically, there is no capacity limitation as the FMC can manage as many address cycles as needed.*

**Table 293. 16-bit PC Card**

| FMC signal name | I/O | Function                                                                                                                                                                                 |
|-----------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A[10:0]         | O   | Address bus                                                                                                                                                                              |
| NIORD           | O   | Output enable for I/O space                                                                                                                                                              |
| NIOWR           | O   | Write enable for I/O space                                                                                                                                                               |
| NREG            | O   | Register signal indicating if access is in Common or Attribute space                                                                                                                     |
| D[15:0]         | I/O | Bidirectional databus                                                                                                                                                                    |
| NCE4_1          | O   | Chip Select 1                                                                                                                                                                            |
| NCE4_2          | O   | Chip Select 2 (indicates if access is 16-bit or 8-bit)                                                                                                                                   |
| NOE             | O   | Output enable in Common and in Attribute space                                                                                                                                           |
| NWE             | O   | Write enable in Common and in Attribute space                                                                                                                                            |
| NWAIT           | I   | PC Card wait input signal to the FMC (memory signal name IORDY)                                                                                                                          |
| INTR            | I   | PC Card interrupt to the FMC (only for PC Cards that can generate an interrupt)                                                                                                          |
| CD              | I   | PC Card presence detection. Active high. If an access is performed to the PC Card banks while CD is low, an AHB error is generated. Refer to <a href="#">Section 37.3: AHB interface</a> |

### 37.6.2 NAND Flash / PC Card supported memories and transactions

*Table 294 shows the supported devices, access modes and transactions. Transactions not allowed (or not supported) by the NAND Flash / PC Card controller are shown in gray.*

**Table 294. Supported memories and transactions**

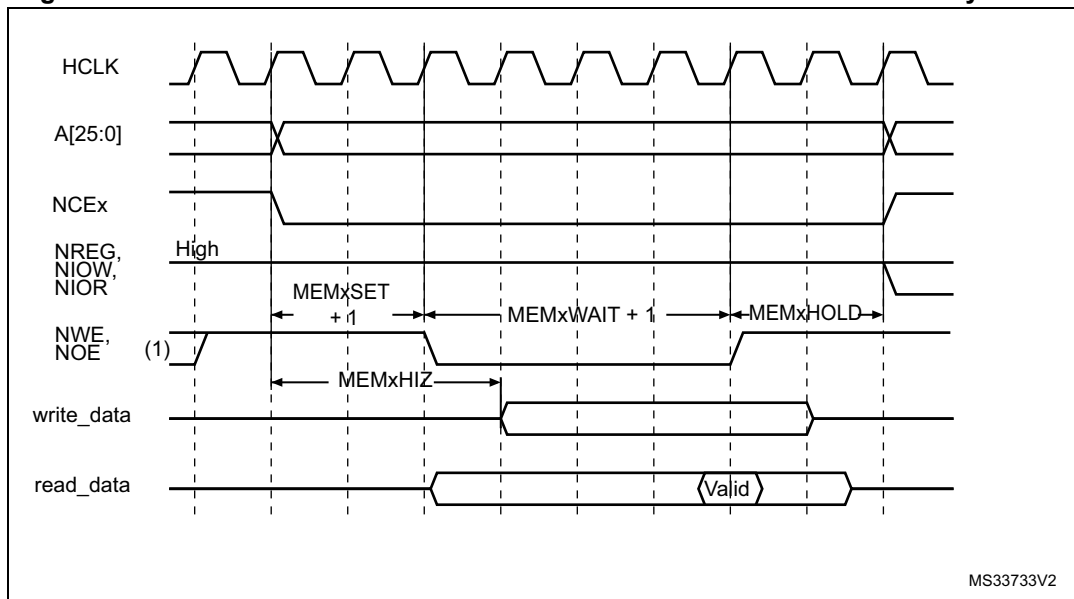
| Device      | Mode         | R/W | AHB data size | Memory data size | Allowed/not allowed | Comments                  |
|-------------|--------------|-----|---------------|------------------|---------------------|---------------------------|
| NAND 8-bit  | Asynchronous | R   | 8             | 8                | Y                   | -                         |
|             | Asynchronous | W   | 8             | 8                | Y                   | -                         |
|             | Asynchronous | R   | 16            | 8                | Y                   | Split into 2 FMC accesses |
|             | Asynchronous | W   | 16            | 8                | Y                   | Split into 2 FMC accesses |
|             | Asynchronous | R   | 32            | 8                | Y                   | Split into 4 FMC accesses |
|             | Asynchronous | W   | 32            | 8                | Y                   | Split into 4 FMC accesses |
| NAND 16-bit | Asynchronous | R   | 8             | 16               | Y                   | -                         |
|             | Asynchronous | W   | 8             | 16               | N                   | -                         |
|             | Asynchronous | R   | 16            | 16               | Y                   | -                         |
|             | Asynchronous | W   | 16            | 16               | Y                   | -                         |
|             | Asynchronous | R   | 32            | 16               | Y                   | Split into 2 FMC accesses |
|             | Asynchronous | W   | 32            | 16               | Y                   | Split into 2 FMC accesses |

### 37.6.3 Timing diagrams for NAND Flash memory and PC Card

Each PC Card/CompactFlash and NAND Flash memory bank is managed through a set of registers:

- Control register: FMC\_PCRx
- Interrupt status register: FMC\_SRx
- ECC register: FMC\_ECCRx
- Timing register for Common memory space: FMC\_PMEMx
- Timing register for Attribute memory space: FMC\_PATTx
- Timing register for I/O space: FMC\_PIOx

Each timing configuration register contains three parameters used to define number of HCLK cycles for the three phases of any PC Card/CompactFlash or NAND Flash access, plus one parameter that defines the timing for starting driving the data bus when a write access is performed. [Figure 476](#) shows the timing parameter definitions for common memory accesses, knowing that Attribute and I/O (only for PC Card) memory space access timings are similar.

**Figure 476. NAND Flash/PC Card controller waveforms for common memory access**

1. NOE remains high (inactive) during write accesses. NWE remains high (inactive) during read accesses.
2. For write accesses, the hold phase delay is (MEMHOLD) x HCLK cycles, while it is (MEMHOLD + 2) x HCLK cycles for read accesses.

### 37.6.4 NAND Flash operations

The command latch enable (CLE) and address latch enable (ALE) signals of the NAND Flash memory device are driven by address signals from the FMC controller. This means that to send a command or an address to the NAND Flash memory, the CPU has to perform a write to a specific address in its memory space.

A typical page read operation from the NAND Flash device requires the following steps:

3. Program and enable the corresponding memory bank by configuring the FMC\_PCRx and FMC\_PMEMx (and for some devices, FMC\_PATTx, see [Section 37.6.5: NAND Flash prewait functionality](#)) registers according to the characteristics of the NAND Flash memory (PWID bits for the data bus width of the NAND Flash, PTYP = 1, PWAITEN = 0 or 1 as needed, see [Section 37.4.2: NAND Flash memory/PC Card address mapping](#) for timing configuration).
4. The CPU performs a byte write to the common memory space, with data byte equal to one Flash command byte (for example 0x00 for Samsung NAND Flash devices). The LE input of the NAND Flash memory is active during the write strobe (low pulse on NWE), thus the written byte is interpreted as a command by the NAND Flash memory. Once the command is latched by the memory device, it does not need to be written again for the following page read operations.
5. The CPU can send the start address (STARTAD) for a read operation by writing four byte (or three for smaller capacity devices), STARTAD[7:0], STARTAD[16:9], STARTAD[24:17] and finally STARTAD[25] (for 64 Mb x 8 bit NAND Flash memories) in the common memory or attribute space. The ALE input of the NAND Flash device is active during the write strobe (low pulse on NWE), thus the written byte are interpreted as the start address for read operations. Using the attribute memory space makes it possible to use a different timing configuration of the FMC, which can be used to

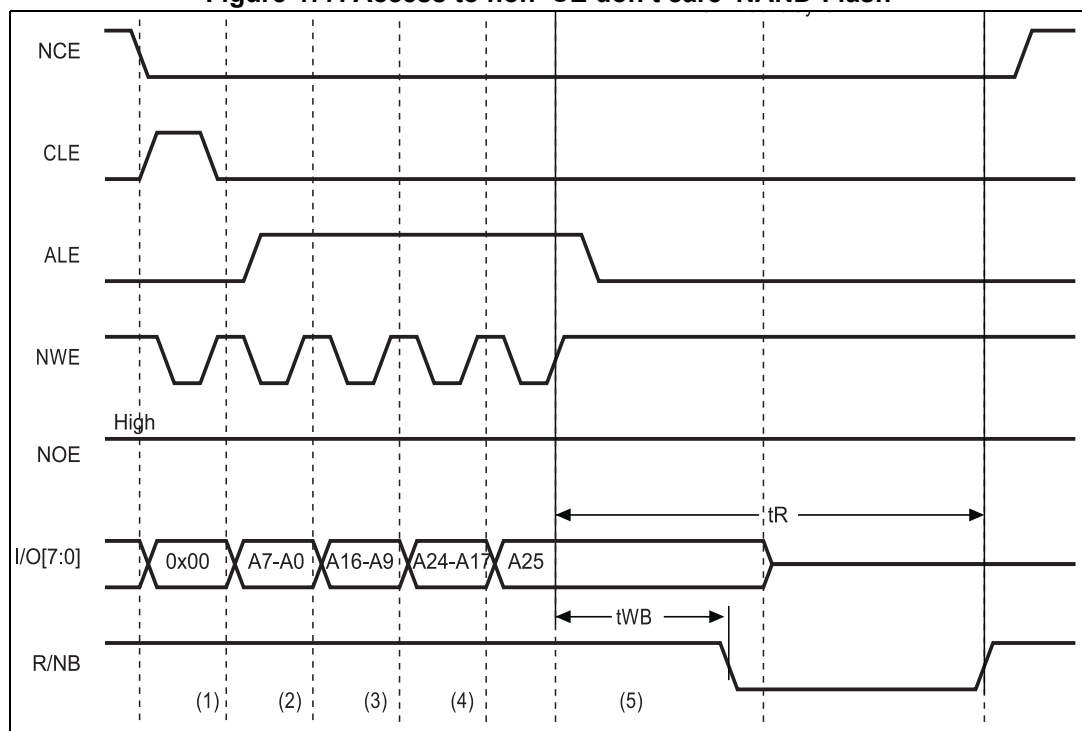
implement the prewait functionality needed by some NAND Flash memories (see details in [Section 37.6.5: NAND Flash prewait functionality](#)).

6. The controller waits for the NAND Flash memory to be ready (R/NB signal high), before starting a new access to the same or another memory bank. While waiting, the controller holds the NCE signal active (low).
7. The CPU can then perform byte read operations from the common memory space to read the NAND Flash page (data field + Spare field) byte by byte.
8. The next NAND Flash page can be read without any CPU command or address write operation. This can be done in three different ways:
  - by simply performing the operation described in step 5
  - a new random address can be accessed by restarting the operation at step 3
  - a new command can be sent to the NAND Flash device by restarting at step 2

### 37.6.5 NAND Flash prewait functionality

Some NAND Flash devices require that, after writing the last part of the address, the controller waits for the R/NB signal to go low. (see [Figure 457](#)).

**Figure 477. Access to non 'CE don't care' NAND-Flash**



1. CPU wrote byte 0x00 at address 0x7001 0000.
2. CPU wrote byte A7~A0 at address 0x7002 0000.
3. CPU wrote byte A16~A9 at address 0x7002 0000.
4. CPU wrote byte A24~A17 at address 0x7002 0000.
5. CPU wrote byte A25 at address 0x7802 0000: FMC performs a write access using FMC\_PATT2 timing definition, where  $ATTHOLD \geq 7$  (providing that  $(7+1) \times HCLK = 112 \text{ ns} > t_{WB \text{ max}}$ ). This guarantees that NCE remains low until R/NB goes low and high again (only requested for NAND Flash memories where NCE is not don't care).

When this functionality is required, it can be ensured by programming the MEMHOLD value to meet the  $t_{WB}$  timing. However CPU read accesses to the NAND Flash memory has a hold delay of  $(MEMHOLD + 2) \times HCLK$  cycles, while CPU write accesses have a hold delay of  $(MEMHOLD) \times HCLK$  cycles.

To cope with this timing constraint, the attribute memory space can be used by programming its timing register with an ATTHOLD value that meets the  $t_{WB}$  timing, and by keeping the MEMHOLD value at its minimum value. The CPU must then use the common memory space for all NAND Flash read and write accesses, except when writing the last address byte to the NAND Flash device, where the CPU must write to the attribute memory space.

### 37.6.6 Computation of the error correction code (ECC) in NAND Flash memory

The FMC PC Card controller includes two error correction code computation hardware blocks, one per memory bank. They reduce the host CPU workload when processing the ECC by software.

These two ECC blocks are identical and associated with Bank 2 and Bank 3. As a consequence, no hardware ECC computation is available for memories connected to Bank 4.

The ECC algorithm implemented in the FMC can perform 1-bit error correction and 2-bit error detection per 256, 512, 1 024, 2 048, 4 096 or 8 192 byte read or written from/to the NAND Flash memory. It is based on the Hamming coding algorithm and consists in calculating the row and column parity.

The ECC modules monitor the NAND Flash data bus and read/write signals (NCE and NWE) each time the NAND Flash memory bank is active.

The ECC operates as follows:

- When accessing NAND Flash memory bank 2 or bank 3, the data present on the D[15:0] bus is latched and used for ECC computation.
- When accessing any other address in NAND Flash memory, the ECC logic is idle, and does not perform any operation. As a result, write operations to define commands or addresses to the NAND Flash memory are not taken into account for ECC computation.

Once the desired number of byte has been read/written from/to the NAND Flash memory by the host CPU, the FMC\_ECCR2/3 registers must be read to retrieve the computed value. Once read, they should be cleared by resetting the ECCEN bit to '0'. To compute a new data block, the ECCEN bit must be set to one in the FMC\_PCR2/3 registers.

To perform an ECC computation:

1. Enable the ECCEN bit in the FMC\_PCR2/3 register.
2. Write data to the NAND Flash memory page. While the NAND page is written, the ECC block computes the ECC value.
3. Read the ECC value available in the FMC\_ECCR2/3 register and store it in a variable.
4. Clear the ECCEN bit and then enable it in the FMC\_PCR2/3 register before reading back the written data from the NAND page. While the NAND page is read, the ECC block computes the ECC value.
5. Read the new ECC value available in the FMC\_ECCR2/3 register.
6. If the two ECC values are the same, no correction is required, otherwise there is an ECC error and the software correction routine returns information on whether the error can be corrected or not.

### 37.6.7 PC Card/CompactFlash operations

#### Address spaces and memory accesses

The FMC supports CompactFlash devices and PC Cards in Memory mode and I/O mode (True IDE mode is not supported).

The CompactFlash and PC Cards are made of 3 memory spaces:

- Common Memory space
- Attribute space
- I/O Memory space

The nCE2 and nCE1 pins (FMC\_NCE4\_2 and FMC\_NCE4\_1 respectively) select the card and indicate whether a byte or a word operation is being performed: nCE2 accesses the odd byte on D15-8 and nCE1 accesses the even byte on D7-0 if A0=0 or the odd byte on D7-0 if A0=1. The full word is accessed on D15-0 if both nCE2 and nCE1 are low.

The memory space is selected by asserting low nOE for read accesses or nWE for write accesses, combined with the low assertion of nCE2/nCE1 and nREG.

- If pin nREG=1 during the memory access, the common memory space is selected
- If pin nREG=0 during the memory access, the attribute memory space is selected

The I/O space is selected by asserting nIORD space for read accesses or nIOWR for write accesses [instead of nOE/nWE for memory space], combined with nCE2/nCE1. Note that nREG must also be asserted low when accessing I/O space.

Three type of accesses are allowed for a 16-bit PC Card:

- Accesses to Common Memory space for data storage can be either 8-bit accesses at even addresses or 16-bit AHB accesses.  
Note that 8-bit accesses at odd addresses are not supported and nCE2 is not driven low. A 32-bit AHB request is translated into two 16-bit memory accesses.
- Accesses to Attribute Memory space where the PC Card stores configuration information are limited to 8-bit AHB accesses at even addresses.  
Note that a 16-bit AHB access is converted into a single 8-bit memory transfer: nCE1 is asserted low, nCE2 is asserted high and only the even byte on D7-D0 are valid. Instead a 32-bit AHB access is converted into two 8-bit memory transfers at even addresses: nCE1 is asserted low, nCE2 is asserted high and only the even byte are valid.
- Accesses to I/O space can be either 8-bit or 16 bit AHB accesses.

Table 295. 16-bit PC-Card signals and access type

| nCE2 | nCE1 | nREG | nOE/nWE | nIORD | A10 | A9 | A7-1 | A0 | Space               | Access type                                    | Allowed/not Allowed |
|------|------|------|---------|-------|-----|----|------|----|---------------------|------------------------------------------------|---------------------|
| 1    | 0    | 1    | 0       | 1     | X   | X  | X-X  | X  | Common Memory Space | Read/Write byte on D7-D0                       | YES                 |
| 0    | 1    | 1    | 0       | 1     | X   | X  | X-X  | X  |                     | Read/Write byte on D15-D8                      | Not supported       |
| 0    | 0    | 1    | 0       | 1     | X   | X  | X-X  | 0  |                     | Read/Write word on D15-D0                      | YES                 |
| X    | 0    | 0    | 0       | 1     | 0   | 1  | X-X  | 0  | Attribute Space     | Read or Write Configuration Registers          | YES                 |
| X    | 0    | 0    | 0       | 1     | 0   | 0  | X-X  | 0  |                     | Read or Write CIS (Card Information Structure) | YES                 |
| 1    | 0    | 0    | 0       | 1     | X   | X  | X-X  | 1  | Attribute Space     | Invalid Read or Write (odd address)            | YES                 |
| 0    | 1    | 0    | 0       | 1     | X   | X  | X-X  | x  |                     | Invalid Read or Write (odd address)            | YES                 |
| 1    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 0  | I/O space           | Read Even Byte on D7-0                         | YES                 |
| 1    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 1  |                     | Read Odd Byte on D7-0                          | YES                 |
| 1    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 0  |                     | Write Even Byte on D7-0                        | YES                 |
| 1    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 1  |                     | Write Odd Byte on D7-0                         | YES                 |
| 0    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 0  |                     | Read Word on D15-0                             | YES                 |
| 0    | 0    | 0    | 1       | 0     | X   | X  | X-X  | 0  |                     | Write word on D15-0                            | YES                 |
| 0    | 1    | 0    | 1       | 0     | X   | X  | X-X  | X  |                     | Read Odd Byte on D15-8                         | Not supported       |
| 0    | 1    | 0    | 1       | 0     | X   | X  | X-X  | X  |                     | Write Odd Byte on D15-8                        | Not supported       |

FMC Bank 4 gives access to those 3 memory spaces as described in [Section 37.4.2: NAND Flash memory/PC Card address mapping](#) and [Table 257: NAND/PC Card memory mapping and timing registers](#).

### Wait feature

The CompactFlash or PC Card may request the FMC to extend the length of the access phase programmed by MEMWAITx/ATTWAITx/IOWAITx bits, asserting the nWAIT signal after nOE/nWE or nIORD/nIOWR activation if the wait feature is enabled through the PWAITEN bit in the FMC\_PCRx register. To detect correctly the nWAIT assertion, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed as follows:

$$xxWAITx \geq 4 + \frac{\text{max\_wait\_assertion\_time}}{\text{HCLK}}$$

where max\_wait\_assertion\_time is the maximum time taken by nWAIT to go low once nOE/nWE or nIORD/nIOWR is low.

After WAIT de-assertion, the FMC extends the WAIT phase for 4 HCLK clock cycles.

## 37.6.8 NAND Flash/PC Card controller registers

### PC Card/NAND Flash control registers 2..4 (FMC\_PCR2..4)

Address offset:  $0x40 + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0x0000 0018

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19         | 18 | 17 | 16       | 15 | 14 | 13 | 12        | 11 | 10 | 9 | 8        | 7 | 6     | 5  | 4         | 3  | 2    | 1  | 0     |    |         |    |          |  |
|----------|----|----|----|----|----|----|----|----|----|----|----|------------|----|----|----------|----|----|----|-----------|----|----|---|----------|---|-------|----|-----------|----|------|----|-------|----|---------|----|----------|--|
| Reserved |    |    |    |    |    |    |    |    |    |    |    | ECCPS[2:0] |    |    | TAR[3:0] |    |    |    | TCLR[3:0] |    |    |   | Reserved |   | ECCEN |    | PWID[1:0] |    | PTYP |    | PBKEN |    | PWAITEN |    | Reserved |  |
|          |    |    |    |    |    |    |    |    |    |    |    |            |    |    |          |    |    |    |           |    |    |   |          |   | rw    | rw | rw        | rw | rw   | rw | rw    | rw | rw      | rw |          |  |

Bits 31:20 Reserved, must be kept at reset value

Bits 19:17 **ECCPS[2:0]**: ECC page size.

Defines the page size for the extended ECC:

000: 256 byte  
 001: 512 byte  
 010: 1024 byte  
 011: 2048 byte  
 100: 4096 byte  
 101: 8192 byte

Bits 16:13 **TAR[3:0]**: ALE to RE delay.

Sets time from ALE low to RE low in number of AHB clock cycles (HCLK).

Time is:  $t_{ar} = (TAR + SET + 2) \times THCLK$  where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)  
 1111: 16 HCLK cycles

*Note: SET is MEMSET or ATTSET according to the addressed space.*

Bits 12:9 **TCLR[3:0]**: CLE to RE delay.

Sets time from CLE low to RE low in number of AHB clock cycles (HCLK).

Time is  $t_{clr} = (TCLR + SET + 2) \times THCLK$  where THCLK is the HCLK clock period

0000: 1 HCLK cycle (default)  
 1111: 16 HCLK cycles

*Note: SET is MEMSET or ATTSET according to the addressed space.*

Bits 8:7 Reserved, must be kept at reset value

Bit 6 **ECCEN**: ECC computation logic enable bit

0: ECC logic is disabled and reset (default after reset),  
 1: ECC logic is enabled.

Bits 5:4 **PWID[1:0]**: Data bus width.

Defines the external memory device width.

00: 8 bits  
 01: 16 bits (default after reset). This value is mandatory for PC Cards.  
 10: reserved, do not use  
 11: reserved, do not use

Bit 3 **PTYP**: Memory type.

Defines the type of device attached to the corresponding memory bank:

- 0: PC Card, CompactFlash, CF+ or PCMCIA
- 1: NAND Flash (default after reset)

Bit 2 **PBKEN**: PC Card/NAND Flash memory bank enable bit.

Enables the memory bank. Accessing a disabled memory bank causes an ERROR on AHB bus

- 0: Corresponding memory bank is disabled (default after reset)
- 1: Corresponding memory bank is enabled

Bit 1 **PWAITEN**: Wait feature enable bit.

Enables the Wait feature for the PC Card/NAND Flash memory bank:

- 0: disabled
- 1: enabled

*Note: For a PC Card, when the wait feature is enabled, the MEMWAITx/ATTWAITx/IOWAITx bits must be programmed to a value as follows:*

$$xxWAITx \geq 4 + \max\_wait\_assertion\_time/HCLK$$

*Where max\_wait\_assertion\_time is the maximum time taken by NWAIT to go low once nOE/nWE or nIORD/nIOWR is low.*

Bit 0 Reserved.

### FIFO status and interrupt register 2..4 (FMC\_SR2..4)

Address offset:  $0x44 + 0x20 * (x-1)$ ,  $x = 2..4$

Reset value: 0x0000 0040

This register contains information about the FIFO status and interrupt. The FMC features a FIFO that is used when writing to memories to transfer up to 16 words of data from the AHB.

This is used to quickly write to the FIFO and free the AHB for transactions to peripherals other than the FMC, while the FMC is draining its FIFO into the memory. One of these register bits indicates the status of the FIFO, for ECC purposes.

The ECC is calculated while the data are written to the memory. To read the correct ECC, the software must consequently wait until the FIFO is empty.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |       |      |      |      |     |     |     |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|-------|------|------|------|-----|-----|-----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5     | 4    | 3    | 2    | 1   | 0   |     |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | FEMPT | IFEN | ILEN | IREN | IFS | ILS | IRS |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | r     | rW   | rW   | rW   | rW  | rW  | rW  |

Bits 31:7 Reserved, must be kept at reset value

Bit 6 **FEMPT**: FIFO empty.

Read-only bit that provides the status of the FIFO

- 0: FIFO not empty
- 1: FIFO empty

Bit 5 **IFEN**: Interrupt falling edge detection enable bit

- 0: Interrupt falling edge detection request disabled
- 1: Interrupt falling edge detection request enabled

- Bit 4 **ILEN**: Interrupt high-level detection enable bit  
 0: Interrupt high-level detection request disabled  
 1: Interrupt high-level detection request enabled
- Bit 3 **IREN**: Interrupt rising edge detection enable bit  
 0: Interrupt rising edge detection request disabled  
 1: Interrupt rising edge detection request enabled
- Bit 2 **IFS**: Interrupt falling edge status  
 The flag is set by hardware and reset by software.  
 0: No interrupt falling edge occurred  
 1: Interrupt falling edge occurred

*Note: This bit is set by programming it to 1 by software.*

- Bit 1 **ILS**: Interrupt high-level status  
 The flag is set by hardware and reset by software.  
 0: No Interrupt high-level occurred  
 1: Interrupt high-level occurred
- Bit 0 **IRS**: Interrupt rising edge status  
 The flag is set by hardware and reset by software.  
 0: No interrupt rising edge occurred  
 1: Interrupt rising edge occurred

*Note: This bit is set by programming it to 1 by software.*

### Common memory space timing register 2..4 (FMC\_PMEM2..4)

Address offset: Address:  $0x48 + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0xFCFC FCFC

Each FMC\_PMEMx ( $x = 2..4$ ) read/write register contains the timing information for PC Card or NAND Flash memory bank x. This information is used to access either the common memory space of the 16-bit PC Card/CompactFlash, or the NAND Flash for command, address write access and data read/write access.

|             |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| MEMHIZ[7:0] |    |    |    |    |    |    |    | MEMHOLD[7:0] |    |    |    |    |    |    |    | MEMWAIT[7:0] |    |    |    |    |    |    |    | MEMSET[7:0] |    |    |    |    |    |    |    |
| rw          | rw | rw | rw | rw | rw | rw | rw | rw           | rw | rw | rw | rw | rw | rw | rw | rw           | rw | rw | rw | rw | rw | rw | rw | rw          | rw | rw | rw | rw | rw | rw | rw |

Bits 31:24 **MEMHIZ[7:0]**: Common memory x data bus Hi-Z time

Defines the number of HCLK clock cycles during which the data bus is kept Hi-Z after the start of a PC Card/NAND Flash write access to common memory space on socket x. This is only valid for write transactions:

0000 0000: 1 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

Bits 23:16 **MEMHOLD[7:0]**: Common memory x hold time

For NAND Flash read accesses to the common memory space, these bits define the number of (HCLK+2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For NAND Flash write accesses to the common memory space, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: reserved

0000 0001: 1 HCLK cycle for write accesses, 3 HCLK cycles for read accesses

1111 1110: 254 HCLK cycle for write accesses, 256 HCLK cycles for read accesses

1111 1111: Reserved.

Bits 15:8 **MEMWAIT[7:0]**: Common memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND Flash read or write access to common memory space on socket x. The duration of command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: reserved

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)

1111 1110: 255 HCLK cycles (+ wait cycle introduced by deasserting NWAIT)

1111 1111: Reserved

Bits 7:0 **MEMSET[7:0]**: Common memory x setup time

Defines the number of HCLK (+1) clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card/NAND Flash read or write access to common memory space on socket x:

0000 0000: 1 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

## Attribute memory space timing registers 2..4 (FMC\_PATT2..4)

Address offset:  $0x4C + 0x20 * (x - 1)$ ,  $x = 2..4$

Reset value: 0xFCFC FCFC

Each FMC\_PATTx ( $x = 2..4$ ) read/write register contains the timing information for PC Card/CompactFlash or NAND Flash memory bank x. It is used for 8-bit accesses to the attribute memory space of the PC Card/CompactFlash or to access the NAND Flash for the last address write access if the timing must differ from that of previous accesses (for Ready/Busy management, refer to [Section 37.6.5: NAND Flash prewait functionality](#)).

|             |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |              |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |
|-------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|--------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|
| 31          | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15           | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7           | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| ATTHIZ[7:0] |    |    |    |    |    |    |    | ATTHOLD[7:0] |    |    |    |    |    |    |    | ATTWAIT[7:0] |    |    |    |    |    |    |    | ATTSET[7:0] |    |    |    |    |    |    |    |
| rW          | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW           | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW |

Bits 31:24 **ATTHIZ[7:0]**: Attribute memory x data bus Hi-Z time

Defines the number of HCLK clock cycles during which the data bus is kept in Hi-Z after the start of a PC CARD/NAND Flash write access to attribute memory space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved.

Bits 23:16 **ATTHOLD[7:0]**: Attribute memory x hold time

For PC Card/NAND Flash read accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles (HCLK +2) clock cycles during which the address is held after the command is deasserted (NWE, NOE).

For PC Card/NAND Flash write accesses to attribute memory space on socket x, these bits define the number of HCLK clock cycles during which the data are held after the command is deasserted (NWE, NOE).

0000 0000: Reserved

0000 0001: 1 HCLK cycle for write access, 3 HCLK cycles for read accesses

1111 1110: 254 HCLK cycle for write access, 256 HCLK cycles for read accesses

1111 1111: Reserved.

Bits 15:8 **ATTWAIT[7:0]**: Attribute memory x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (NWE, NOE), for PC Card/NAND Flash read or write access to attribute memory space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: reserved

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1110: 255 HCLK cycles (+ wait cycle introduced by the card deasserting NWAIT)

1111 1111: Reserved

Bits 7:0 **ATTSET[7:0]**: Attribute memory x setup time

Defines the number of HCLK (+1) clock cycles to set up address before the command assertion (NWE, NOE), for PC CARD/NAND Flash read or write access to attribute memory space on socket x:

0000 0000: 1 HCLK cycle

1111 1110: 255 HCLK cycles

1111 1111: Reserved

**I/O space timing register 4 (FMC\_PIO4)**

Address offset: 0xB0

Reset value: 0xFCFCFCFC

The FMC\_PIO4 read/write registers contain the timing information used to access the I/O space of the 16-bit PC Card/CompactFlash.

|            |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |             |    |    |    |    |    |    |    |            |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|-------------|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23          | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15          | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7          | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| IOHIZ[7:0] |    |    |    |    |    |    |    | IOHOLD[7:0] |    |    |    |    |    |    |    | IOWAIT[7:0] |    |    |    |    |    |    |    | IOSET[7:0] |    |    |    |    |    |    |    |
| rW         | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW | rW          | rW | rW | rW | rW | rW | rW | rW | rW         | rW | rW | rW | rW | rW | rW | rW |

**Bits 31:24 IOHIZ[7:0]:** I/O x data bus Hi-Z time

Defines the number of HCLK clock cycles during which the data bus is kept in Hi-Z after the start of a PC Card write access to I/O space on socket x. Only valid for write transaction:

0000 0000: 0 HCLK cycle

1111 1111: 255 HCLK cycles

**Bits 23:16 IOHOLD[7:0]:** I/O x hold time

Defines the number of HCLK clock cycles during which the address is held (and data for write access) after the command deassertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: reserved

0000 0001: 1 HCLK cycle

1111 1111: 255 HCLK cycles

**Bits 15:8 IOWAIT[7:0]:** I/O x wait time

Defines the minimum number of HCLK (+1) clock cycles to assert the command (SMNWE, SMNOE), for PC Card read or write access to I/O space on socket x. The duration for command assertion is extended if the wait signal (NWAIT) is active (low) at the end of the programmed value of HCLK:

0000 0000: reserved, do not use this value

0000 0001: 2 HCLK cycles (+ wait cycle introduced by deassertion of NWAIT)

1111 1111: 256 HCLK cycles (+ wait cycle introduced by the Card deasserting NWAIT)

**Bits 7:0 IOSET[7:0]:** I/O x setup time

Defines the number of HCLK (+1) clock cycles to set up the address before the command assertion (NWE, NOE), for PC Card read or write access to I/O space on socket x:

0000 0000: 1 HCLK cycle

1111 1111: 256 HCLK cycles

**ECC result registers 2/3 (FMC\_ECCR2/3)**

Address offset:  $0x54 + 0x20 * (x - 1)$ ,  $x = 2$  or  $3$

Reset value: 0x0000 0000

These registers contain the current error correction code value computed by the ECC computation modules of the FMC controller (one module per NAND Flash memory bank). When the CPU reads the data from a NAND Flash memory page at the correct address (refer to [Section 37.6.6: Computation of the error correction code \(ECC\) in NAND Flash memory](#)), the data read/written from/to the NAND Flash memory are processed automatically by the ECC computation module. When X byte have been read (according to the ECCPS field in the FMC\_PCRx registers), the CPU must read the computed ECC value from the FMC\_ECCx registers. It then verifies if these computed parity data are the same as the parity value recorded in the spare area, to determine whether a page is valid, and, to correct it otherwise. The FMC\_ECCRx registers should be cleared after being read by setting the ECCEN bit to '0'. To compute a new data block, the ECCEN bit must be set to '1'.

|           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31        | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| ECC[31:0] |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r         |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

Bits 31:0 **ECC[31:0]**: ECC result

This field contains the value computed by the ECC computation logic. [Table 296](#) describes the contents of these bit fields.

**Table 296. ECC result relevant bits**

| ECCPS[2:0] | Page size in byte | ECC bits  |
|------------|-------------------|-----------|
| 000        | 256               | ECC[21:0] |
| 001        | 512               | ECC[23:0] |
| 010        | 1024              | ECC[25:0] |
| 011        | 2048              | ECC[27:0] |
| 100        | 4096              | ECC[29:0] |
| 101        | 8192              | ECC[31:0] |

## 37.7 SDRAM controller

### 37.7.1 SDRAM controller main features

The main features of the SDRAM controller are the following:

- Two SDRAM banks with independent configuration
- 8-bit, 16-bit, 32-bit data bus width
- 13-bits Address Row, 11-bits Address Column, 4 internal banks: 4x16Mx32bit (256 MB), 4x16Mx16bit (128 MB), 4x16Mx8bit (64 MB)
- Word, half-word, byte access
- SDRAM clock can be HCLK/2 or HCLK/3
- Automatic row and bank boundary management
- Multibank ping-pong access
- Programmable timing parameters
- Automatic Refresh operation with programmable Refresh rate
- Self-refresh mode
- Power-down mode
- SDRAM power-up initialization by software
- CAS latency of 1,2,3
- Cacheable Read FIFO with depth of 6 lines x32-bit (6 x14-bit address tag)

### 37.7.2 SDRAM External memory interface signals

At startup, the SDRAM I/O pins used to interface the FMC SDRAM controller with the external SDRAM devices must be configured by the user application. The SDRAM controller I/O pins which are not used by the application, can be used for other purposes.

**Table 297. SDRAM signals**

| SDRAM signal | I/O type | Description                                                            | Alternate function |
|--------------|----------|------------------------------------------------------------------------|--------------------|
| SDCLK        | O        | SDRAM clock                                                            | -                  |
| SDCKE[1:0]   | O        | SDCKE0: SDRAM Bank 1 Clock Enable<br>SDCKE1: SDRAM Bank 2 Clock Enable | -                  |
| SDNE[1:0]    | O        | SDNE0: SDRAM Bank 1 Chip Enable<br>SDNE1: SDRAM Bank 2 Chip Enable     | -                  |
| A[12:0]      | O        | Address                                                                | FMC_A[12:0]        |
| D[31:0]      | I/O      | Bidirectional data bus                                                 | FMC_D[31:0]        |
| BA[1:0]      | O        | Bank Address                                                           | FMC_A[15:14]       |
| NRAS         | O        | Row Address Strobe                                                     | -                  |
| NCAS         | O        | Column Address Strobe                                                  | -                  |
| SDNWE        | O        | Write Enable                                                           | -                  |
| NBL[3:0]     | O        | Output Byte Mask for write accesses<br>(memory signal name: DQM[3:0])  | FMC_NBL[3:0]       |

### 37.7.3 SDRAM controller functional description

All SDRAM controller outputs (signals, address and data) change on the falling edge of the memory clock (FMC\_SDCLK).

#### SDRAM initialization

The initialization sequence is managed by software. If the two banks are used, the initialization sequence must be generated simultaneously to Bank 1 and Bank 2 by setting the Target Bank bits CTB1 and CTB2 in the FMC\_SDCMR register:

1. Program the memory device features into the FMC\_SDCRx register. The SDRAM clock frequency, RBURST and RPIPE must be programmed in the FMC\_SDCR1 register.
2. Program the memory device timing into the FMC\_SDTRx register. The TRP and TRC timings must be programmed in the FMC\_SDTR1 register.
3. Set MODE bits to '001' and configure the Target Bank bits (CTB1 and/or CTB2) in the FMC\_SDCMR register to start delivering the clock to the memory (SDCKE is driven high).
4. Wait during the prescribed delay period. Typical delay is around 100  $\mu$ s (refer to the SDRAM datasheet for the required delay after power-up).
5. Set MODE bits to '010' and configure the Target Bank bits (CTB1 and/or CTB2) in the FMC\_SDCMR register to issue a "Precharge All" command.
6. Set MODE bits to '011', and configure the Target Bank bits (CTB1 and/or CTB2) as well as the number of consecutive Auto-refresh commands (NRFS) in the FMC\_SDCMR register. Refer to the SDRAM datasheet for the number of Auto-refresh commands that should be issued. Typical number is 8.
7. Configure the MRD field according to your SDRAM device, set the MODE bits to '100', and configure the Target Bank bits (CTB1 and/or CTB2) in the FMC\_SDCMR register to issue a "Load Mode Register" command in order to program the SDRAM. In particular:
  - a) The CAS latency must be selected following configured value in FMC\_SDCR1/2 registers
  - b) The Burst Length (BL) of 1 must be selected by configuring the M[2:0] bits to 000 in the mode register (refer to the SDRAM datasheet). If the Mode Register is not the same for both SDRAM banks, this step has to be repeated twice, once for each bank, and the Target Bank bits set accordingly.
8. Program the refresh rate in the FMC\_SDRTR register  
The refresh rate corresponds to the delay between refresh cycles. Its value must be adapted to SDRAM devices.
9. For mobile SDRAM devices, to program the extended mode register it should be done once the SDRAM device is initialized: First, a dummy read access should be performed while BA1=1 and BA=0 (refer to SDRAM address mapping section for BA[1:0] address mapping) in order to select the extended mode register instead of Load mode register and then program the needed value.

At this stage the SDRAM device is ready to accept commands. If a system reset occurs during an ongoing SDRAM access, the data bus might still be driven by the SDRAM device. Therefore the SDRAM device must be first reinitialized after reset before issuing any new access by the NOR Flash/PSRAM/SRAM or NAND Flash/PC Card controller.

**Note:** *If two SDRAM devices are connected to the FMC, all the accesses performed at the same time to both devices by the Command Mode register (Load Mode Register and Self-refresh*

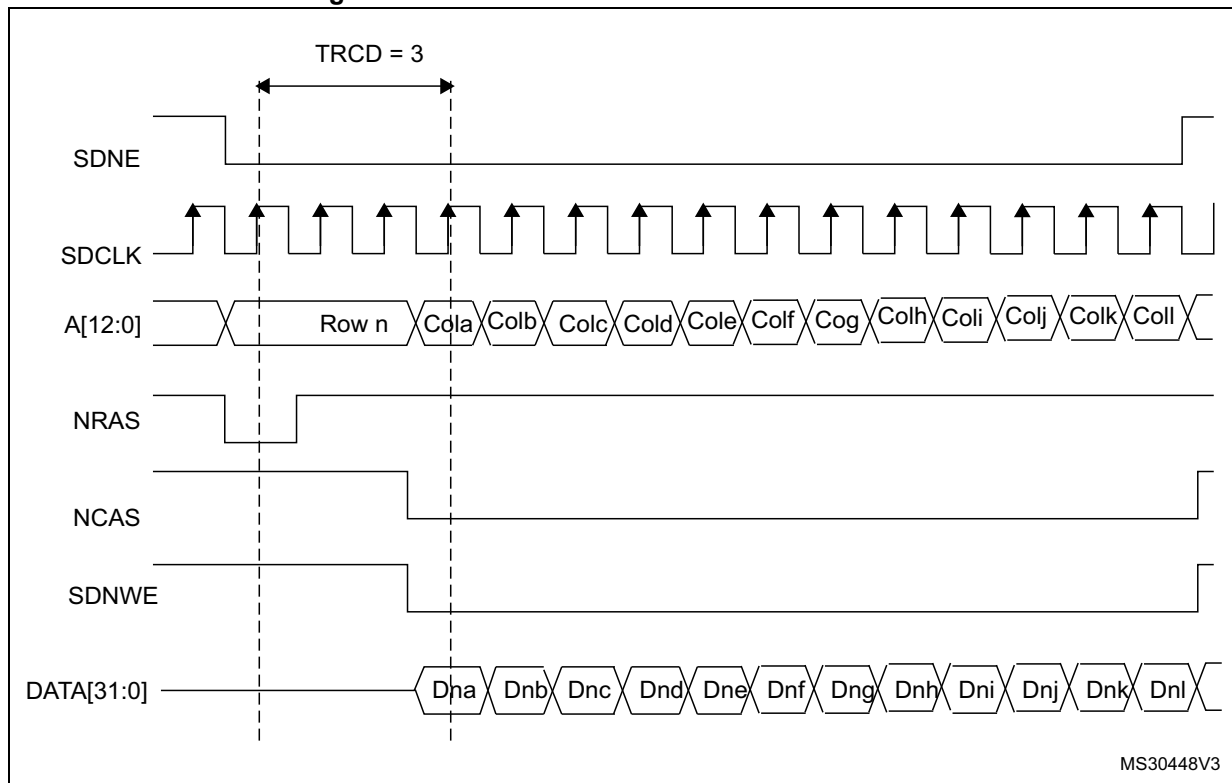
commands) are issued using the timing parameters configured for SDRAM Bank 1 (TMRD, TRAS and TXSR timings) in the FMC\_SDTR1 register.

### SDRAM controller write cycle

The SDRAM controller accepts single and burst write requests and translates them into single memory accesses. In both cases, the SDRAM controller keeps track of the active row for each bank to be able to perform consecutive write accesses to different banks (Multibank ping-pong access).

Before performing any write access, the SDRAM bank write protection must be disabled by clearing the WP bit in the FMC\_SDCRx register.

**Figure 478. Burst write SDRAM access waveforms**



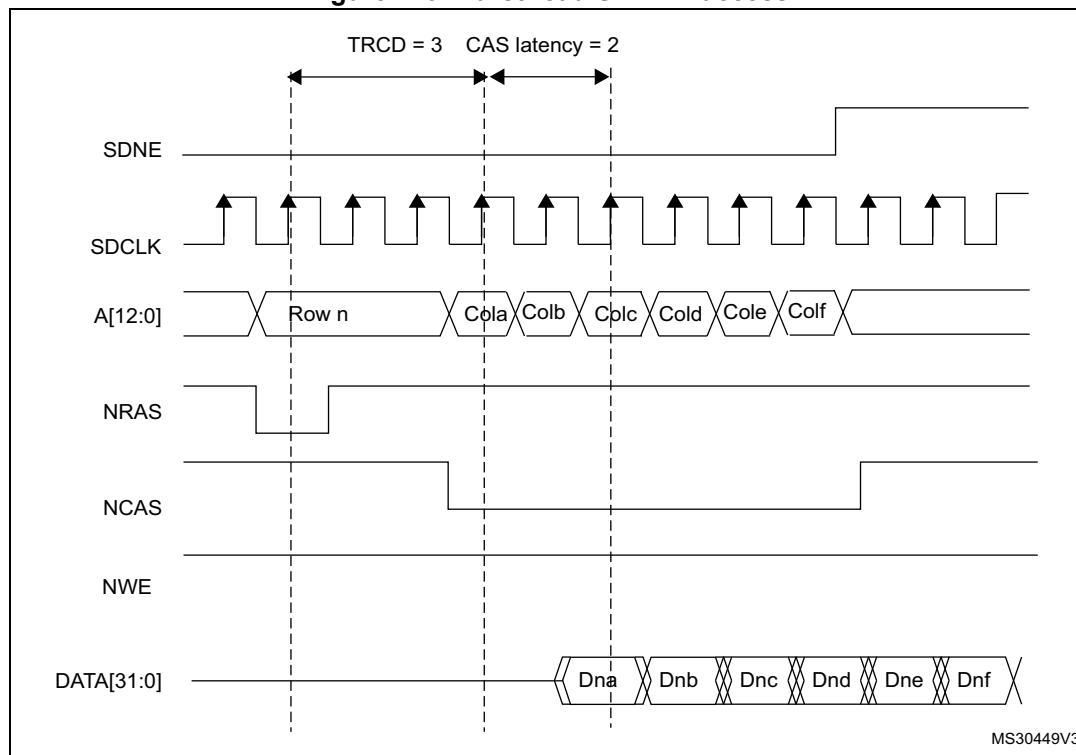
The SDRAM controller always checks the next access.

- If the next access is in the same row or in another active row, the write operation is carried out,
- if the next access targets another row (not active), the SDRAM controller generates a precharge command, activates the new row and initiates a write command.

### SDRAM controller read cycle

The SDRAM controller accepts single and burst read requests and translates them into single memory accesses. In both cases, the SDRAM controller keeps track of the active row in each bank to be able to perform consecutive read accesses in different banks (Multibank ping-pong access).

**Figure 479. Burst read SDRAM access**



The FMC SDRAM controller features a Cacheable read FIFO (6 lines x 32 bits). It is used to store data read in advance during the CAS latency period and during the RPIPE delay. The following formula is applied:

$$\text{Number of anticipated data} = \text{CAS latency} + 1 + (\text{RPIPE delay}) / 2$$

The RBURST bit must be set in the FMC\_SDCR1 register to anticipate the next read access.

Example:

- CAS latency = 3, RPIPE delay = 0: 4 data (not committed) are stored in the FIFO.
- CAS latency = 3, RPIPE delay = 2: 5 data (not committed) are stored in the FIFO.

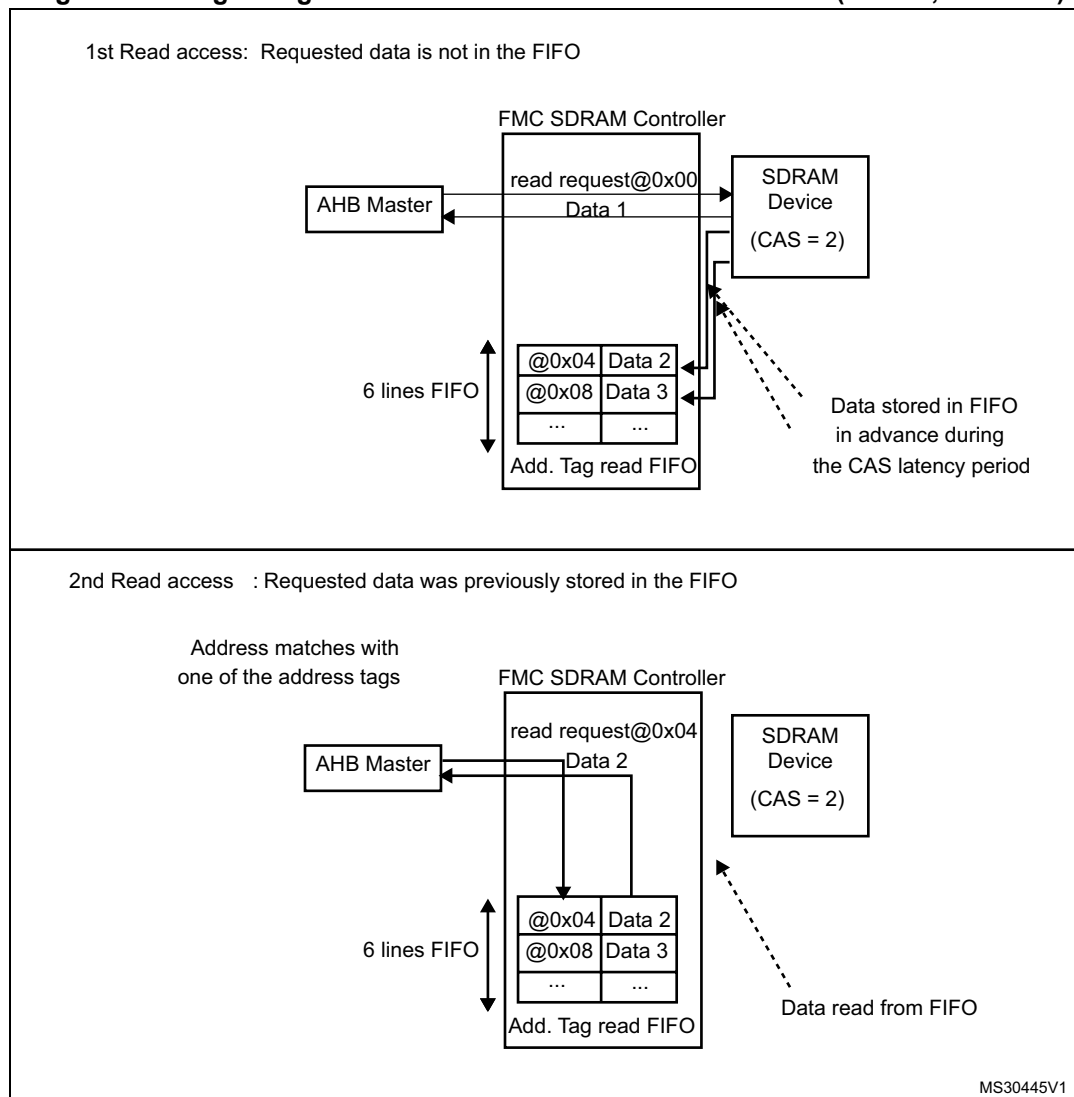
The read FIFO features a 14-bit address tag to each line to identify its content: 11 bits for the column address, 2 bits to select the internal bank and the active row, and 1 bit to select the SDRAM device

When the end of the row is reached in advance during an AHB burst read, the data read in advance (not committed) are not stored in the read FIFO. For single read access, data are correctly stored in the FIFO.

Each time a read request occurs, the SDRAM controller checks:

- If the address matches one of the address tags, data are directly read from the FIFO and the corresponding address tag/ line content is cleared and the remaining data in the FIFO are compacted to avoid empty lines.
- Otherwise, a new read command is issued to the memory and the FIFO is updated with new data. If the FIFO is full, the older data are lost.

**Figure 480. Logic diagram of Read access with RBURST bit set (CAS=2, RPIPE=0)**



During a write access or a Precharge command, the read FIFO is flushed and ready to be filled with new data.

After the first read request, if the current access was not performed to a row boundary, the SDRAM controller anticipates the next read access during the CAS latency period and the RPIPE delay (if configured). This is done by incrementing the memory address. The following condition must be met:

- RBURST control bit should be set to '1' in the FMC\_SDCR1 register.

The address management depends on the next AHB request:

- Next AHB request is sequential (AHB Burst)  
In this case, the SDRAM controller increments the address.
- Next AHB request is not sequential
  - If the new read request targets the same row or another active row, the new address is passed to the memory and the master is stalled for the CAS latency period, waiting for the new data from memory.
  - If the new read request does not target an active row, the SDRAM controller generates a Precharge command, activates the new row, and initiates a read command.

If the RURST is reset, the read FIFO is not used.

### Row and bank boundary management

When a read or write access crosses a row boundary, if the next read or write access is sequential and the current access was performed to a row boundary, the SDRAM controller executes the following operations:

1. Precharge of the active row,
2. Activation of the new row
3. Start of a read/write command.

At a row boundary, the automatic activation of the next row is supported for all columns and data bus width configurations.

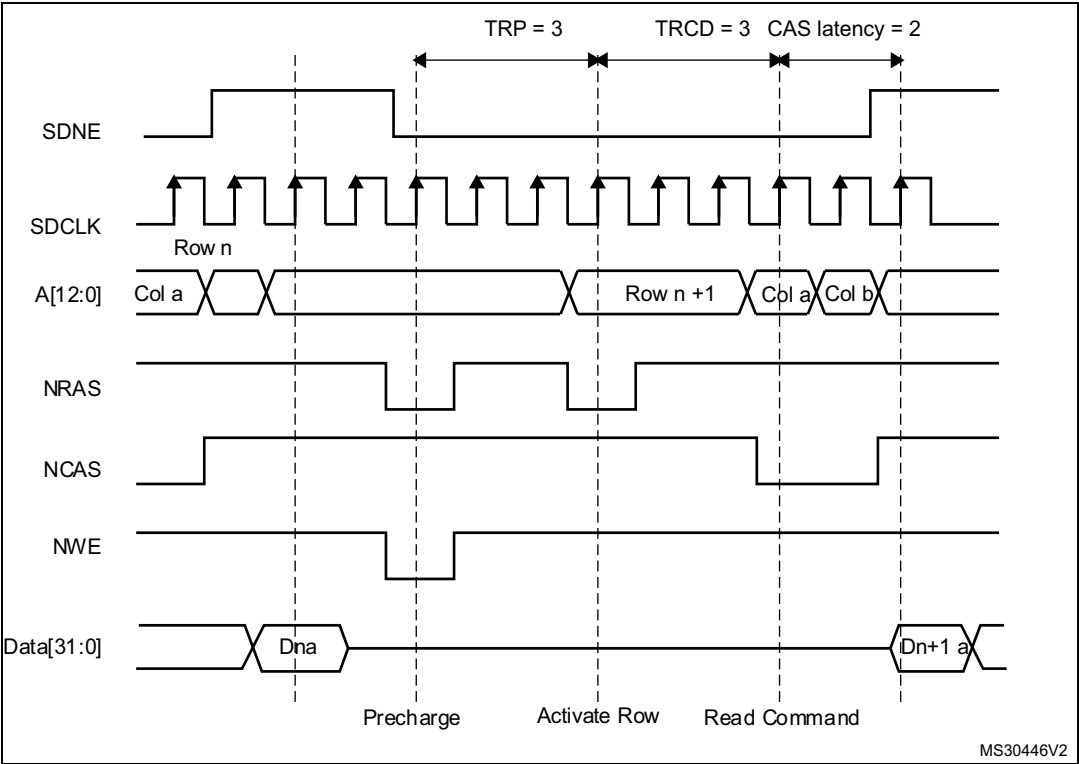
If necessary, the SDRAM controller inserts additional clock cycles between the following commands:

- Between Precharge and Active commands to match TRP parameter (only if the next access is in a different row in the same bank),
- Between Active and Read commands to match the TRCD parameter.

These parameters are defined into the FMC\_SDTRx register.

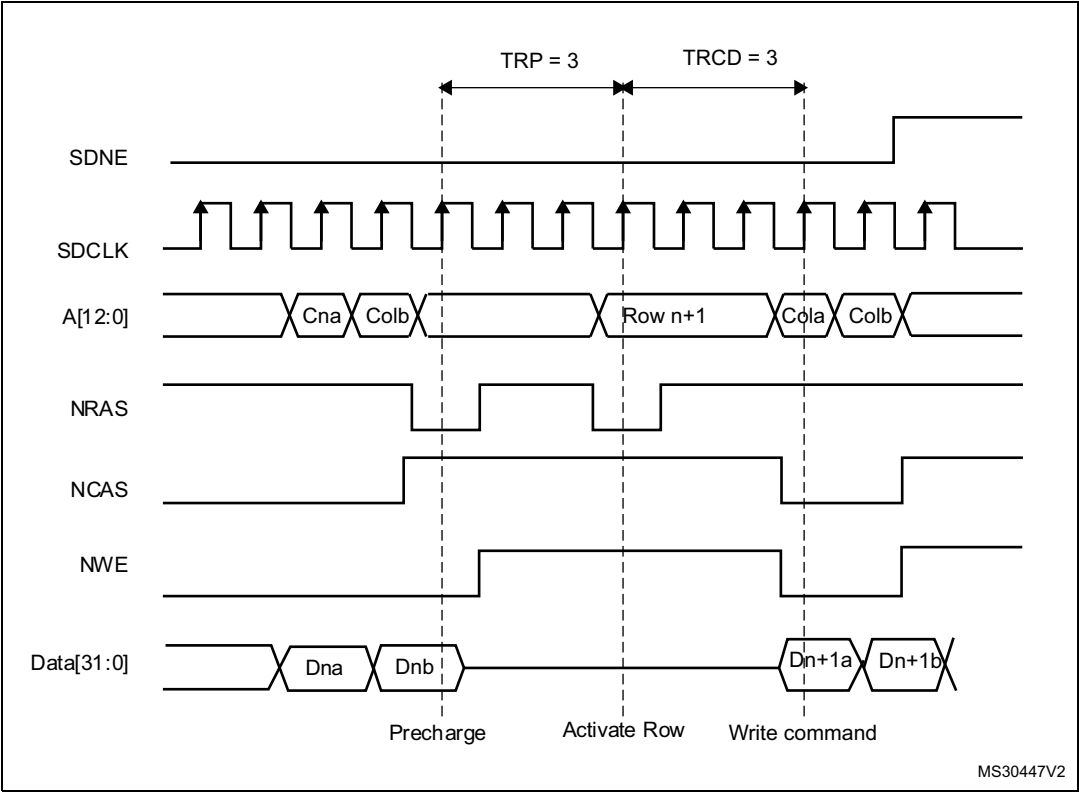
Refer to [Figure 481](#) and [Figure 482](#) for read and burst write access crossing a row boundary.

Figure 481. Read access crossing row boundary



MS30446V2

Figure 482. Write access crossing row boundary



MS30447V2

If the next access is sequential and the current access crosses a bank boundary, the SDRAM controller activates the first row in the next bank and initiates a new read/write command. Two cases are possible:

- If the current bank is not the last one, the active row in the new bank must be precharged. At a bank boundary, the automatic activation of the next row is supported for all rows/columns and data bus width configuration.
- For 13-bit row address, 11-bit column address, 4 internal banks and bus width 32-bit SDRAM memories, if the current bank is the last one and the selected SDRAM device is connected to Bank 1, the SDRAM controller continues to read/write from the second SDRAM device (assuming it has been initialized):
  - a) The SDRAM controller activates the first row (after precharging the active row, if there is already an active row in the first internal bank, and initiates a new read/write command.
  - b) If the first row is already activated, the SDRAM controller just initiates a read/write command.

*Note: At bank boundary, if the current bank is the last one, the automatic activation of the next row is supported only when addressing 13-bit rows, 11-bit columns, 4 internal banks and 32-bit data bus SDRAM devices. Otherwise, the SDRAM address range is violated and an AHB error is generated.*

### SDRAM controller refresh cycle

The Auto-refresh command is used to refresh the SDRAM device content. The SDRAM controller periodically issues auto-refresh commands. An internal counter is loaded with the COUNT value in the register FMC\_SDRTR. This value defines the number of memory clock cycles between the refresh cycles (refresh rate). When this counter reaches zero, an internal pulse is generated.

If a memory access is ongoing, the auto-refresh request is delayed. However, if the memory access and the auto-refresh requests are generated simultaneously, the auto-refresh request takes precedence.

If the memory access occurs during an auto-refresh operation, the request is buffered and processed when the auto-refresh is complete.

If a new auto-refresh request occurs while the previous one was not served, the RE (Refresh Error) bit is set in the Status register. An Interrupt is generated if it has been enabled (REIE = '1').

If SDRAM lines are not in idle state (not all row are closed), the SDRAM controller generates a PALL (Precharge ALL) command before the auto-refresh.

If the Auto-refresh command is generated by the FMC\_SDCMR Command Mode register (Mode bits = '011'), a PALL command (Mode bits = '010') must be issued first.

### 37.7.4 Low power modes

Two low power modes are available:

- Self-refresh mode  
The auto-refresh cycles are performed by the SDRAM device itself to retain data without external clocking.
- Power-down mode  
The auto-refresh cycles are performed by the SDRAM controller.

#### Self-refresh mode

This mode is selected by setting the MODE bits to '101' and by configuring the Target Bank bits (CTB1 and/or CTB2) in the FMC\_SDCMR register.

The SDRAM clock stops running after a TRAS delay and the internal refresh timer stops counting only if one of the following conditions is met:

- A Self-refresh command is issued to both devices
- One of the devices is not activated (SDRAM bank is not initialized).

Before entering Self-Refresh mode, the SDRAM controller automatically issues a PALL command.

If the Write data FIFO is not empty, all data are sent to the memory before activating the Self-refresh mode and the BUSY status flag remains set.

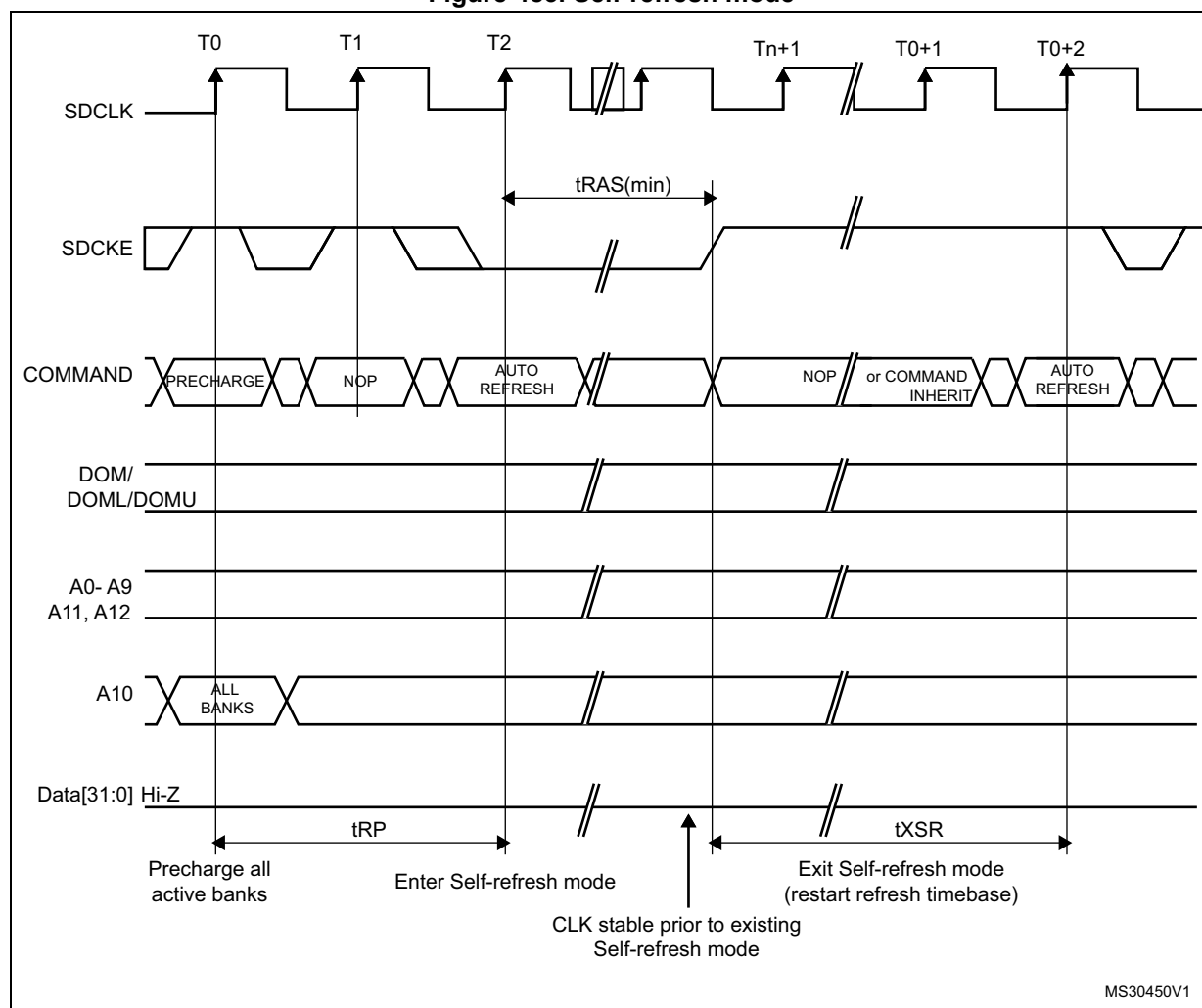
In Self-refresh mode, all SDRAM device inputs become don't care except for SDCKE which remains low.

The SDRAM device must remain in Self-refresh mode for a minimum period of time of TRAS and can remain in Self-refresh mode for an indefinite period beyond that. To guarantee this minimum period, the BUSY status flag remains high after the Self-refresh activation during a TRAS delay.

As soon as an SDRAM device is selected, the SDRAM controller generates a sequence of commands to exit from Self-refresh mode. After the memory access, the selected device remains in Normal mode.

To exit from Self-refresh, the MODE bits must be set to '000' (Normal mode) and the Target Bank bits (CTB1 and/or CTB2) must be configured in the FMC\_SDCMR register.

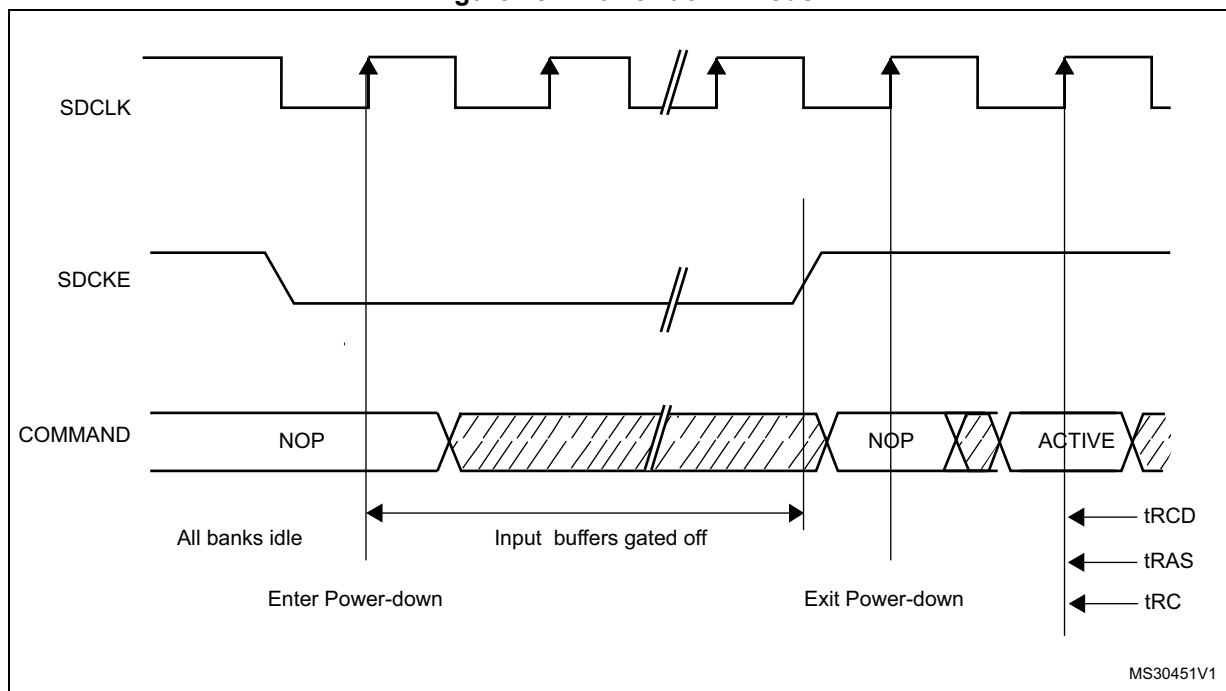
Figure 483. Self-refresh mode



### Power-down mode

This mode is selected by setting the MODE bits to '110' and by configuring the Target Bank bits (CTB1 and/or CTB2) in the FMC\_SDCMR register.

Figure 484. Power-down mode



MS30451V1

If the Write data FIFO is not empty, all data are sent to the memory before activating the Power-down mode.

As soon as an SDRAM device is selected, the SDRAM controller exits from the Power-down mode. After the memory access, the selected SDRAM device remains in Normal mode.

During Power-down mode, all SDRAM device input and output buffers are deactivated except for the SDCKE which remains low.

The SDRAM device cannot remain in Power-down mode longer than the refresh period and cannot perform the Auto-refresh cycles by itself. Therefore, the SDRAM controller carries out the refresh operation by executing the operations below:

1. Exit from Power-down mode and drive the SDCKE high
2. Generate the PALL command only if a row was active during Power-down mode
3. Generate the auto-refresh command
4. Drive SDCKE low again to return to Power-down mode.

To exit from Power-down mode, the MODE bits must be set to '000' (Normal mode) and the Target Bank bits (CTB1 and/or CTB2) must be configured in the FMC\_SDCMR register.

### 37.7.5 SDRAM controller registers

#### SDRAM Control registers 1,2 (FMC\_SDCR1,2)

Address offset:  $0x140 + 4 * (x - 1)$ ,  $x = 1, 2$

Reset value: 0x0000 02D0

This register contains the control parameters for each SDRAM memory bank

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14         | 13 | 12     | 11         | 10 | 9  | 8        | 7  | 6  | 5         | 4  | 3       | 2  | 1       | 0  |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|----|--------|------------|----|----|----------|----|----|-----------|----|---------|----|---------|----|
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RPIPE[1:0] |    | RBURST | SDCLK[1:0] |    | WP | CAS[1:0] |    | NB | MWID[1:0] |    | NR[1:0] |    | NC[1:0] |    |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rW         | rW | rW     | rW         | rW | rW | rW       | rW | rW | rW        | rW | rW      | rW | rW      | rW |

Bits 31:15 Reserved, must be kept at reset value

Bits 14:13 **RPIPE[1:0]**: Read pipe

These bits define the delay, in HCLK clock cycles, for reading data after CAS latency.

- 00: No HCLK clock cycle delay
- 01: One HCLK clock cycle delay
- 10: Two HCLK clock cycle delay
- 11: reserved, do not use

*Note: The corresponding bits in the FMC\_SDCR2 register are read only.*

Bit 12 **RBURST**: Burst read

This bit enables burst read mode. The SDRAM controller anticipates the next read commands during the CAS latency and stores data in the Read FIFO.

- 0: single read requests are not managed as bursts
- 1: single read requests are always managed as bursts

Note: The corresponding bit in the FMC\_SDCR2 register is don't care.

Bits 11:10 **SDCLK[1:0]**: SDRAM clock configuration

These bits define the SDRAM clock period for both SDRAM banks and allow disabling the clock before changing the frequency. In this case the SDRAM must be re-initialized.

- 00: SDCLK clock disabled
- 01: Reserved
- 10: SDCLK period = 2 x HCLK periods
- 11: SDCLK period = 3 x HCLK periods

*Note: The corresponding bits in the FMC\_SDCR2 register are read only.*

Bit 9 **WP**: Write protection

This bit enables write mode access to the SDRAM bank.

- 0: Write accesses allowed
- 1: Write accesses ignored

Bits 8:7 **CAS[1:0]**: CAS Latency

This bits sets the SDRAM CAS latency in number of memory clock cycles

- 00: reserved, do not use.
- 01: 1 cycle
- 10: 2 cycles
- 11: 3 cycles

Bit 6 **NB**: Number of internal banks

This bit sets the number of internal banks.

0: Two internal Banks

1: Four internal Banks

Bits 5:4 **MWID[1:0]**: Memory data bus width.

These bits define the memory device width.

00: 8 bits

01: 16 bits

10: 32 bits

11: reserved, do not use.

Bits 3:2 **NR[1:0]**: Number of row address bits

These bits define the number of bits of a row address.

00: 11 bit

01: 12 bits

10: 13 bits

11: reserved, do not use.

Bits 1:0 **NC[1:0]**: Number of column address bits

These bits define the number of bits of a column address.

00: 8 bits

01: 9 bits

10: 10 bits

11: 11 bits.

**Note:** *Before modifying the RBURST or RPIPE settings or disabling the SDCLK clock, the user must first send a PALL command to make sure ongoing operations are complete.*

### SDRAM Timing registers 1,2 (FMC\_SDTR1,2)

Address offset:  $0x148 + 4 * (x - 1)$ ,  $x = 1,2$

Reset value: 0x0FFF FFFF

This register contains the timing parameters of each SDRAM bank

|          |    |    |    |           |    |    |    |          |    |    |    |          |    |    |    |          |    |    |    |           |    |    |    |           |    |    |    |           |    |    |    |
|----------|----|----|----|-----------|----|----|----|----------|----|----|----|----------|----|----|----|----------|----|----|----|-----------|----|----|----|-----------|----|----|----|-----------|----|----|----|
| 31       | 30 | 29 | 28 | 27        | 26 | 25 | 24 | 23       | 22 | 21 | 20 | 19       | 18 | 17 | 16 | 15       | 14 | 13 | 12 | 11        | 10 | 9  | 8  | 7         | 6  | 5  | 4  | 3         | 2  | 1  | 0  |
| Reserved |    |    |    | TRCD[3:0] |    |    |    | TRP[3:0] |    |    |    | TWR[3:0] |    |    |    | TRC[3:0] |    |    |    | TRAS[3:0] |    |    |    | TXSR[3:0] |    |    |    | TMRD[3:0] |    |    |    |
| rW       | rW | rW | rW | rW        | rW | rW | rW | rW       | rW | rW | rW | rW       | rW | rW | rW | rW       | rW | rW | rW | rW        | rW | rW | rW | rW        | rW | rW | rW | rW        | rW | rW | rW |

Bits 31:28 Reserved, must be kept at reset value

Bits 27:24 **TRCD[3:0]**: Row to column delay

These bits define the delay between the Activate command and a Read/Write command in number of memory clock cycles.

0000: 1 cycle.

0001: 2 cycles

....

1111: 16 cycles

Bits 23:20 **TRP[3:0]**: Row precharge delay

These bits define the delay between a Precharge command and another command in number of memory clock cycles. The TRP timing is only configured in the FMC\_SDTR1 register. If two SDRAM devices are used, the TRP must be programmed with the timing of the slowest device.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

*Note: The corresponding bits in the FMC\_SDTR2 register are don't care.*

Bits 19:16 **TWR[3:0]**: Recovery delay

These bits define the delay between a Write and a Precharge command in number of memory clock cycles.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

*Note: TWR must be programmed to match the write recovery time ( $t_{WR}$ ) defined in the SDRAM datasheet, and to guarantee that:*

$$TWR \geq TRAS - TRCD \text{ and } TWR \geq TRC - TRCD - TRP$$

*Example: TRAS= 4 cycles, TRCD= 2 cycles. So, TWR >= 2 cycles. TWR must be programmed to 0x1.*

*If two SDRAM devices are used, the FMC\_SDTR1 and FMC\_SDTR2 must be programmed with the same TWR timing corresponding to the slowest SDRAM device.*

Bits 15:12 **TRC[3:0]**: Row cycle delay

These bits define the delay between the Refresh command and the Activate command, as well as the delay between two consecutive Refresh commands. It is expressed in number of memory clock cycles. The TRC timing is only configured in the FMC\_SDTR1 register. If two SDRAM devices are used, the TRC must be programmed with the timings of the slowest device.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

*Note: TRC must match the TRC and TRFC (Auto Refresh period) timings defined in the SDRAM device datasheet.*

*Note: The corresponding bits in the FMC\_SDTR2 register are don't care.*

Bits 11:8 **TRAS[3:0]**: Self refresh time

These bits define the minimum Self-refresh period in number of memory clock cycles.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

Bits 7:4 **TXSR[3:0]**: Exit Self-refresh delay

These bits define the delay from releasing the Self-refresh command to issuing the Activate command in number of memory clock cycles.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

Bits 3:0 **TMRD[3:0]**: Load Mode Register to Active

These bits define the delay between a Load Mode Register command and an Active or Refresh command in number of memory clock cycles.

0000: 1 cycle

0001: 2 cycles

....

1111: 16 cycles

**Note:** *If two SDRAM devices are connected, all the accesses performed simultaneously to both devices by the Command Mode register (Load Mode Register and Self-refresh commands) are issued using the timing parameters configured for Bank 1 (TMRD, TRAS and TXSR timings) in the FMC\_SDTR1 register.*

*The TRP and TRC timings are only configured in the FMC\_SDTR1 register. If two SDRAM devices are used, the TRP and TRC timings must be programmed with the timings of the slowest device.*

### SDRAM Command Mode register (FMC\_SDCMR)

Address offset: 0x150

Reset value: 0x0000 0000

This register contains the command issued when the SDRAM device is accessed. This register is used to initialize the SDRAM device, and to activate the Self-refresh and the Power-down modes. As soon as the MODE field is written, the command is issued only to one or to both SDRAM banks according to CTB1 and CTB2 command bits. This register is the same for both SDRAM banks.

| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21        | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9         | 8  | 7  | 6  | 5    | 4    | 3         | 2  | 1  | 0  |
|----------|----|----|----|----|----|----|----|----|----|-----------|----|----|----|----|----|----|----|----|----|----|----|-----------|----|----|----|------|------|-----------|----|----|----|
| Reserved |    |    |    |    |    |    |    |    |    | MRD[11:0] |    |    |    |    |    |    |    |    |    |    |    | NRFS[3:0] |    |    |    | CTB1 | CTB2 | MODE[2:0] |    |    |    |
|          |    |    |    |    |    |    |    |    |    | rW        | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW | rW        | rW | rW | rW | rW   | rW   | rW        | rW | rW | rW |

Bits 31:22 Reserved, must be kept at reset value

Bits 21:9 **MRD[11:0]**: Mode Register definition

This 13-bit field defines the SDRAM Mode Register content. The Mode Register is programmed using the Load Mode Register command.

Bits 8:5 **NRFS[3:0]**: Number of Auto-refresh

These bits define the number of consecutive Auto-refresh commands issued when MODE = '011'.

0000: 1 Auto-refresh cycle

0001: 2 Auto-refresh cycles

....

1110: 15 Auto-refresh cycles

1111: Reserved

Bit 4 **CTB1**: Command Target Bank 1

This bit indicates whether the command is issued to SDRAM Bank 1 or not.

0: Command not issued to SDRAM Bank 1

1: Command issued to SDRAM Bank 1

Bit 3 **CTB2**: Command Target Bank 2

This bit indicates whether the command is issued to SDRAM Bank 2 or not.

- 0: Command not issued to SDRAM Bank 2
- 1: Command issued to SDRAM Bank 2

Bits 2:0 **MODE[2:0]**: Command mode

These bits define the command issued to the SDRAM device.

- 000: Normal Mode
- 001: Clock Configuration Enable
- 010: PALL ("All Bank Precharge") command
- 011: Auto-refresh command
- 100: Load Mode Register
- 101: Self-refresh command
- 110: Power-down command
- 111: Reserved

*Note: When a command is issued, at least one Command Target Bank bit ( CBT1 or CBT2) must be set. If both banks are used, the commands must be issued to the two banks at the same time by setting the CBT1 and CBT2 bits.*

### SDRAM Refresh Timer register (FMC\_SDRTR)

Address offset: 0x154

Reset value: 0x0000 0000

This register sets the refresh rate in number of SDCLK clock cycles between the refresh cycles by configuring the Refresh Timer Count value.

$$\text{Refresh rate} = (\text{SDRAM refresh rate} \times \text{SDRAM clock frequency}) - 20$$

$$\text{SDRAM refresh rate} = \text{SDRAM refresh period} / \text{Number of rows}$$

### Example

$$\text{SDRAM refresh rate} = 64 \text{ ms} / (8196 \text{ rows}) = 7.81 \mu\text{s}$$

where 64 ms is the SDRAM refresh period.

$$7.81 \mu\text{s} \times 60 \text{ MHz} = 468.6$$

The refresh rate must be increased by 20 SDRAM clock cycles (as in the above example) to obtain a safe margin if an internal refresh request occurs when a read request has been accepted. It corresponds to a COUNT value of '0000111000000' (448).

This 13-bit field is loaded into a timer which is decremented using the SDRAM clock. This timer generates a refresh pulse when zero is reached. The COUNT value must be set at least to 41 SDRAM clock cycles.

As soon as the FMC\_SDRTR register is programmed, the timer starts counting. If the value programmed in the register is '0', no refresh is carried out. This register must not be reprogrammed after the initialization procedure to avoid modifying the refresh rate.

Each time a refresh pulse is generated, this 13-bit COUNT field is reloaded into the counter.

If a memory access is in progress, the Auto-refresh request is delayed. However, if the memory access and Auto-refresh requests are generated simultaneously, the Auto-refresh takes precedence. If the memory access occurs during a refresh operation, the request is buffered to be processed when the refresh is complete.

This register common to SDRAM bank 1 and bank 2.

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |    |             |    |    |    |    |    |    |    |    |    |    |    |     |   |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|----|-------------|----|----|----|----|----|----|----|----|----|----|----|-----|---|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14   | 13 | 12          | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0   |   |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | REIE |    | COUNT[12:0] |    |    |    |    |    |    |    |    |    |    |    | CRE |   |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    | rw   | rw | rw          | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw | rw  | w |

Bits 31: 15 Reserved, must be kept at reset value

Bit 14 **REIE**: RES Interrupt Enable

0: Interrupt is disabled

1: An Interrupt is generated if RE = 1

Bits 13:1 **COUNT[12:0]**: Refresh Timer Count

This 13-bit field defines the refresh rate of the SDRAM device. It is expressed in number of memory clock cycles. It must be set at least to 41 SDRAM clock cycles (0x29).

$COUNT = (SDRAM \text{ refresh rate} \times SDRAM \text{ clock frequency}) - 20$

$SDRAM \text{ refresh rate} = SDRAM \text{ refresh period} / \text{Number of rows}$

Bit 0 **CRE**: Clear Refresh error flag

This bit is used to clear the Refresh Error Flag (RE) in the Status Register.

0: no effect

1: Refresh Error flag is cleared

**Note:** *The programmed COUNT value must not be equal to the sum of the following timings:  $TWR + TRP + TRC + TRCD + 4$  memory clock cycles.*

SDRAM Status register (FMC\_SDSR)

Address offset: 0x158

Reset value: 0x0000 0000

|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |      |   |             |   |             |   |    |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|------|---|-------------|---|-------------|---|----|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5    | 4 | 3           | 2 | 1           | 0 |    |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | BUSY |   | MODES2[1:0] |   | MODES1[1:0] |   | RE |
|          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   | r    | r | r           | r | r           | r | r  |

Bits 31:5 Reserved, must be kept at reset value

Bit 5 **BUSY**: Busy status

This bit defines the status of the SDRAM controller after a Command Mode request

0: SDRAM Controller is ready to accept a new request

1; SDRAM Controller is not ready to accept a new request

Bits 4:3 **MODES2[1:0]**: Status Mode for Bank 2

This bit defines the Status Mode of SDRAM Bank 2.

00: Normal Mode

01: Self-refresh mode

10: Power-down mode

Bits 2:1 **MODES1[1:0]**: Status Mode for Bank 1

This bit defines the Status Mode of SDRAM Bank 1.

00: Normal Mode

01: Self-refresh mode

10: Power-down mode

Bit 0 **RE**: Refresh error flag

0: No refresh error has been detected

1: A refresh error has been detected

An interrupt is generated if REIE = 1 and RE = 1

## 37.8 FMC register map

The following table summarizes the FMC registers.

**Table 298. FMC register map**

| Offset | Register  | 31       | 30          | 29 | 28 | 27          | 26 | 25          | 24 | 23           | 22 | 21          | 20 | 19       | 18          | 17          | 16         | 15          | 14     | 13          | 12      | 11      | 10      | 9        | 8        | 7         | 6         | 5         | 4     | 3     | 2 | 1 | 0 |
|--------|-----------|----------|-------------|----|----|-------------|----|-------------|----|--------------|----|-------------|----|----------|-------------|-------------|------------|-------------|--------|-------------|---------|---------|---------|----------|----------|-----------|-----------|-----------|-------|-------|---|---|---|
| 0x00   | FMC_BCR1  | Reserved |             |    |    |             |    |             |    |              |    |             |    | CCLKEN   | CBURSTRW    | CPSIZE[2:0] | ASYNCPWAIT | EXTMOD      | WAITEN | WREN        | WAITCFG | WRAPMOD | WAITPOL | BURSTEN  | Reserved | FACCEN    | MWID[1:0] | MTYP[1:0] | MUXEN | MBKEN |   |   |   |
| 0x08   | FMC_BCR2  | Reserved |             |    |    |             |    |             |    |              |    |             |    | CBURSTRW | CPSIZE[2:0] | ASYNCPWAIT  | EXTMOD     | WAITEN      | WREN   | WAITCFG     | WRAPMOD | WAITPOL | BURSTEN | Reserved | FACCEN   | MWID[1:0] | MTYP[1:0] | MUXEN     | MBKEN |       |   |   |   |
| 0x10   | FMC_BCR3  | Reserved |             |    |    |             |    |             |    |              |    |             |    | CBURSTRW | CPSIZE[2:0] | ASYNCPWAIT  | EXTMOD     | WAITEN      | WREN   | WAITCFG     | WRAPMOD | WAITPOL | BURSTEN | Reserved | FACCEN   | MWID[1:0] | MTYP[1:0] | MUXEN     | MBKEN |       |   |   |   |
| 0x18   | FMC_BCR4  | Reserved |             |    |    |             |    |             |    |              |    |             |    | CBURSTRW | CPSIZE[2:0] | ASYNCPWAIT  | EXTMOD     | WAITEN      | WREN   | WAITCFG     | WRAPMOD | WAITPOL | BURSTEN | Reserved | FACCEN   | MWID[1:0] | MTYP[1:0] | MUXEN     | MBKEN |       |   |   |   |
| 0x04   | FMC_BTR1  | Res.     | ACCMOD[1:0] |    |    | DATLAT[3:0] |    | CLKDIV[3:0] |    | BUSTURN[3:0] |    | DATAST[7:0] |    |          |             |             |            | ADDHLD[3:0] |        | ADDSET[3:0] |         |         |         |          |          |           |           |           |       |       |   |   |   |
| 0x0C   | FMC_BTR2  | Res.     | ACCMOD[1:0] |    |    | DATLAT[3:0] |    | CLKDIV[3:0] |    | BUSTURN[3:0] |    | DATAST[7:0] |    |          |             |             |            | ADDHLD[3:0] |        | ADDSET[3:0] |         |         |         |          |          |           |           |           |       |       |   |   |   |
| 0x14   | FMC_BTR3  | Res.     | ACCMOD[1:0] |    |    | DATLAT[3:0] |    | CLKDIV[3:0] |    | BUSTURN[3:0] |    | DATAST[7:0] |    |          |             |             |            | ADDHLD[3:0] |        | ADDSET[3:0] |         |         |         |          |          |           |           |           |       |       |   |   |   |
| 0x1C   | FMC_BTR4  | Res.     | ACCMOD[1:0] |    |    | DATLAT[3:0] |    | CLKDIV[3:0] |    | BUSTURN[3:0] |    | DATAST[7:0] |    |          |             |             |            | ADDHLD[3:0] |        | ADDSET[3:0] |         |         |         |          |          |           |           |           |       |       |   |   |   |
| 0x104  | FMC_BWTR1 | Res.     | ACCMOD[1:0] |    |    | Res.        |    |             |    | BUSTURN[3:0] |    | DATAST[7:0] |    |          |             |             |            | ADDHLD[3:0] |        | ADDSET[3:0] |         |         |         |          |          |           |           |           |       |       |   |   |   |

Table 298. FMC register map (continued)

| Offset | Register   | 31          | 30 | 29          | 28 | 27        | 26 | 25           | 24       | 23 | 22           | 21        | 20 | 19           | 18          | 17         | 16       | 15        | 14       | 13          | 12        | 11        | 10       | 9           | 8         | 7         | 6    | 5       | 4         | 3       | 2        | 1   | 0   |
|--------|------------|-------------|----|-------------|----|-----------|----|--------------|----------|----|--------------|-----------|----|--------------|-------------|------------|----------|-----------|----------|-------------|-----------|-----------|----------|-------------|-----------|-----------|------|---------|-----------|---------|----------|-----|-----|
| 0x10C  | FMC_BWTR2  | Res.        |    | ACCMOD[1:0] |    | Res.      |    |              |          |    | BUSTURN[3:0] |           |    |              | DATAST[7:0] |            |          |           |          | ADDHLD[3:0] |           |           |          | ADDSET[3:0] |           |           |      |         |           |         |          |     |     |
| 0x104  | FMC_BWTR3  | Res.        |    |             |    | Res.      |    |              |          |    | BUSTURN[3:0] |           |    |              | DATAST[7:0] |            |          |           |          | ADDHLD[3:0] |           |           |          | ADDSET[3:0] |           |           |      |         |           |         |          |     |     |
| 0x10C  | FMC_BWTR4  | Res.        |    |             |    | Res.      |    |              |          |    | BUSTURN[3:0] |           |    |              | DATAST[7:0] |            |          |           |          | ADDHLD[3:0] |           |           |          | ADDSET[3:0] |           |           |      |         |           |         |          |     |     |
| 0x60   | FMC_PCR2   | Reserved    |    |             |    |           |    |              |          |    |              |           |    | ECCPS[2:0]   |             |            | TAR[3:0] |           |          | TCLR[3:0]   |           |           | Res.     | ECCEN       | PWID[1:0] |           |      | PTYP    | PBKEN     | PWAITEN | Reserved |     |     |
| 0x80   | FMC_PCR3   | Reserved    |    |             |    |           |    |              |          |    |              |           |    | ECCPS[2:0]   |             |            | TAR[3:0] |           |          | TCLR[3:0]   |           |           | Res.     | ECCEN       | PWID[1:0] |           |      | PTYP    | PBKEN     | PWAITEN | Reserved |     |     |
| 0xA0   | FMC_PCR4   | Reserved    |    |             |    |           |    |              |          |    |              |           |    | ECCPS[2:0]   |             |            | TAR[3:0] |           |          | TCLR[3:0]   |           |           | Res.     | ECCEN       | PWID[1:0] |           |      | PTYP    | PBKEN     | PWAITEN | Reserved |     |     |
| 0x64   | FMC_SR2    | Reserved    |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          |             |           |           |          |             |           | FEMPT     |      | IFEN    | ILEN      | IREN    | IFS      | ILS | IRS |
| 0x84   | FMC_SR3    | Reserved    |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          |             |           |           |          |             |           | FEMPT     |      | IFEN    | ILEN      | IREN    | IFS      | ILS | IRS |
| 0xA4   | FMC_SR4    | Reserved    |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          |             |           |           |          |             |           | FEMPT     |      | IFEN    | ILEN      | IREN    | IFS      | ILS | IRS |
| 0x68   | FMC_PMEM2  | MEMHIZ[7:0] |    |             |    |           |    | MEMHOLD[7:0] |          |    |              |           |    | MEMWAIT[7:0] |             |            |          |           |          | MEMSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x88   | FMC_PMEM3  | MEMHIZ[7:0] |    |             |    |           |    | MEMHOLD[7:0] |          |    |              |           |    | MEMWAIT[7:0] |             |            |          |           |          | MEMSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0xA8   | FMC_PMEM4  | MEMHIZ[7:0] |    |             |    |           |    | MEMHOLD[7:0] |          |    |              |           |    | MEMWAIT[7:0] |             |            |          |           |          | MEMSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x6C   | FMC_PATT2  | ATTHIZ[7:0] |    |             |    |           |    | ATTHOLD[7:0] |          |    |              |           |    | ATTWAIT[7:0] |             |            |          |           |          | ATTSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x8C   | FMC_PATT3  | ATTHIZ[7:0] |    |             |    |           |    | ATTHOLD[7:0] |          |    |              |           |    | ATTWAIT[7:0] |             |            |          |           |          | ATTSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0xAC   | FMC_PATT4  | ATTHIZ[7:0] |    |             |    |           |    | ATTHOLD[7:0] |          |    |              |           |    | ATTWAIT[7:0] |             |            |          |           |          | ATTSET[7:0] |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0xB0   | FMC_PIO4   | IOHIZ[7:0]  |    |             |    |           |    | IOHOLD[7:0]  |          |    |              |           |    | IOWAIT[7:0]  |             |            |          |           |          | IOSET[7:0]  |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x74   | FMC_ECCR2  | ECC[31:0]   |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          |             |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x94   | FMC_ECCR3  | ECC[31:0]   |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          |             |           |           |          |             |           |           |      |         |           |         |          |     |     |
| 0x140  | FMC_SDCR_1 | Reserved    |    |             |    |           |    |              |          |    |              |           |    |              |             | RPIPE[1:0] |          | RBURST    | CLK[1:0] |             | WP        | CAS[1:0]  |          | NB          | MWID[1:0] |           |      | NR[1:0] |           | NC[1:0] |          |     |     |
| 0x144  | FMC_SDCR_2 | Reserved    |    |             |    |           |    |              |          |    |              |           |    |              |             |            |          |           |          | CLK[1:0]    |           | WP        | CAS[1:0] |             | NB        | MWID[1:0] |      |         | NR[1:0]   |         | NC[1:0]  |     |     |
| 0x148  | FMC_SDTR1  | Reserved    |    |             |    | TRCD[3:0] |    |              | TRP[3:0] |    |              | TWR[3:0]  |    |              | TRC[3:0]    |            |          | TRAS[3:0] |          |             | TXSR[3:0] |           |          | TMRD[3:0]   |           |           |      |         |           |         |          |     |     |
| 0x14C  | FMC_SDTR2  | Reserved    |    |             |    | TRCD[3:0] |    |              | TRP[3:0] |    |              | TWR[3:0]  |    |              | TRC[3:0]    |            |          | TRAS[3:0] |          |             | TXSR[3:0] |           |          | TMRD[3:0]   |           |           |      |         |           |         |          |     |     |
| 0x150  | FMC_SDCMR  | Reserved    |    |             |    |           |    |              |          |    |              | MRD[12:0] |    |              |             |            |          |           |          |             |           | NRFS[3:0] |          |             |           |           | CTB1 | CTB2    | MODE[2:0] |         |          |     |     |

Table 298. FMC register map (continued)

| Offset | Register  | 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16   | 15          | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7    | 6           | 5 | 4           | 3 | 2  | 1   | 0 |
|--------|-----------|----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------|-------------|----|----|----|----|----|---|---|------|-------------|---|-------------|---|----|-----|---|
| 0x154  | FMC_SDRTR | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    | REIE | COUNT[12:0] |    |    |    |    |    |   |   |      |             |   |             |   |    | CRE |   |
| 0x158  | FMC_SDSR  | Reserved |    |    |    |    |    |    |    |    |    |    |    |    |    |    |      |             |    |    |    |    |    |   |   | BUSY | MODES2[1:0] |   | MODES1[1:0] |   | RE |     |   |

## 38 Debug support (DBG)

This section applies to the whole STM32F4xx family, unless otherwise specified.

## 38.1 Overview

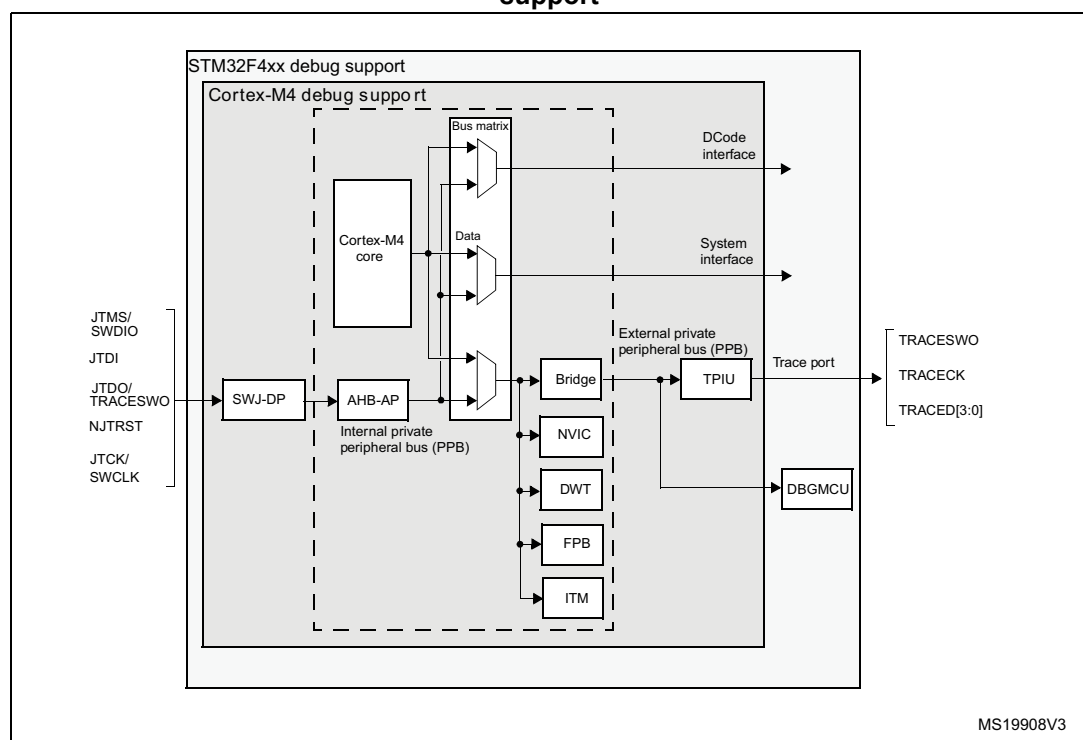
The STM32F4xx are built around a Cortex®-M4 with FPU core, which contains hardware extensions for advanced debugging features. The debug extensions allow the core to be stopped either on a given instruction fetch (breakpoint), or on data access (watchpoint). When stopped, the core's internal state and the system's external state may be examined. Once examination is complete, the core and the system may be restored and program execution resumed.

The debug features are used by the debugger host when connecting to and debugging the STM32F4xx MCUs.

Two interfaces for debug are available:

- Serial wire
- JTAG debug port

**Figure 485. Block diagram of STM32 MCU and Cortex®-M4 with FPU-level debug support**



**Note:** The debug features embedded in the Cortex®-M4 with FPU core are a subset of the Arm® CoreSight Design Kit.

The Arm® Cortex®-M4 with FPU core provides integrated on-chip debug support. It is comprised of:

- SWJ-DP: Serial wire / JTAG debug port
- AHP-AP: AHB access port
- ITM: Instrumentation trace macrocell
- FPB: Flash patch breakpoint
- DWT: Data watchpoint trigger
- TPUI: Trace port unit interface (available on larger packages, where the corresponding pins are mapped)
- ETM: Embedded Trace Macrocell (available on larger packages, where the corresponding pins are mapped)

It also includes debug features dedicated to the STM32F4xx:

- Flexible debug pinout assignment
- MCU debug box (support for low-power modes, control over peripheral clocks, etc.)

*Note:* For further information on debug functionality supported by the Arm® Cortex®-M4 with FPU core, refer to the Cortex®-M4 with FPU -r0p1 Technical Reference Manual and to the CoreSight Design Kit-r0p1 TRM (see [Section 38.2](#)).

## 38.2 Reference Arm® documentation

- Cortex®-M4 with FPU r0p1 Technical Reference Manual (TRM)  
(see Related documents on page 1)
- Arm® Debug Interface V5
- Arm® CoreSight Design Kit revision r0p1 Technical Reference Manual

## 38.3 SWJ debug port (serial wire and JTAG)

The core of the STM32F4xx integrates the Serial Wire / JTAG Debug Port (SWJ-DP). It is an Arm® standard CoreSight debug port that combines a JTAG-DP (5-pin) interface and a SW-DP (2-pin) interface.

- The JTAG Debug Port (JTAG-DP) provides a 5-pin standard JTAG interface to the AHP-AP port.
- The Serial Wire Debug Port (SW-DP) provides a 2-pin (clock + data) interface to the AHP-AP port.

In the SWJ-DP, the two JTAG pins of the SW-DP are multiplexed with some of the five JTAG pins of the JTAG-DP.

Figure 486. SWJ debug port

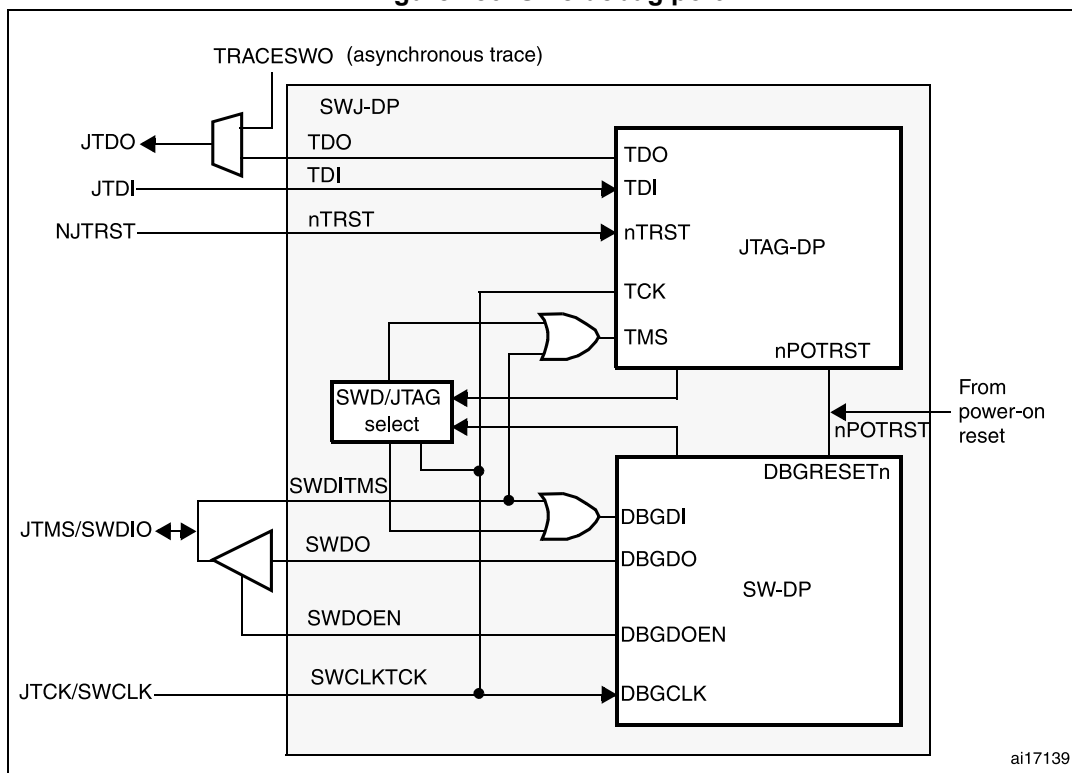


Figure 486 shows that the asynchronous TRACE output (TRACESWO) is multiplexed with TDO. This means that the asynchronous trace can only be used with SW-DP, not JTAG-DP.

### 38.3.1 Mechanism to select the JTAG-DP or the SW-DP

By default, the JTAG-Debug Port is active.

If the debugger host wants to switch to the SW-DP, it must provide a dedicated JTAG sequence on TMS/TCK (respectively mapped to SWDIO and SWCLK) which disables the JTAG-DP and enables the SW-DP. This way it is possible to activate the SWDP using only the SWCLK and SWDIO pins.

This sequence is:

1. Send more than 50 TCK cycles with TMS (SWDIO) = 1
2. Send the 16-bit sequence on TMS (SWDIO) = 0111100111100111 (MSB transmitted first)
3. Send more than 50 TCK cycles with TMS (SWDIO) = 1

## 38.4 Pinout and debug port pins

The STM32F4xx MCUs are available in various packages with different numbers of available pins. As a result, some functionality (ETM) related to pin availability may differ between packages.

### 38.4.1 SWJ debug port pins

Five pins are used as outputs from the STM32F4xx for the SWJ-DP as *alternate functions* of general-purpose I/Os. These pins are available on all packages.

**Table 299. SWJ debug port pins**

| SWJ-DP pin name | JTAG debug port |                          | SW debug port |                                    | Pin assignment |
|-----------------|-----------------|--------------------------|---------------|------------------------------------|----------------|
|                 | Type            | Description              | Type          | Debug assignment                   |                |
| JTMS/SWDIO      | I               | JTAG Test Mode Selection | IO            | Serial Wire Data Input/Output      | PA13           |
| JTCK/SWCLK      | I               | JTAG Test Clock          | I             | Serial Wire Clock                  | PA14           |
| JTDI            | I               | JTAG Test Data Input     | -             | -                                  | PA15           |
| JTDO/TRACESWO   | O               | JTAG Test Data Output    | -             | TRACESWO if async trace is enabled | PB3            |
| NJTRST          | I               | JTAG Test nReset         | -             | -                                  | PB4            |

### 38.4.2 Flexible SWJ-DP pin assignment

After RESET (SYSRESETn or PORESETn), all five pins used for the SWJ-DP are assigned as dedicated pins immediately usable by the debugger host (note that the trace outputs are not assigned except if explicitly programmed by the debugger host).

However, the STM32F4xx MCUs offers the possibility of disabling some or all of the SWJ-DP ports and so, of releasing the associated pins for general-purpose IO (GPIO) usage. For more details on how to disable SWJ-DP port pins, please refer to [Section 8.3.2: I/O pin multiplexer and mapping](#).

**Table 300. Flexible SWJ-DP pin assignment**

| Available debug ports                         | SWJ IO pin assigned |                     |             |            |              |
|-----------------------------------------------|---------------------|---------------------|-------------|------------|--------------|
|                                               | PA13 / JTMS / SWDIO | PA14 / JTCK / SWCLK | PA15 / JTDI | PB3 / JTDO | PB4 / NJTRST |
| Full SWJ (JTAG-DP + SW-DP) - Reset State      | X                   | X                   | X           | X          | X            |
| Full SWJ (JTAG-DP + SW-DP) but without NJTRST | X                   | X                   | X           | X          |              |
| JTAG-DP Disabled and SW-DP Enabled            | X                   | X                   |             |            |              |
| JTAG-DP Disabled and SW-DP Disabled           | Released            |                     |             |            |              |

### 38.4.3 Internal pull-up and pull-down on JTAG pins

It is necessary to ensure that the JTAG input pins are not floating since they are directly connected to flip-flops to control the debug mode features. Special care must be taken with the SWCLK/TCK pin which is directly connected to the clock of some of these flip-flops.

To avoid any uncontrolled IO levels, the device embeds internal pull-ups and pull-downs on the JTAG input pins:

- NJTRST: Internal pull-up
- JTDI: Internal pull-up
- JTMS/SWDIO: Internal pull-up
- TCK/SWCLK: Internal pull-down

Once a JTAG IO is released by the user software, the GPIO controller takes control again. The reset states of the GPIO control registers put the I/Os in the equivalent state:

- NJTRST: AF input pull-up
- JTDI: AF input pull-up
- JTMS/SWDIO: AF input pull-up
- JTCK/SWCLK: AF input pull-down
- JTDO: AF output floating

The software can then use these I/Os as standard GPIOs.

*Note: The JTAG IEEE standard recommends to add pull-ups on TDI, TMS and nTRST but there is no special recommendation for TCK. However, for JTCK, the device needs an integrated pull-down.*

*Having embedded pull-ups and pull-downs removes the need to add external resistors.*

### 38.4.4 Using serial wire and releasing the unused debug pins as GPIOs

To use the serial wire DP to release some GPIOs, the user software must change the GPIO (PA15, PB3 and PB4) configuration mode in the GPIO\_MODER register. This releases PA15, PB3 and PB4 which now become available as GPIOs.

When debugging, the host performs the following actions:

- Under system reset, all SWJ pins are assigned (JTAG-DP + SW-DP).
- Under system reset, the debugger host sends the JTAG sequence to switch from the JTAG-DP to the SW-DP.
- Still under system reset, the debugger sets a breakpoint on vector reset.
- The system reset is released and the Core halts.
- All the debug communications from this point are done using the SW-DP. The other JTAG pins can then be reassigned as GPIOs by the user software.

*Note:* For user software designs, note that:

*To release the debug pins, remember that they are first configured either in input-pull-up (nTRST, TMS, TDI) or pull-down (TCK) or output tristate (TDO) for a certain duration after reset until the instant when the user software releases the pins.*

*When debug pins (JTAG or SW or TRACE) are mapped, changing the corresponding IO pin configuration in the IOPORT controller has no effect.*

## 38.5 STM32F4xx JTAG TAP connection

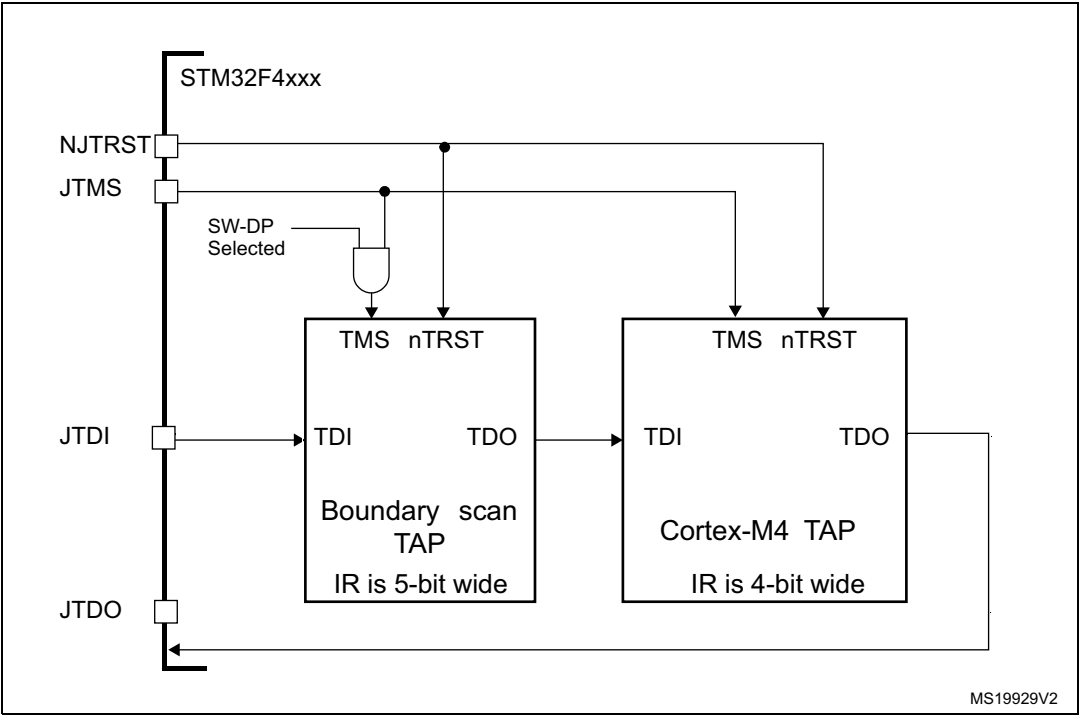
The STM32F4xx MCUs integrate two serially connected JTAG TAPs, the boundary scan TAP (IR is 5-bit wide) and the Cortex®-M4 with FPU TAP (IR is 4-bit wide).

To access the TAP of the Cortex®-M4 with FPU for debug purposes:

1. First, it is necessary to shift the BYPASS instruction of the boundary scan TAP.
2. Then, for each IR shift, the scan chain contains 9 bits (=5+4) and the unused TAP instruction must be shifted in using the BYPASS instruction.
3. For each data shift, the unused TAP, which is in BYPASS mode, adds 1 extra data bit in the data scan chain.

*Note:* **Important:** Once Serial-Wire is selected using the dedicated Arm® JTAG sequence, the boundary scan TAP is automatically disabled (JTMS forced high).

Figure 487. JTAG TAP connections



## 38.6 ID codes and locking mechanism

There are several ID codes inside the STM32F4xx MCUs. ST strongly recommends tools designers to lock their debuggers using the MCU DEVICE ID code located in the external PPB memory map at address 0xE0042000.

### 38.6.1 MCU device ID code

The STM32F4xx MCUs integrate an MCU ID code. This ID identifies the ST MCU part-number and the die revision. It is part of the DBG\_MCU component and is mapped on the external PPB bus (see [Section 38.16](#)). This code is accessible using the JTAG debug port (four to five pins) or the SW debug port (two pins) or by the user software. It is even accessible while the MCU is under system reset.

Only the DEV\_ID[11:0] must be used for identification by the debugger/programmer tools.

#### DBGMCU\_IDCODE

Address: 0xE004 2000

Only 32-bits access supported. Read-only.

|              |    |    |    |              |    |    |    |    |    |    |    |    |    |    |    |
|--------------|----|----|----|--------------|----|----|----|----|----|----|----|----|----|----|----|
| 31           | 30 | 29 | 28 | 27           | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| REV_ID[15:0] |    |    |    |              |    |    |    |    |    |    |    |    |    |    |    |
| r            | r  | r  | r  | r            | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15           | 14 | 13 | 12 | 11           | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| Reserved     |    |    |    | DEV_ID[11:0] |    |    |    |    |    |    |    |    |    |    |    |
|              |    |    |    | r            | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:16 **REV\_ID[15:0]** Revision identifier

This field indicates the revision of the device.

STM32F405xx/07xx and STM32F415xx/17xx devices:

0x1000 = Revision A

0x1001 = Revision Z

0x1003 = Revision 1

0x1007 = Revision 2

0x100F = Revision Y and 4

0x101F = Revision 5 and 6

STM32F42xxx and STM32F43xxx devices:

0x1000 = Revision A

0x1003 = Revision Y

0x1007 = Revision 1

0x2001 = Revision 3

0x2003 = Revision 4, 5 and B

Bits 15:12 Reserved, must be kept at reset value.

Bits 11:0 **DEV\_ID[11:0]**: Device identifier (STM32F405xx/07xx and STM32F415xx/17xx)

The device ID is 0x413.

Bits 11:0 **DEV\_ID[11:0]**: Device identifier (STM32F42xxx and STM32F43xxx)

The device ID is 0x419

## 38.6.2 Boundary scan TAP

### JTAG ID code

The TAP of the STM32F4xx BSC (boundary scan) integrates a JTAG ID code equal to .

- 0x06413041 for STM32F405xx/07xx and STM32F415xx/17xx devices
- 0x06419041 for STM32F42xxx and STM32F43xxx devices

## 38.6.3 Cortex<sup>®</sup>-M4 with FPU TAP

The TAP of the Arm<sup>®</sup> Cortex<sup>®</sup>-M4 with FPU integrates a JTAG ID code. This ID code is the Arm<sup>®</sup> default one and has not been modified. This code is only accessible by the JTAG Debug Port, it is 0x4BA00477 (corresponds to Cortex<sup>®</sup>-M4 with FPU r0p1, see [Section 38.2](#)).

## 38.6.4 Cortex<sup>®</sup>-M4 with FPU JEDEC-106 ID code

The Arm<sup>®</sup> Cortex<sup>®</sup>-M4 with FPU integrates a JEDEC-106 ID code. It is located in the 4 KB ROM table mapped on the internal PPB bus at address 0xE00FF000\_0xE00FFFFF.

This code is accessible by the JTAG Debug Port (4 to 5 pins) or by the SW Debug Port (two pins) or by the user software.

## 38.7 JTAG debug port

A standard JTAG state machine is implemented with a 4-bit instruction register (IR) and five data registers (for full details, refer to the Cortex<sup>®</sup>-M4 with FPU r0p1 *Technical Reference Manual (TRM)*, for references, see [Section 38.2](#)).

**Table 301. JTAG debug port data registers**

| IR(3:0) | Data register       | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1111    | BYPASS<br>[1 bit]   | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1110    | IDCODE<br>[32 bits] | ID CODE<br>0x4BA00477 (Arm <sup>®</sup> Cortex <sup>®</sup> -M4 with FPU r0p1 ID Code)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1010    | DPACC<br>[35 bits]  | Debug port access register<br>This initiates a debug port and allows access to a debug port register.<br>– When transferring data IN:<br>Bits 34:3 = DATA[31:0] = 32-bit data to transfer for a write request<br>Bits 2:1 = A[3:2] = 2-bit address of a debug port register.<br>Bit 0 = RnW = Read request (1) or write request (0).<br>– When transferring data OUT:<br>Bits 34:3 = DATA[31:0] = 32-bit data which is read following a read request<br>Bits 2:0 = ACK[2:0] = 3-bit Acknowledge:<br>010 = OK/FAULT<br>001 = WAIT<br>OTHER = reserved<br>Refer to <a href="#">Table 302</a> for a description of the A[3:2] bits |

Table 301. JTAG debug port data registers (continued)

| IR(3:0) | Data register      | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1011    | APACC<br>[35 bits] | <p>Access port access register</p> <p>Initiates an access port and allows access to an access port register.</p> <ul style="list-style-type: none"> <li>– When transferring data IN: <ul style="list-style-type: none"> <li>Bits 34:3 = DATA[31:0] = 32-bit data to shift in for a write request</li> <li>Bits 2:1 = A[3:2] = 2-bit address (sub-address AP registers).</li> <li>Bit 0 = RnW= Read request (1) or write request (0).</li> </ul> </li> <li>– When transferring data OUT: <ul style="list-style-type: none"> <li>Bits 34:3 = DATA[31:0] = 32-bit data which is read following a read request</li> <li>Bits 2:0 = ACK[2:0] = 3-bit Acknowledge: <ul style="list-style-type: none"> <li>010 = OK/FAULT</li> <li>001 = WAIT</li> <li>OTHER = reserved</li> </ul> </li> </ul> </li> </ul> <p>There are many AP registers (see AHB-AP) addressed as the combination of:</p> <ul style="list-style-type: none"> <li>– The shifted value A[3:2]</li> <li>– The current value of the DP SELECT register</li> </ul> |
| 1000    | ABORT<br>[35 bits] | <p>Abort register</p> <ul style="list-style-type: none"> <li>– Bits 31:1 = Reserved</li> <li>– Bit 0 = DAPABORT: write 1 to generate a DAP abort.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Table 302. 32-bit debug port registers addressed through the shifted value A[3:2]

| Address | A[3:2] value | Description                                                                                                                                                                                                                                                                                                                                                     |
|---------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x0     | 00           | Reserved, must be kept at reset value.                                                                                                                                                                                                                                                                                                                          |
| 0x4     | 01           | <p>DP CTRL/STAT register. Used to:</p> <ul style="list-style-type: none"> <li>– Request a system or debug power-up</li> <li>– Configure the transfer operation for AP accesses</li> <li>– Control the pushed compare and pushed verify operations.</li> <li>– Read some status flags (overrun, power-up acknowledges)</li> </ul>                                |
| 0x8     | 10           | <p>DP SELECT register: Used to select the current access port and the active 4-words register window.</p> <ul style="list-style-type: none"> <li>– Bits 31:24: APSEL: select the current AP</li> <li>– Bits 23:8: reserved</li> <li>– Bits 7:4: APBANKSEL: select the active 4-words register window on the current AP</li> <li>– Bits 3:0: reserved</li> </ul> |
| 0xC     | 11           | <p>DP RDBUFF register: Used to allow the debugger to get the final result after a sequence of operations (without requesting new JTAG-DP operation)</p>                                                                                                                                                                                                         |

## 38.8 SW debug port

### 38.8.1 SW protocol introduction

This synchronous serial protocol uses two pins:

- SWCLK: clock from host to target
- SWDIO: bidirectional

The protocol allows two banks of registers (DPACC registers and APACC registers) to be read and written to.

Bits are transferred LSB-first on the wire.

For SWDIO bidirectional management, the line must be pulled-up on the board (100 K $\Omega$  recommended by Arm®).

Each time the direction of SWDIO changes in the protocol, a turnaround time is inserted where the line is not driven by the host nor the target. By default, this turnaround time is one bit time, however this can be adjusted by configuring the SWCLK frequency.

### 38.8.2 SW protocol sequence

Each sequence consist of three phases:

1. Packet request (8 bits) transmitted by the host
2. Acknowledge response (3 bits) transmitted by the target
3. Data transfer phase (33 bits) transmitted by the host or the target

**Table 303. Packet request (8-bits)**

| Bit | Name   | Description                                                                      |
|-----|--------|----------------------------------------------------------------------------------|
| 0   | Start  | Must be “1”                                                                      |
| 1   | APnDP  | 0: DP Access<br>1: AP Access                                                     |
| 2   | RnW    | 0: Write Request<br>1: Read Request                                              |
| 4:3 | A[3:2] | Address field of the DP or AP registers (refer to <a href="#">Table 302</a> )    |
| 5   | Parity | Single bit parity of preceding bits                                              |
| 6   | Stop   | 0                                                                                |
| 7   | Park   | Not driven by the host. Must be read as “1” by the target because of the pull-up |

Refer to the Cortex®-M4 with FPU r0p1 *TRM* for a detailed description of DPACC and APACC registers.

The packet request is always followed by the turnaround time (default 1 bit) where neither the host nor target drive the line.

Table 304. ACK response (3 bits)

| Bit  | Name | Description                        |
|------|------|------------------------------------|
| 0..2 | ACK  | 001: FAULT<br>010: WAIT<br>100: OK |

The ACK Response must be followed by a turnaround time only if it is a READ transaction or if a WAIT or FAULT acknowledge has been received.

Table 305. DATA transfer (33 bits)

| Bit   | Name           | Description                       |
|-------|----------------|-----------------------------------|
| 0..31 | WDATA or RDATA | Write or Read data                |
| 32    | Parity         | Single parity of the 32 data bits |

The DATA transfer must be followed by a turnaround time only if it is a READ transaction.

### 38.8.3 SW-DP state machine (reset, idle states, ID code)

The State Machine of the SW-DP has an internal ID code which identifies the SW-DP. It follows the JEP-106 standard. This ID code is the default Arm<sup>®</sup> one and is set to **0x2BA01477** (corresponding to Cortex<sup>®</sup>-M4 with FPU r0p1).

*Note:*

*Note that the SW-DP state machine is inactive until the target reads this ID code.*

- The SW-DP state machine is in RESET STATE either after power-on reset, or after the DP has switched from JTAG to SWD or after the line is high for more than 50 cycles
- The SW-DP state machine is in IDLE STATE if the line is low for at least two cycles after RESET state.
- After RESET state, it is **mandatory** to first enter into an IDLE state AND to perform a READ access of the DP-SW ID CODE register. Otherwise, the target issues a FAULT acknowledge response on another transactions.

Further details of the SW-DP state machine can be found in the Cortex<sup>®</sup>-M4 with FPU r0p1 *TRM* and the *CoreSight Design Kit r0p1 TRM*.

### 38.8.4 DP and AP read/write accesses

- Read accesses to the DP are not posted: the target response can be immediate (if ACK=OK) or can be delayed (if ACK=WAIT).
- Read accesses to the AP are posted. This means that the result of the access is returned on the next transfer. If the next access to be done is NOT an AP access, then the DP-RDBUFF register must be read to obtain the result.  
The READOK flag of the DP-CTRL/STAT register is updated on every AP read access or RDBUFF read request to know if the AP read access was successful.
- The SW-DP implements a write buffer (for both DP or AP writes), that enables it to accept a write operation even when other transactions are still outstanding. If the write buffer is full, the target acknowledge response is "WAIT". With the exception of

IDCODE read or CTRL/STAT read or ABORT write which are accepted even if the write buffer is full.

- Because of the asynchronous clock domains SWCLK and HCLK, two extra SWCLK cycles are needed after a write transaction (after the parity bit) to make the write effective internally. These cycles must be applied while driving the line low (IDLE state). This is particularly important when writing the CTRL/STAT for a power-up request. If the next transaction (requiring a power-up) occurs immediately, it fails.

### 38.8.5 SW-DP registers

Access to these registers are initiated when APnDP=0

**Table 306. SW-DP registers**

| A[3:2] | R/W        | CTRLSEL bit of SELECT register | Register     | Notes                                                                                                                                                                                                                                                                                            |
|--------|------------|--------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00     | Read       | -                              | IDCODE       | The manufacturer code is not set to ST code. <b>0x2BA01477</b> (identifies the SW-DP)                                                                                                                                                                                                            |
| 00     | Write      | -                              | ABORT        | -                                                                                                                                                                                                                                                                                                |
| 01     | Read/Write | 0                              | DP-CTRL/STAT | Purpose is to: <ul style="list-style-type: none"> <li>request a system or debug power-up</li> <li>configure the transfer operation for AP accesses</li> <li>control the pushed compare and pushed verify operations.</li> <li>read some status flags (overrun, power-up acknowledges)</li> </ul> |
| 01     | Read/Write | 1                              | WIRE CONTROL | Purpose is to configure the physical serial port protocol (like the duration of the turnaround time)                                                                                                                                                                                             |
| 10     | Read       | -                              | READ RESEND  | Enables recovery of the read data from a corrupted debugger transfer, without repeating the original AP transfer.                                                                                                                                                                                |
| 10     | Write      | -                              | SELECT       | The purpose is to select the current access port and the active 4-words register window                                                                                                                                                                                                          |
| 11     | Read/Write | -                              | READ BUFFER  | This read buffer is useful because AP accesses are posted (the result of a read AP request is available on the next AP transaction). This read buffer captures data from the AP, presented as the result of a previous read, without initiating a new transaction                                |

### 38.8.6 SW-AP registers

Access to these registers are initiated when APnDP=1

There are many AP registers (see AHB-AP) addressed as the combination of:

- The shifted value A[3:2]
- The current value of the DP SELECT register

## 38.9 AHB-AP (AHB access port) - valid for both JTAG-DP and SW-DP

### Features:

- System access is independent of the processor status.
- Either SW-DP or JTAG-DP accesses AHB-AP.
- The AHB-AP is an AHB master into the Bus Matrix. Consequently, it can access all the data buses (Dcode Bus, System Bus, internal and external PPB bus) but the ICode bus.
- Bitband transactions are supported.
- AHB-AP transactions bypass the FPB.

The address of the 32-bits AHB-AP registers are 6-bits wide (up to 64 words or 256 bytes) and consists of:

- c) Bits [7:4] = the bits [7:4] APBANKSEL of the DP SELECT register
- d) Bits [3:2] = the 2 address bits of A[3:2] of the 35-bit packet request for SW-DP.

The AHB-AP of the Cortex®-M4 with FPU includes 9 x 32-bits registers:

**Table 307. Cortex®-M4 with FPU AHB-AP registers**

| Address offset | Register name                  | Notes                                                                                                                         |
|----------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 0x00           | AHB-AP Control and Status Word | Configures and controls transfers through the AHB interface (size, hprot, status on current transfer, address increment type) |
| 0x04           | AHB-AP Transfer Address        | -                                                                                                                             |
| 0x0C           | AHB-AP Data Read/Write         | -                                                                                                                             |
| 0x10           | AHB-AP Banked Data 0           | Directly maps the 4 aligned data words without rewriting the Transfer Address register.                                       |
| 0x14           | AHB-AP Banked Data 1           |                                                                                                                               |
| 0x18           | AHB-AP Banked Data 2           |                                                                                                                               |
| 0x1C           | AHB-AP Banked Data 3           |                                                                                                                               |
| 0xF8           | AHB-AP Debug ROM Address       | Base Address of the debug interface                                                                                           |
| 0xFC           | AHB-AP ID register             | -                                                                                                                             |

Refer to the Cortex®-M4 with FPU *r0p1 TRM* for further details.

## 38.10 Core debug

Core debug is accessed through the core debug registers. Debug access to these registers is by means of the *Advanced High-performance Bus* (AHB-AP) port. The processor can access these registers directly over the internal *Private Peripheral Bus* (PPB).

It consists of 4 registers:

**Table 308. Core debug registers**

| Register | Description                                                                                                                                                                                            |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DHCSR    | The 32-bit Debug Halting Control and Status register<br>This provides status information about the state of the processor enable core debug halt and step the processor                                |
| DCRSR    | The 17-bit Debug Core register Selector register:<br>This selects the processor register to transfer data to or from.                                                                                  |
| DCRDR    | The 32-bit Debug Core register Data register:<br>This holds data for reading and writing registers to and from the processor selected by the DCRSR (Selector) register.                                |
| DEMCR    | The 32-bit Debug Exception and Monitor Control register:<br>This provides Vector Catching and Debug Monitor Control. This register contains a bit named <b>TRCENA</b> which enable the use of a TRACE. |

*Note:* **Important:** these registers are not reset by a system reset. They are only reset by a power-on reset.

Refer to the Cortex®-M4 with FPU r0p1 TRM for further details.

To Halt on reset, it is necessary to:

- enable the bit0 (VC\_CORRESET) of the Debug and Exception Monitor Control register
- enable the bit0 (C\_DEBUGEN) of the Debug Halting Control and Status register.

### 38.11 Capability of the debugger host to connect under system reset

The reset system of the STM32F4xx MCU comprises the following reset sources:

- POR (power-on reset) which asserts a RESET at each power-up.
- Internal watchdog reset
- Software reset
- External reset

The Cortex<sup>®</sup>-M4 with FPU differentiates the reset of the debug part (generally PORRESETn) and the other one (SYSRESETn)

This way, it is possible for the debugger to connect under System Reset, programming the Core Debug registers to halt the core when fetching the reset vector. Then the host can release the system reset and the core immediately halts without having executed any instructions. In addition, it is possible to program any debug features under System Reset.

*Note: It is highly recommended for the debugger host to connect (set a breakpoint in the reset vector) under system reset.*

### 38.12 FPB (Flash patch breakpoint)

The FPB unit:

- implements hardware breakpoints
- patches code and data from code space to system space. This feature gives the possibility to correct software bugs located in the Code Memory Space.

The use of a Software Patch or a Hardware Breakpoint is exclusive.

The FPB consists of:

- 2 literal comparators for matching against literal loads from Code Space and remapping to a corresponding area in the System Space.
- 6 instruction comparators for matching against instruction fetches from Code Space. They can be used either to remap to a corresponding area in the System Space or to generate a Breakpoint Instruction to the core.

## 38.13 DWT (data watchpoint trigger)

The DWT unit consists of four comparators. They are configurable as:

- a hardware watchpoint or
- a trigger to an ETM or
- a PC sampler or
- a data address sampler

The DWT also provides some means to give some profiling informations. For this, some counters are accessible to give the number of:

- Clock cycle
- Folded instructions
- Load store unit (LSU) operations
- Sleep cycles
- CPI (clock per instructions)
- Interrupt overhead

## 38.14 ITM (instrumentation trace macrocell)

### 38.14.1 General description

The ITM is an application-driven trace source that supports *printf* style debugging to trace *Operating System* (OS) and application events, and emits diagnostic system information. The ITM emits trace information as packets which can be generated as:

- **Software trace.** Software can write directly to the ITM stimulus registers to emit packets.
- **Hardware trace.** The DWT generates these packets, and the ITM emits them.
- **Time stamping.** Timestamps are emitted relative to packets. The ITM contains a 21-bit counter to generate the timestamp. The Cortex<sup>®</sup>-M4 with FPU clock or the bit clock rate of the *Serial Wire Viewer* (SWV) output clocks the counter.

The packets emitted by the ITM are output to the TPIU (Trace Port Interface Unit). The formatter of the TPIU adds some extra packets (refer to TPIU) and then output the complete packets sequence to the debugger host.

The bit TRCEN of the Debug Exception and Monitor Control register must be enabled before programming or using the ITM.

### 38.14.2 Time stamp packets, synchronization and overflow packets

Time stamp packets encode time stamp information, generic control and synchronization. It uses a 21-bit timestamp counter (with possible prescalers) which is reset at each time stamp packet emission. This counter can be either clocked by the CPU clock or the SWV clock.

A synchronization packet consists of 6 bytes equal to 0x80\_00\_00\_00\_00\_00 which is emitted to the TPIU as 00 00 00 00 00 80 (LSB emitted first).

A synchronization packet is a timestamp packet control. It is emitted at each DWT trigger.

For this, the DWT must be configured to trigger the ITM: the bit CYCCNTENA (bit0) of the DWT Control register must be set. In addition, the bit2 (SYNCENA) of the ITM Trace Control register must be set.

*Note: If the SYNENA bit is not set, the DWT generates Synchronization triggers to the TPIU which sends only TPIU synchronization packets and not ITM synchronization packets.*

An overflow packet consists is a special timestamp packets which indicates that data has been written but the FIFO was full.

**Table 309. Main ITM registers**

| Address                | Register                     | Details                                                                                                                                                   |
|------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| @E0000FB0              | ITM lock access              | Write 0xC5ACCE55 to unlock Write Access to the other ITM registers                                                                                        |
| @E0000E80              | ITM trace control            | Bits 31-24 = Always 0                                                                                                                                     |
|                        |                              | Bits 23 = Busy                                                                                                                                            |
|                        |                              | Bits 22-16 = 7-bits ATB ID which identifies the source of the trace data.                                                                                 |
|                        |                              | Bits 15-10 = Always 0                                                                                                                                     |
|                        |                              | Bits 9:8 = TSPrescale = Time Stamp Prescaler                                                                                                              |
|                        |                              | Bits 7-5 = Reserved                                                                                                                                       |
|                        |                              | Bit 4 = SWOENA = Enable SWV behavior (to clock the timestamp counter by the SWV clock).                                                                   |
|                        |                              | Bit 3 = DWTENA: Enable the DWT Stimulus                                                                                                                   |
|                        |                              | Bit 2 = SYNCENA: this bit must be to 1 to enable the DWT to generate synchronization triggers so that the TPIU can then emit the synchronization packets. |
|                        |                              | Bit 1 = TSENA (Timestamp Enable)                                                                                                                          |
|                        |                              | Bit 0 = ITMENA: Global Enable Bit of the ITM                                                                                                              |
| @E0000E40              | ITM trace privilege          | Bit 3: mask to enable tracing ports31:24                                                                                                                  |
|                        |                              | Bit 2: mask to enable tracing ports23:16                                                                                                                  |
|                        |                              | Bit 1: mask to enable tracing ports15:8                                                                                                                   |
|                        |                              | Bit 0: mask to enable tracing ports7:0                                                                                                                    |
| @E0000E00              | ITM trace enable             | Each bit enables the corresponding Stimulus port to generate trace.                                                                                       |
| @E0000000-<br>E000007C | Stimulus port registers 0-31 | Write the 32-bits data on the selected Stimulus Port (32 available) to be traced out.                                                                     |

### Example of configuration

To output a simple value to the TPIU:

- Configure the TPIU and assign TRACE I/Os by configuring the DBGMCU\_CR (refer to [Section 38.17.2](#) and [Section 38.16.3](#))
- Write 0xC5ACCE55 to the ITM Lock Access register to unlock the write access to the ITM registers
- Write 0x00010005 to the ITM Trace Control register to enable the ITM with Sync enabled and an ATB ID different from 0x00
- Write 0x1 to the ITM Trace Enable register to enable the Stimulus Port 0
- Write 0x1 to the ITM Trace Privilege register to unmask stimulus ports 7:0
- Write the value to output in the Stimulus Port register 0: this can be done by software (using a printf function)

## 38.15 ETM (Embedded Trace Macrocell™)

### 38.15.1 ETM general description

The ETM enables the reconstruction of program execution. Data are traced using the Data Watchpoint and Trace (DWT) component or the Instruction Trace Macrocell (ITM) whereas instructions are traced using the Embedded Trace Macrocell (ETM).

The ETM transmits information as packets and is triggered by embedded resources. These resources must be programmed independently and the trigger source is selected using the Trigger Event register (0xE0041008). An event could be a simple event (address match from an address comparator) or a logic equation between 2 events. The trigger source is one of the fourth comparators of the DWT module, The following events can be monitored:

- Clock cycle matching
- Data address matching

For more informations on the trigger resources refer to [Section 38.13](#).

The packets transmitted by the ETM are output to the TPIU (Trace Port Interface Unit). The formatter of the TPIU adds some extra packets (refer to [Section 38.17](#)) and then outputs the complete packet sequence to the debugger host.

### 38.15.2 ETM signal protocol and packet types

This part is described in the chapter 7 ETMv3 Signal Protocol of the Arm® IHI 0014N document.

### 38.15.3 Main ETM registers

For more information on registers refer to the chapter 3 of the Arm® IHI 0014N specification.

**Table 310. Main ETM registers**

| Address    | Register                 | Details                                                                                                |
|------------|--------------------------|--------------------------------------------------------------------------------------------------------|
| 0xE0041FB0 | ETM Lock Access          | Write 0xC5ACCE55 to unlock the write access to the other ETM registers.                                |
| 0xE0041000 | ETM Control              | This register controls the general operation of the ETM, for instance how tracing is enabled.          |
| 0xE0041010 | ETM Status               | This register provides information about the current status of the trace and trigger logic.            |
| 0xE0041008 | ETM Trigger Event        | This register defines the event that controls trigger.                                                 |
| 0xE004101C | ETM Trace Enable Control | This register defines which comparator is selected.                                                    |
| 0xE0041020 | ETM Trace Enable Event   | This register defines the trace enabling event.                                                        |
| 0xE0041024 | ETM Trace Start/Stop     | This register defines the traces used by the trigger source to start and stop the trace, respectively. |

### 38.15.4 ETM configuration example

To output a simple value to the TPIU:

- Configure the TPIU and enable the I/O\_TRACEN to assign TRACE I/Os in the STM32F4xx debug configuration register.
- Write 0xC5AC CE55 to the ETM Lock Access register to unlock the write access to the ITM registers
- Write 0x0000 1D1E to the ETM control register (configure the trace)
- Write 0x0000 406F to the ETM Trigger Event register (define the trigger event)
- Write 0x0000 006F to the ETM Trace Enable Event register (define an event to start/stop)
- Write 0x0200 0000x0000 0001 to the ETM Trace Start/stop register (enable the trace)
- Write 0x0000191E to the ETM Control register (end of configuration)

## 38.16 MCU debug component (DBGMCU)

The MCU debug component helps the debugger provide support for:

- Low-power modes
- Clock control for timers, watchdog, I2C and bxCAN during a breakpoint
- Control of the trace pins assignment

### 38.16.1 Debug support for low-power modes

To enter low-power mode, the instruction WFI or WFE must be executed.

The MCU implements several low-power modes which can either deactivate the CPU clock or reduce the power of the CPU.

The core does not allow FCLK or HCLK to be turned off during a debug session. As these are required for the debugger connection, during a debug, they must remain active. The MCU integrates special means to allow the user to debug software in low-power modes.

For this, the debugger host must first set some debug configuration registers to change the low-power mode behavior:

- In Sleep mode, DBG\_SLEEP bit of DBGMCU\_CR register must be previously set by the debugger. This feeds HCLK with the same clock that is provided to FCLK (system clock previously configured by the software).
- In Stop mode, the bit DBG\_STOP must be previously set by the debugger. This enables the internal RC oscillator clock to feed FCLK and HCLK in STOP mode.

### 38.16.2 Debug support for timers, watchdog, bxCAN and I<sup>2</sup>C

During a breakpoint, it is necessary to choose how the counter of timers and watchdog must behave:

- They can continue to count inside a breakpoint. This is usually required when a PWM is controlling a motor, for example.
- They can stop to count inside a breakpoint. This is required for watchdog purposes.

For the bxCAN, the user can choose to block the update of the receive register during a breakpoint.

For the I<sup>2</sup>C, the user can choose to block the SMBUS timeout during a breakpoint.

For timers having complementary outputs, when the counter is stopped (DBG\_TIMx\_STOP = 1), the outputs are disabled (as if the MOE bit was reset) for safety purposes.

### 38.16.3 Debug MCU configuration register

This register allows the configuration of the MCU under DEBUG. This concerns:

- Low-power mode support
- Timer and watchdog counter support
- bxCAN communication support
- Trace pin assignment

This DBGMCU\_CR is mapped on the External PPB bus at address 0xE0042004

It is asynchronously reset by the PORESET (and not the system reset). It can be written by the debugger under system reset.

If the debugger host does not support these features, it is still possible for the user software to write to these registers.

#### DBGMCU\_CR register

Address: 0xE004 2004

Only 32-bit access supported

POR Reset: 0x0000 0000 (not reset by system reset)

|          |    |    |    |    |    |    |    |                  |    |            |          |    |             |          |           |
|----------|----|----|----|----|----|----|----|------------------|----|------------|----------|----|-------------|----------|-----------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23               | 22 | 21         | 20       | 19 | 18          | 17       | 16        |
| Reserved |    |    |    |    |    |    |    |                  |    |            |          |    |             |          |           |
| 15       | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7                | 6  | 5          | 4        | 3  | 2           | 1        | 0         |
| Reserved |    |    |    |    |    |    |    | TRACE_MODE [1:0] |    | TRACE_IOEN | Reserved |    | DBG_STANDBY | DBG_STOP | DBG_SLEEP |
|          |    |    |    |    |    |    |    | rw               | rw | rw         |          |    | rw          | rw       | rw        |

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:5 **TRACE\_MODE[1:0] and TRACE\_IOEN**: Trace pin assignment control

- With TRACE\_IOEN=0:
  - TRACE\_MODE=xx: TRACE pins not assigned (default state)
- With TRACE\_IOEN=1:
  - TRACE\_MODE=00: TRACE pin assignment for Asynchronous mode
  - TRACE\_MODE=01: TRACE pin assignment for Synchronous mode with a TRACEDATA size of 1
  - TRACE\_MODE=10: TRACE pin assignment for Synchronous mode with a TRACEDATA size of 2
  - TRACE\_MODE=11: TRACE pin assignment for Synchronous mode with a TRACEDATA size of 4

Bits 4:3 Reserved, must be kept at reset value.

Bit 2 **DBG\_STANDBY**: Debug Standby mode

0: (FCLK=Off, HCLK=Off) The whole digital part is unpowered.

From software point of view, exiting from Standby is identical than fetching reset vector (except a few status bit indicated that the MCU is resuming from Standby)

1: (FCLK=On, HCLK=On) In this case, the digital part is not unpowered and FCLK and HCLK are provided by the internal RC oscillator which remains active. In addition, the MCU generate a system reset during Standby mode so that exiting from Standby is identical than fetching from reset

Bit 1 **DBG\_STOP**: Debug Stop mode

0: (FCLK=Off, HCLK=Off) In STOP mode, the clock controller disables all clocks (including HCLK and FCLK). When exiting from STOP mode, the clock configuration is identical to the one after RESET (CPU clocked by the 8 MHz internal RC oscillator (HSI)). Consequently, the software must reprogram the clock controller to enable the PLL, the Xtal, etc.

1: (FCLK=On, HCLK=On) In this case, when entering STOP mode, FCLK and HCLK are provided by the internal RC oscillator which remains active in STOP mode. When exiting STOP mode, the software must reprogram the clock controller to enable the PLL, the Xtal, etc. (in the same way it would do in case of DBG\_STOP=0)

Bit 0 **DBG\_SLEEP**: Debug Sleep mode

0: (FCLK=On, HCLK=Off) In Sleep mode, FCLK is clocked by the system clock as previously configured by the software while HCLK is disabled.

In Sleep mode, the clock controller configuration is not reset and remains in the previously programmed state. Consequently, when exiting from Sleep mode, the software does not need to reconfigure the clock controller.

1: (FCLK=On, HCLK=On) In this case, when entering Sleep mode, HCLK is fed by the same clock that is provided to FCLK (system clock as previously configured by the software).

### 38.16.4 Debug MCU APB1 freeze register (DBGMCU\_APB1\_FZ)

The DBGMCU\_APB1\_FZ register is used to configure the MCU under Debug. It concerns APB1 peripherals. It is mapped on the external PPB bus at address 0xE004 2008.

The register is asynchronously reset by the POR (and not the system reset). It can be written by the debugger under system reset.

Address : 0xE004 2008

Only 32-bits access are supported.

Power-on reset (POR): 0x0000 0000 (not reset by system reset)

[illegible]

Bits 31:27    Reserved, must be kept at reset value.

Bit 26 **DBG CAN2 STOP**: Debug CAN2 stopped when Core is halted

0: Same behavior as in normal mode  
1: The CAN2 receive registers are frozen

Bit 25 **DBG\_CAN1\_STOP**: Debug CAN2 stopped when Core is halted

- 0: Same behavior as in normal mode
- 1: The CAN2 receive registers are frozen

Bit 24    Reserved, must be kept at reset value.

Bit 23 **DBG I2C3 SMBUS TIMEOUT:** SMBUS timeout mode stopped when Core is halted

0: Same behavior as in normal mode  
1: The SMBUS timeout is frozen

Bit 22 **DBG\_I2C2\_SMBUS\_TIMEOUT**: SMBUS timeout mode stopped when Core is halted

0: Same behavior as in normal mode  
1: The SMBUS timeout is frozen

Bit 21 **DBG I2C1 SMBUS TIMEOUT**: SMBUS timeout mode stopped when Core is halted

0: Same behavior as in normal mode  
1: The SMBUS timeout is frozen

Bit 20:13 Reserved, must be kept at reset value.

Bit 12 **DBG\_IWDG\_STOP**: Debug independent watchdog stopped when core is halted  
 0: The independent watchdog counter clock continues even if the core is halted  
 1: The independent watchdog counter clock is stopped when the core is halted

Bit 11 **DBG\_WWDG\_STOP**: Debug Window Watchdog stopped when Core is halted  
 0: The window watchdog counter clock continues even if the core is halted  
 1: The window watchdog counter clock is stopped when the core is halted

Bit 10 **DBG\_RTC\_STOP**: RTC stopped when Core is halted  
 0: The RTC counter clock continues even if the core is halted  
 1: The RTC counter clock is stopped when the core is halted

Bit 9 Reserved, must be kept at reset value.

Bits 8:0 **DBG\_TIMx\_STOP**: TIMx counter stopped when core is halted (x=2..7, 12..14)  
 0: The clock of the involved timer counter is fed even if the core is halted  
 1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

### 38.16.5 Debug MCU APB2 Freeze register (DBGMCU\_APB2\_FZ)

The DBGMCU\_APB2\_FZ register is used to configure the MCU under Debug. It concerns APB2 peripherals.

This register is mapped on the external PPB bus at address 0xE004 200C

It is asynchronously reset by the POR (and not the system reset). It can be written by the debugger under system reset.

Address: 0xE004 200C

Only 32-bit access is supported.

POR: 0x0000 0000 (not reset by system reset)

|          |    |    |    |    |    |    |    |    |    |    |    |    |                |                |               |
|----------|----|----|----|----|----|----|----|----|----|----|----|----|----------------|----------------|---------------|
| 31       | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18             | 17             | 16            |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    | DBG_TIM11_STOP | DBG_TIM10_STOP | DBG_TIM9_STOP |
|          |    |    |    |    |    |    |    |    |    |    |    |    | rw             | rw             | rw            |
| 15       | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2              | 1              | 0             |
| Reserved |    |    |    |    |    |    |    |    |    |    |    |    | DBG_TIM8_STOP  | DBG_TIM1_STOP  |               |
|          |    |    |    |    |    |    |    |    |    |    |    |    | rw             | rw             |               |

Bits 31:19 Reserved, must be kept at reset value.

Bits 18:16 **DBG\_TIMx\_STOP**: TIMx counter stopped when core is halted (x=9..11)  
 0: The clock of the involved timer counter is fed even if the core is halted  
 1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

Bits 15: Reserved, must be kept at reset value.

Bit 1 **DBG\_TIM8\_STOP**: TIM8 counter stopped when core is halted

0: The clock of the involved timer counter is fed even if the core is halted

1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

Bit 0 **DBG\_TIM1\_STOP**: TIM1 counter stopped when core is halted

0: The clock of the involved timer counter is fed even if the core is halted

1: The clock of the involved timer counter is stopped and the outputs are disabled when the core is halted

## 38.17 TPIU (trace port interface unit)

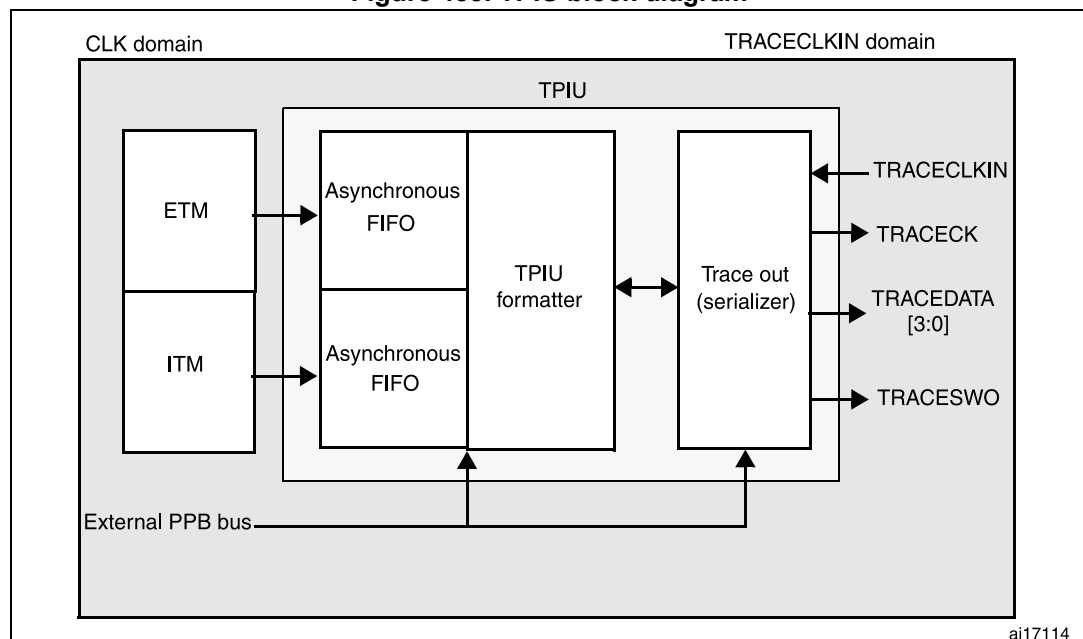
### 38.17.1 Introduction

The TPIU acts as a bridge between the on-chip trace data from the ITM and the ETM.

The output data stream encapsulates the trace source ID, that is then captured by a *trace port analyzer* (TPA).

The core embeds a simple TPIU, especially designed for low-cost debug (consisting of a special version of the CoreSight TPIU).

**Figure 488. TPIU block diagram**



### 38.17.2 TRACE pin assignment

- **Asynchronous mode**  
The asynchronous mode requires 1 extra pin and is available on all packages. It is only available if using Serial Wire mode (not in JTAG mode).

**Table 311. Asynchronous TRACE pin assignment**

| TPUI pin name | Trace synchronous mode |                         | STM32F4xx pin assignment |
|---------------|------------------------|-------------------------|--------------------------|
|               | Type                   | Description             |                          |
| TRACESWO      | O                      | TRACE Async Data Output | PB3                      |

- **Synchronous mode**  
The synchronous mode requires from 2 to 6 extra pins depending on the data trace size and is only available in the larger packages. In addition it is available in JTAG mode and in Serial Wire mode and provides better bandwidth output capabilities than asynchronous trace.

**Table 312. Synchronous TRACE pin assignment**

| TPUI pin name | Trace synchronous mode |                                              | STM32F4xxpin assignment |
|---------------|------------------------|----------------------------------------------|-------------------------|
|               | Type                   | Description                                  |                         |
| TRACECK       | O                      | TRACE Clock                                  | PE2                     |
| TRACED[3:0]   | O                      | TRACE Sync Data Outputs<br>Can be 1, 2 or 4. | PE[6:3]                 |

#### TPUI TRACE pin assignment

By default, these pins are NOT assigned. They can be assigned by setting the TRACE\_IOEN and TRACE\_MODE bits in the **MCU Debug component configuration register**. This configuration has to be done by the debugger host.

In addition, the number of pins to assign depends on the trace configuration (asynchronous or synchronous).

- **Asynchronous mode:** 1 extra pin is needed
- **Synchronous mode:** from 2 to 5 extra pins are needed depending on the size of the data trace port register (1, 2 or 4):
  - TRACECK
  - TRACED(0) if port size is configured to 1, 2 or 4
  - TRACED(1) if port size is configured to 2 or 4
  - TRACED(2) if port size is configured to 4
  - TRACED(3) if port size is configured to 4

To assign the TRACE pin, the debugger host must program the bits TRACE\_IOEN and TRACE\_MODE[1:0] of the Debug MCU configuration register (DBGMCU\_CR). By default the TRACE pins are not assigned.

This register is mapped on the external PPB and is reset by the PORESET (and not by the SYSTEM reset). It can be written by the debugger under SYSTEM reset.

Table 313. Flexible TRACE pin assignment

| DBGMCU_CR register |                  | Pins assigned for:       | TRACE IO pin assigned   |              |                 |                           |                 |                 |
|--------------------|------------------|--------------------------|-------------------------|--------------|-----------------|---------------------------|-----------------|-----------------|
| TRACE_IOEN         | TRACE_MODE [1:0] |                          | PB3 /JTDO/ TRACESWO     | PE2/ TRACECK | PE3 / TRACED[0] | PE4 / TRACED[1]           | PE5 / TRACED[2] | PE6 / TRACED[3] |
| 0                  | XX               | No Trace (default state) | Released <sup>(1)</sup> | -            |                 |                           |                 |                 |
| 1                  | 00               | Asynchronous Trace       | TRACESWO                | -            | -               | Released (usable as GPIO) |                 |                 |
| 1                  | 01               | Synchronous Trace 1 bit  | Released <sup>(1)</sup> | TRACECK      | TRACED[0]       | -                         | -               | -               |
| 1                  | 10               | Synchronous Trace 2 bit  |                         | TRACECK      | TRACED[0]       | TRACED[1]                 | -               | -               |
| 1                  | 11               | Synchronous Trace 4 bit  |                         | TRACECK      | TRACED[0]       | TRACED[1]                 | TRACED[2]       | TRACED[3]       |

1. When Serial Wire mode is used, it is released. But when JTAG is used, it is assigned to JTDO.

**Note:** By default, the TRACECLKIN input clock of the TPIU is tied to GND. It is assigned to HCLK two clock cycles after the bit TRACE\_IOEN has been set.

The debugger must then program the Trace Mode by writing the PROTOCOL[1:0] bits in the SPP\_R (Selected Pin Protocol) register of the TPIU.

- PROTOCOL=00: Trace Port Mode (synchronous)
- PROTOCOL=01 or 10: Serial Wire (Manchester or NRZ) Mode (asynchronous mode). Default state is 01

It then also configures the TRACE port size by writing the bits [3:0] in the CPSPS\_R (Current Sync Port Size register) of the TPIU:

- 0x1 for 1 pin (default state)
- 0x2 for 2 pins
- 0x8 for 4 pins

### 38.17.3 TPUI formatter

The formatter protocol outputs data in 16-byte frames:

- seven bytes of data
- eight bytes of mixed-use bytes consisting of:
  - 1 bit (LSB) to indicate it is a DATA byte ('0') or an ID byte ('1').
  - 7 bits (MSB) which can be data or change of source ID trace.
- one byte of auxiliary bits where each bit corresponds to one of the eight mixed-use bytes:
  - if the corresponding byte was a data, this bit gives bit0 of the data.
  - if the corresponding byte was an ID change, this bit indicates when that ID change takes effect.

*Note:* Refer to the Arm® CoreSight Architecture Specification v1.0 (Arm® IHI 0029B) for further information

### 38.17.4 TPUI frame synchronization packets

The TPUI can generate two types of synchronization packets:

- The Frame Synchronization packet (or Full Word Synchronization packet)  
It consists of the word: 0x7F\_FF\_FF\_FF (LSB emitted first). This sequence can not occur at any other time provided that the ID source code 0x7F has not been used.  
It is output periodically **between** frames.  
In continuous mode, the TPA must discard all these frames once a synchronization frame has been found.
- The Half-Word Synchronization packet  
It consists of the half word: 0x7F\_FF (LSB emitted first).  
It is output periodically **between or within** frames.  
These packets are only generated in continuous mode and enable the TPA to detect that the TRACE port is in IDLE mode (no TRACE to be captured). When detected by the TPA, it must be discarded.

### 38.17.5 Transmission of the synchronization frame packet

There is no Synchronization Counter register implemented in the TPIU of the core. Consequently, the synchronization trigger can only be generated by the **DWT**. Refer to the registers DWT Control register (bits SYNCTAP[11:10]) and the DWT Current PC Sampler Cycle Count register.

The TPUI Frame synchronization packet (0x7F\_FF\_FF\_FF) is emitted:

- after each TPIU reset release. This reset is synchronously released with the rising edge of the TRACECLKIN clock. This means that this packet is transmitted when the TRACE\_IOEN bit in the DBGMCU\_CFG register is set. In this case, the word 0x7F\_FF\_FF\_FF is not followed by any formatted packet.
- at each DWT trigger (assuming DWT has been previously configured). Two cases occur:
  - If the bit SYNENA of the ITM is reset, only the word 0x7F\_FF\_FF\_FF is emitted without any formatted stream which follows.
  - If the bit SYNENA of the ITM is set, then the ITM synchronization packets follow (0x80\_00\_00\_00\_00\_00), formatted by the TPUI (trace source ID added).

### 38.17.6 Synchronous mode

The trace data output size can be configured to 4, 2 or 1 pin: TRACED(3:0)

The output clock is output to the debugger (TRACECK)

Here, TRACECLKIN is driven internally and is connected to HCLK only when TRACE is used.

*Note:* In this synchronous mode, it is not required to provide a stable clock frequency.

The TRACE I/Os (including TRACECK) are driven by the rising edge of TRACCLKIN (equal to HCLK). Consequently, the output frequency of TRACECK is equal to HCLK/2.

### 38.17.7 Asynchronous mode

This is a low cost alternative to output the trace using only 1 pin: this is the asynchronous output pin TRACESWO. Obviously there is a limited bandwidth.

TRACESWO is multiplexed with JTDO when using the SW-DP pin. This way, this functionality is available in all STM32F4xx packages.

This asynchronous mode requires a constant frequency for TRACECLKIN. For the standard UART (NRZ) capture mechanism, 5% accuracy is needed. The Manchester encoded version is tolerant up to 10%.

### 38.17.8 TRACECLKIN connection inside the STM32F4xx

In the STM32F4xx, this TRACECLKIN input is internally connected to HCLK. This means that when in asynchronous trace mode, the application is restricted to use to time frames where the CPU frequency is stable.

**Note:** ***Important:** when using asynchronous trace: it is important to be aware that:*

*The default clock of the STM32F4xx MCUs is the internal RC oscillator. Its frequency under reset is different from the one after reset release. This is because the RC calibration is the default one under system reset and is updated at each system reset release.*

*Consequently, the trace port analyzer (TPA) must not enable the trace (with the TRACE\_IOEN bit) under system reset, because a Synchronization Frame Packet is issued with a different bit time than trace packets which are transmitted after reset release.*

### 38.17.9 TPIU registers

The TPIU APB registers can be read and written only if the bit TRCENA of the Debug Exception and Monitor Control register (DEMCR) is set. Otherwise, the registers are read as zero (the output of this bit enables the PCLK of the TPIU).

**Table 314. Important TPIU registers**

| Address    | Register              | Description                                                                                                                                                                                                                             |
|------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xE0040004 | Current port size     | Allows the trace port size to be selected:<br>Bit 0: Port size = 1<br>Bit 1: Port size = 2<br>Bit 2: Port size = 3, not supported<br>Bit 3: Port Size = 4<br>Only 1 bit must be set. By default, the port size is one bit. (0x00000001) |
| 0xE00400F0 | Selected pin protocol | Allows the Trace Port Protocol to be selected:<br>Bit1:0=<br>00: Sync Trace Port Mode<br>01: Serial Wire Output - manchester (default value)<br>10: Serial Wire Output - NRZ<br>11: reserved                                            |

Table 314. Important TPIU registers (continued)

| Address    | Register                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0xE0040304 | Formatter and flush control | <p>Bits 31-9 = always '0<br/>           Bit 8 = TrgIn = always '1 to indicate that triggers are indicated<br/>           Bits 7-4 = always 0<br/>           Bits 3-2 = always 0<br/>           Bit 1 = EnFCont. In Sync Trace mode (Select_Pin_Protocol register bit1:0=00), this bit is forced to '1: the formatter is automatically enabled in continuous mode. In asynchronous mode (Select_Pin_Protocol register bit1:0 &lt;&gt; 00), this bit can be written to activate or not the formatter.<br/>           Bit 0 = always 0</p> <p>The resulting default value is 0x102</p> <p><b>Note:</b> In synchronous mode, because the TRACECTL pin is not mapped outside the chip, the formatter is always enabled in continuous mode -this way the formatter inserts some control packets to identify the source of the trace packets).</p> |
| 0xE0040300 | Formatter and flush status  | Not used in Cortex®-M4 with FPU, always read as 0x00000008                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 38.17.10 Example of configuration

- Set the bit TRCENA in the Debug Exception and Monitor Control register (DEMCR)
- Write the TPIU Current Port Size register to the desired value (default is 0x1 for a 1-bit port size)
- Write TPIU Formatter and Flush Control register to 0x102 (default value)
- Write the TPIU Select Pin Protocol to select the sync or async mode. Example: 0x2 for async NRZ mode (UART like)
- Write the DBGMCU control register to 0x20 (bit IO\_TRACEN) to assign TRACE I/Os for async mode. A TPIU Sync packet is emitted at this time (FF\_FF\_FF\_7F)
- Configure the ITM and write the ITM Stimulus register to output a value

## 38.18 DBG register map

The following table summarizes the Debug registers.

**Table 315. DBG register map and reset values**

| Addr.       | Register                   | 31       | 30 | 29 | 28 | 27            | 26 | 25            | 24 | 23       | 22 | 21                     | 20 | 19                     | 18 | 17                     | 16 | 15            | 14 | 13       | 12 | 11 | 10 | 9               | 8 | 7          | 6 | 5        | 4 | 3             | 2 | 1             | 0 |               |   |              |   |                |   |                |   |                |   |               |   |               |   |               |   |               |   |               |   |               |   |
|-------------|----------------------------|----------|----|----|----|---------------|----|---------------|----|----------|----|------------------------|----|------------------------|----|------------------------|----|---------------|----|----------|----|----|----|-----------------|---|------------|---|----------|---|---------------|---|---------------|---|---------------|---|--------------|---|----------------|---|----------------|---|----------------|---|---------------|---|---------------|---|---------------|---|---------------|---|---------------|---|---------------|---|
| 0xE004 2000 | DBGMCU_IDCODE              | REV_ID   |    |    |    |               |    |               |    |          |    |                        |    |                        |    |                        |    | Reserved      |    | DEV_ID   |    |    |    |                 |   |            |   |          |   |               |   |               |   |               |   |              |   |                |   |                |   |                |   |               |   |               |   |               |   |               |   |               |   |               |   |
|             | Reset value <sup>(1)</sup> | X        | X  | X  | X  | X             | X  | X             | X  | X        | X  | X                      | X  | X                      | X  | X                      | X  |               |    |          |    |    |    | X               | X | X          | X | X        | X | X             | X | X             | X | X             |   |              |   |                |   |                |   |                |   |               |   |               |   |               |   |               |   |               |   |               |   |
| 0xE004 2004 | DBGMCU_CR                  | Reserved |    |    |    |               |    |               |    |          |    |                        |    | DBG_TIM7_STOP          |    | Reserved               |    | Reserved      |    |          |    |    |    | TRACE_MODE[1:0] |   | TRACE_IOEN |   | Reserved |   |               |   | DBG_STANDBY   |   | DBG_STOP      |   | DBG_SLEEP    |   |                |   |                |   |                |   |               |   |               |   |               |   |               |   |               |   |               |   |
|             | Reset value                |          |    |    |    |               |    |               |    |          |    |                        |    | 0                      | 0  |                        |    |               |    |          |    |    |    | 0               | 0 | 0          | 0 |          |   |               |   | 0             | 0 | 0             | 0 | 0            | 0 | 0              | 0 | 0              | 0 | 0              | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 |
| 0xE004 2008 | DBGMCU_APB1_FZ             | Reserved |    |    |    | DBG_CAN2_STOP |    | DBG_CAN1_STOP |    | Reserved |    | DBG_I2C3_SMBUS_TIMEOUT |    | DBG_I2C2_SMBUS_TIMEOUT |    | DBG_I2C1_SMBUS_TIMEOUT |    | Reserved      |    |          |    |    |    |                 |   |            |   |          |   | DBG_IWDG_STOP |   | DBG_WWDG_STOP |   | Reserved      |   | DBG_RTC_STOP |   | DBG_TIM14_STOP |   | DBG_TIM13_STOP |   | DBG_TIM12_STOP |   | DBG_TIM7_STOP |   | DBG_TIM6_STOP |   | DBG_TIM5_STOP |   | DBG_TIM4_STOP |   | DBG_TIM3_STOP |   | DBG_TIM2_STOP |   |
|             | Reset value                |          |    |    |    | 0             | 0  | 0             | 0  | 0        | 0  | 0                      | 0  | 0                      | 0  | 0                      | 0  |               |    |          |    |    |    |                 |   |            |   |          |   | 0             | 0 | 0             | 0 | 0             | 0 | 0            | 0 | 0              | 0 | 0              | 0 | 0              | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 |
| 0xE004 200C | DBGMCU_APB2_FZ             | Reserved |    |    |    |               |    |               |    |          |    |                        |    | DBG_TIM11_STOP         |    | DBG_TIM10_STOP         |    | DBG_TIM9_STOP |    | Reserved |    |    |    |                 |   |            |   |          |   |               |   | DBG_TIM8_STOP |   | DBG_TIM1_STOP |   |              |   |                |   |                |   |                |   |               |   |               |   |               |   |               |   |               |   |               |   |
|             | Reset value                |          |    |    |    |               |    |               |    |          |    |                        |    | 0                      | 0  | 0                      | 0  | 0             | 0  |          |    |    |    |                 |   |            |   |          |   |               |   | 0             | 0 | 0             | 0 | 0            | 0 | 0              | 0 | 0              | 0 | 0              | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 | 0             | 0 |

1. The reset value is product dependent. For more information, refer to [Section 38.6.1: MCU device ID code](#).

## 39 Device electronic signature

The electronic signature is stored in the Flash memory area. It can be read using the JTAG/SWD or the CPU. It contains factory-programmed identification data that allow the user firmware or other external devices to automatically match its interface to the characteristics of the STM32F4xx microcontrollers.

### 39.1 Unique device ID register (96 bits)

The unique device identifier is ideally suited:

- for use as serial numbers (for example USB string serial numbers or other end applications)
- for use as security keys in order to increase the security of code in Flash memory while using and combining this unique ID with software cryptographic primitives and protocols before programming the internal Flash memory
- to activate secure boot processes, etc.

The 96-bit unique device identifier provides a reference number which is unique for any device and in any context. These bits can never be altered by the user.

The 96-bit unique device identifier can also be read in single bytes/half-words/words in different ways and then be concatenated using a custom algorithm.

**Base address: 0x1FFF 7A10**

Address offset: 0x00

Read only = 0xFFFF XXXX where X is factory-programmed

|           |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|-----------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 31        | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| UID(31:0) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r         | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 31:0 **UID(31:0)**: X and Y coordinates on the wafer

Address offset: 0x04

Read only = 0xFFFF XXXX where X is factory-programmed

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| UID(63:48) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| UID(47:32) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:8 **UID(63:40):** LOT\_NUM[23:0]  
Lot number (ASCII encoded).

Bits 7:0 **UID(39:32):** WAF\_NUM[7:0]  
Wafer number (ASCII encoded).

Address offset: 0x08

Read only = 0xFFFF XXXX where X is factory-programmed

|            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 31         | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| UID(95:80) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |
| 15         | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 0  |
| UID(79:64) |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| r          | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  | r  |

Bits 31:0 **UID(95:64):** LOT\_NUM[55:24]  
Lot number (ASCII encoded).

## 39.2 Flash size

Base address: 0x1FFF 7A22

Address offset: 0x00

Read only = 0xFFFF where X is factory-programmed

|        |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 15     | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| F_SIZE |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| r      | r  | r  | r  | r  | r  | r | r | r | r | r | r | r | r | r | r |

Bits 15:0 **F\_ID(15:0):** Flash memory size  
This bitfield indicates the size of the device Flash memory expressed in Kbytes.  
As an example, 0x0400 corresponds to 1024 Kbytes.

## 40 Important security notice

The STMicroelectronics group of companies (ST) places a high value on product security, which is why the ST product(s) identified in this documentation may be certified by various security certification bodies and/or may implement our own security measures as set forth herein. However, no level of security certification and/or built-in security measures can guarantee that ST products are resistant to all forms of attacks. As such, it is the responsibility of each of ST's customers to determine if the level of security provided in an ST product meets the customer needs both in relation to the ST product alone, as well as when combined with other components and/or software for the customer end product or application. In particular, take note that:

- ST products may have been certified by one or more security certification bodies, such as Platform Security Architecture ([www.psacertified.org](http://www.psacertified.org)) and/or Security Evaluation standard for IoT Platforms ([www.trustcb.com](http://www.trustcb.com)). For details concerning whether the ST product(s) referenced herein have received security certification along with the level and current status of such certification, either visit the relevant certification standards website or go to the relevant product page on [www.st.com](http://www.st.com) for the most up to date information. As the status and/or level of security certification for an ST product can change from time to time, customers should re-check security certification status/level as needed. If an ST product is not shown to be certified under a particular security standard, customers should not assume it is certified.
- Certification bodies have the right to evaluate, grant and revoke security certification in relation to ST products. These certification bodies are therefore independently responsible for granting or revoking security certification for an ST product, and ST does not take any responsibility for mistakes, evaluations, assessments, testing, or other activity carried out by the certification body with respect to any ST product.
- Industry-based cryptographic algorithms (such as AES, DES, or MD5) and other open standard technologies which may be used in conjunction with an ST product are based on standards which were not developed by ST. ST does not take responsibility for any flaws in such cryptographic algorithms or open technologies or for any methods which have been or may be developed to bypass, decrypt or crack such algorithms or technologies.
- While robust security testing may be done, no level of certification can absolutely guarantee protections against all attacks, including, for example, against advanced attacks which have not been tested for, against new or unidentified forms of attack, or against any form of attack when using an ST product outside of its specification or intended use, or in conjunction with other components or software which are used by customer to create their end product or application. ST is not responsible for resistance against such attacks. As such, regardless of the incorporated security features and/or any information or support that may be provided by ST, each customer is solely responsible for determining if the level of attacks tested for meets their needs, both in relation to the ST product alone and when incorporated into a customer end product or application.
- All security features of ST products (inclusive of any hardware, software, documentation, and the like), including but not limited to any enhanced security features added by ST, are provided on an "AS IS" BASIS. AS SUCH, TO THE EXTENT PERMITTED BY APPLICABLE LAW, ST DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, unless the applicable written and signed contract terms specifically provide otherwise.

## 41 Revision history

**Table 316. Document revision history**

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-Sep-2011 | 1       | Initial release.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 19-Oct-2012 | 2       | <p>Updated reference documents and added Table 1: Applicable products on cover page.</p> <p><b>MEMORY:</b><br/>Updated Section 2: Memory and bus architecture.</p> <p><b>PWR:</b><br/>Updated VDDA and VREF+ decoupling capacitor in Figure 7: Power supply overview.<br/>VOSRDY bit changed to read-only in Section 5.4.3: PWR power control/status register (PWR_CSR).<br/>Removed VDDA in Section 5.2.3: Programmable voltage detector (PVD) and remove VDDA in PVDO bit description (Section 5.4.3: PWR power control/status register (PWR_CSR)).</p> <p><b>RCC:</b><br/>Updated Figure 20: Simplified diagram of the reset circuit and minimum reset pulse duration guaranteed by pulse generator restricted to internal reset sources.</p> <p><b>GPIOs:</b><br/>Updated Section 8.3.1: General-purpose I/O (GPIO).</p> <p><b>DMA:</b><br/>Updated direct mode description in Section 10.2: DMA main features.<br/>Updated direct mode description in Section : Memory-to-peripheral mode, and Section 10.3.12: FIFO/: Direct mode.<br/>Updated register access in Section 10.5: DMA registers.<br/>Modified Stream2 /Channel 2 in Table 42: DMA1 request mapping.<br/>Added note related to EN bit in Section 10.5.5: DMA stream x configuration register (DMA_SxCR) (x = 0..7). Updated definition of NDT[15:0] bits in Section 10.5.6: DMA stream x number of data register (DMA_SxNDTR) (x = 0..7).</p> <p><b>Interrupts:</b><br/>Updated number of maskable interrupts to 82 in Section 12.1.1: NVIC features.<br/>Updated Section 12.2: External interrupt/event controller (EXTI).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Oct-2012 | 2<br>(continued) | <p><b>ADC:</b><br/>           Changed ADCCLK frequency to 30 MHz in Section 13.5: Channel-wise programmable sampling timee.<br/>           Added recovery from ADC sequence in Section 13.8.1: Using the DMA and Section 13.8.2: Managing a sequence of conversions without using the DMA.<br/>           Updated AWDIE in Section 13.13.2: ADC control register 1 (ADC_CR1). Added read and write access in Section 13.13: ADC registers.</p> <p><b>Advanced control timers (TIM1 and TIM8):</b><br/>           Updated 16-bit prescaler range in Section 17.2: TIM1 and TIM8 main features.<br/>           Updated OC1 block diagram in Figure 114: Output stage of capture/compare channel (channel 1 to 3).<br/>           Updated update event generation in Upcounting mode and Downcounting mode in Section 17.3.2: Counter modes and Section 17.3.3: Repetition counter.<br/>           Updated bits that control the dead-time generation in Section 17.3.11: Complementary outputs and dead-time insertion.<br/>           Updated ways to generate a break in Section 17.3.12: Using the break function.<br/>           Changed OCxREF to ETR in the example given in Section 17.3.13: Clearing the OCxREF signal on an external event and changed OCREF_CLR to ETRF in Figure 124: Clearing TIMx OCxREF.<br/>           Updated configuration for example of counter operation in encoder interface mode in Section 17.3.16: Encoder interface mode.<br/>           Added register access in Section 17.4: TIM1 and TIM8 registers.<br/>           Changed definition of ARR[15:0] bits in Section 17.4.12: TIM1 and TIM8 auto-reload register (TIMx_ARR).<br/>           Updated BKE definition in Section 17.4.18: TIM1 and TIM8 break and dead-time register (TIMx_BDTR).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Oct-2012 | 2<br>(continued) | <p><b>General purpose timers (TIM2 to TIM5):</b><br/> Removed all references to “repetition counter”.<br/> Added Figure 134: General-purpose timer block diagram.<br/> Updated 16-bit prescaler range in Section 18.2: TIM2 to TIM5 main features.<br/> External clock mode 2 ETR restricted to TIM2 to TIM4 in<br/> Section 18.3.3: Clock selection and Section 18.3.6: PWM input mode.<br/> Updated Section 18.3.9: PWM mode and Section 18.3.11: Clearing the OCxREF<br/> signal on an external event.<br/> Updated Figure 174: Master/Slave timer example to change ITR1 to<br/> ITR0.<br/> Updated read and write access to registers in Section 18.4: TIM2 to TIM5<br/> registers.<br/> Restored bits 15 to 8 of TIMx_SMCR as well as Table 98: TIMx internal trigger<br/> connection in Section 14.4.3.<br/> Removed note 1 related to OC1M bits in Section 18.4.13: TIMx capture/compare<br/> register 1 (TIMx_CCR1).<br/> Updated TIMx_CCER bit description for TIM2 to TIM5 in<br/> Section 18.4.9: TIMx capture/compare enable register (TIMx_CCER).</p> <p><b>General purpose timers (TIM9 to TIM14):</b><br/> Updated 16-bit prescaler range in Section 19.2.1: TIM9/TIM12 main features and<br/> Section 19.2.2: TIM10/TIM11 and TIM13/TIM14 main features.<br/> Updated Figure 181: General-purpose timer block diagram (TIM10/11/13/14)) to<br/> remove TRGO trigger controller output.<br/> Added register access in Section 19.4: TIM9 and TIM12 registers<br/> and Section 19.5: TIM10/11/13/14 registers.</p> <p><b>Basic timers (TIM6 and TIM7):</b><br/> Removed all references to “repetition counter”.<br/> Updated 16-bit prescaler range in Section 20.2: TIM6 and TIM7 main features.</p> <p><b>HASH:</b><br/> Updated Section 25.3.1: Duration of the processing.</p> <p><b>RNG:</b><br/> Updated Section 24.1: RNG introduction.</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Oct-2012 | 2<br>(continued) | <p><b>RTC:</b><br/> Updated Figure 237: RTC block diagram.<br/> Added formula to compute fck_apre in Figure 26.3.1: Clock and prescalers.<br/> Updated Section 26.3.9: RTC reference clock detection.<br/> Updated Section : RTC register write protection.<br/> Added RTC_SSR shadow register in Section 26.3.6: Reading the calendar.<br/> Updated description of DC[4:0] bits in Section 26.6.7: RTC calibration register (RTC_CALIBR).<br/> Renamed RTC_BKxR into RTC_BKPxR in Table 121: RTC register map and reset values.<br/> Added power-on reset value and changed reset value to system reset value in Section 26.6.11: RTC sub second register (RTC_SSR).<br/> Updated definition of ALARMOUTTYPE in Section 26.6.17: RTC tamper and alternate function configuration register (RTC_TAFCR).</p> <p><b>I2C:</b><br/> Modified Section 27.3.8: DMA requests.<br/> Updated bit 14 description in Section 27.6.3: I2C Own address register 1 (I2C_OAR1)).<br/> Updated definition of PE bit and note related to SWRST bit; moved note related to STOP bit to the whole register in Section 27.6.1: I2C Control register 1 (I2C_CR1).</p> <p><b>USART:</b><br/> Section 30.6.6: Control register 3 (USART_CR3): removed notes related to UART5 in DMAT and DMAR description.<br/> Updated TTable 142: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 Hz, oversampling by 16 and<br/> Table 143: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 MHz, oversampling by 8.</p> <p><b>SPI/I2S:</b><br/> Updated Section 28.1: SPI introduction.<br/> Changed I2S simplex communication/mode to half-duplex communication/mode.<br/> Updated flags in reception/transmission modes in Section 28.2.2: I2S features.<br/> Added Frame error flag in Table 128: I2S interrupt requests.<br/> Added register access in Section 28.5: SPI and I2S registers.<br/> Updated ERRIE definition in Section 28.5.2: SPI control register 2 (SPI_CR2).<br/> Renamed TIFRFE to FRE and definition updated in Section 28.5.3: SPI status register (SPI_SR).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Oct-2012 | 2<br>(continued) | <p><b>SDIO:</b><br/>Updated value and description for bits [45:40] and [7:1] in Table 176: R4 response.<br/>Updated value at bits [45:40] in Table 178: R5 response.</p> <p><b>CAN:</b><br/>Updated Figure 335: Dual CAN block diagram.<br/>Modified definition of CAN2SB bits in Section : CAN filter master register (CAN_FMR).<br/>Added register access in Section 32.9: CAN registers</p> <p><b>ETHERNET:</b><br/>Updated standard for precision networked clock synchronization in Section 33.1: Ethernet introduction and Section 33.2.1: MAC core features.<br/>Updated CR bit definition in Section : Ethernet MAC MII address register (ETH_MACMIIAR).<br/>Replace RTPR by PM bit in Table 192: Source address filtering.</p> <p><b>USB OTG FS</b><br/>Updated remote wake-up signaling bit and the resume interrupt in Section : Suspended state.<br/>Added peripheral register access in Section 34.16: OTG_FS control and status registers.<br/>Updated INEPTXSA description in OTG_FS_DIEPTXFx.<br/>Changed PHYSEL from bit 7 to bit 6 of the OTG_FS_GUSBCFG register.</p> <p><b>USB OTG HS</b><br/>Updated remote wake-up signaling bit and the resume interrupt in Section : Suspended state.<br/>Added peripheral register access in Section 35.12: OTG_HS control and status registers.<br/>Updated OTG_HS_CID reset value.<br/>Updated INEPTXSA description in OTG_HS_DIEPTXFx.<br/>Updated FSLSPCS for LS host mode, added PHYSEL in Section : OTG_HS host configuration register (OTG_HS_HCFG).<br/>Renamed PHYSEL into PHSEL and changed from bit 7 to bit 6 of the OTG_HS_GUSBCFG register.<br/>Updated OTG_HS_DIEPEACHMSK1 and OTG_HS_DOEPEACHMSK1 reset values.</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Oct-2012 | 2<br>(continued) | <p><b>FSMC:</b></p> <p>Updated step b) in Section 36.3.1: Supported memories and transactions.</p> <p>Updated Table 196: FSMC_BTRx bit fields.</p> <p>Changed Clock divide ration min in Table 246: Programmable NAND/PC Card access parameters.</p> <p>Updated case of synchronous accesses in Section 36.5: NOR Flash/PSRAM controller.</p> <p>Changed minimum value for ADDSET to 0 in Table 203, Table 206, Table 207, Table 209, and Table 210.</p> <p>Move note from Figure 437: Mode1 write accesses and Figure 436: Mode1 read accesses. Move note from Figure 439: ModeA write accesses to Figure 438: ModeA read accesses.</p> <p>Updated Section : WAIT management in asynchronous accesses.</p> <p>Added register access in Section 36.5.6: NOR/PSRAM control registers and Section 36.6.2: NAND Flash / PC Card supported memories and transactions.</p> <p>Removed caution note in Section 36.6.1: External memory interface signalss.</p> <p>Updated Table 249: 16-bit PC Card.</p> <p>Updated step 3 in Section 36.6.4: NAND Flash operations.</p> <p>Updated Figure 455: Access to non 'CE don't care' NAND-Flash and note below in Section 36.6.5: NAND Flash prewait functionality.</p> <p>Updated access to I/O Space in Section 36.6.7: PC Card/CompactFlash operationss. Updated Table 251: 16-bit PC-Card signals and access type. Updated BUSTURN bit definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4)). Changed bits 16 to 19 to BUSTURN in Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4)</p> <p><b>DEBUG:</b></p> <p>Updated Section 38.4.3: Internal pull-up and pull-down on JTAG pins.</p> <p><b>Electronic signature</b></p> <p>Updated Section 24: Device electronic signature introduction.</p> <p>Updated REV_ID[15:0] to add revision Z in Section 24.1: Unique device ID register (96 bits).</p> <p>Updated address and example in Section 24.2: Flash size.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13-Nov-2012 | 3       | <p>Added STM32F42x and STM32F43x devices.</p> <p>Removed reference du Flash programming manual on cover page. Added Section 2.3.2: Flash memory overview and Section 3: Embedded Flash memory interface.</p> <p>Change RTC_50Hz into RTC_REFIN in Section 8.3.2: I/O pin multiplexer and mapping. Modified RTC alternate function naming in Section 8: General-purpose I/Os (GPIO) and Section 26: Real-time clock (RTC).</p> <p>Updated max. input frequency in Section 26.3.1: Clock and prescalers.</p> <p>Changed bit access type from 'rw' to 'w' and bit description updated in Section 10.5.3: DMA low interrupt flag clear register (DMA_LIFCR) and Section 10.5.4: DMA high interrupt flag clear register (DMA_HIFCR).</p> <p>Updated Figure 18: Frequency measurement with TIM5 in Input capture mode.</p> <p>Updated Section : Signals synchronization in Section 36: Flexible static memory controller (FSMC)</p> <p>Section 34: USB on-the-go full-speed (OTG_FS): updated Section Figure 389.: USB host-only connection, Section : VBUS valid, and Section : Host detection of a peripheral connection.</p> <p>Section 35: USB on-the-go high-speed (OTG_HS): updated Section : VBUS valid, and Section : Detection of peripheral connection by the host.</p>                                                                                                                                                    |
| 19-Feb-2013 | 4       | <p>Updated Section 2: Memory and bus architecture.</p> <p>Updated Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices, and Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices. Updated Table 4: Memory mapping vs. Boot mode/physical remap. Updated Figure 5: Sequential 32-bit instruction execution. removed note 1 from Table 13: Maximum program/erase parallelism.</p> <p><b>PWR:</b></p> <p>Updated Figure 7: Power supply overview.</p> <p>Updated Section 5.1.3: Voltage regulator.</p> <p>Added ADCDC1 bit in Section 5.5.1: PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx.</p> <p><b>SYSCFG:</b></p> <p>Added ADCxDC2 bit in Section 8.2.3: SYSCFG peripheral mode configuration register (SYSCFG_PMC) for STM32F42xxx and STM32F43xxx.</p> <p><b>ADC:</b></p> <p>Updated Section 13.9.3: Interleaved mode, Section 13.9.4: Alternate trigger mode, and Section 13.9.5: Combined regular/injected simultaneous mode to describe case of interrupted conversion.</p> <p>Updated Section : Temperature sensor, VREFINT and VBAT internal channels, Section 13.10: Temperature sensor, and Section 13.11: Battery charge monitoring.</p> <p><b>RTC:</b></p> <p>Updated BKP[31:0] bit description in Section 26.6.20: RTC backup registers (RTC_BKPxR).</p> <p><b>I2C:</b></p> <p>Updated Section 27.3.5: Programmable noise filter.</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19-Feb-2013 | 4<br>(continued) | <p><b>FSMC:</b></p> <p>Updated write FIFO size in Section 36.1: FSMC main features.</p> <p>Updated Figure 434: FSMC block diagram.</p> <p>Updated Section 36.5.4: NOR Flash/PSRAM controller asynchronous transactions.</p> <p>Modified differences between Mode B and mode 1 in Section : Mode 2/B - NOR Flash.</p> <p>Modified differences between Mode C and mode 1 in Section : Mode C - NOR Flash - OE toggling.</p> <p>Modified differences between Mode D and mode 1 in Section : Mode D - asynchronous access with extended address.</p> <p>Updated NWAIT signal in Figure 449: Asynchronous wait during a read access, Figure 450: Asynchronous wait during a write access, Figure 451: Wait configurations, Figure 452: Synchronous multiplexed read mode - NOR, PSRAM (CRAM), and Figure 453: Synchronous multiplexed write mode - PSRAM (CRAM).</p> <p>Updated Table 195 to Table 214.</p> <p>Updated Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4).</p> <p><b>DEBUG</b></p> <p>Updated Figure 485: Block diagram of STM32 MCU and Cortex®-M4 with FPU-level debug support.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-Sep-2013 | 5       | <p>Added STM32F429xx and STM32F439xx part numbers.</p> <p>Replaced FSMC by FMC added Chrom-ART Accelerator, LCD-TFT and SAI interface.</p> <p>Updated Figure 2: System architecture for STM32F42xxx and STM32F43xxx devices.</p> <p><b>PWR:</b></p> <p>Updated Section 5.2.2: Brownout reset (BOR).</p> <p>Added note related to CSS enabling in Entering Stop mode sections in Section 5.3.4: Stop mode (STM32F405xx/07xx and STM32F415xx/17xx) and Section 5.3.5: Stop mode (STM32F42xxx and STM32F43xxx). Updated Stop mode entry in Table 27 and Table 29.</p> <p>Updated WUF bit definition in PWR_CSR registers. Changed CWUF and CSBF access type to 'w' in PWR_CR register.</p> <p><b>RCC:</b> Updated LSEBYP bit definition in RCC_BDCR register.</p> <p><b>GPIOs:</b></p> <p>Updated description of OSPEEDR bits. Removed frequency value in description of OSPEEDR bits. Corrected typos: "IDRy[15:0]" replaced with "IDRy" in "GPIOx_IDR" register, "ODRy[15:0]" replaced with "ODRy" in "GPIOx_ODR" register and "OTy[1:0]" replaced with "OTy" in "GPIOx_OTYPER" register.</p> <p><b>DCMI:</b> Updated Section 15.4: DCMI clocks.</p> <p><b>IWDG:</b> Corrected Figure 213: Independent watchdog block diagram.</p> <p><b>RTC:</b></p> <p>Replaced all occurrences of "power-on reset" with "backup domain reset". Added caution note under Table 121: RTC register map and reset values. Changed SHPF bit type to 'r' in Section 26.6.4: RTC initialization and status register (RTC_ISR)..</p> <p><b>SPI:</b> Updated definition of ERRIE bit in Section 28.5.2: SPI control register 2 (SPI_CR2).</p> <p><b>UART:</b></p> <p>Updated Section 30.3.8: LIN (local interconnection network) mode.</p> <p>Removed note in Section 30.3.13: Continuous communication using DMA.</p> <p><b>ETHERNET:</b></p> <p>Modified ETH_MACA0HR (and ETH_DMABMR reset values.</p> <p>Updated definitions of TSTS bit in ETH_MACSR, and TSTTR in ETH_PTPTSSR.</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-Sep-2013 | 5<br>(continued) | <p><b>USB OTG-FS:</b><br/>Removed note related to VDD range limitation below Figure 387: OTG A-B device connection and Figure 388: USB peripheral-only connection.</p> <p><b>FSMC:</b><br/>Updated Table 229, Table 232, Table 235, Table 239.<br/>Replaced all occurrences of DATALAT by DATLAT and SRAM/CRAM by SRAM/PSRAM in the whole section.<br/>Updated Section 36.1: FSMC main features. Changed bits 27 to 20 of FSMC_BWTR1..4 to reserved.<br/>Updated Section 36.6.7: PC Card/CompactFlash operations.<br/>Updated WREN bit in Table 231, Table 232, Table 233, Table 236, Table 239, Table 242, Table 245, and Table 249.<br/>Updated Section 36.5.4: NOR Flash/PSRAM controller asynchronous transactions, Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4), Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4).<br/>Updated definition of PWID in Section : PC Card/NAND Flash control registers 2..4 (FSMC_PCR2..4).</p> <p><b>FMC:</b><br/>Updated TRDC definition in Section : SDRAM Timing registers 1,2 (FMC_SDTR1,2).</p> <p><b>DEBUG:</b> updated Figure 487: JTAG TAP connections.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03-Feb-2014 | 6       | <p>Added note related to over-drive mode unavailable in 1.8 to 2.1 V VDD range in Section 3.5.1: Relation between CPU clock frequency and Flash memory read time.</p> <p>Updated maximum CPU frequency in Section 3.5.2: Adaptive real-time memory accelerator (ART Accelerator™).</p> <p><b>PWR:</b><br/>Updated Run mode/ over-drive mode in Section 5.1.4: Voltage regulator for STM32F42xxx and STM32F43xxx.</p> <p><b>RCC for STM32F42/43xx:</b><br/>Changed APB1/2 and AHB maximum frequencies.xw</p> <p><b>GPIOs:</b><br/>Updated Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx.</p> <p><b>DMA:</b><br/>Updated Section 10.3.7: Pointer incrementation and Section 10.3.11: Single and burst transfers..</p> <p><b>INTERRUPTS AND EVENTS:</b><br/>Updated Table 62: Vector table for STM32F42xxx and STM32F43xxx.</p> <p><b>ADC:</b><br/>Updated Section 13.3.10: Discontinuous mode/Section : Regular group.</p> <p><b>DCMI:</b><br/>Updated Section 15.5.2: DCMI physical interface.</p> <p><b>LTDC:</b><br/>Updated resolution in note below Figure 82: LCD-TFT Synchronous timings.</p> <p><b>TIM1 and 8:</b><br/>Added note related to IC1F in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1).</p> <p><b>TIM2 to 5:</b><br/>Updated note related to IC1F in Section 18.4.7: TIMx capture/compare mode register 1 (TIMx_CCMR1).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03-Feb-2014 | 6<br>(continued) | <p><b>TIM9 to 14:</b><br/>Updated note related to IC1F in Section 19.5.5: TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1).</p> <p><b>RTC:</b><br/>Updated Section 26.3.11: RTC smooth digital calibration.<br/>Changed ALRBIE to ALRBE (bit 9) in Section 26.6.3: RTC control register (RTC_CR).</p> <p><b>I2C:</b><br/>Introduced Sm (standard mode) and Fm (fast mode) acronyms.</p> <p><b>FSMC:</b><br/>Updated BUSTURN definition in Table 245: FSMC_BTRx bit fields.</p> <p><b>FMC:</b><br/>Added Mobile LPDDR SDRAM.<br/>Updated Section : SDRAM initialization and Section : SDRAM controller read cycle and Figure 476: NAND Flash/PC Card controller waveforms for common memory access.<br/>Updated Section : SRAM/NOR-Flash chip-select control registers 1..4 (FMC_BCR1..4), Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4), Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4), Section : SDRAM Timing registers 1,2 (FMC_SDTR1,2) and Section : SDRAM Refresh Timer register (FMC_SDRTR).<br/>Removed mention “default valeur after reset” in Section : Common memory space timing register 2..4 (FMC_PMEM2..4), Section : Attribute memory space timing registers 2..4 (FMC_PATT2..4), and Section : I/O space timing register 4 (FMC_PIO4).<br/>Updated BUSTURN definition in Table 288: FMC_BTRx bit fields.<br/>Updated REV_ID bits in Section 38.6.1: MCU device ID code.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-May-2014 | 7       | <p><b>Embedded Flash memory interface:</b><br/>Updated Section : Physical remap in STM32F42xxx and STM32F43xxx. Updated bank 2 selection in Section 2.4: Boot configuration. Updated notes related to MERx and SER bits in Section : Mass Erase. Updated Section 3.7.5: Proprietary code readout protection (PCROP). Updated FLASH_OPTCR register reset value for STM32F42/43xx in Section 3.9.10: Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx and Section 3.9.11: Flash option control register (FLASH_OPTCR1) for STM32F42xxx and STM32F43xxx.</p> <p><b>RCC (STM32F42/43xx):</b><br/>Updated PPLN caution note in Section 6.3.2: RCC PLL configuration register (RCC_PLLCFGR)</p> <p><b>SYSCFG</b><br/>Updated MEM_MODE in Section 9.3.1: SYSCFG memory remap register (SYSCFG_MEMRMP)</p> <p><b>LTDC:</b><br/>Changed resolution do XGA (1024x768) in Section 16.2: LTDC main features, Section 16.4.1: LTDC Global configuration parameters, and updated Section 16.7.3: LTDC Active Width Configuration Register (LTDC_AWCR).</p> <p><b>RTC</b><br/>Added note in Section 26.3.14: Calibration clock output.</p> <p><b>TIMER 1/8:</b><br/>Removed note related to IC1F bits in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1),</p> <p><b>TIM2 to 5:</b><br/>Replaced IC2S by CC2S.<br/>Updated Figure 161: Output stage of capture/compare channel (channel 1).<br/>Removed note related to IC1F bits in Section 18.4.7: TIMx capture/compare mode register 1 (TIMx_CCMR1).</p> <p><b>TIM9 to 14:</b><br/>Removed note related to IC1F bits in Section 19.5.5: TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1).</p> <p><b>USB OTG-HS:</b><br/>Updated DSPD definition in Section : OTG_HS device configuration register (OTG_HS_DCFG).</p> <p><b>FSMC</b><br/>Updated DATLAT bits definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15-May-2014 | 7<br>(continued) | <b>FMC</b><br>Updated Figure 474: Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM). Updated DATLAT bits definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4).<br>Updated FMC_BWTRx register address offsets in Table 297: FMC register map.<br><br><b>DEBUG</b><br>Added revision code '3' in Section : DBGMCU_IDCODE. |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14-Oct-2014 | 8       | <p><b>Memory and bus architecture:</b><br/>Updated Table 3: Memory mapping vs. Boot mode/physical remap in STM32F405xx/07xx and STM32F415xx/17xx and Table 4: Memory mapping vs. Boot mode/physical remap in STM32F42xxx and STM32F43xxx.</p> <p><b>RCC (STM32F40/41xx) and RCC (STM32F42/43xx):</b><br/>Removed all references to Flash programming manual. Changed RCC_AHB1LPENR, RCC_APB1LPENR, RCC_APB2LPENR, RCC_PLLI2SCFGR and RCC_APB2LPENR reset values.<br/>Updated access type to “r” for bits 24 to 31 in RCC_CSR.</p> <p><b>GPIOs:</b><br/>Updated Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx.</p> <p><b>IWDG</b><br/>Update note in Table 107: Min/max IWDG timeout period (in ms) at 32 kHz (LSI).</p> <p><b>CRYPTO and HASH</b><br/>Removed STM32F405/407xx and STM32F42xx from the whole sections.</p> <p>Removed STM32F405/407xx and STM32F42xx from the whole section.</p> <p><b>TIM10/11/13/14</b><br/>Added TIMx_DIER description in Section 19.5: TIM10/11/13/14 registers.</p> <p><b>ETHERNET:</b><br/>Updated Table 187: Clock range.</p> <p><b>USB OTG FS:</b><br/>Removed TRDT formula in Section 34.17.7: Worst case response time and added Table 203: TRDT values.</p> <p><b>USB OTG HS:</b><br/>Removed TRDT formula in Section 35.13.8: Worst case response time and added Table 213: TRDT values.</p> <p><b>FSMC:</b><br/>Updated EXTMOD definition in Section : SRAM/NOR-Flash chip-select control registers 1..4 (FSMC_BCR1..4).<br/>Updated ADDSET definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4).</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14-Oct-2014 | 8<br>(continued) | <b>FMC:</b><br>Modified step 7 in Section : SDRAM initialization.<br>Modified SDRAM refresh rate equations and example in Section : SDRAM Refresh Timer register (FMC_SDRTR) and updated definition of COUNT bits.<br>Updated EXTMOD definition in Section : SRAM/NOR-Flash chip-select control registers 1..4 (FMC_BCR1..4).<br>Updated ADDSET definition in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4). |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16-Mar-2015 | 9       | <p><b>PWR:</b><br/> Updated Section 5.1.2: Battery backup domain.<br/> Updated Table 23: Low-power mode summary to add Return from ISR as entry condition.<br/> Added Section : Entering low-power mode and Section : Exiting low-power mode.<br/> Updated Section : Entering Sleep mode, Section : Exiting Sleep mode, Table 24: Sleep-now entry and exit and Table 25: Sleep-on-exit entry and exit.<br/> Updated Section : Entering Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx), Section : Exiting Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx) and Table 27: Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx). Updated Section : Entering Stop mode (STM32F42xxx and STM32F43xxx), Section : Exiting Stop mode (STM32F42xxx and STM32F43xxx) and Table 29: Stop mode entry and exit (STM32F42xxx and STM32F43xxx).<br/> Updated Section : Entering Standby mode, Section : Exiting Standby mode and Table 30: Standby mode entry and exit.</p> <p><b>RCC:</b><br/> Updated bits 24 to 31 access type in Section 7.3.21: RCC clock control &amp; status register (RCC_CSR).</p> <p><b>GPIOs:</b><br/> Added port A reset value in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K).</p> <p><b>DMA:</b><br/> Update FTH[1:0] description in Section 10.5.10: DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7).</p> <p><b>TIM2/5:</b><br/> Register format changed to 32 bits instead of 16 in Section 18.4.10: TIMx counter (TIMx_CNT) and Section 18.4.12: TIMx auto-reload register (TIMx_ARR).</p> <p><b>TIM9 to 14:</b><br/> Updated Table 101: TIMx internal trigger connection</p> <p><b>WWDG:</b><br/> Updated Figure 214: Watchdog block diagram and Section 22.4: How to program the watchdog timeout.<br/> Updated Figure 215: Window watchdog timing diagram</p> <p><b>RNG:</b><br/> Replaced PLL48CLK by RNG_CLK in the whole section.</p> |

Table 316. Document revision history (continued)

| Date        | Version          | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16-Mar-2015 | 9<br>(continued) | <p><b>I2C2:</b><br/>Updated <code>FREQ[5:0]</code> description in Section 27.6.2: I2C Control register 2 (<code>I2C_CR2</code>).</p> <p><b>USART:</b><br/>Removed note related to <code>RXNEIE</code> in Section : Reception using DMA</p> <p><b>FSMC:</b><br/>Updated Figure 474: Synchronous multiplexed read mode waveforms - NOR, PSRAM (CRAM).</p> <p><b>USB OTG FS</b><br/>Updated Table 203: TRDT values</p> <p><b>FMC</b><br/>Updated <code>FMC_NL</code> in Figure 456: FMC block diagram.<br/>Updated 'Memory wait' and 'Memory data bus high-z' parameters in Table 289: Programmable NAND Flash/PC Card access parameters.<br/>Updated Section : Common memory space timing register 2..4 (<code>FMC_PMEM2..4</code>).<br/>Updated Figure 476: NAND Flash/PC Card controller waveforms for common memory access.</p> <p><b>DEBUG:</b><br/>Updated <code>REV_ID[15:0]</code> and JTAG ID code in Section 38.6.1: MCU device ID code and Section 38.6.2: Boundary scan TAP, respectively</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28-Jul-2015 | 10      | <p><b>Embedded Flash memory interface</b><br/>Updated Section 3.7.5: Proprietary code readout protection (PCROP),</p> <p><b>Power controller (PWR)</b><br/>Added the last sentence in Subsection: Entering low-power mode of Section 5.3: Low-power modes,<br/>Added the bullet points about the interrupt in mode entry in Table 24: Sleep-now entry and exit, Table 25: Sleep-on-exit entry and exit, Table 27: Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx), Table 29: Stop mode entry and exit (STM32F42xxx and STM32F43xxx)<br/>Added the last point to Mode entry, on return from ISR in Table 30: Standby mode entry and exit,<br/>Added the note in Section: Entering sleep mode in Section 5.3.3: Sleep mode.</p> <p><b>General-purpose I/Os (GPIO)</b><br/>Updated OSPEED[1:0] definition of GPIOx_OSPEEDR register in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K)</p> <p><b>LCD-TFT Controller (LTDC)</b><br/>Corrected the bit field for WHSTPOS in the second bullet point in Section: Window in Section 16.4.2: Layer programmable parameters.</p> <p><b>Advanced-control timers (TIM1&amp;TIM8)</b><br/>Added the note in Section 17.3.20: Timer synchronization,<br/>Updated ETF[3:0] description in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR),<br/>Updated IC1F[3:0] description in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1),<br/>Added the note to MMS2 bit description in Section 17.4.8: TIM1 and TIM8 capture/compare mode register 2 (TIMx_CCMR2),<br/>Added the note to SMS[2:0] bit description in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR).</p> <p><b>General-purpose timers (TIM2 to TIM5)</b><br/>Added the note in Section 18.3.15: Timer synchronization,<br/>Updated SMS[2:0] description in Section 18.4.3: TIMx slave mode control register (TIMx_SMCR),<br/>Added the note to MMS2 bit description in Section 18.4.2: TIMx control register 2 (TIMx_CR2),<br/>Added the note to SMS[2:0] bit description in Section 18.4.3: TIMx slave mode control register (TIMx_SMCR).</p> |

Table 316. Document revision history (continued)

| Date        | Version           | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28-Jul-2015 | 10<br>(Continued) | <p><b>General-purpose timers (TIM9 to TIM14)</b><br/> Added the note in Section 19.3.12: Timer synchronization (TIM9/12),<br/> Added the note to MMS2 bit description,<br/> Added the note to SMS[2:0] bit description in Section 19.4.2: TIM9/12 slave mode control register (TIMx_SMCR).</p> <p><b>Window watchdog (WWDG)</b><br/> Updated.Figure 214: Watchdog block diagram</p> <p><b>Controller area network (bxCAN)</b><br/> Replaced tCAN with tq,</p> <p><b>Flexible static memory controller (FSMC)</b><br/> Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 36.5.5: Synchronous transactions,<br/> Updated MEMHIZx, MEMHOLDx, MEMSETx bit field descriptions for FSMC_PME2..4 register in Section 36.5.5: Synchronous transactions,<br/> Updated ATTSET, ATTHOLD, ATTHIZ bit field descriptions for FSMC_PATT2..4 register in Section 36.5.5: Synchronous transactions,<br/> Updated IRS and IFS bit descriptions for FMC_SR2..4 in Section 36.5.5: Synchronous transactions,<br/> Renamed ADDSET as ADDSET[3:0] and MTYP as MTYP[1:0],<br/> Addition of CPSIZE in FSMC_BCRx bit fields in Table 226: FSMC_BCRx bit fields, Table 228: FSMC_BCRx bit fields, Table 231: FSMC_BCRx bit fields, Table 234: FSMC_BCRx bit fields, Table 237: FSMC_BCRx bit fields, Table 240: FSMC_BCRx bit fields, Table 242: FSMC_BCRx bit fields,<br/> Added CPIZE[2:0] in FMC_BCR1...4 registers in ,Section 36.5.6: NOR/PSRAM control registers Section NOR/PSRAM control re<br/> Added CPSIZE[2:0] for FMC_BCRx registers in Section 36.6.9: FSMC register map.</p> |

Table 316. Document revision history (continued)

| Date        | Version           | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28-Jul-2015 | 10<br>(Continued) | <p><b>Flexible memory controller (FMC)</b></p> <p>Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 37.5.5: Synchronous transactions,</p> <p>Updated BUSTURN bit field description for FMC_BTR1..4 register in Section 37.5.6: NOR/PSRAM controller registers,</p> <p>Updated MEMHIZx, MEMHOLDx, MEMSETx bit field descriptions for FMC_PME2..4 register in Section 37.6.8: NAND Flash/PC Card controller registers,</p> <p>Updated ATTSET, ATTHOLD, ATTHIZ bit field descriptions for FMC_PATT2..4 register in Section 37.6.8: NAND Flash/PC Card controller registers,</p> <p>Updated IRS and IFS bit descriptions for FMC_SR2..4 in Section 37.6.8: NAND Flash/PC Card controller registers,</p> <p>Updated the section SDRAM initialization with the last item in the numbered list in Section 37.7.5: SDRAM controller registers,</p> <p>Renamed ADDSET as ADDSET[3:0] and MTYP as MTYP[1:0],</p> <p>Addition of CPSIZE in Table 269: FMC_BCRx bit fields, Table 271: FMC_BCRx bit fields, Table 274: FMC_BCRx bit fields, Table 277: FMC_BCRx bit fields, Table 280: FMC_BCRx bit fields, Table 283: FMC_BCRx bit fields, Table 285: FMC_BCRx bit fields, Table 287: FMC_BCRx bit fields,</p> <p>Added the paragraph about Cross boundary page for Cellular RAM 1.5 in Section 37.5.5: Synchronous transactions,</p> <p>Added CPIZE[2:0] in FMC_BCR1...4 registers in Section 37.5.6: NOR/PSRAM controller registers,</p> <p>Added CPSIZE[2:0] for FMC_BCRx registers in Section 37.8: FMC register map.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20-Oct-2015 | 11      | <p><b>Reset and clock controller (RCC)</b><br/>Updated STM32F405/407/415/417xx Figure 21: Clock tree.<br/>Updated</p> <p><b>General purpose I/O (GPIOs)</b><br/>Changed definition of OSPEEDR bits in Section 8.4.3: GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K).</p> <p><b>LCD-TFT display controller (LTDC):</b><br/>Changed LRDC_IER into LTDC_IER in Section 16.5: LTDC interrupts.<br/>Updated AHBP[11:0], AAV[11:0 and TOTALW[11:0 in Table 92: LTDC register map and reset values.</p> <p><b>Controller area network (bxCAN):</b><br/>Updated Section 32.3.4: Acceptance filters and Section 32.7.4: Identifier filtering.</p> <p><b>Flexible static memory controller (FSMC)</b><br/>Updated BUSTURN description in Section : SRAM/NOR-Flash write timing registers 1..4 (FSMC_BWTR1..4) and Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4)<br/>Updated note related to IRS and IFS bits in Section : FIFO status and interrupt register 2..4 (FSMC_SR2..4).</p> <p><b>Flexible memory controller (FMC)</b><br/>Updated paragraph related to the cacheable read FIFO in Section : SDRAM controller read cycle.<br/>Updated BUSTURN description in Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4) and Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4).<br/>Updated note related to IRS and IFS bits in Section : FIFO status and interrupt register 2..4 (FMC_SR2..4).</p> <p><b>Real-time clock (RTC2)</b><br/>Updated WUCKSEL prescaler input in Figure 237: RTC block diagram.<br/>Updated 3rd step in Section : Programming the wakeup timer.<br/>Updated WUTWF bit definition in Section 26.6.4: RTC initialization and status register (RTC_ISR).</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17-May-2016 | 12      | <p><b>Flash memory interface</b><br/> Removed note related to boot from Bank 2 in Section 2.4: Boot configuration.<br/> Updated notes in Section 3.7.3: Read protection (RDP).<br/> Changed number of LATENCY bits in Section 3.9.2: Flash access control register (FLASH_ACR) for STM32F42xxx and STM32F43xxx<br/> In Table 9: 1 Mbyte dual bank Flash memory organization (STM32F42xxx and STM32F43xxx): updated sector 19 size and option bytes (bank 2) address range.</p> <p><b>Power control (PWR)</b><br/> Removed reference to low-power mode in Section 5.1.4: Voltage regulator for STM32F42xxx and STM32F43xxx, Section : Entering Stop mode (STM32F42xxx and STM32F43xxx) and Section : Exiting Stop mode (STM32F42xxx and STM32F43xxx).</p> <p><b>Analog-to-digital converter (ADC)</b><br/> Added note related to ADC_HTR and ADC_LTR register programming in Section 13.13.7: ADC watchdog higher threshold register (ADC_HTR) and Section 13.13.8: ADC watchdog lower threshold register (ADC_LTR).</p> <p><b>Chrom-Art Accelerator™ controller (DMA2D)</b><br/> Updated Section 11.3.12: DMA2D transfer control (start, suspend, abort and completion).<br/> Section 11.5.8: DMA2D foreground PFC control register (DMA2D_FGPFCCR): updated START bit access type<br/> Section 11.5.10: DMA2D background PFC control register (DMA2D_BGPFCCR): updated START bit access and description.</p> <p><b>LCD-TFT controller (LTDC)</b><br/> Updated Section 16.3.2: LTDC reset and clocks.<br/> Modified LCD_DE description in Table 89: LCD-TFT pins and signal interface.<br/> Modified Section 16.7.15: LTDC Layerx Window Horizontal Position Configuration Register (LTDC_LxWHPCR) (where x=1..2) and Section 16.7.16: LTDC Layerx Window Vertical Position Configuration Register (LTDC_LxWVPCR) (where x=1..2).</p> <p><b>General-purpose timers (TIM2 to TIM5)</b><br/> Updated Section 18.4.11: TIMx prescaler (TIMx_PSC).</p> <p><b>General-purpose timers (TIM9 to TIM14)</b><br/> Added OPM bit in Section 19.5.1: TIM10/11/13/14 control register 1 (TIMx_CR1).<br/> Updated Section 19.4.9: TIM9/12 prescaler (TIMx_PSC) and Section 19.5.8: TIM10/11/13/14 prescaler (TIMx_PSC).</p> |

Table 316. Document revision history (continued)

| Date        | Version           | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17-May-2016 | 12<br>(continued) | <p><b>General-purpose timers (TIM6 and TIM7)</b><br/>Updated Section 20.4.7: TIM6 and TIM7 prescaler (TIMx_PSC).</p> <p><b>Real-time clock (RTC)</b><br/>Updated conditions for running under System reset in Section 26.3.7: Resetting the RTC.<br/>Updated Section 26.3.14: Calibration clock output.<br/>Added note related to TSE in Section 26.6.3: RTC control register (RTC_CR).<br/>Updated caution note related to TAMP1TRG in Section 26.6.17: RTC tamper and alternate function configuration register (RTC_TAFCR) register.</p> <p><b>Universal synchronous asynchronous receiver transmitter (USART)</b><br/>Replaced all occurrences of nCTS by CTS, nRTS by RTS and SCLK by CK.</p> <p><b>Flexible static memory controller (FSMC)</b><br/>Updated Section 36.3: AHB interface.<br/>Added note related to the hold phase delay below Figure 454: NAND/PC Card controller timing for common memory access.<br/>Updated Section 36.6.5: NAND Flash prewait functionality.<br/>Updated BUSTURN description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FSMC_BTR1..4).<br/>Updated MEMHOLDx in Section : Common memory space timing register 2..4 (FSMC_PMEM2..4) and ATTHOLD in Section : Attribute memory space timing registers 2..4 (FSMC_PATT2..4).</p> <p><b>Flexible memory controller (FMC)</b><br/>Updated Section 37.3: AHB interface.<br/>Added note related to the hold phase delay below Figure 476: NAND Flash/PC Card controller waveforms for common memory access.<br/>Updated Section 37.6.5: NAND Flash prewait functionality.<br/>Updated BUSTURN description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4).<br/>Updated MEMHOLDx in Section : Common memory space timing register 2..4 (FMC_PMEM2..4) and ATTHOLD in Section : Attribute memory space timing registers 2..4 (FMC_PATT2..4).</p> <p><b>Debug (DBG)</b><br/>Updated value to be programmed to the ETM Trace Start/stop register to enable the trace in Section 38.15.4: ETM configuration example.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20-Sep-2016 | 13      | <p><b>Analog-to-digital converter (ADC)</b><br/>Updated DMA mode 1 and DMA mode 3 description in Section 13.9: Multi ADC mode.</p> <p><b>LCD-TFT controller</b><br/>Updated values to be programmed to LTDC_SSCR in Section : Example of Synchronous timings configuration<br/>Updated Section 16.4.2: Layer programmable parameters/Windowing.</p> <p><b>Advanced-control timers (TIM1 and TIM8)</b><br/>Updated Section 17.3.21: Debug mode.<br/>Extended Section 17.4.20: TIM1 and TIM8 DMA address for full transfer (TIMx_DMAR) to 32 bits.<br/>Updated Table 95: Output control bits for complementary OCx and OCxN channels with break feature output state for MOE = 0.<br/>Updated TIM1 and TIM8 auto-reload register (TIMx_ARR) reset value.<br/>Updated TIMx_CCR1/2/3/4 description when CC1 channel is configured as inputs and changed bit access type to rw/ro.</p> <p><b>General-purpose timers (TIM2 to TIM5)</b><br/>Updated TIMx auto-reload register (TIMx_ARR) reset value.<br/>Updated TIMx_CCR1/2/3/4 description when CC1 channel is configured as inputs and changed bit access type to rw/ro.</p> <p><b>General-purpose timers (TIM9 to TIM14)</b><br/>Updated TIM9/12 auto-reload register (TIMx_ARR) and TIM10/11/13/14 auto-reload register (TIMx_ARR) reset value.<br/>Updated TIMx_CCR1 description when CC1 channel is configured as inputs and changed bit access type to rw/ro.</p> <p><b>Basic timers (TIM6 to TIM7)</b><br/>Updated TIM6 and TIM7 auto-reload register (TIMx_ARR).</p> <p><b>Secure digital input/output interface (SDIO)</b><br/>Updated Section 31.1: SDIO main features up to 50 MHz.<br/>Updated Section 31.3: SDIO functional description SDIO_CK description.<br/>Updated note removing 48 MHz in Section 31.9.1: SDIO power control register (SDIO_POWER), Section 31.9.2: SDI clock control register (SDIO_CLKCR), Section 31.9.4: SDIO command register (SDIO_CMD) and Section 31.9.9: SDIO data control register (SDIO_DCTRL).</p> |

Table 316. Document revision history (continued)

| Date        | Version           | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20-Sep-2016 | 13<br>(continued) | <b>FMC</b><br>Update BUSTURN bit description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4) and Section : SRAM/NOR-Flash write timing registers 1..4 (FMC_BWTR1..4).<br><b>Debug support</b><br>Specified behavior of timers with complementary outputs in Section 38.16.2: Debug support for timers, watchdog, bxCAN and I2C.<br>Updated DBG_TIMx_STOP bit description in Section 38.16.4: Debug MCU APB1 freeze register (DBGMCU_APB1_FZ) and Section 38.16.4: Debug MCU APB1 freeze register (DBGMCU_APB1_FZ).<br><b>Electronic signature</b><br>Updated Section 24.1: Unique device ID register (96 bits).                                                                                                      |
| 21-Apr-2017 | 14                | Updated:<br>– Section 5.5.2: PWR power control/status register (PWR_CSR) for STM32F42xxx and STM32F43xxx<br>– Section 6.3.14: RCC APB2 peripheral clock enable register (RCC_APB2ENR)<br>– Section 14.3.5: DAC output voltage<br>– Section 38.6.1: MCU device ID code<br>– Figure 237: RTC block diagram<br>Deleted:<br>– Section 7.3.15: RCC APB2 peripheral clock enable register(RCC_APB2ENR)                                                                                                                                                                                                                                                                                                                                                 |
| 18-Jul-2017 | 15                | Updated:<br>– Section 3.9.10: Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx<br>– OTG_FS USB configuration register (OTG_FS_GUSBCFG)<br>– Table 142: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 Hz, oversampling by 16 and Table 143: Error calculation for programmed baud rates at fPCLK = 42 MHz or fPCLK = 84 MHz, oversampling by 8.                                                                                                                                                                                                                                                                                                                                          |
| 23-Apr-2018 | 16                | Updated:<br>– Section 30.6.1: Status register (USART_SR)<br>– Section 34.16.4: Device-mode registers<br>– Section 34.17.6: Operational model<br>– Section 35.12.4: Device-mode registers<br>– Section 34: USB on-the-go full-speed (OTG_FS)<br>– Table 199: Host-mode control and status registers (CSRs)<br>– Table 205: OTG_FS register map and reset values<br>– Table 210: Device-mode control and status registers<br>– Table 215: OTG_HS register map and reset values<br>Added:<br>– Figure 412: SOF trigger output to TIM2 ITR1 connection<br>– RXOLNY register changed from SPI_CR2 to SPI_CR1 in Section 28.3.4: Configuring the SPI for half-duplex communication and Unidirectional receive-only procedure (BIDIMODE=0 and RXONLY=1) |

**Table 316. Document revision history (continued)**

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07-Jun-2018 | 17      | Updated: <ul style="list-style-type: none"><li>– Figure 16: Clock tree (STM32F42xxx an STM32F43xxx) and Figure 21: Clock tree (STM32F405xx/07xx and STM32F415xx/17xx)</li><li>– Figure 27: Selecting an alternate function on STM32F42xxx and STM32F43xxx</li><li>– Table 61: Vector table for STM32F405xx/07xx and STM32F415xx/17xx and Table 62: Vector table for STM32F42xxx and STM32F43xxx</li><li>– Section 29.17.5: SAI xInterrupt mask register (SAI_xIM) where x is A or B</li><li>– Section 38.6.1: MCU device ID code</li></ul> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25-Feb-2019 | 18      | <p><b>Section 6: Reset and clock control for STM32F42xxx and STM32F43xxx (RCC)</b><br/>Updated OTGHSULPILPEN bit description in RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR) and OTGHSULPIEN bit description in RCC AHB1 peripheral clock register (RCC_AHB1ENR).</p> <p><b>Section 7: Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC):</b><br/>Updated RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR) reset value.<br/>Updated OTGHSULPILPEN bit description in RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR) and OTGHSULPIEN bit description in RCC AHB1 peripheral clock enable register (RCC_AHB1ENR).</p> <p><b>Section 13: Analog-to-digital converter (ADC)</b><br/>Update Section : Dual ADC mode.</p> <p><b>Section 17: Advanced-control timers (TIM1 and TIM8)</b><br/>Updated Figure 113: Capture/compare channel 1 main circuit.<br/>Figure 19: General-purpose timers (TIM9 to TIM14)<br/>Updated Figure 194: Capture/compare channel 1 main circuit.</p> <p><b>Section 34: USB on-the-go full-speed (OTG_FS)</b><br/>Updated Section : SETUP and OUT data transfers and Updated Section : IN data transfers. Modified Table 200: Device-mode control and status registers.<br/>Section 35: USB on-the-go high-speed (OTG_HS)<br/>Updated Table 208: Core global control and status registers (CSRs) and Table 210: Device-mode control and status registers.<br/>Updated Section : OTG_HS device IN endpoint transmit FIFO size register (OTG_HS_DIEPTXFx) (x = 1..5, where x is the FIFO_number), Section : OTG device endpoint-x control register (OTG_HS_DIEPCTLx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x control register (OTG_HS_DOEPCTLx) (x = 1..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x interrupt register (OTG_HS_DIEPINTx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x interrupt register (OTG_HS_DOEPINTx) (x = 0..5, where x = Endpoint_number), Section : OTG_HS device endpoint-x DMA address register (OTG_HS_DIEPDMAx / OTG_HS_DOEPDMAx) (x = 0..5, where x = Endpoint_number).<br/>Updated Section : SETUP and OUT data transfers and Section : IN data transfers.</p> <p><b>Section 38: Debug support (DBG)</b><br/>Updated REV_ID in DBGMCU_CR register.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25-Feb-2021 | 19      | <p>Updated:</p> <ul style="list-style-type: none"> <li>– Section 2: Memory and bus architecture:</li> <li>– Figure 1: System architecture for STM32F405xx/07xx and STM32F415xx/17xx devices</li> <li>– Figure 2: System architecture for STM32F42xxx and STM32F43xxx devices</li> <li>– Section 3: Embedded Flash memory interface:</li> <li>– Table 16: Description of the option bytes (STM32F405xx/07xx and STM32F415xx/17xx)</li> <li>– Table 17: Description of the option bytes (STM32F42xxx and STM32F43xxx)</li> <li>– Section 12: Interrupts and events:</li> <li>– Table 61: Vector table for STM32F405xx/07xx and STM32F415xx/17xx</li> <li>– Table 62: Vector table for STM32F42xxx and STM32F43xxx</li> <li>– Section 13: Analog-to-digital converter (ADC):</li> <li>– Section 13.3.5: Continuous conversion mode</li> <li>– Section 13.8.1: Using the DMA</li> <li>– Section 13.10: Temperature sensor</li> <li>– Section 17: Advanced-control timers (TIM1 and TIM8):</li> <li>– Figure 86: Advanced-control timer block diagram</li> <li>– Section 17.4.9: TIM1 and TIM8 capture/compare enable register (TIMx_CCER)</li> <li>– Section 27: Inter-integrated circuit (I2C) interface:</li> <li>– Section 27.6.2: I2C Control register 2 (I2C_CR2)</li> <li>– Section 27.6.8: I2C Clock control register (I2C_CCR)</li> <li>– Section 31: Secure digital input/output interface (SDIO):</li> <li>– Section 31.9.2: SDI clock control register (SDIO_CLKCR)</li> <li>– Section 32: Controller area network (bxCAN):</li> <li>– Section 32.4.1: Initialization mode</li> <li>– Section 33: Ethernet (ETH): media access control (MAC) with DMA controller:</li> <li>– Section : TxDMA operation: default (non-OSF) mode</li> <li>– Section : Normal Tx DMA descriptors</li> <li>– Section 35: USB on-the-go high-speed (OTG_HS):</li> <li>– Section : Host port power</li> <li>– Section : OTG_HS core interrupt register (OTG_HS_GINTSTS)</li> <li>– Section : OTG_HS all endpoints interrupt mask register (OTG_HS_DAINTRMSK)</li> <li>– Table 215: OTG_HS register map and reset values</li> <li>– Section 35.13.2: Host initialization</li> <li>– Figure 416: Transmit FIFO write task</li> <li>– Figure 417: Receive FIFO read task</li> <li>– Figure 426: Receive FIFO packet read in slave mode</li> <li>– Figure 427: Processing a SETUP packet</li> <li>– Section 38: Debug support (DBG):</li> <li>– Section : DBGMCU_IDCODE</li> </ul> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05-Feb-2024 | 20      | <p><b>Section : Introduction:</b></p> <ul style="list-style-type: none"> <li>– Mentioned that the microcontrollers include ST state-of-the-art patented technology</li> <li>– Added errata sheets in the list of reference documents.</li> </ul> <p><b>Section 2: Memory and bus architecture</b></p> <ul style="list-style-type: none"> <li>– <i>Section 2.3.1: Embedded SRAM</i>, updated the SRAM that can be accessed through System or I-Code/D-Code bus.</li> </ul> <p><b>Section 3.4: Embedded flash memory in STM32F42xxx and STM32F43xxx</b></p> <ul style="list-style-type: none"> <li>– Updated Bank 2 section 14 base address in <i>Table 12: Flash module - 2 Mbyte dual bank organization (STM32F42xxx and STM32F43xxx)</i>.</li> <li>– Specified that dual bank organization is not available on 512 Kbyte devices, and added <i>Table 16: 512 Kbyte single bank flash memory organization (STM32F42xxx and STM32F43xxx)</i>.</li> <li>– Updated <i>Section 3.8.2: Program/erase parallelism</i>.</li> <li>– In <i>Section 3.9.3: Read protection (RDP)</i>, added note concerning RDP when debugger is connected through JTAG/SWD.</li> </ul> <p><b>Section 5: Power controller (PWR)</b></p> <ul style="list-style-type: none"> <li>– Updated <i>Figure 12: Power-on reset/power-down reset waveform</i>.</li> <li>– Updated no external battery use case in <i>Section 5.1.2: Battery backup domain</i>.</li> <li>– Updated DBP bit description in <i>PWR power control register (PWR_CR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> and <i>PWR power control register (PWR_CR)</i> for <i>STM32F42xxx</i> and <i>STM32F43xxx</i>.</li> <li>– Updated BRE bit description in <i>PWR power control/status register (PWR_CSR)</i> for <i>STM32F405xx/07xx</i> and <i>STM32F415xx/17xx</i> and <i>PWR power control/status register (PWR_CSR)</i> for <i>STM32F42xxx</i> and <i>STM32F43xxx</i>.</li> </ul> <p><b>Section 6: Reset and clock control for STM32F42xxx and STM32F43xxx and NA(RCC)</b></p> <ul style="list-style-type: none"> <li>– Updated <i>Section 6.1.1: System reset</i> and <i>Section 6.1.3: Backup domain reset</i>.</li> <li>– Updated ethernet PTP clock in <i>Figure 16: Clock tree</i>.</li> <li>– Added note regarding backup domain reset in BDRST bit of <i>RCC Backup domain control register (RCC_BDCR)</i>.</li> <li>– Added register reset values in <i>Table 38: RCC register map and reset values for STM32F42xxx and STM32F43xxx</i></li> </ul> <p><b>Section 7: Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC)</b></p> <ul style="list-style-type: none"> <li>– Updated <i>Section 7.1.1: System reset</i> and <i>Section 7.1.3: Backup domain reset</i>.</li> <li>– Updated ethernet PTP clock in <i>Figure 21: Clock tree</i></li> <li>– Added note regarding backup domain reset in BDRST bit of <i>RCC Backup domain control register (RCC_BDCR)</i>.</li> <li>– Added register reset values in <i>Table 39: RCC register map and reset values for STM32F405xx/07xx and STM32F415xx/17xx</i></li> </ul> |

Table 316. Document revision history (continued)

| Date        | Version           | Changes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05-Feb-2024 | 20<br>(continued) | <p><b>Section 10: DMA controller (DMA)</b><br/>Updated DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7) address offset.</p> <p><b>Section 12: Interrupts and events</b><br/>Changed Pending register (EXTI_PR) reset value to 0x0000 0000.</p> <p><b>Section 17: Advanced-control timers (TIM1 and TIM8)</b><br/>Updated Section 17.3.7: PWM input mode.<br/>Updated SMS in Section 17.4.3: TIM1 and TIM8 slave mode control register (TIMx_SMCR).<br/>Updated OC1PE in Section 17.4.7: TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1).</p> <p><b>Section 22: Window watchdog (WWDG)</b><br/>Updated <math>t_{WWDG}</math> equation in Section 22.4: How to program the watchdog timeout.</p> <p><b>Section 26: Real-time clock (RTC)</b><br/>– Updated HSE clock in Figure 237: RTC block diagram (NA devices).<br/>– Updated Section 26.3.6: Reading the calendar.</p> <p><b>Section 29: Serial audio interface (SAI)</b><br/>– In the whole section, replaced TDM by free protocol mode.<br/>– In Section 29.18.4: SAI x frame configuration register (SAI_XFRCR) where x is A or B, specified that FRL[7:0] must be configured when the audio block is disabled.</p> <p><b>Section 34: USB on-the-go full-speed (OTG_FS)</b><br/>– Updated Figure 392: Device-mode FIFO address mapping and AHB FIFO access mapping.<br/>– Modified OTG_FS USB configuration register (OTG_FS_GUSBCFG) OTG_FS_GUSBCFG reset value.</p> <p><b>Section 33: Ethernet (ETH): media access control (MAC) with DMA controller</b><br/>Updated bits that control the checksum in Section : Transmit checksum offload</p> <p><b>Section 38: Debug support (DBG)</b><br/>Removed note on APB bridge write buffer after Table 302: Flexible SWJ-DP pin assignment<br/>Updated REV_ID[15:0] in Section : DBGMCU_IDCODE</p> <p><b>Section 39: Device electronic signature</b><br/>Updated Section 39.1: Unique device ID register (96 bits)</p> <p>Added Section 40: Important security notice.</p> |

Table 316. Document revision history (continued)

| Date        | Version | Changes                                                                                                                                                                                                           |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07-Jun-2024 | 21      | Updated the note in Bit[11:0] description of <a href="#">Section 27.6.8: I<sup>2</sup>C Clock control register (I2C_CCR)</a> .<br>Fixed typo in <a href="#">Section 12.1.3: Interrupt and exception vectors</a> . |

# Index

## A

|           |     |
|-----------|-----|
| ADC_CCR   | 430 |
| ADC_CDR   | 433 |
| ADC_CR1   | 419 |
| ADC_CR2   | 421 |
| ADC_CSR   | 429 |
| ADC_DR    | 428 |
| ADC_HTR   | 424 |
| ADC_JDRx  | 428 |
| ADC_JOFRx | 424 |
| ADC_JSQR  | 427 |
| ADC_LTR   | 425 |
| ADC_SMPR1 | 423 |
| ADC_SMPR2 | 423 |
| ADC_SQR1  | 425 |
| ADC_SQR2  | 426 |
| ADC_SQR3  | 426 |
| ADC_SR    | 418 |

## C

|            |          |
|------------|----------|
| CAN_BTR    | 1109     |
| CAN_ESR    | 1108     |
| CAN_FA1R   | 1119     |
| CAN_FFA1R  | 1119     |
| CAN_FiRx   | 1120     |
| CAN_FM1R   | 1118     |
| CAN_FMR    | 1117     |
| CAN_FS1R   | 1118     |
| CAN_IER    | 1106     |
| CAN_MCR    | 1100     |
| CAN_MSR    | 1102     |
| CAN_RDHxR  | 1116     |
| CAN_RDLxR  | 1116     |
| CAN_RDTxR  | 1115     |
| CAN_RF0R   | 1105     |
| CAN_RF1R   | 1106     |
| CAN_RlRx   | 1114     |
| CAN_TDHxR  | 1113     |
| CAN_TDLxR  | 1113     |
| CAN_TDTxR  | 1112     |
| CAN_TlRx   | 1111     |
| CAN_TSR    | 1103     |
| CRC_DR     | 115      |
| CRC_IDR    | 115      |
| CRYP_CR    | 751, 753 |
| CRYP_DIN   | 757      |
| CRYP_DMACR | 759      |

|            |     |
|------------|-----|
| CRYP_DOUT  | 758 |
| CRYP_IMSCR | 759 |
| CRYP_IV0LR | 763 |
| CRYP_IV0RR | 763 |
| CRYP_IV1LR | 764 |
| CRYP_IV1RR | 764 |
| CRYP_K0LR  | 761 |
| CRYP_K0RR  | 761 |
| CRYP_K1LR  | 762 |
| CRYP_K1RR  | 762 |
| CRYP_K2LR  | 762 |
| CRYP_K2RR  | 762 |
| CRYP_K3LR  | 762 |
| CRYP_K3RR  | 763 |
| CRYP_MISR  | 760 |
| CRYP_RISR  | 760 |
| CRYP_SR    | 756 |

## D

|                |      |
|----------------|------|
| DAC_CR         | 448  |
| DAC_DHR12L1    | 452  |
| DAC_DHR12L2    | 453  |
| DAC_DHR12LD    | 454  |
| DAC_DHR12R1    | 451  |
| DAC_DHR12R2    | 453  |
| DAC_DHR12RD    | 454  |
| DAC_DHR8R1     | 452  |
| DAC_DHR8R2     | 453  |
| DAC_DHR8RD     | 455  |
| DAC_DOR1       | 455  |
| DAC_DOR2       | 455  |
| DAC_SR         | 456  |
| DAC_SWTRIGR    | 451  |
| DBGMCU_APB1_FZ | 1708 |
| DBGMCU_APB2_FZ | 1709 |
| DBGMCU_CR      | 1706 |
| DBGMCU_IDCODE  | 1693 |
| DCMI_CR        | 470  |
| DCMI_CWSIZE    | 480  |
| DCMI_CWSTRT    | 480  |
| DCMI_DR        | 481  |
| DCMI_ESCR      | 478  |
| DCMI_ESUR      | 479  |
| DCMI_ICR       | 477  |
| DCMI_IER       | 475  |
| DCMI_MIS       | 476  |
| DCMI_RIS       | 474  |
| DCMI_SR        | 473  |

|                  |     |
|------------------|-----|
| DMA_HIFCR .....  | 330 |
| DMA_HISR .....   | 329 |
| DMA_LIFCR .....  | 330 |
| DMA_LISR .....   | 328 |
| DMA_SxCR .....   | 331 |
| DMA_SxFCR .....  | 336 |
| DMA_SxM0AR ..... | 335 |
| DMA_SxM1AR ..... | 335 |
| DMA_SxNDTR ..... | 334 |
| DMA_SxPAR .....  | 335 |

**E**

|                     |      |
|---------------------|------|
| ETH_DMABMR .....    | 1226 |
| ETH_DMACHRBAR ..... | 1239 |
| ETH_DMACHRDR .....  | 1238 |
| ETH_DMACHTBAR ..... | 1238 |
| ETH_DMACHTDR .....  | 1238 |
| ETH_DMAIER .....    | 1235 |
| ETH_DMAMFBOCR ..... | 1237 |
| ETH_DMAOMR .....    | 1232 |
| ETH_DMARDLAR .....  | 1228 |
| ETH_DMARPDR .....   | 1228 |
| ETH_DMARSWTR .....  | 1237 |
| ETH_DMASR .....     | 1229 |
| ETH_DMATDLAR .....  | 1229 |
| ETH_DMATPDR .....   | 1227 |
| ETH_MACA0HR .....   | 1208 |
| ETH_MACA0LR .....   | 1209 |
| ETH_MACA1HR .....   | 1209 |
| ETH_MACA1LR .....   | 1210 |
| ETH_MACA2HR .....   | 1210 |
| ETH_MACA2LR .....   | 1211 |
| ETH_MACA3HR .....   | 1212 |
| ETH_MACA3LR .....   | 1212 |
| ETH_MACCR .....     | 1194 |
| ETH_MACDBGRR .....  | 1205 |
| ETH_MACFCR .....    | 1201 |
| ETH_MACFFR .....    | 1197 |
| ETH_MACHTHR .....   | 1198 |
| ETH_MACHTLR .....   | 1199 |
| ETH_MACIMR .....    | 1208 |
| ETH_MACMIAR .....   | 1199 |
| ETH_MACMIIDR .....  | 1200 |
| ETH_MACPMTCSR ..... | 1204 |
| ETH_MACRWUFR .....  | 1203 |
| ETH_MACSR .....     | 1207 |
| ETH_MACVLANTR ..... | 1202 |
| ETH_MMCCR .....     | 1213 |
| ETH_MMCRAECR .....  | 1218 |
| ETH_MMCRFCECR ..... | 1217 |
| ETH_MMCRGUFCR ..... | 1218 |

|                       |      |
|-----------------------|------|
| ETH_MMCRIMR .....     | 1215 |
| ETH_MMCRIR .....      | 1213 |
| ETH_MMCTGFCR .....    | 1217 |
| ETH_MMCTGFMSCCR ..... | 1217 |
| ETH_MMCTGFSCCR .....  | 1216 |
| ETH_MMCTIMR .....     | 1216 |
| ETH_MMCTIR .....      | 1214 |
| ETH_PTPPPSCR .....    | 1225 |
| ETH_PTPSSIR .....     | 1221 |
| ETH_PTPTSAR .....     | 1223 |
| ETH_PTPTSCR .....     | 1218 |
| ETH_PTPTSHR .....     | 1221 |
| ETH_PTPTSHUR .....    | 1222 |
| ETH_PTPTSLR .....     | 1222 |
| ETH_PTPTSLUR .....    | 1223 |
| ETH_PTPTSSR .....     | 1224 |
| ETH_PTPTTHR .....     | 1224 |
| ETH_PTPTTLR .....     | 1224 |
| EXTI_EMR .....        | 387  |
| EXTI_FTSR .....       | 388  |
| EXTI_IMR .....        | 387  |
| EXTI_PR .....         | 389  |
| EXTI_RTSR .....       | 388  |
| EXTI_SWIER .....      | 389  |

**F**

|                      |               |
|----------------------|---------------|
| FLITF_FCR .....      | 104, 106      |
| FLITF_FKEYR .....    | 101           |
| FLITF_FOPTCR .....   | 107, 109, 111 |
| FLITF_FOPTKEYR ..... | 101           |
| FLITF_FSR .....      | 102-103       |
| FSMC_BCR1..4 .....   | 1580, 1643    |
| FSMC_BTR1..4 .....   | 1583, 1645    |
| FSMC_BWTR1..4 .....  | 1586, 1649    |
| FSMC_PCR2..4 .....   | 1596          |
| FSMC_PMEM2..4 .....  | 1598          |
| FSMC_SR2..4 .....    | 1597          |

**G**

|                     |     |
|---------------------|-----|
| GPIOx_AFRH .....    | 289 |
| GPIOx_AFRL .....    | 288 |
| GPIOx_BSRR .....    | 287 |
| GPIOx_IDR .....     | 286 |
| GPIOx_LCKR .....    | 287 |
| GPIOx_MODER .....   | 284 |
| GPIOx_ODR .....     | 286 |
| GPIOx_OSPEEDR ..... | 285 |
| GPIOx_OTYPER .....  | 284 |
| GPIOx_PUPDR .....   | 285 |

**H**

|                 |          |
|-----------------|----------|
| HASH_CR .....   | 785, 788 |
| HASH_CSRx ..... | 797      |
| HASH_DIN .....  | 791      |
| HASH_HR0 .....  | 793      |
| HASH_HR1 .....  | 793-794  |
| HASH_HR2 .....  | 793-794  |
| HASH_HR3 .....  | 794      |
| HASH_HR4 .....  | 794      |
| HASH_IMR .....  | 795      |
| HASH_SR .....   | 796      |
| HASH_STR .....  | 792      |

**I**

|                 |     |
|-----------------|-----|
| I2C_CCR .....   | 873 |
| I2C_CR1 .....   | 863 |
| I2C_CR2 .....   | 865 |
| I2C_DR .....    | 868 |
| I2C_OAR1 .....  | 867 |
| I2C_OAR2 .....  | 867 |
| I2C_SR1 .....   | 868 |
| I2C_SR2 .....   | 871 |
| I2C_TRISE ..... | 874 |
| IWDG_KR .....   | 713 |
| IWDG_PR .....   | 713 |
| IWDG_RLR .....  | 714 |
| IWDG_SR .....   | 714 |

**O**

|                         |      |
|-------------------------|------|
| OTG_FS_CID .....        | 1293 |
| OTG_FS_DAIN'T .....     | 1310 |
| OTG_FS_DAIN'TMSK .....  | 1311 |
| OTG_FS_DCFG .....       | 1305 |
| OTG_FS_DCTL .....       | 1306 |
| OTG_FS_DIEPCTL0 .....   | 1312 |
| OTG_FS_DIEPCTLx .....   | 1314 |
| OTG_FS_DIEPEMPMSK ..... | 1312 |
| OTG_FS_DIEPINTx .....   | 1321 |
| OTG_FS_DIEPMSK .....    | 1308 |
| OTG_FS_DIEPTSIZ0 .....  | 1323 |
| OTG_FS_DIEPTSIZx .....  | 1326 |
| OTG_FS_DIEPTXF0 .....   | 1291 |
| OTG_FS_DIEPTXFx .....   | 1294 |
| OTG_FS_DOEPCTL0 .....   | 1317 |
| OTG_FS_DOEPCTLx .....   | 1318 |
| OTG_FS_DOEPINTx .....   | 1322 |
| OTG_FS_DOEPMSK .....    | 1309 |
| OTG_FS_DOEPTSIZ0 .....  | 1325 |
| OTG_FS_DOEPTSIZx .....  | 1327 |
| OTG_FS_DSTS .....       | 1307 |

|                           |      |
|---------------------------|------|
| OTG_FS_DTXFSTSx .....     | 1327 |
| OTG_FS_DVBUSDIS .....     | 1311 |
| OTG_FS_DVBUSPULSE .....   | 1311 |
| OTG_FS_GAHBCFG .....      | 1277 |
| OTG_FS_GCCFG .....        | 1292 |
| OTG_FS_GINTMSK .....      | 1286 |
| OTG_FS_GINTSTS .....      | 1282 |
| OTG_FS_GOTGCTL .....      | 1274 |
| OTG_FS_GOTGINT .....      | 1275 |
| OTG_FS_GRSTCTL .....      | 1280 |
| OTG_FS_GRXFSIZ .....      | 1290 |
| OTG_FS_GRXSTSP .....      | 1289 |
| OTG_FS_GRXSTSR .....      | 1289 |
| OTG_FS_GUSBCFG .....      | 1278 |
| OTG_FS_HAINT .....        | 1297 |
| OTG_FS_HAINTMSK .....     | 1298 |
| OTG_FS_HCCHARx .....      | 1301 |
| OTG_FS_HCFG .....         | 1295 |
| OTG_FS_HCINTMSKx .....    | 1303 |
| OTG_FS_HCINTx .....       | 1302 |
| OTG_FS_HCTSIZx .....      | 1304 |
| OTG_FS_HFIR .....         | 1295 |
| OTG_FS_HFNUM .....        | 1296 |
| OTG_FS_HNPTXFSIZ .....    | 1291 |
| OTG_FS_HNPTXSTS .....     | 1291 |
| OTG_FS_HPRT .....         | 1298 |
| OTG_FS_HPTXFSIZ .....     | 1294 |
| OTG_FS_HPTXSTS .....      | 1296 |
| OTG_FS_PCGCCTL .....      | 1328 |
| OTG_HS_CID .....          | 1432 |
| OTG_HS_DAIN'T .....       | 1453 |
| OTG_HS_DAIN'TMSK .....    | 1454 |
| OTG_HS_DCFG .....         | 1446 |
| OTG_HS_DCTL .....         | 1448 |
| OTG_HS_DEACHINT .....     | 1457 |
| OTG_HS_DEACHINTMSK .....  | 1458 |
| OTG_HS_DIEPCTLx .....     | 1460 |
| OTG_HS_DIEPDMAx .....     | 1474 |
| OTG_HS_DIEPEACHMSK1 ..... | 1458 |
| OTG_HS_DIEPEMPMSK .....   | 1457 |
| OTG_HS_DIEPINTx .....     | 1467 |
| OTG_HS_DIEPMSK .....      | 1451 |
| OTG_HS_DIEPTSIZ0 .....    | 1470 |
| OTG_HS_DIEPTSIZx .....    | 1472 |
| OTG_HS_DIEPTXFx .....     | 1433 |
| OTG_HS_DOEPCTL0 .....     | 1463 |
| OTG_HS_DOEPCTLx .....     | 1464 |
| OTG_HS_DOEPDMAx .....     | 1474 |
| OTG_HS_DOEPEACHMSK1 ..... | 1459 |
| OTG_HS_DOEPINTx .....     | 1469 |
| OTG_HS_DOEPMSK .....      | 1452 |
| OTG_HS_DOEPTSIZ0 .....    | 1471 |

|                   |      |
|-------------------|------|
| OTG_HS_DOEPTSIZE  | 1473 |
| OTG_HS_DSTS       | 1450 |
| OTG_HS_DTHRCTL    | 1456 |
| OTG_HS_DTXFSTSx   | 1473 |
| OTG_HS_DVBUSDIS   | 1454 |
| OTG_HS_DVBUSPULSE | 1455 |
| OTG_HS_GAHBCFG    | 1414 |
| OTG_HS_GCCFG      | 1431 |
| OTG_HS_GINTMSK    | 1425 |
| OTG_HS_GINTSTS    | 1421 |
| OTG_HS_GNPTXFSIZ  | 1430 |
| OTG_HS_GNPTXSTS   | 1430 |
| OTG_HS_GOTGCTL    | 1411 |
| OTG_HS_GOTGINT    | 1412 |
| OTG_HS_GRSTCTL    | 1418 |
| OTG_HS_GRXFSIZ    | 1429 |
| OTG_HS_GRXSTSP    | 1428 |
| OTG_HS_GRXSTSR    | 1428 |
| OTG_HS_GUSBCFG    | 1415 |
| OTG_HS_HAINT      | 1437 |
| OTG_HS_HAINTMSK   | 1437 |
| OTG_HS_HCCHARx    | 1440 |
| OTG_HS_HCDMAx     | 1446 |
| OTG_HS_HCFG       | 1433 |
| OTG_HS_HCINTMSKx  | 1444 |
| OTG_HS_HCINTx     | 1443 |
| OTG_HS_HCSPLTx    | 1442 |
| OTG_HS_HCTSIZx    | 1445 |
| OTG_HS_HFIR       | 1435 |
| OTG_HS_HFNUM      | 1435 |
| OTG_HS_HPRT       | 1438 |
| OTG_HS_HPTXFSIZ   | 1432 |
| OTG_HS_HPTXSTS    | 1436 |
| OTG_HS_PCGCCTL    | 1475 |
| OTG_HS_TX0FSIZ    | 1430 |

**P**

|         |          |
|---------|----------|
| PWR_CR  | 142, 146 |
| PWR_CSR | 143, 149 |

**R**

|               |          |
|---------------|----------|
| RCC_AHB1ENR   | 182, 244 |
| RCC_AHB1LPENR | 191, 252 |
| RCC_AHB1RSTR  | 172, 235 |
| RCC_AHB2ENR   | 184, 246 |
| RCC_AHB2LPENR | 194, 254 |
| RCC_AHB2RSTR  | 175, 238 |
| RCC_AHB3ENR   | 185, 247 |
| RCC_AHB3LPENR | 195, 255 |
| RCC_AHB3RSTR  | 176, 239 |
| RCC_APB1ENR   | 185, 247 |

|               |                         |
|---------------|-------------------------|
| RCC_APB1LPENR | 195, 256                |
| RCC_APB1RSTR  | 176, 239                |
| RCC_APB2ENR   | 189, 250                |
| RCC_APB2LPENR | 199, 259                |
| RCC_APB2RSTR  | 180, 242                |
| RCC_BDCR      | 201, 261                |
| RCC_CFGR      | 167, 230                |
| RCC_CIR       | 169, 232                |
| RCC_CR        | 163, 226                |
| RCC_CSR       | 202, 262                |
| RCC_PLLCFGR   | 165, 205, 208, 228, 265 |
| RCC_SSCGR     | 204, 264                |
| RNG_CR        | 772                     |
| RNG_DR        | 773                     |
| RNG_SR        | 772                     |
| RTC_ALRMAR    | 828                     |
| RTC_ALRMBR    | 829                     |
| RTC_ALRMBSSR  | 838                     |
| RTC_BKxR      | 839                     |
| RTC_CALIBR    | 827                     |
| RTC_CALR      | 833                     |
| RTC_CR        | 821                     |
| RTC_DR        | 820                     |
| RTC_ISR       | 823                     |
| RTC_PRER      | 826                     |
| RTC_SHIFTR    | 831                     |
| RTC_SSR       | 830                     |
| RTC_TR        | 819                     |
| RTC_TSDR      | 832                     |
| RTC_TSSSR     | 833                     |
| RTC_TSTR      | 831                     |
| RTC_WPR       | 830                     |
| RTC_WUTR      | 826                     |

**S**

|              |      |
|--------------|------|
| SDIO_CLKCR   | 1064 |
| SDIO_DCOUNT  | 1070 |
| SDIO_DCTRL   | 1069 |
| SDIO_DLEN    | 1068 |
| SDIO_DTIMER  | 1067 |
| SDIO_FIFO    | 1077 |
| SDIO_FIFOCNT | 1076 |
| SDIO_ICR     | 1072 |
| SDIO_MASK    | 1074 |
| SDIO_POWER   | 1063 |
| SDIO_RESPCMD | 1066 |
| SDIO_RESPx   | 1067 |
| SDIO_STA     | 1071 |
| SPI_CR1      | 919  |
| SPI_CR2      | 921  |
| SPI_CRCPR    | 924  |

|                 |          |
|-----------------|----------|
| SPI_DR          | 923      |
| SPI_I2SCFGR     | 925      |
| SPI_I2SPR       | 927      |
| SPI_RXCR        | 924      |
| SPI_SR          | 922      |
| SPI_TXCR        | 925      |
| SYS_CFG_EXTICR1 | 294, 300 |
| SYS_CFG_EXTICR2 | 294, 301 |
| SYS_CFG_EXTICR3 | 295, 301 |
| SYS_CFG_EXTICR4 | 296, 302 |
| SYS_CFG_MEMRMP  | 292, 297 |

## T

|            |                         |
|------------|-------------------------|
| TIM2_OR    | 649                     |
| TIM5_OR    | 650                     |
| TIMx_ARR   | 645, 685, 695, 709      |
| TIMx_BDTR  | 586                     |
| TIMx_CCER  | 579, 643, 684, 694      |
| TIMx_CCMR1 | 575, 639, 681, 691      |
| TIMx_CCMR2 | 577, 642                |
| TIMx_CCR1  | 584, 646, 686, 696      |
| TIMx_CCR2  | 585, 646, 686           |
| TIMx_CCR3  | 585, 647                |
| TIMx_CCR4  | 586, 647                |
| TIMx_CNT   | 583, 645, 685, 695, 708 |
| TIMx_CR1   | 564, 630, 675, 689, 705 |
| TIMx_CR2   | 565, 632, 707           |
| TIMx_DCR   | 588, 648                |
| TIMx_DIER  | 570, 635, 677, 690, 707 |
| TIMx_DMAR  | 589, 649                |
| TIMx_EGR   | 573, 638, 680, 691, 708 |
| TIMx_PSC   | 583, 645, 685, 695, 709 |
| TIMx_RCR   | 584                     |
| TIMx_SMCR  | 568, 633, 676           |
| TIMx_SR    | 572, 636, 679, 690, 708 |

## U

|            |      |
|------------|------|
| USART_BRR  | 1013 |
| USART_CR1  | 1013 |
| USART_CR2  | 1016 |
| USART_CR3  | 1017 |
| USART_DR   | 1013 |
| USART_GTPR | 1020 |
| USART_SR   | 1010 |

## W

|          |     |
|----------|-----|
| WWDG_CFR | 721 |
| WWDG_CR  | 720 |
| WWDG_SR  | 721 |

**IMPORTANT NOTICE – READ CAREFULLY**

STMicroelectronics NV and its subsidiaries ("ST") reserve the right to make changes, corrections, enhancements, modifications, and improvements to ST products and/or to this document at any time without notice. Purchasers should obtain the latest relevant information on ST products before placing orders. ST products are sold pursuant to ST's terms and conditions of sale in place at the time of order acknowledgment.

Purchasers are solely responsible for the choice, selection, and use of ST products and ST assumes no liability for application assistance or the design of purchasers' products.

No license, express or implied, to any intellectual property right is granted by ST herein.

Resale of ST products with provisions different from the information set forth herein shall void any warranty granted by ST for such product.

ST and the ST logo are trademarks of ST. For additional information about ST trademarks, refer to [www.st.com/trademarks](http://www.st.com/trademarks). All other product or service names are the property of their respective owners.

Information in this document supersedes and replaces information previously supplied in any prior versions of this document.

© 2024 STMicroelectronics – All rights reserved